**DE LA SALLE UNIVERSITY - MANILA**
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# Personal Finance Informatics for Single Working Adults

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Department of Information Technology
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CAP-IT0

by

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# CHAPTER 1

## 1.0 Research Background

### 1.0.1 Introduction

#### 1.0.1.1 Personal Finance

Personal finance encompasses two main components. The term "finance" pertains to money management or the science behind handling finances, while "personal" refers to individuals. In essence, personal finance entails the strategic management of financial resources by individuals or families, encompassing budgeting, savings, and expenditure decisions, while also considering potential future financial risks. Additionally, personal finance involves the proactive allocation and progression of funds toward future goals while also mitigating risks (Herrala et al., 2023)(Alag et al., 2020). It is an individual activity that allows individuals to manage finances for the purpose of effectively saving money to achieve lifetime goals such as buying one's own house, car, or paying off debts (Windfried, 2017)(Herrala et al., 2023).

Personal finance generally refers to the way a person obtains, budgets, spends, and saves their money while being concerned with future goals and financial risks. The main objective around personal finance is to achieve financial independence and security so life goals are achievable while worrying less about emergencies and expenses (Windfried, 2017).

Performing one's own finances has an impact towards a person's life and future. According to the Personal Financial Planning Theory, planning and managing personal finances leads to a happy life as this would allow them to comfortably make decisions on any financial circumstances while also achieving their financial goals and freedom. Thus, it is in the best interest of individuals to learn personal finance (Alag et al., 2020).

#### 1.0.1.2 Personal Informatics

Changing behavior, developing consciousness, or fulfilling curiosity may begin with gaining a better understanding of oneself and routines. To help them track their routines, feelings, and actions and gain this self-awareness, people often turn to technology for assistance (Epstein, 2018). Personal informatics refers to the practice of individuals collecting, managing, and reflecting on personal data to gain insights into their behaviors, habits, and overall well-being. It involves the use of various tools and technologies to track and analyze data related to different aspects of one's life, such as physical activity, sleep patterns, diet, productivity, and mood. The goal of personal informatics is to help individuals make informed decisions and take actions to improve their lives based on the insights derived from their data.

According to Li, Dey, and Forlizzi (2010), personal informatics systems can be understood through a stage-based model that encompasses five key stages: Preparation, Collection, Integration, Reflection, and Action. In the Preparation stage, users decide what type of information to collect and what tools to use for data collection. This stage involves setting goals, selecting appropriate metrics, and choosing suitable tracking methods. Personal Informatics involves gathering personal data for self-reflection and self-knowledge, aiming to enhance individuals' lives through renewed self-understanding (Rapp & Tirassa, 2017). Historically, fields like medicine utilized self-monitoring systems, later adopted for private use, as seen with home weight scales (Rapp & Cena, 2016). Today, technologies like sensors enable the tracking of diverse personal data, from glucose levels to sleep patterns, fostering self-reflection and understanding (Rapp & Tirassa, 2017). These systems gained traction around 2013, emphasizing behavior tracking for self-reflection, with recent development focusing on consumer-level adoption (Rapp & Cena, 2016). However, research on advanced system support, leveraging intelligent computing techniques, remains limited despite the potential for richer data collection (Ohlin & Olsson, 2015).

### 1.0.2 Background of the Study

As everyone journeys through life, the simplicity of childhood gives way to a web of increasing complexities and responsibilities. Initially, as children, people were sheltered from financial concerns, nurtured by parents, and spent days in carefree play. However, as people mature, the weight of responsibilities gradually settles upon their shoulders. Education, personal care, and environmental awareness become primary concerns, though often supported by parental or mentor figures. For many youths, the gravity of these responsibilities may not fully register, as parental support often cushions the financial burden. Yet, as they age, the landscape of obligations shifts. Adult life introduces new challenges: homeownership, entrepreneurship, and the heavy needs of dependents. Alongside the joys and satisfactions of each life stage, complexity grows, often intertwined with the state of finances. Personal finance emerges as a critical thread weaving through the fabric of lives. It encompasses the strategic management of monetary resources, from budgeting and savings to investment decisions and risk mitigation. By mastering personal finance, individuals can navigate financial landscapes with greater ease, striving towards goals of financial independence and security. The impact of personal finance reaches beyond mere monetary concerns, influencing overall well-being and life satisfaction. Through proactive planning and management, individuals can steer their financial destiny, pursuing their aspirations while safeguarding against unforeseen challenges (Windfried, 2017).

Financial stress is intricately linked to an individual's overall financial well-being, serving as a key determinant of their quality of life (Mukherjee et al, 2017). This stress, deemed a significant psycho-social strain, significantly impacts various aspects of daily life, including work productivity and personal relationships (Bailey et al, 1998). The association between financial stress and financial behaviors has been investigated, with measures such as the number of financial stressors encountered. These stressors encompass difficulties like inability to purchase clothing, discuss financial matters, pay utilities, save for emergencies, handle medical bills, maintain a car, and manage relationships affected by financial concerns. Research indicates a negative correlation between financial stress and positive financial practices, while it correlates positively with detrimental financial behaviors. Individuals experiencing higher levels of financial stress tend to save less regularly, perceive poorer financial management skills, and engage more frequently in negative financial behaviors like paying interest, making minimum payments, issuing checks with insufficient funds, and regretting purchases (Lim et al, 2017).

Recognizing the significant impact of financial stress on individuals' well-being underscores the paramount importance of attaining financial stability in today's society. Effective personal finance management, covering income, expenses, savings, and investments, is crucial for prosperity, stability, and overall quality of life. Additionally, skillful financial management is essential for navigating economic uncertainties, achieving goals, and laying a solid foundation for the future (Qian, 2023). This solidifies the importance of financial planning and management, encompassing prudent decision-making regarding income, expenses, and investments (Safira et al, 2023).

Amidst the fast-paced changes in today's economy and rising living costs due to inflation affecting various aspects like education and healthcare, personal financial planning has become essential for working adults. Having sufficient personal financial planning enables individuals to efficiently manage their finances according to their goals and objectives (Xian, 2022). This signifies how Personal finance decisions represent a critical aspect of individuals' lives, encompassing a wide array of economic choices expected to become more diverse and complex with the proliferation of financial products. These decisions, ranging from day-to-day money management to long-term planning for milestones like retirement, remain significant across different life stages (Hershey et al, 2015). Financial responsibilities evolve with age and life events, transitioning from basic tasks in early adulthood to more complex decisions related to family, homeownership, and retirement planning in middle and older adulthood (Hershey et al, 2015). Thus, by promoting financial awareness, individuals can make wiser lifestyle choices, enhance financial management practices, and prioritize savings for increased

self-sufficiency in daily expenses (Safira et al, 2023). Enhanced financial awareness fosters better decision-making and improves later-life quality. This awareness facilitates more efficient money management, saving, and investing. It encompasses knowledge, skills, and confidence essential for sound financial decisions. However, concerning deficiencies exist in financial awareness among both youth and adults. Recognizing its significance is vital for consumers to invest wisely and manage budgets effectively, as well as for market stability and proper functioning (Smalej et al, 2022).

Financial behavior, which encompasses actions related to managing money including spending, saving, and budgeting, is directly influenced by financial awareness (Acharya & Poudel, 2023; Muat et al, 2024). These behaviors range from short-term actions like saving for specific events to long-term commitments such as retirement savings (Muat et al, 2024). Effective management of these aspects is increasingly crucial for financial well-being. Despite challenges faced by young adults in making sound financial choices, those with better self financial understanding and management tend to demonstrate improved decision-making. Positive financial behaviors contribute to increased satisfaction with financial well-being (Acharya & Poudel, 2023). Financial habits such as budgeting, saving, spending patterns, and credit management significantly impact the financial well-being of individuals, with direct and indirect relationships existing between financial behavior and stability (Acharya & Poudel, 2023).

#### 1.0.2.1 Personal Finance in the Philippines

Handling personal finance is a significant activity for every adult. This activity involves paying bills, budgeting, organizing taxes, paying debts, saving money, and investing where adults are commonly concerned in managing (Di Domenico et al, 2022). Personal finance encompasses the strategic management of an individual's financial resources that involves sensible decision making for income allocation, expenditure prioritization, and risk mitigation to achieve better financial well-being (Panos et al., 2020). Financial well-being is the state of a person where they fully capitalize the joy of life while satisfying their financial obligations and feeling financially secure of their future situation (Kulshreshtha et al, 2023) (Brzozowski & Visano, 2023) (Ferrer, 2017). However, having low financial well-being exerts significant adverse effects on both psychological and physical health. Individuals experiencing this dilemma often report diminished confidence, increased stress levels, and reduced productivity in the workplace. Moreover, financial strain is associated with heightened absenteeism, delays, and difficulties maintaining concentration, further aggravating the challenges individuals face in navigating their daily lives (Ferrer, 2017). The financial situations Filipinos currently face raises the amount of anxiety and uncertainty around their lives causing more stress and worry to handle (Renamae et al, 2023). According to the Inquirer from

2021, 7 out of 10 Filipinos are burdened with debt, resulting with the Philippines being the most stressed country in Asia-Pacific when managing personal finances.

However, according to the 2021 Financial Inclusion survey conducted by Bangko Sentral ng Pilipinas, 35 million Filipino adults, which is only 45% of the total respondents, utilizes a personal budget. Furthermore, despite the increase of Filipino owning savings accounts, the percentage of adults with savings dropped to 37% in 2021 from 53% in 2019. However, this is the result of the reduced income and increased expenses during the COVID-19 pandemic.

There were 28.9 million savers in 2021 compared to 38.6 million in 2019, a 9.7 million decline. This was in line with the decrease of account users who used their accounts to save in 2021, from 76% in 2019 to 56% in 2021. Among the 37% identified savers, their reasons for savings, 81% responded that they allocate their money for emergencies while there were 43% savers that want to achieve financial goals, 41% to cover their expenses, and 12% preparing for retirement.

#### 1.0.2.2 Personal Informatics

Personal informatics refers to the use of interactive technologies and data-driven approaches to help individuals monitor, analyze, and reflect on various aspects of their lives, including behaviors, habits, and thoughts. This field aims to empower individuals to gain self-understanding, make informed decisions, and ultimately improve their quality of life through the insights derived from personal data.

Ian Li, Yevgeniy Medynskiy, Jon Froehlich, and Jakob Eg Larsen (2012) discuss the increasing prevalence and accessibility of personal informatics systems, which offer users the ability to track diverse dimensions of their daily activities, such as physical exercise, sleep patterns, dietary intake, and emotional well-being. These systems typically involve the use of sensors, mobile applications, and online platforms to collect and analyze data related to individual behaviors and experiences.

Moreover, personal informatics tools are designed to facilitate self-reflection and behavior change by providing users with visualizations, feedback mechanisms, and personalized insights based on their data. By leveraging these features, individuals can gain deeper insights into their behaviors, identify patterns, set goals, and track progress over time, thereby fostering positive changes in their habits and routines.

The field of personal informatics also encompasses broader societal implications, as aggregated data from these systems can contribute to public health research, urban

planning, and policy-making efforts. By harnessing the collective insights generated by personal informatics data, researchers and policymakers can gain a better understanding of population-level trends, identify areas for intervention, and design targeted strategies to improve community well-being.

#### 1.0.2.3 Data Gathered from Survey

##### 1.0.2.3.1 Financial Well-Being

Based on the question in *Appendix B2.1,* the survey results indicate that a significant portion of single working adults rate their financial well-being as neutral (3). However, according to *Appendix B2.2* results suggest that a majority of single working adults are very dissatisfied or dissatisfied with their current level of financial stability which is related to their confidence in financial situations, connected to financial well-being as satisfying personal financial obligations. With 49% of respondents choosing options 1 and 2 (very dissatisfied or dissatisfied), it indicates a significant portion of individuals are experiencing financial insecurity or dissatisfaction. Furthermore, based on *Appendix B2.7,* 53% of respondents feel stress in managing their finances. The top 3 factors that contribute to their stress is their unplanned expenses, low income, and poor financial planning. The prevalence of stress among respondents when managing finances, largely attributed to unplanned expenses, low income, and poor financial planning, indicates a significant barrier to achieving financial well-being.

##### 1.0.2.3.2 Financial Planning

Based on *Appendix B3.14*, the distribution of responses illustrates varying levels of confidence in individuals' ability to handle unexpected expenses or financial emergencies. While a majority express some degree of confidence, a notable proportion (12%) indicate lower confidence in their capabilities. This suggests potential challenges in preparing for unforeseen financial circumstances and underscores the importance of bolstering emergency fund reserves and implementing robust financial planning strategies. Strengthening confidence in managing unexpected expenses can provide individuals with greater financial resilience and peace of mind.

In *Appendix B3.31*, the data reveals insights into the frequency of individuals exceeding their budget due to unplanned expenses. A significant percentage (43%) report often exceeding their budget, indicating difficulties in managing unexpected financial obligations effectively. This highlights the importance of incorporating buffer allowances within budgetary frameworks to accommodate unforeseen expenses and mitigate the risk of budgetary overruns. Additionally, cultivating proactive financial planning habits, such as regularly reviewing and adjusting budgets, can help individuals better anticipate and manage unplanned expenditures, ultimately fostering greater financial stability and preparedness.

#### 1.0.2.3.3 Financial Decision Making

Based on *Appendix B5.26, t*he distribution of responses indicates that a significant portion of single working adults perceive a lack of control over their expenses, with 49% expressing dissatisfaction. This suggests challenges in making an informed financial decision, potentially characterized by difficulties in budgeting, prioritizing spending, and managing financial resources effectively. Furthermore, *Appendix B5.29* results indicate that a significant proportion of respondents (62%) report making purchases without prior planning, this suggests a tendency towards impulsive spending behaviors among a notable portion of single working adults. Such behavior can negatively impact financial decision-making by potentially leading to overspending and difficulty in adhering to budgetary constraints.

#### 1.0.2.3.4 Financial Management

Based on *Appendix B4.20*, the data reveals a mix of satisfaction levels regarding individuals' monthly savings, with a notable proportion (9%) expressing dissatisfaction. This indicates potential challenges in achieving optimal savings levels, highlighting the importance of refining financial planning strategies to enhance savings adequacy. Additionally, *Appendix B5.27* shows that while a significant portion of individuals review their expenses to identify potential areas for savings or cutbacks, a considerable percentage (6%) never engage in this practice, while 21% rarely do so. Regular expense review is essential for optimizing spending habits and promoting better financial management. Moreover, *Appendix B5.30* indicates that although many individuals track their expenses against their budget occasionally, a notable percentage (28%) display inconsistency or lack of engagement in this practice. Consistent tracking of expenses against budget is vital for maintaining financial discipline and achieving long-term financial goals. These findings underscore the importance of individuals cultivating proactive financial management habits to bolster savings habits and overall financial well-being.

#### 1.0.2.3.5 Financial Awareness

Based on *Appendix B4.22*, the data indicates a range of behaviors regarding the frequency of reviewing savings progress among respondents. A significant proportion (27%) rarely engage in this practice, while a substantial portion (35%) review their savings progress occasionally. However, there is also a notable percentage (14%) who consistently review their savings progress. This variability suggests that some individuals may not actively monitor their savings growth, potentially missing opportunities for optimization or adjustment of savings strategies. In contrast, those who regularly review their savings progress may benefit from greater awareness and control over their financial trajectory.

In *Appendix B6.41*, the distribution of responses concerning confidence in understanding different investment options shows a diverse range of sentiments among respondents. While the majority express some degree of confidence, a considerable proportion (29.2%) report feeling only moderately confident. Additionally, 8.3% indicate low confidence in their understanding of investment options. This indicates a potential need for improved financial education or access to resources to enhance individuals' understanding of investment opportunities. Strengthening confidence in investment knowledge can empower individuals to make informed decisions regarding wealth accumulation and long-term financial planning, ultimately contributing to their overall financial well-being.

#### 1.0.2.3.6 Financial Behavior

Based on *Appendix B3.15,* the data indicates that 32% of the respondents are inconsistent when it comes to reviewing their progress towards reaching their financial goals. Thus, they are inconsistent in tracking and monitoring their financial goals which contributes to poor financial behavior. Additionally, according to *Appendix B4.21*, the responses regarding confidence in consistently saving a portion of income among single working adults suggests a mixed landscape of financial behaviors. The significant percentage (28% and 9% respectively) suggests a notable portion of respondents feel confident or very confident in their ability to save regularly. However, the individuals that chose not 1 and 2 indicating not being confident (20% and 23% respectively) suggests that a substantial segment of single working adults may experience uncertainty or lack confidence in their saving capabilities, potentially indicating challenges in their financial behavior in saving money. Finally, from *Appendix B5.26*, the significant percentage selecting unsatisfied 3 (31%) indicates a substantial portion express

dissatisfaction with their expense control. Furthermore, 18% feel very unsatisfied with their expense control, potentially indicating challenges in adhering to their financial habits, affecting their financial behavior.

#### 1.0.2.3.7 Self Understanding

Based on *Appendix B5.31,* responses indicate that a significant portion of respondents (66%) frequently exceed their budget due to unplanned expenses. This suggests a lack of self-understanding and self-awareness regarding their spending habits and the importance of budgeting effectively. Furthermore, according on *Appendix B5.28,* the response distribution indicates that a substantial portion of respondents may have limited alignment between their expenses and financial goals and priorities, with only 29% expressing strong alignment (options 4 and 5). This suggests a potential lack of self-understanding regarding the connection between spending habits and overarching financial objectives.

#### 1.0.2.3.8 Overspending and Impulsive Spending

The survey results of *Appendix B7.56* reveal a multitude of challenges individuals face when trying to adhere to a budget, notably including unexpected emergencies, impulsive buying or overspending, and difficulty estimating expenses. These findings suggest that overspending and impulsive buying are often aggravated by unforeseen financial circumstances and a lack of accurate expense projections. Individuals may struggle to resist impulse purchases or curb excessive spending, particularly when faced with unexpected expenses or difficulties in accurately estimating their financial obligations. Meanwhile the results from *Appendix B5.31* indicate varying degrees of frequency with which individuals exceed their budget due to unplanned expenses. The significant portion of respondents (43%) often suggests that a considerable proportion experience frequent occurrences of overspending due to unforeseen expenses, potentially indicating challenges in budget management. Additionally, the presence of 23% very often implies a notable portion facing regular impulse spending, further underscoring difficulties in adhering to financial plans and potential impulsive buying behaviors.

#### 1.0.2.3.9 Tracking Expenses

*Appendix B7.56* suggests that tracking expenses poses a significant challenge for many single working adults, with 56.4% indicating difficulty in this aspect and 57.4% in difficulty in estimating their own expenses. Furthermore, their responses on *Appendix B5.31* indicate that a significant proportion of respondents (66%) frequently exceed their budget due to unplanned expenses. In

terms of tracking expenses, the data suggests that a notable portion of single working adults may encounter challenges in this area, with 56.4% reporting difficulty. This highlights a potential correlation between difficulties in tracking expenses and exceeding budget limits due to unplanned expenses with potential gaps in their ability to monitor and control their spending habits effectively, which could hinder their efforts to stick to a budget. This results underscoring the importance of improving expense monitoring practices which could be beneficial in helping these individuals gain better control over their finances to enhance budget adherence.

#### 1.0.2.3.10 Budgeting Techniques

*Appendix B7.46* indicates a diverse range of preferences among users. The prevalence of spreadsheet usage (30.9%) suggests that many individuals prioritize customization and flexibility in their budgeting approach, leveraging the versatile features of spreadsheet software. However, the significant percentage of users still opting for pen and paper (25.5%) hints at a preference for simplicity and tangibility in tracking finances. The popularity of mobile apps (26.6%) signifies a growing trend towards digital solutions, offering convenience and real-time access to budgeting tools. Notably, the proportion of users abstaining from any specific method (16%) may suggest a segment of individuals either not actively managing their finances or employing informal strategies.

## 1.1. Research Problem

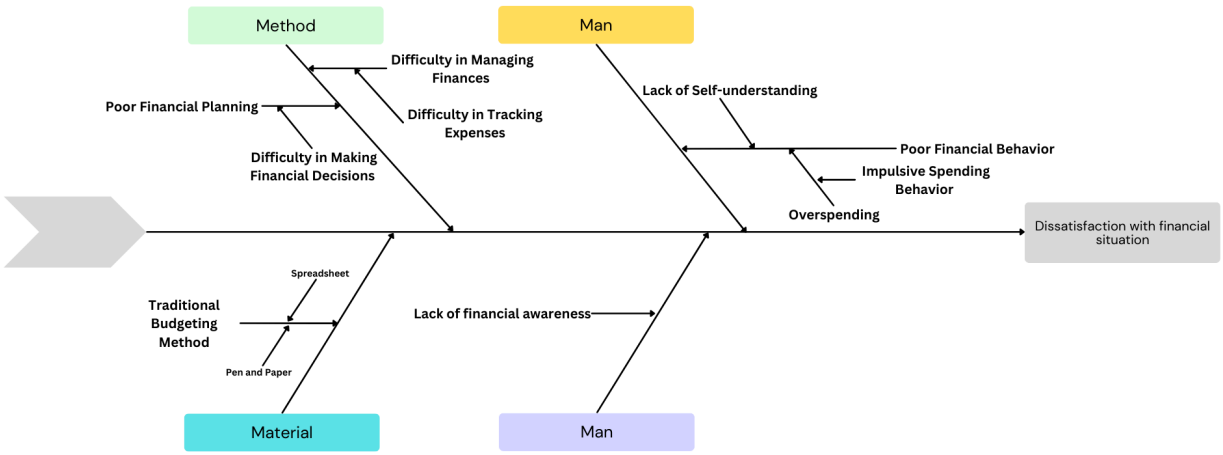


*Figure 1.1.1 Ishikawa Diagram*

### 1.1.1 General Problem

Dissatisfaction with one's financial situation, an indicator of one's financial well-being, is a pressing concern affecting single Filipino working adults across various socioeconomic backgrounds. Poor financial planning aggravates this situation, leaving individuals ill-prepared to handle financial challenges and plan for the future effectively. Under financial planning, the difficulty in making sound financial decisions compounds the stress, as individuals struggle to prioritize and allocate their financial resources wisely. Additionally, challenges in managing finances, coupled with a lack of financial awareness, further contribute to feelings of unease and discontentment, making it difficult for individuals to navigate their financial circumstances confidently. Furthermore, poor financial behavior, driven by a lack of self-understanding and impulse-driven spending tendencies, perpetuates the cycle of financial stress, often resulting in overspending and hindering efforts to achieve greater financial well-being.

### 1.1.2 Specific Problem

#### 1.1.2.1 Poor Financial Planning

The increasing complexity of personal financial planning poses a significant challenge for single working adults in today's rapidly evolving economic landscape. With inflation impacting essential expenses, coupled with the globalization of financial markets, and their spending habits, navigating financial decisions has become increasingly difficult for them. Despite recognizing the paramount importance of attaining financial stability, single working adults often face barriers in acquiring the necessary financial behavior and capability to effectively manage their finances.

##### 1.1.2.1.1 Difficulty in Making Financial Decisions

The difficulty in making sound financial decisions persists among single working adults due to a combination of limited financial awareness and ineffective financial behavior. Many individuals lack comprehensive knowledge and understanding of financial concepts, leading to challenges in budgeting, saving, and investing. Additionally, without proper financial planning strategies in place, single working adults struggle to manage their finances, further worsening their financial stress. Moreover, ingrained financial behaviors, such as their level of control on their impulse spending or a lack of savings discipline, can hinder their ability to make prudent financial choices.

#### 1.1.2.2 Difficulty in Managing Finances

For many single working adults, managing personal finances proves to be a difficult task. They often find themselves exceeding their budgets, lacking confidence in handling unexpected expenses, and struggling with impulsive spending habits. Furthermore, a significant portion of this demographic expresses dissatisfaction with the amount they are able to save. In addition, many single working adults express dissatisfaction with their level of control over expenses. Single working adults face increased financial responsibilities, further complicating budgeting and saving efforts. Those with lower income encounter difficulties covering basic expenses, leading to financial stress and hindering savings efforts. These challenges underscore the pervasive difficulties experienced by single working adults in achieving financial stability and highlight the need for targeted interventions and support mechanisms to address these pressing concerns.

#### 1.1.2.2.1 Difficulty in Tracking Expenses

Many individuals face challenges in effectively tracking their expenses, leading to a lack of clarity and control over their financial situation. Despite efforts to manage budgets and allocate resources appropriately, the difficulty in consistently recording and monitoring expenses hampers their ability to make informed financial decisions. This struggle not only impacts personal financial management but also impedes progress towards achieving long-term financial goals.

### 1.1.2.3 Lack of Financial Awareness

Financial awareness is crucial, reflecting an individual's capacity to evaluate their financial decision-making skills. High Financial Awareness individuals make informed choices, understanding their financial strengths and weaknesses, managing spending habits, and planning for the future. Conversely, low Financial Awareness individuals face challenges, resulting in poor financial decisions, overspending. The lack of financial awareness among single working adults presents a significant challenge, hindering their ability to effectively manage their finances and plan for the future. Without sufficient understanding of key financial concepts and strategies, individuals in this demographic may struggle to budget effectively, save for emergencies, and make informed investment decisions. This lack of awareness can lead to financial stress, poor financial outcomes, and limited opportunities for long-term financial growth.

### 1.1.2.4 Poor Financial Behavior

Poor financial behavior among single working adults poses a substantial challenge, impeding their financial stability and overall well-being. Many individuals in this demographic exhibit behaviors such as overspending, failing to save adequately, and making impulsive financial decisions. These behaviors can lead to financial stress, debt and missed opportunities for wealth accumulation and long-term financial planning.

#### 1.1.2.4.1 Lack of Self-understanding

The lack of self-understanding is a critical factor contributing to poor financial behavior among single working adults. Without a clear understanding of their own financial habits, attitudes, and motivations, individuals may struggle to identify and address problematic behaviors such as overspending, failure to save adequately, and impulsive decision-making. This lack of self-awareness can lead to ongoing financial challenges, including financial stress and missed opportunities for long-term financial security. Furthermore, without a deep understanding of their own financial tendencies, individuals may find it difficult

to develop effective strategies for improving their financial well-being and achieving their long-term financial goals.

#### 1.1.2.4.2 Overspending

Overspending is a challenge among working individuals, leading to financial instability, stressful management of personal finances, and hindered progress towards financial goals. This behavior often results from impulsive purchasing, inadequate budgeting, and a lack of financial discipline, ultimately undermining individuals' financial well-being and long-term financial security.

##### 1.1.2.4.2.1 Impulsive Spending Behavior

Impulsive spending behavior, characterized by unplanned and emotionally-driven purchases without consideration of alternatives, is prevalent among consumers. This behavior is deeply rooted in emotional decision-making, where consumers rely on emotions such as joy, love, and fear rather than careful evaluation of options. Stressful circumstances can aggravate impulsive spending tendencies, leading individuals to disregard financial constraints. Young adults, particularly those with limited financial resources, often struggle with impulsive spending behaviors, influenced by factors like emotional satisfaction. Consumerism further perpetuates impulsive spending, as individuals seek satisfaction through material acquisitions, yet remain unsatisfied and continue the cycle of consumption.

### 1.1.2.5 Traditional Budgeting Method

Traditional budgeting methods, such as pen and paper or manual spreadsheet tracking, often lack the accuracy and flexibility required to effectively manage modern-day financial complexities. These methods can be time-consuming and offer limited capabilities for real-time monitoring and analysis. Consequently, individuals relying solely on traditional budgeting methods may struggle to maintain financial discipline, optimize spending, and adapt to changing economic conditions. Thus, there is a pressing need to explore innovative solutions that bridge the gap between traditional methods and digital advancements, enhancing the effectiveness and accessibility of budget management for diverse user needs.

#### 1.1.2.5.1 Spreadsheet Method

The spreadsheet method, commonly employed for budgeting purposes, offers certain advantages such as customizable templates and the ability to perform complex calculations. However, its utility may be limited by factors such as accessibility and real-time tracking. While spreadsheets provide a structured format for organizing financial data, users may find it cumbersome to input information manually and lack the convenience of automatic updates. Furthermore, spreadsheets may not be easily accessible on-the-go, making it challenging for individuals to track expenses or receive alerts for overspending promptly.

#### 1.1.2.5.2 Pen and Paper Method

The pen and paper method comes with inherent limitations, notably in terms of convenience and durability. Paper records can easily be misplaced, damaged, or lost, leading to potential inaccuracies and disruptions in financial tracking. Moreover, maintaining meticulous records manually may prove time-consuming and prone to human error, detracting from the efficiency of budget management efforts. Despite its simplicity, the pen and paper method may lack the adaptability and accessibility required for effective financial planning in today's fast-paced digital landscape.

## 1.2 Conceptual Framework

In order to address the problems identified, a conceptual framework has been formulated to address said problems. The framework as shown in Figure 1.1 indicates the modules which includes the users, problems, concepts, input, features, and opportunities it is involved with. Each module will be discussed separately in the succeeding sections.

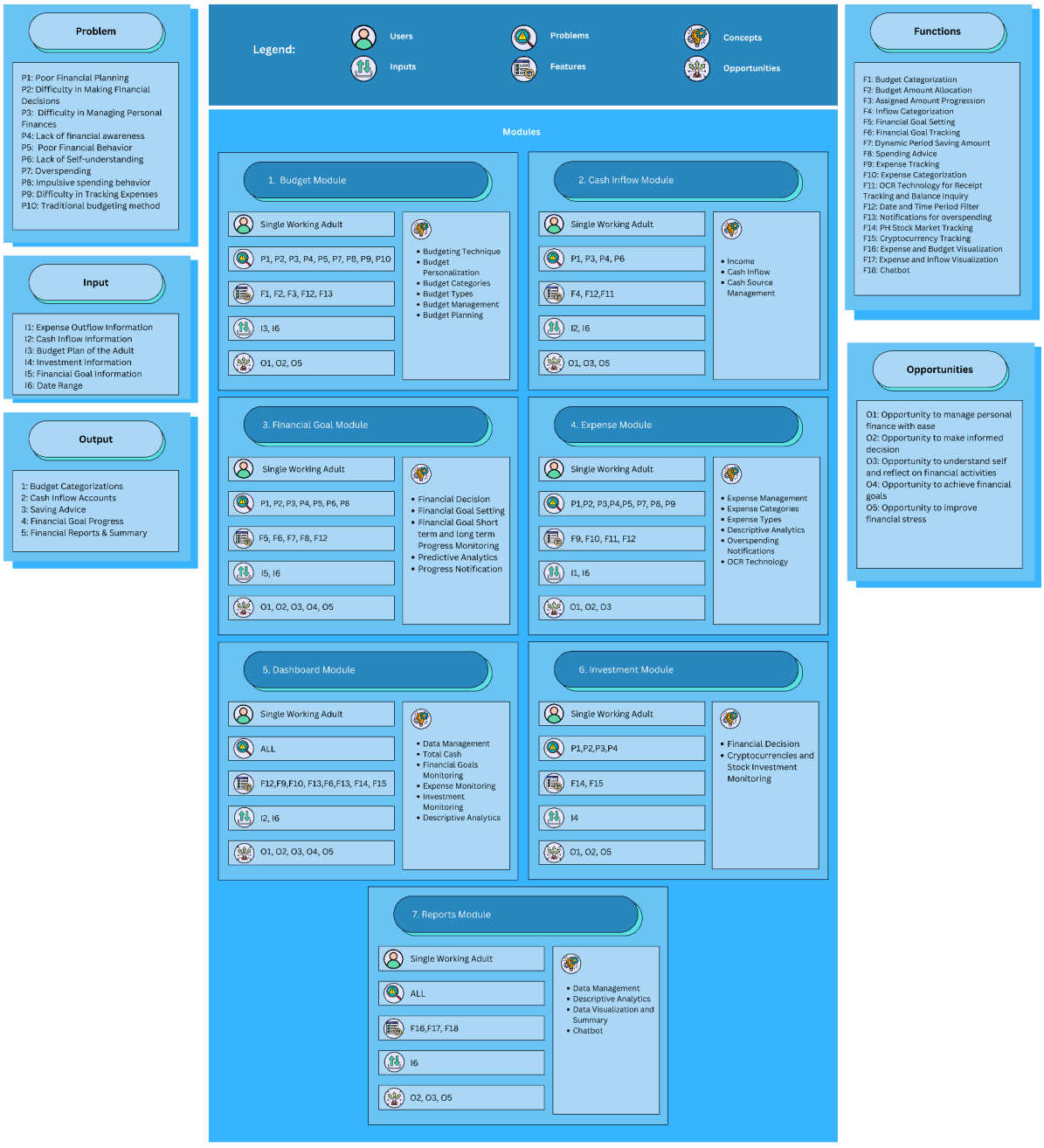


*Figure 1.1: Conceptual Framework*

### 1.2.1 Budget Module

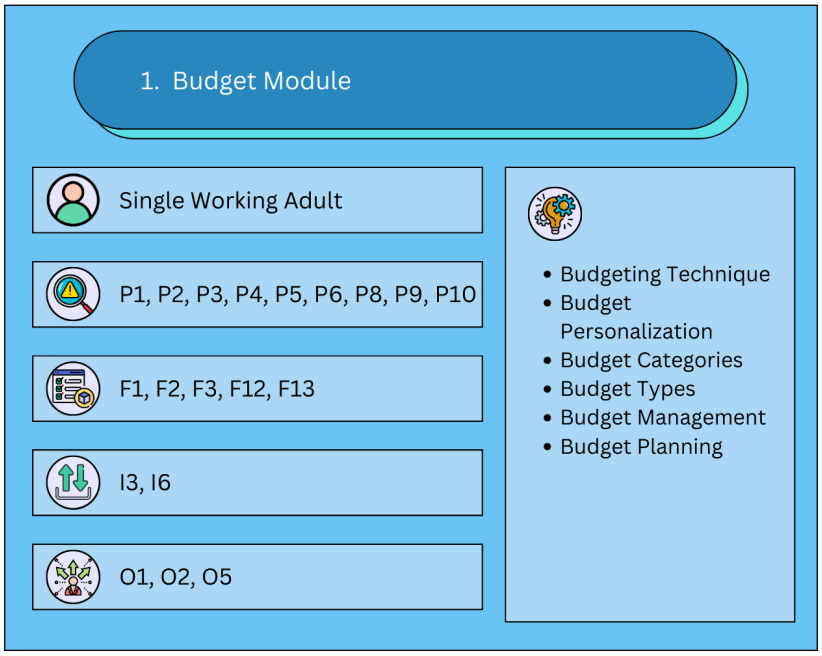


*Figure 1.2: Budget Module*

The Budget module will be available to users in this system. This is where the adult can set up a budget plan which aims to aid single working adults in planning out their budgets based on their situation and empower users to take charge of their finances. This module will allow users to set their desired budget allocation amount on different categories of specific types such as their bills, needs, and wants which promotes budget planning. Along with each category is a specified time period they need to save for it. This module will display progress bars on how close they are in reaching the limit of budget in these categories. In each category, it will display the amount they planned, the amount they spend, and the amount remaining they can spend. It will also recommend the daily amount they can spend on that category which will be computed based on the monthly period of the budget. Users will be notified if they are near in reaching the planned amount allocated to that category or near spending over the planned amount. Once users reach over the limit of their allocated amount for a category, they will be alerted of overspending, promoting awareness and understanding of their financial patterns and adhering to set their budgets. By providing tools for budget creation and monitoring, users can establish realistic spending limits and monitor their adherence to these budgets, identifying areas of potential overspending.

The budget categories defined by users in the budget module will be integrated with the expense module since the categories used in the expense module will be based on the categories from the budget module when they record their expenses. Additionally, by integrating with financial goal-setting features, users can align their financial aspirations based on the budget categories to ensure the attainability of their financial goals. Furthermore, the reports module will also utilize the categories from this module for data management and visualization in identifying spending trends and areas for potential savings.

The budget module's structured approach aids in establishing clear financial goals and resource allocation, combating poor financial planning. Real-time insights provided by the module in regard to their spending habits empower informed decision-making and self-understanding. Progress bars and alerts on approaching budget limits help users align spending with financial goals and allocated amounts for each category, addressing decision-making challenges, lack of financial awareness, and potential overspending. Through budget creation and monitoring tools, the module assists users in managing finances effectively, addressing the difficulty associated with managing it. Displaying planned versus actual expenditure aids in tracking adherence to budget limits and optimizing financial resources, easing personal finance management. Additionally, alerts on nearing or exceeding budget limits discourage overspending and impulsive behavior. By promoting accountability and discipline, the module fosters healthy financial habits, combating poor financial behavior. With this module, users can gain greater control over their financial situation, alleviating stress associated with personal finances. Additionally, this module also addresses the tedious process of applying traditional budgeting methods like pen and paper. This is because through the system budget categorization is more organized and spending of users can be tracked by the system not allowing the user to overspend but rather follow the allocated budget for expenditures.

### 1.2.2 Cash Inflow Module

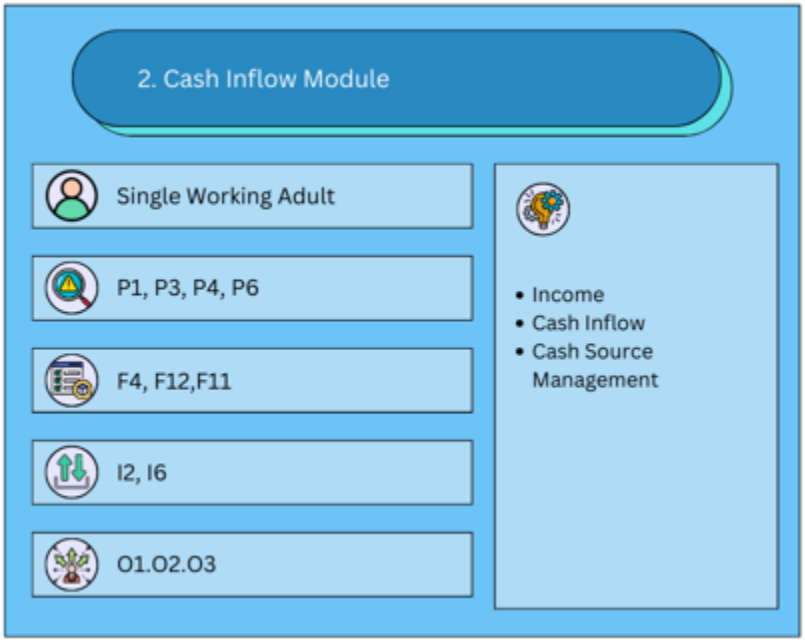


*Figure 1.3: Cash Inflow Module*

The Cash Inflow Module will be available for users. This is where the user can input information about their earnings in the form of accounts. The module will allow users to either manually input earnings or input earnings through the Optical Character Recognition. The Optical Character Recognition Technology that will be utilized will be able to scan the ATM balance inquiries of deposits to reduce the need for repeatedly manually inputting earnings. Additionally, the user can either create a new account which will allow the user to name the account and input the existing balance of that account. But if they have already created an existing account they have an option to add a cash inflow. The user can choose already existing accounts where they can add an amount they have gained and input the source where this earning came from. The total cash within these accounts will all be added to provide the user with the total cash that he/she has gained.

This module is integrated with the dashboard module so that the user can easily access adding an inflow whenever he/she has an earned money that needs to be added to his/her accounts. Additionally, he/she can also access the Optical Character Recognition Technology for scanning ATM balance inquiries to also update the total amount of his/her savings. This module will also be integrated with the expense module because the expenses of the user will be deducted from the total amount of cash of the user from different sources of their inflow. This

module will also be integrated with the budget module because this module will limit the amount of budget that can be allocated for the categories of expense of the user. Furthermore, this module will also be integrated with the financial goal module since the total amount of cash within the accounts will determine whether the targeted amount for the financial goal is achievable. Additionally, this module will also be integrated with the reports module because the graphs within this module will compare the cash inflow amount of the user with the expenses of the user.

This module tackles poor financial planning of users, because through this module they can see and track their sources of earnings and the total of cash stored within their accounts. Thus, helping the user in planning out their budget that will be allocated to their expenses. Additionally, it also addresses difficulty in managing personal finances, since the user can already track and monitor the amount of money that is gained on the different accounts they will be able to budget properly and save properly in accordance with their total earnings. Furthermore, this module also addresses users' lack of financial awareness because the user can reflect on the list of the different accounts where their earnings come from and the total cash within those accounts making them aware of their financial situation. Lastly, this also addresses lack of self-understanding because through the list of the different accounts they can also reflect and make informed decisions whether or not to budget on a certain category of expense.

### 1.2.3 Financial Goal Module

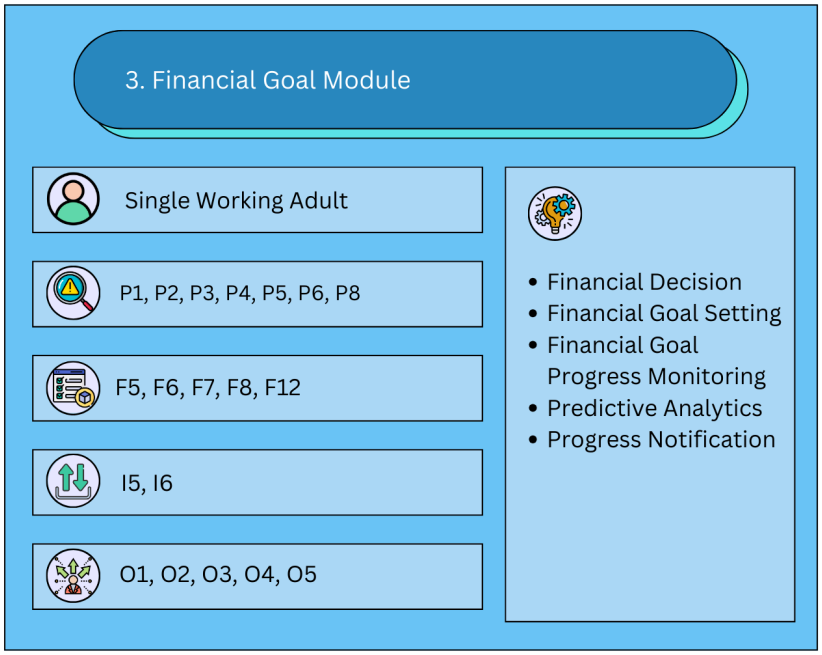


*Figure 1.4: Financial Goal Module*

The Financial Goal Module is where single adults will identify and set their financial goals. It offers them the opportunity to make their own financial decisions. The user will specify their financial goal, the target amount, their preferred frequency of allocation, and their supposed target date of achieving it. With those inputs, the system will suggest an amount the user must save based on their preferred frequency. However, based on the cash sources of the user integrated with the cash inflow module, their current budget plan integrated with the budget module, and their expense pattern from the expense module, using predictive analytics, the system will calculate whether the inputted target date is feasible, realistic, and attainable to achieve by the user. In case the system identifies they can't, this module will have the saving advice feature that will recommend an adjusted certain amount allocation based on their preferred frequency to the user so that it would be feasible in their current financial situation. It addresses the concern of having uninformed financial decisions as this module guides users towards setting goals that are aligned with their financial capabilities.

Moreover, along with setting goals, is the ability to keep track of the users progress towards their financial goals. This module will visualize whether the user is on track or not of achieving the financial goal by displaying a progress bar of the financial goal of the user. Visual aids such as progress bars will offer intuitive insights towards goal attainment, keeping users motivated and engaged. Lastly, it will alert and notify users, based on the frequency, to allocate

the amount specified during goal setting towards their financial goal to keep users informed and on track, such as reminding about progress milestones which serves to keep users engaged with their financial objectives and accountable for their actions. In case they failed to allocate the specified amount, it will stack the missing amount with the upcoming amount, based on preferred frequency, of the financial goal.

This module tackles poor financial planning by guiding users to set specific, achievable goals aligned with their financial capabilities and addresses difficulty in making financial decisions by allowing users to make their own decisions while providing personalized saving advice based on their financial situation. Furthermore, it helps users manage personal finances by tracking progress towards goals, ensuring they stay organized with savings and spending. Moreover, it combats lack of financial awareness by increasing users' awareness through regular notifications and visual aids like progress bars. It also addresses poor financial behavior and impulsive spending by promoting better financial habits through goal setting and motivating them to save. Lastly, it tackles lack of self-understanding by guiding users to set realistic goals that align with their financial situation.

### 1.2.4 Expense Module

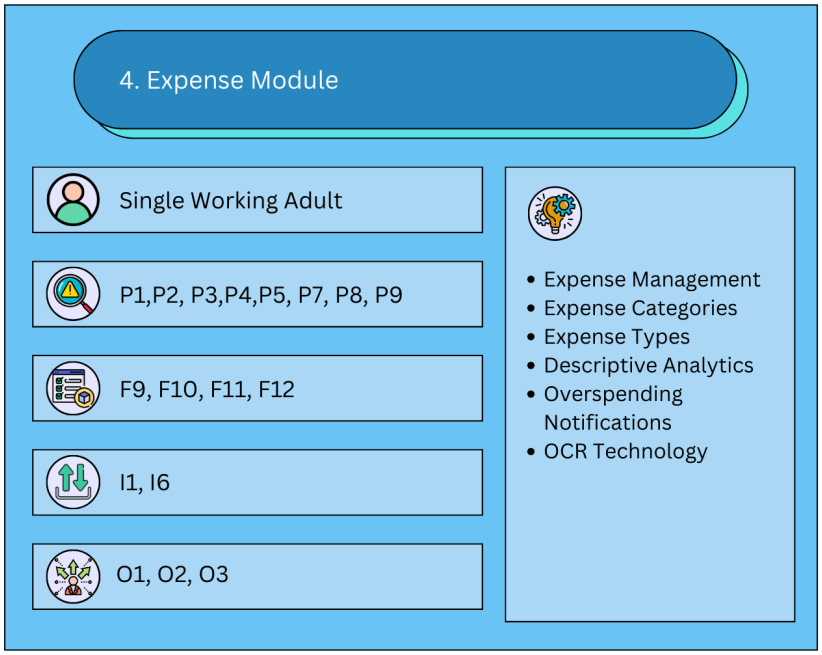


*Figure 1.5: Expense Tracking Module*

The Expense Module will be available for the users. This is where the user can input information about their expenditures. The module will allow the single working adult to select the category of the expense based on the categories identified from the budget module. The categories will vary depending on users but these categories are classified by their types such as wants, needs, and bills so users are aware of their priorities. Based on the frequency indicated in categories specially for bills, users will receive notifications so they will be reminded to pay them. Along with the category of expense is the source of cash inflow the expense will be deducted from. Users will be prompted to include the total amount of the expense. Optical Character Recognition Technology will be utilized on receipts to obtain the total expense to reduce the troublesome need of typing expenses repeatedly.

The module will notify users daily, by default, at the end of the day, to input their remaining cash at hand. This is also where the cash inflow module will work with the expense module to compare the remaining cash at hand and the current amount in the system. If the remaining cash at hand is not the same as the expected total amount of cash the user should have at the moment, this module will calculate the difference between the two. The module will then prompt the user to allocate the remaining missing expenses to achieve balance in their expenses.

The expense module is mainly integrated in the dashboard module, since this is where the expense module will be located. Additionally, the module will be integrated with the budget module since the category the expense is basing from is the category identified in the budget module. If the expense exceeds the budgeted amount, the user would receive a notification or prompt informing them of the overspending in that category. Through a progress bar, it will indicate whether the user is within budget for each category. This could be represented using color-coding progress bars to easily visualize spending relative to the budget. It will offer suggestions for adjusting the budget if the user consistently exceeds their allocated amounts in certain categories. These suggestions could be based on spending trends and patterns identified through in the Reports Module. It will give users insight into their spending habits in the form of visualizations. The Cash inflow module is also integrated in this module as a choice of cash inflow source to deduct their expenses from.

This module tackles poor financial planning by providing a platform for users to input and categorize their expenditures, helping them understand where their money is going and make adjustments accordingly. It addresses difficulty in making financial decisions by integrating with the budget module, allowing users to compare their actual expenses with their budgeted amounts and receive notifications if they overspend in certain categories. By incorporating reminders for bill payments and utilizing Optical Character Recognition Technology to streamline the expense input process, reducing the burden of manual entry, makes finance management less struggling. It also addresses lack of financial awareness by offering visualizations and progress bars to

indicate spending relative to budgeted amounts, empowering users to track and control their expenses more effectively. Additionally, it looks into poor financial behavior by providing suggestions for adjusting budgets based on spending trends and patterns identified through the Reports Module, encouraging users to develop better spending habits. Furthermore, it tackles overspending and impulsive spending by alerting users when they exceed budgeted amounts and offering suggestions for corrective action. Lastly, it addresses difficulty in tracking expenses by integrating with the cash inflow module to ensure users maintain balance between their income and expenses, prompting them to allocate missing expenses to achieve financial equilibrium.

### 1.2.5 Dashboard Module

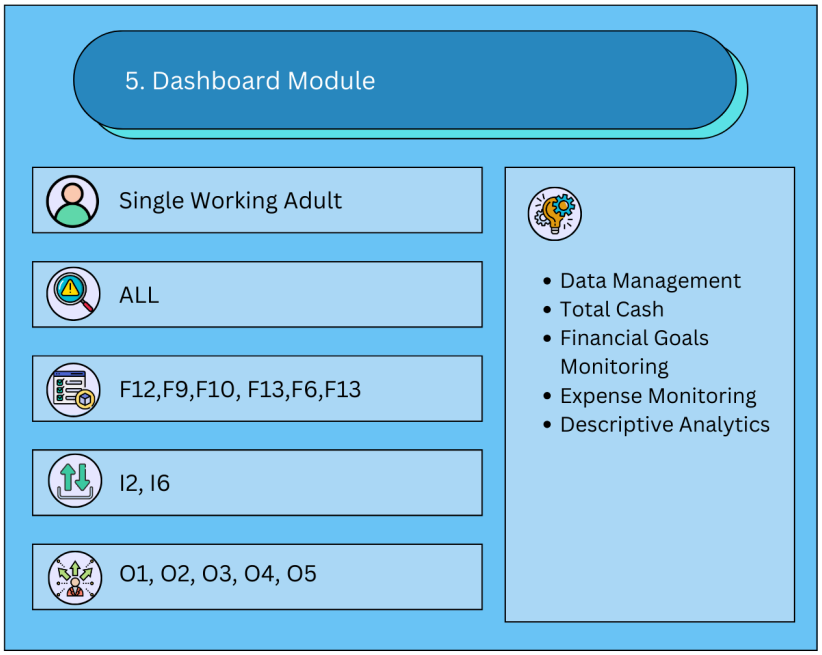


Figure 1.6: Dashboard Module

The Dashboard Module serves as the central activity of finances within the system. This module acts as a pivotal tool, providing users with comprehensive insights and reports to monitor, analyze, and reflect on their financial activities which gives users the opportunity to track and understand their spending habits effectively. This module will display the total cash the user has at hand based on the source of cash in the cash inflow module. It will also have a segment integrated with the financial goal setting module wherein financial goal near completion or designated time period to save money will be displayed so that the users are aware of their goals and accomplishments. Expense module will be mainly integrated in the dashboard since this is the module where users can input their expenses,It will be integrated as well with the

budget module which will display budget categories that are progressing near being the planned amount. Additionally, stock or crypto investments from the investment module will be displayed in terms of price movements. A quick action button will be available in this module for quick access of inserting expense which will have OCR option in it. The module incorporates alerts and notifications to keep users engaged and informed. Whether it's a reminder about upcoming expenses, deviations from budget targets, or progress milestones towards financial goals, these prompts serve to keep users accountable and proactive in managing their personal finances.

Poor financial planning is tackled through its provision of comprehensive insights, reports, and budget tracking tools, aiding users in creating and adjusting financial plans. Difficulty in making financial decisions and managing personal finances is addressed by integrating financial goal setting, enabling users to prioritize objectives and make informed decisions aligned with their aspirations and by seamlessly integrating expense input and budget tracking features, facilitating effective management of financial transactions. Furthermore, Lack of financial awareness and poor financial behavior is alleviated through the module's provision of insights into spending habits and financial behaviors. Lack of self-understanding is tackled by providing visibility into financial goals and accomplishments, fostering a deeper understanding of personal financial objectives and progress. Overspending and impulsive spending behavior are addressed by promoting awareness of budget deviations and integrating quick action buttons for expense input, encouraging mindful spending habits. Lastly, difficulty in tracking expenses (P9) is mitigated by offering user-friendly expense tracking features and OCR options, facilitating seamless management of financial transactions.

### 1.2.6 Investment Module

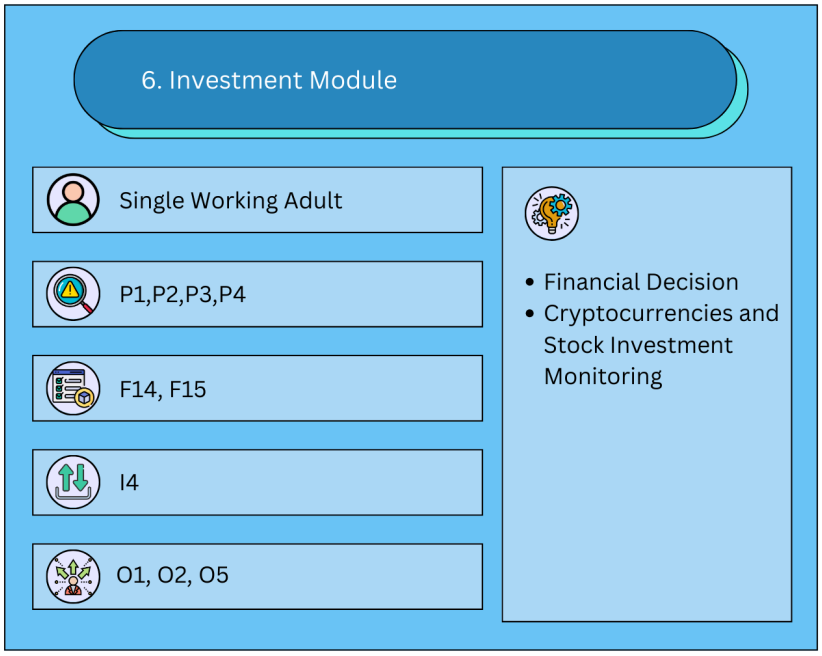


*Figure 1.7: Investment Tracking Module*

The investment module allows users to input details about their stock and cryptocurrencies investments. This module aims to help single working adults in monitoring their investments and make them aware of the performance of their current investments. The user can input details such as stock or cryptocurrency they invested on, the purchase price, quantity they own, and date of purchase. The system validates this data and calculates performance over time, considering factors like value changes, and overall return which helps users assess and track their investment choices. The investment tracking module will be integrated with the dashboard to display stock or crypto value of their investment to help users visualize and track their investment activity and performance. This enables users to analyze their investments and make informed decisions about their strategy and allocation.

This module tackles poor financial planning by providing users with a platform to monitor their investments and assess their performance over time, enabling them to adjust their financial strategies accordingly. Secondly, it addresses difficulty in making financial decisions by offering users insights into the performance of their investments, helping them make informed decisions about their investment choices and allocation. Furthermore, it helps users manage personal finances by integrating investment tracking with the dashboard, allowing users to visualize and track their investment activity and performance alongside other financial metrics.

Moreover, it also looks into lack of financial awareness by providing users with real-time information on the value changes and overall return of their investments, increasing their awareness and understanding of their investment portfolio.

### 1.2.7 Reports Module

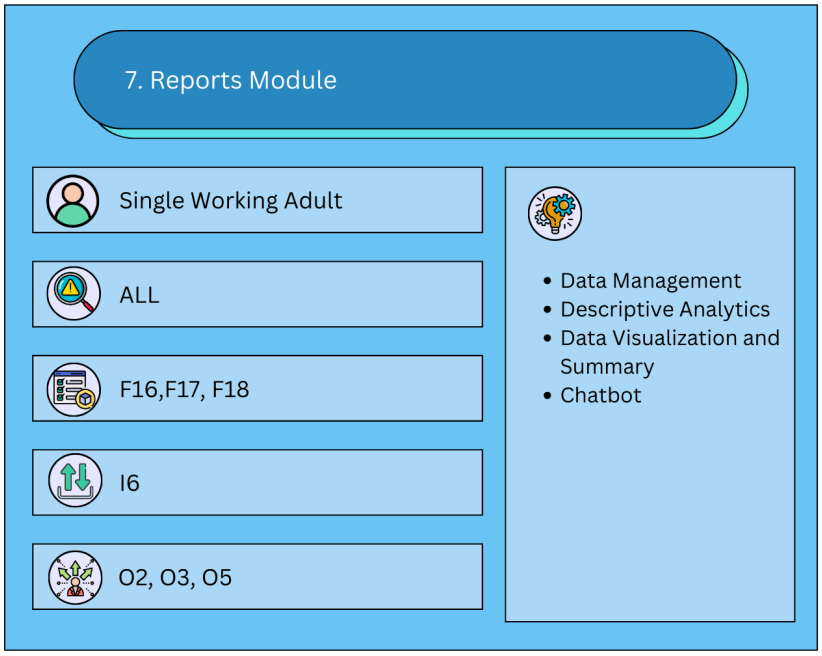


*Figure 1.7: Reports Module*

The reports module is where users are able to view their financial data for analysis and reflection to understand themselves further. This module integrates their financial data to make sense of their personal finances for single working adults. It integrates the amount of inflow compared to their outflow, along with their comparison of their planned budget in each category with their actual expense under that category whether they are spending beyond their means. Depending on the expense records of the user compared to their planned budget, this module will signify to the user that they have saved a certain amount or o have overspent in that category which allows them to realize that they might need to cut back expenses under that category or adjust their budget plan accordingly. Users can adjust the date range of their choice to check their history and progress whether they have improved or not. This module will include a chatbot to provide a simple summary of user finances for their reflection to understand themselves further. Moreover, to be able to retrieve the necessary financial data, this module is connected with the cash inflow, expense, and budget module in order to visualize the data with graphs and comparisons.

The Reports Module effectively addresses various financial challenges faced by single working adults. Poor financial planning, difficulty in making financial decisions, and managing personal finances is mitigated by providing users with insights into their financial data, enabling them to analyze their inflow and outflow, compare their planned budget with actual expenses, and make adjustments accordingly. Lack of financial awareness, poor financial behavior, and lack of self-understanding is addressed by visualizing financial data through graphs and comparisons, enhancing users' understanding of their financial behaviors and habits. Overspending and impulsive spending behavior are addressed through clear indicators of spending beyond budget limits, encouraging users to practice mindful spending habits. Lastly, Difficulty in tracking expenses is alleviated by integrating expense records and budget comparisons, providing users with a comprehensive overview of their financial transactions and allowing for easy tracking and analysis.

#### 1.2.7.1 Chatbot

Upon initiating a conversation, users are presented with a series of predefined buttons representing different facets of their financial status. These buttons cover various aspects such as "Total Inflow," "Expenses," "Budget Comparisons," "Financial Goals," and "Progress Charts.". Users navigate through the chatbot's options by selecting the relevant buttons corresponding to their desired queries or actions. For instance, to view their total inflow for the month, users simply click on the "Total Inflow" button. Once a button is selected, the chatbot retrieves the corresponding financial data from integrated modules, including cash inflow, expense records, budget comparisons, and progress charts. After retrieving the data, the chatbot generates a summary related to the selected category. For example, if the user clicked on the "Expenses" button, the chatbot might provide a breakdown of their expenses by category along with any budget variances or overspending alerts. By limiting user interaction to buttons, the chatbot streamlines the process of accessing and understanding financial information, making it more accessible and intuitive for single working adults. This button-based interface eliminates the need for typing queries, ensuring a user-friendly experience conducive to effective financial management. In essence, the chatbot in the Reports Module serves as a virtual financial assistant, empowering users to gain insights into their financial status, make informed decisions, and take proactive steps towards achieving their financial goals.

## 1.3 Design Architecture

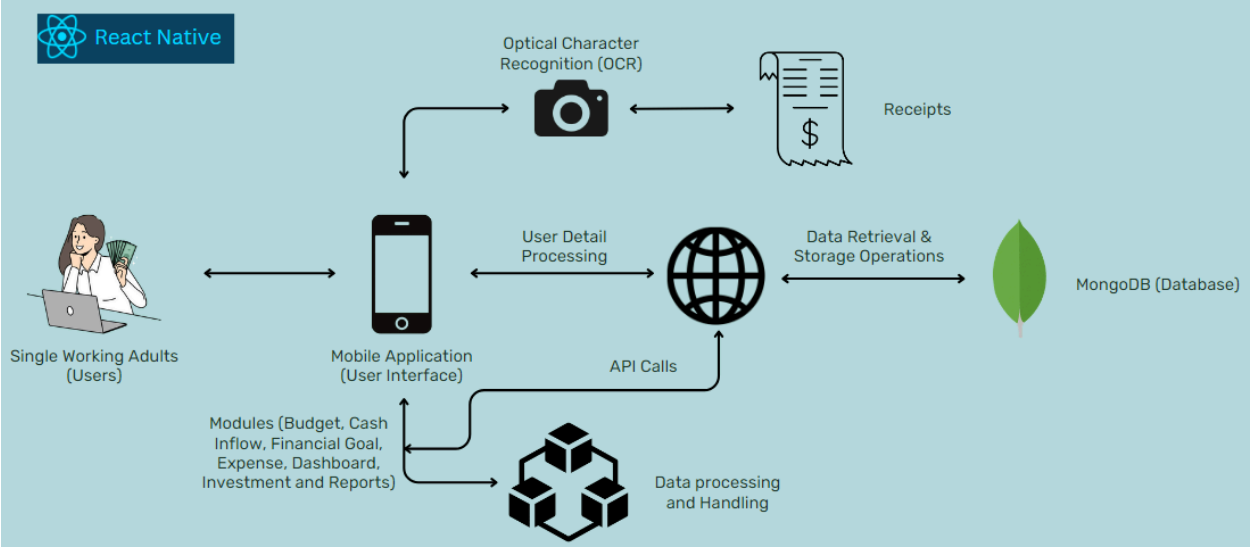


Figure 1.8: Design Architecture

The design architecture of the platform will employ separate components to ensure efficiency upon integration of the system. The system will use OCR to help with the entry of data. The system will also utilize a simple user interface for personal use of single working adults, including its functionalities that will be used in Budget, Cash Inflow, Financial Goal, Expense, Dashboard, Investment and Reports modules, integrated with personal informatics. The system will also use API calls to retrieve necessary data. The system will be developed using React Native and will run on mobile devices. It will be available offline and online through mobile applications. For the backend MongoDB will handle the database and will be deployed in the google play store.

## 1.4 Objectives

### 1.4.1 General Objective

To develop a Personal Finance Management System with integrated personal informatics and optical character recognition to help with data entry that would aid adults in managing their finances and achieve their financial goals due to the stress they feel in doing so because of their financial situation and impulsive spending habits.

### 1.4.2 Specific Objectives

Specifically, this proposal aims to:

- Assess the level of financial situation among single working adults

- Identify the primary challenges and barriers faced by single working adults in managing their personal finances and financial well-being
- Research on the most appropriate analytical algorithms for assessing financial goals progress
- Design and implement a Personal Finance Management System with Personal Informatics and optical character recognition tailored to the needs of single working adults
- Conduct a system testing and user acceptance testing to ensure that the requirement conform with target users

## 1.5 Significance of the Study

The significance of the study is to attempt a solution on the current behavior of spending through introducing a mobile application that helps personal finances through personal informatics.

### 1.5.1 Significance for Single Working Adults

Managing finances provides the benefit of reducing having too much debt, income going towards unnecessary expenses, and paying interests. In this process, individuals realize which are important and which are not worth their income, becoming self-aware about their life. Furthermore, they develop a sense of financial security, prioritizing their goals over material goods (Alag, et al. 2020).

In the study about Personal Financial Management Among Private Employees, personal finance activities have reduced the stress of employees from unexpected expenses. Furthermore, it made it more possible to achieve their goals and invest for their future. They never had to rely on others for extra money when they don't have enough money since through these activities, they have determined that they have enough money for their needs. They have reduced their debts and increased their investment, improving the overall flow of their money (Alag, et al. 2020).

### 1.5.2 Significance for the Academic Community

The significance of the study for the academic community extends to providing valuable knowledge and practical insights into personal finance management that can benefit students and educators. By delving into the complexities of financial behaviors, budgeting strategies, and investment tracking methods, the study offers students a comprehensive understanding of real-world financial challenges and effective management techniques. Through related literature, practical examples, and theoretical frameworks, students can gain practical skills and knowledge that are directly applicable to their personal financial situations and future careers. Furthermore,

educators can leverage the findings and methodologies presented in this study to enrich their curriculum, develop engaging course materials, and facilitate interactive learning experiences that fosters finance and responsible money management among students. By incorporating these insights into finance,business,economics, or related courses, educators can empower students with the tools and knowledge they need to navigate complex financial landscapes, make informed decisions, and achieve their financial goals. In summary, the study serves as a valuable resource for academia, enabling students to develop essential life skills, educators to enhance their teaching practices, and institutions to fulfill their mission of preparing students for success in an increasingly complex financial world.

### 1.5.3 Significance for Other Researchers

The significance of this study to researchers lies in its potential to contribute to the ongoing research in the field of personal finance. By examining the challenges faced by single working adults in managing their finances and exploring the effectiveness of various budgeting strategies and financial tools, researchers can gain valuable insights into the factors influencing financial decision-making and outcomes among this demographic. Furthermore, the development and implementation of a personal finance system with integrated personal informatics provide researchers with a practice framework for testing and evaluating new methodologies,algorithms, and technologies aimed at improving decision-making and outcomes in finance among single working adults.

Researchers can further the findings of this study by conducting additional research to validate the effectiveness of the proposed personal finance system in diverse settings and populations. This could involve longitudinal studies to assess the long-term impact of the system on financial behaviors, outcomes, and overall financial well-being among single working adults. Additionally, researchers can explore the scalability and adaptability of the system to different cultural contexts, socioeconomic backgrounds, and technological infrastructures to ensure its widespread applicability and accessibility. Researchers can collaborate with industry partners,policymakers, and other stakeholders to integrate the findings of this study into broader initiatives aimed at financial literacy, inclusion , and empowerment among single working adults globally. By building upon the insights and methodologies presented in this study, researchers can advance our understanding of personal finance and contribute to the development of innovative solutions to address the evolving needs and challenges of today's single working adults in managing their finances immediately.

### 1.5.4 Significance for the Proponents

The significance of this study for the proponents lies in its potential to empower them with the necessary tools and knowledge to effectively manage their personal finances. By developing a mobile application that integrates budgeting, goal-setting, and financial tracking

functionalities, proponents can gain better control over their financial situation, irrespective of their income level or prior financial literacy. This application serves as a practical resource for proponents to set clear financial goals, track their progress, and make informed decisions about their spending and saving habits. Furthermore, the system's adaptability to individual preferences and circumstances ensures that it can cater to the diverse needs and goals of its users. Ultimately, by equipping proponents with the means to manage their finances effectively, the mobile application can contribute to their financial well-being, stability, and long-term success.

The study's significance also extends beyond mere financial management by providing an opportunity for proponents to apply lessons learned from other courses. For instance, individuals with backgrounds in software development, user experience design, or data analytics can gain valuable hands-on experience by contributing to the development and optimization of the platform. By bridging the gap between theoretical knowledge and practical application, the study enables the proponents to enhance their skills, expand their expertise, and cultivate a holistic understanding of financial management within a technological context. This interdisciplinary approach not only enriches the learning experience but also equips proponents with a versatile skill set that is applicable across various domains and industries, thereby fostering their personal and professional development.

## 1.6 Scope and Limitations

### 1.6.1 Process Scope & Limitations

The application will involve the process of personal financing for single working adults. Processes that are required of the single working adult is setting up new accounts for the cash inflow module and allocating budgets to different categories of expenditures on a given date. Furthermore, the adult can also set up their financial goals through the system and track and monitor their progress as they continue to allocate a targeted amount of money. Thus, performing financial decision making as they achieve their financial goals. This will only be limited to saving money, saving for emergency funds, and saving for an item. Financial activities will also be performed by the user through following the short term goals set by the system. Single working adults will be tasked to save a specified amount of money that will contribute to the long term goal of the single working adult. Other financial decisions that the single working adult will make is deciding the best financial action when assessing and understanding their current financial status. This can be in terms of their expenditures, cash inflow, and investments that can be viewed through visualizations.

### 1.6.2 User Scope & Limitations

The study targets Filipino single working adults ranging from ages 18–65 who are currently experiencing heightened concerns about their financial well-being. These individuals

should be unmarried and not part of a family unit when it comes to managing personal finances. These working adults should have a source of income from their jobs and currently have financial responsibilities to fulfill (e.g. budgeting, paying bills, saving, investing, etc.). Their income level doesn't matter as long as the users would want to improve their financial well-being due to their current circumstances. They are expected to have a smart mobile device and know how to operate them to help in using the app. But this will only be available to working adults that have an Android device due to the limitations of the programming language that will be used to develop the system.

### 1.6.3 Data and Information Scope & Limitations

The data that will be gathered and processed by the application will be limited to single working adults' input, which includes personal information like name, username, and password. This will be gathered through user registration. Financial data will also be collected by the system. Some of these data include expense, cash inflow, investment, budget plan, financial goal information, and date range.

The financial data that will be collected from single working adults will be further processed by the application to create meaningful insights and information about the user's financial activities. Which further helps single working adults understand themselves, reflect on their situation, and make informed decisions in budgeting for their financial well-being.

The data gathered from users will be stored in an online database named MongoDB, which will be accessed by the mobile application and presented to the users of the application. The financial data (cash inflow and expense) that will come from the user can either be manually inputted by the user or the user can utilize the Optical Character Recognition Technology for efficient inputting of data. Due to limited API’s available and applicable in the Philippines, the investment module will only track crypto and stocks readily made available from the available API.

# CHAPTER 2

## 2.0 Review of Related Concepts

### 2.0.1 Personal Informatics

The term “Personal” refers to concerning oneself as an individual person privately without being concerned about being a member of a group or public. The term “informatics” refers to the branch of study about information in regards to its structure, properties, meaning and how it is stored and processed (Dudley, 2018). Similarly, Walters (2014) defines informatics as the science that looks into information and how it is gathered, stored, retrieved, and classified.

Personal Informatics gathers personal relevant information from people for the purpose of self-reflection and self-knowledge. This concept highlights the primary objective of Personal informatics towards the users and that it is to enhance their life based on the renewed self-understanding collected by the system (Rapp & Tirassa, 2017) (Dudley, 2018)(Cuttone et al, 2014).

An effective personal informatics system aids individuals in collecting relevant personal information and using it for their personal reflection and understanding (Kersten-van Dijk et al, 2017) (Dudley, 2018). People have been utilizing personal informatics systems for self interest and collecting information that would improve their personal dealings. They have their own varying reasons for obtaining personal information about themselves like improving current health conditions. It shows their motivation in finding meaning in their own personal life and interest in bringing change to their lives (Dudley, 2018).

Personal Informatics with the help of current technologies such as sensors and portable devices open new opportunities towards helping people in gathering personal data to help them gain self-reflections and further understanding (Cuttone et al, 2014) (Rapp & Tirassa, 2017) (Kersten-van Dijk et al, 2017). Varying personal data such as glucose level, steps count, food, sleep, and even places that a person has visited can be tracked and gathered by ubiquitous technologies. The primary purpose of gathering this data is to help individuals understand themselves better and find the necessary reflection to influence their lives towards a better goal (Rapp & Cena, 2016).

With its further development and various data available for tracking, PI has the means of supporting people in their objectives to make sense about themselves. Furthermore, personal informatics data is not just limited to an area of a person's life where that PI data is available for tracking. There are also different areas of people's lives where that data can affect and be useful to them. Also, sometimes the obvious data considered for tracking in an area, like health, in PI Systems might not reveal much value towards users. Instead, tracking external factors unrelated

to that area might help identify hidden relationships with other variables and offer useful insights (Dudley, 2018).

In present time, personal informatics is designed to focus on gathering personal data and displaying post-activity statistics for personal viewing. The most prominent example of its application is on automation with automated data collection from tasks performed manually before (Rapp & Cena, 2016). There is a lack of research on complicated types of system support that draw inspiration from intelligent computing techniques. Considering that personal informatics is a field where individuals can voluntarily give richer data, this presents a compelling argument for the application of intelligent computers (Ohlin & Olsson, 2015).

A fast and increasing development of Personal Informatics Systems is in the context of finances, tracking incomes and expenditures through applications. Development around smart appliances includes sensing devices that promote awareness in terms of consumption around fixed expenses such as water, gas, or electricity which influences their decision in energy-saving (Dudley, 2018).

### 2.0.2 Personal Finance Management, Financial Informatics, Self-Monitoring, and Feedback Interventions

Personal Financial Management is the act of achieving financial independence through responsibly handling financial situations (Munohsamy, 2015). It addresses financial management in every part of life. Budgeting, saving, investing, managing debt, and other personal finance-related activities enable us to effectively manage the finances of our household and assist each person reach their financial goals (Munohsamy, 2015)(Yogasnumurti et al., 2021). Essentially, personal financial management refers to the systematic approach of managing income and distributing costs by means of an elaborate financial plan. A methodical approach to using income is to learn how to track the money that comes in and adjust how it is used to meet expenses (Munohsamy, 2015).

Meanwhile, financial informatics, as similarly defined by Flood (2009) and Walters (2014) is the structure and behavior of systems in terms of managing financial data such as gathering, processing, communicating, manipulating, and classifying such data. Walters (2014) further defines Financial informatics as the intersection of finance and computer science. He further explains that under the modern financial industry, Financial informatics is a crucial discipline.

Besides personal management systems and finance informatics, Personal Informatics is also related with two other techniques that are available in changing a person’s behavior. These

techniques are self-monitoring and feedback interventions. The act of observing and recording one's own action, thoughts, or emotion is considered as self-monitoring. This standalone intervention technique is widely used by people for observing and recording behavior due to the reactive effects it offers which often changes their behavior in a good way. The phenomenon where facilitating a behavior creates a major change in that behavior is called reactivity. Self-monitoring is a component of personal informatics systems where it supports and enhances tracking personal behaviors (Kersten-van Dijk et al, 2017).

The second similar technique is the feedback intervention. In this technique, users receive feedback about their monitored behavior that they want to change for good. An example are students receiving feedback about their performance which in return improves their study. The effects of traditional feedback interventions have been examined well, while having technology applied isn't sufficient yet. In these studies, the effectiveness of feedback intervention is evident. However, the effects identified were more unfavorable rather than helpful(Kersten-van Dijk et al, 2017).

There are many explanations on how these two techniques have effects on behavior. For instance, in self-monitoring, its effects can be explained by Bandura's self-regulation theory where monitored behaviors are compared with goals and by doing so will set it on the right track. It is an operant conditioning where a negative progress towards the goal is identified as punishment and discouraged, while the opposite is seen as reward which reinforces the behavior. Another explanation of the effectiveness of these techniques partly may also come from the observer effect where being observed by self or by technology engages people to stick with social norms. In the perspective of the dual-process model, these two techniques prevent the automatic habit decision making of people allowing them to have a rational process in choosing their actions, changing their behaviors. In the perspective of self-improvement hypothesis, these two can offer precise and useful information for altering the focused behavior. However, the effectiveness they offer in changing behavior may not come from data-driven insights (Kersten-van Dijk et al, 2017).

### 2.0.3 Personal Informatics Models

There are two existing models proposed for Personal Informatics. The stage based model by Li et al.(2010) and the lived informatics model by Epstein et al. (2015). Both models, regardless of their difference in perspective of Personal Informatics, support the self-improvement theory of gathering and inspecting personal data that is necessary for self-insight, and facilitating behavior changes (Kersten-van Dijk et al, 2017).

#### 2.0.3.1 Stage Based Personal Informatics Model

Li et al. (2010) model about Personal Informatics systems shows a process that has five stages which are preparation stage, collection stage, integration stage, reflection stage and actions stage.

##### 2.0.3.1.1 Preparation Stage

This is the starting stage before users can collect information that is personal and related to what they need. It is concerned with the motivation of people in regards to collecting personal information which allows them to determine what information they need to record and how(Ohlin & Olsson, 2015) (Li et al, 2010).

###### 2.0.3.1.1.1 Preparation Barriers

Identifying the information to collect and tool to use for collection is a barrier in the preparation stage. A problem occurs when the tool chosen for collection gathers inadequate information. A change in tool is needed as action which brings the issue of data exports and integration due to having different systems and formats. Not all systems support data exporting and even if they do, it is not ensured that all systems have conforming data format. This concern could result in abandoning previous data (Li et al, 2010).

##### 2.0.3.1.2 Collection Stage

The collection stage is when the users start collecting and gathering information about themselves related to the motivation identified in the preparation stage (Ohlin & Olsson, 2015. Users are expected to place inputs of their personal information with varying frequency of collection depending on the motivation. It could be several times a day, weekly, or even monthly (Li et al, 2010).

###### 2.0.3.1.2.1 Collection Barriers

In terms of collection stage, most problems are related to the tools being used to collect information. Other barriers are related to users' shortcomings such as forgetting, time-constraint, or lack of motivation in collecting information. Some are data-related wherein the collected data may be collected under subjective estimation with no proper basis (Li et al, 2010).

#### 2.0.3.1.3 Integration Stage

The Integration stage is where the collected personal information is being processed from being raw data to a formed result. Useful information is prepared, put together, and reorganized to be used for reflection. This process can be both long and short, depending on how much effort a user will do in preparing the collected information for the next stage. This stage usually goes together with the collection stage so that applicable information is easily accessed for analysis (Ohlin & Olsson, 2015)(Li et al, 2010).

##### 2.0.3.1.3.1 Integration Barriers

The barriers of integration stage is that it becomes a wall from moving towards the reflection stage from the collection stage. Many people encounter problems in their collected data when it comes from multiple inputs and reflection data come from multiple outputs. The format of collected and for reflection data does not match which confuses users (Li et al, 2010).

#### 2.0.3.1.4 Reflection Stage

The Reflection stage is where the data from the previous stage is used for analysis and for users to reflect on for their self improvement. It involves analyzing the information sometimes with visualization for better understanding. It could be a short-term reflection where users can immediately reflect on the information or long-term wherein extensive self-reflection with week-long data gathering is required to identify patterns or trends before making an assessment on how to change behavior (Ohlin & Olsson, 2015)(Li et al, 2010). In personal informatics systems, there are six kinds of questions that must be reflected upon by users: 1) Status, on their current situation 2) History, on their situation in the past 3) Goals, on their situation that they aim for 4) Discrepancy, on their status versus their goals 5) Context, on the what affects their status, and 6) Factors on what other attributes are related.

##### 2.0.3.1.4.1 Maintenance and Discovery

Under the reflection stage are two subtypes. Li et al made changes in their later work, based on a transtheoretical model of behavior change. The discovery phase is where some users, while using PI, identify potential new actions in changing behavior toward their goal. While the maintenance phase is where other users, instead of encountering the discovery phase, may have been following an existing but improved

behavioral routine that ensures behavior changes rather than having a new action routine (Kersten-van Dijk et al, 2017).

**2.0.3.1.4.2 Reflection Barriers**

The barrier of this stage is that users could not understand themselves further. These challenges emerge because of lack of time or difficulties assessing, exploring, and interpreting their information (Li et al, 2010).

**2.0.3.1.5 Action Stage**

This is the last stage where people act upon the newly found understanding of themselves based on the previous stage. They use what they have reflected on to track their progress toward their goal and set their next step and decision toward the path that would help them achieve it (Ohlin & Olsson, 2015)(Li et al, 2010).

**2.0.3.2 Lived Informatics Perspective**

A different conceptual model of tracking people's everyday activities is the lived informatics model. This model goes further on the conceptual understanding of tracking technology usage of people. It emphasizes the value of designing tracking goals that extend outside modifying behavior, of tracking simultaneously with insight, and of providing help for people that experience lapses in their tracking (Epstein, 2018).

Epstein, based on surveys and interviews with self-trackers, argued that PI systems don't necessarily need to stick on the division of the stages of Li et al. (2010) stage-based model. Thus, the lived informatics perspective was suggested as an alternative model of PI. This model looks deeper into the lapses and interruptions of tracking activities, while also emphasizing that the structure of collecting and reflecting on data is flexible together rather than strictly separate stages (Kersten-van Dijk et al, 2017).

**2.0.3.2.1 Lived Informatics Process**

The process of lived informatics model includes deciding to track, selecting tracking tools, then moves to tracking and acting which is considered as the collection, integrations, and reflection stages of Li et al.'s model, and then lapsing in tracking but is subjected to resume (Epstein, 2018).

#### 2.0.3.2.1.1 Deciding

A part of Li et al's preparation stage that Epstein considered as important is the deciding stage. It refers to the decision of people to track their personal data for varying purposes such as sharing and looking into each other's. This stage can either be a starting point for tracking or a returning point after previously having experience tracking before (Epstein, 2018).

#### 2.0.3.2.1.2 Selecting

A second part that Epstein divided from the preparation stage of the stage based model is the selecting stage wherein after deciding to track, people recognize the need for a tool for tracking. Deciding and Selecting are commonly coupled together as they sometimes happen at the same time. Selecting tools may depend on the preference and subjective decision of the user such as looking into its features, aesthetics, or benefits.

#### 2.0.3.2.1.3 Tracking and Acting

In this stage, while tracking and acting, people start to learn about themselves and make the necessary adjustments while still collecting and integrating their personal data. Epstein defines this process as the ongoing process of collecting, integrating, and reflecting which he did not separate into continuous stages as he believes that they happen together.

#### 2.0.3.2.1.4 Lapsing

A key stage being looked further in this model is the stage wherein a person stops at actively tracking themselves is called lapsing. Others see this act as a temporary break of tracking while others permanently stop in doing so. Lapsing is categorized into four which is forgetting, upkeep, skipping, and suspending (Epstein, 2018).

#### 2.0.3.2.1.5 Resuming

Forgetting to track can be considered as short-term lapse which should be followed by resumption of tracking. However, there are cases wherein people completely stop from doing so. Although, in some cases, there are those that return within months which is considered a longer lapse. In this case they may resume either integrating or reflecting with the data they left behind, then deciding later if additional collection is needed and changing tools for tracking rather than resuming straight back to collecting additional data with the same tool (Epstein, 2018).

### 2.0.4 Goal Setting

Setting a goal provides individuals a specific target to work hard for and achieve, usually with a limited timeframe. It is an effective way for individuals to have clear decisions and steps towards achieving that goal. Goals can be anything from being a target for a physical need, personal values, or physical objects. In personal informatics, goal setting has been widely used as a tool in its design to motivate and understand the behavior of individuals. For instance, physical activity trackers are used for counting steps which is helpful in motivating people to attain healthy well-being. Further research developments on how tracking can best support individuals with the process of goal setting, measuring, and tracking have expanded the understanding from quantifiable goals to qualitative and eudaimonic goals (Ekhtiar et al., 2023).

## 2.1 Gaps Existing in Other Personal Informatics Systems

According to (Carlhenricsdotter, 2022) and (Cuttone et al, 2014), many researchers identified that challenges arise with integrating and interpreting data. Among these challenges, one of them shows that collecting data and creating clear and understandable visualization of personal data is difficult and makes tracking redundant. Another challenge is around understanding statistical data. Many users find it uncomfortable in analyzing and interpreting such data for self-reflection by themselves in personal informatics systems. Understanding such data is important in Personal Informatics since it is fundamental for individuals to reflect on data being tracked (Carlhenricsdotter, 2022)(Cuttone et al, 2014). Being able to reflect from it provides individuals an idea of their everyday lives such as their patterns, behavior, or habits which they can make changes and answer with their own actions. Thus, they suggested that the design of self-tracking should move from retrieving information to being available for self reflection. However, PI systems provided no suggestion on actions according to user data which implies barriers for both reflection and action stage, depriving users on how to make necessary changes. While advances have been made in devices for collecting and quantifying data, the process of analyzing and providing feedback to users fell behind. Useful insights may be included into the feedback system of the future. As a result, the presentation and analysis of the data are integrated, for instance to produce actionable insights that should be enhanced for upcoming iterations of self-tracking devices.

## 2.2 Recommendations for Personal Informatics Systems

Many designs have been recommended to help users in understanding themselves with the data they are tracking and for them to reflect upon it. Among these is using charts and visualizations with comparative measures which have been prevalent in many PI systems since many find it easy to interpret data like identifying habits, behavior, or patterns when using these designs. The comparison of data with one's own average or of the general population's

measurement allows users to quickly understand which of their areas might need action. Another recommended form of design is implementing the use of avatars which knocks on the user's empathy helping evoke reflection through visual characters. Using informal language instead of formal also helps users understand data (Carlhenricsdotter, 2022)(Cuttone et al, 2014).

In terms of visualization, Cuttone et al. further elaborated designs that would aid users in understanding the information provided for them. They recommended making data interpretable at a Glance. People wish to get the answers to their questions as quickly and easily as possible. Data visualizations are therefore designed to be easily interpreted at a glance. Another recommendation is to enable pattern analysis of time series data. There are two main types of patterns to look for: 1) periodic patterns such as hourly, weekly, or yearly and global trends, which looks at how variables increase or decrease over a period of time.

Kersten-van Dijk et al ( 2017) suggested that PI systems should support users throughout all the stages since current PI systems are insufficient in terms of doing so in other stages. For instance, is the transition of PI systems from contemplation to preparation and action for users. Another is to focus on self-efficacy which is an important factor of a successful change in behavior, suggesting to optimize PI systems in boosting an individual's confidence in their ability to change.

## 2.3 Predictive Analytics

Predictive analytics, a subset of advanced analytics, utilizes statistical techniques, data mining, machine learning, and artificial intelligence to forecast future events by analyzing current and historical data (Kumar & Garg, 2018). By examining various data types such as economic indicators and unstructured data like texts and images, predictive analytics identifies risks, opportunities, and relationships among factors, enhancing decision-making accuracy (Halladay, 2013). Technological advancements have facilitated the development of powerful predictive models capable of handling vast datasets. Extracting actionable insights from this data is crucial for establishing trust and reliability in predictive analytics.

### 2.3.1 Predictive Analytics Goals

The objectives in predictive analytics may range from capital allocation to risk avoidance, expense reduction, or customer retention. After defining the prediction goal, a suitable model is selected to develop theories about future variable performance and suggest corrective actions using statistical techniques. These theories, represented by formulas or algorithms, are based on analyzing historical data to understand variable performance. Through these formulas, the model correlates variable performance with changes in internal or external

factors, validated through statistical methods like Monte Carlo analysis. Ultimately, predictive analytics aims to inform decision-making by predicting future outcomes and identifying strategies to achieve desired goals (Halladay, 2013).

### 2.3.2 Predictive Analytics Process

#### 2.3.2.1 Requirement Collection

Before constructing a predictive model, it is essential to specify the intended prediction objective and the type of knowledge expected to be acquired through the prediction process (Kumar & Garg, 2018).

#### 2.3.2.2 Data Collection

Once the requirements of the client organization are understood, the analyst gathers datasets necessary for developing the predictive model, possibly from various sources. This could include a comprehensive list of customers interacting with the company's products, presented either in structured or unstructured formats. The analyst then verifies the collected data at the client's site (Kumar & Garg, 2018).

#### 2.3.2.3 Data Analysis and Massaging

Data analysts process collected data, converting unstructured data into a structured format for analysis and model utilization. Subsequently, the quality of the structured data is assessed (Kumar & Garg, 2018).

#### 2.3.2.4 Statistics, Machine Learning

The predictive analytics process utilizes various statistical and machine learning techniques. Key methods include probability theory and regression analysis, along with popular machine learning tools such as artificial neural networks, decision trees, and support vector machines (Kumar & Garg, 2018).

#### 2.3.2.5 Predictive Modeling

During this phase, a model is constructed using statistical and machine learning methods alongside a sample dataset. Subsequently, the model is evaluated against a test dataset, typically a subset of the main dataset, to assess its validity. If successful, the model is deemed suitable (Kumar & Garg, 2018).

#### 2.3.2.6 Prediction and Monitoring

Following successful prediction tests, the model is implemented at the client's site for daily predictions and decision-making. It generates results and reports for managerial use. Continuous monitoring is conducted to verify the accuracy and reliability of the model's predictions (Kumar & Garg, 2018).

### 2.3.3 Predictive Algorithms

#### 2.3.3.1 Linear Regression

Regression analysis is a tool that focuses on the relationship between variables and using them to make predictions. Simple linear regression is establishing a linear relationship between two variables which only uses one variable in predicting another. Multiple linear regression is when more than two variables are needed to make predictions. Supposedly, there should be no limit in terms of independent variables for analysis under MLR. The equation of simple linear regression can be used and be applied for multiple dependent variables through this way (Nugus, 2009). $Y = A + A_1X_1 + A_2X_2 + \ldots + A_nX_n$

#### 2.3.3.2 Time Series Analysis

Time series analysis is a statistical technique that predicts a value through analyzing trends or past movement of the observed object (Nugus, 2009)(Jaramillo, 2024). This technique identifies the patterns of consumption in different variations such as daily, weekly, or seasonal. Through time series analysis, there can be an efficient way of managing finances as this can forecast the consumption demand of a person (Jaramillo, 2024). The set of recorded observations of the analysis over time is a time series. Time series is a set of observed sequential data recorded at regular intervals over a period of time which is analyzed to identify shifting trends. This analysis utilizes linear regression and uses time, in this series, as the independent variable. Seasonal fluctuations, which can influence regression results, can cause adjustments in the data (Nugus, 2009).

#### 2.3.3.3 Random Forest Regression

Random forest is an algorithm that produces multiple decision trees. Decision trees that expand connected decisions (Alam, 2023) and act as an estimator to connect with other parallel tree models, allowing the RF model to integrate the trees to produce its output (Zhang, et al., 2023) through a majority vote (Yu, et al., 2023). The algorithm begins by creating K bootstrap samples, favoring a higher K value. For each sample, a decision tree model is constructed, but only a randomly selected subset of variables (p-variables) is considered for node splitting. The subset size (p) is much smaller than the total number of variables (t) in the dataset. The splitting variable at each node is chosen based on the highest information gain among the selected p-variables. This process repeats recursively for each child node, growing the tree without pruning until further growth is impractical. Finally, predictions for new data points are made by aggregating the class labels with the highest votes across all generated trees (Kadre & Konasani, 2021).

### 2.3.4 Rule-Based System

Reasoning in deductive learning involves the system using rules provided by a domain expert to analyze individual cases. Rule-based reasoning relies on facts and rules to solve problems, with deductive reasoning inferring conclusions from known information based on formal logic rules. The system utilizes user information, like transactions and lifestyle, to suggest spending habits. Two main inference methods are forward chaining, starting from given facts and using rules to produce more facts, and backward chaining (Althnian, 2021).

A rule-based system comprises a rule set, working memory, and an inference engine. The rule set contains rules, the working memory stores facts and intermediate results, and the inference engine performs forward chaining. The process involves matching rules with known facts, selecting a rule, and firing it if the antecedents match the assertions. Conclusions are then added to the working memory, with inference cycles continuing until no updates are made (Althnian, 2021).

### 2.3.5 Saving Advice

This is a feature that enables users to assess if they can achieve a particular savings goal. Users input the desired savings amount (AMOUNT) and the duration (DUR). The system evaluates these inputs based on the user's financial status and determines if the goal is attainable. If not, the system suggests alternative feasible plans by adjusting the duration (DUR) or reducing the savings amount (AMOUNT) (Althnian, 2021).

The monthly saved amount (MAMOUNT) is calculated as the total saved amount (AMOUNT) divided by the duration (DUR) (Althnian, 2021).

$$MAMOUNT = \frac{AMOUNT}{DUR}$$

Where:
AMOUNT = the target amount of money to be saved, entered by the user (Althnian, 2021).
DUR = the target duration of time in months (Althnian, 2021).

The available user's balance (BALANCE) from which the MAMOUNT will be deducted from is calculated as the subtraction of the total commitments from the current balance (Althnian, 2021).

$$BALANCE = INCOME - COMMIT$$

Where:

INCOME = The user's total income (Althnian, 2021).
COMMIT = the total amount of the user's total saving plans, current periodical expenses, budgets, and debts (Althnian, 2021).

The values MAMOUNT and BALANCE will be compared as shown in the saving advice rule set in the table below:

| Rule | IF | THEN |
|---|---|---|
| **R1** | $\frac{MAMOUNT}{BALANCE} < 1$ | The plan is feasible |
| **R2** | $\frac{MAMOUNT}{BALANCE} >= 1$ | The plan is not feasible. An alternative plans is suggested with longer duration and less MAMOUNT that fits the user's current situation. |

*Figure 2.1: Saving Advice Rule Set*

### 2.3.6 Spending Advice

The spending advice feature allows users to determine whether they should make a purchase. Users input the item's price (PRICE) and indicate its necessity. The system evaluates whether the purchase is advisable based on predefined rules. It calculates the available balance (ABALANCE) by subtracting total commitments (COMMIT) from the current balance (BALANCE) associated with the user's account (Althnian, 2021).

Where:
BALANCE = The current balance of the user (Althnian, 2021).
COMMIT = The total amount of the user's total saving plans, current periodical expenses, budgets, and debts. The system then reasons about whether the item should be purchased, according to the spending advice rule set table (Althnian, 2021).

| Rule | IF | THEN |
|---|---|---|
| **R1** | PRICE < ABALANCE | The user affords the purchase<br>&<br>Calculate percentage of the price to the user's total income<br>PRICE_INCOME_PER = (PRICE/INCOME) *100<br>&<br>Calculate percentage of the price to the user's available balance<br>PRICE_ ABALANCE _PER = $\frac{PRICE}{ABALANCE} * 100$ |
| **R2** | PRICE > ABALANCE | The user cannot afford the purchase<br>&<br>Calculate percentage of the unaffordable portion of price to the user's total income<br>PRICE_INCOME_PER= $\frac{PRICE-ABALANCE}{INCOME} * 100$ |
| **R3** | The user cannot afford the purchase<br>&<br>the item is necessary | The purchase is needed<br>&<br>An emergency budget is created by allowing the user to reduce amount allotted to one or more future commitment for the current month, i.e. periodic, budget, or saving plan. |
| **R4** | The user cannot afford the purchase &<br>the item is not necessary | The purchase is troublesome |
| **R5** | The user affords the purchase &<br>the item is necessary | The purchase is advisable |
| **R6** | The user affords the purchase &<br>the item is not necessary &<br>PRICE_INCOME_PER >10 | The purchase is not worth it |

*Figure 2.2: Spending Advice Rule Set*

## 2.4 Optical Character Recognition

The manual handling of receipts can be cumbersome and prone to human error and bias when entering data into an expense database. This process consumes valuable time and effort that could be better utilized elsewhere. Transitioning towards automation offers a more efficient solution (Neng, 2022). One key step in this transition is the conversion of scanned receipts into electronic data through Optical Character Recognition (OCR). By recognizing and extracting text

information from scanned documents, OCR streamlines processes across various sectors, including banking, legal, business, and healthcare (Kumar et al., 2020; Garcia & Claour, 2021; Neng, 2022). OCR is extensively utilized in practical commercial tasks such as recognizing business cards, vehicle number plates, and handwritten text on paper, as well as in sectors like banking, legal, business, and healthcare for automating processes and digitizing documents. (Garcia & Claour, 2021;Neng, 2022). However, receipt OCR outperforms other classical OCR tasks due to potential issues with scan quality, leading businesses to rely on human oversight for key information extraction in this area (Neng, 2022).

### 2.4.1 Preprocessing

Implementing OCR technology presents challenges that must be addressed proactively. Preprocessing images before conversion is recommended to improve character and word-level precision. This involves techniques like skew alteration, lighting correction, binarization, histogram equalization, and noise reduction to tackle issues such as skewness, blurring, and tilting (Neng, 2022; Garcia & Claour, 2021)

Achieving a high character recognition rate involves several steps in converting text from paper to digital format: (1) segmenting to separate text from the background, (2) normalizing to eliminate unnecessary data while preserving important information, (3) extracting features using methods like neural networks or Euler numbers, (4) classifying inputs using techniques such as Support Vector Machine or statistical methods, and (5) post processing, which includes manual verification through spell checking or content matching (Garcia & Claour, 2021).

### 2.4.2 Tools & Techniques

OCR involves multiple sub-processes such as text localization, character segmentation, and character recognition (Kumar et al., 2022). Classical image processing techniques have been suggested to enhance the clarity of character blocks for extraction, with a 99% character-level accuracy implying a 1 in 100 character uncertainty, while a 99.9% accuracy indicates 1 in 1000 characters being uncertain. Notably, Tesseract stands out as the top open-source option among five available OCR tools, which also include Azure Computer Vision API, Google Drive REST API, Text Fairy, and ABBYY FineReader (Neng, 2022).

OCR systems utilize various techniques, including Convolutional Neural Networks (CNNs) for character classification and predominantly for single character recognition in images. To accurately detect arbitrary character sequences, recurrent neural networks (RNNs), particularly long short-term memory (LSTM) networks, are commonly employed. Words are formed by grouping text lines into blobs and analyzing them for consistent text proportions. Text lines are then segmented into words based on the spacing between them. Recognition occurs in two stages: firstly, each word is recognized individually, and then perfect words are passed to an

adaptive classifier for training data. The input image is processed line by line, feeding into the LSTM model (Kumar et al, 2022).

### 2.4.3 Tesseract & OpenCV

Tesseract, an optical character recognition engine compatible with various operating systems, employs CNN and LSTM for precise text extraction from images. However, OCR engines like Tesseract experience reduced performance with images containing noise or shadows. Training models on blurred text images enhances accuracy and reduces error rates in such scenarios. Utilizing OpenCV, an open-source library for machine learning and computer vision, facilitates image preprocessing to mitigate noise. This involves steps such as determining regions of interest, cropping images, noise reduction, thresholding, dilation, erosion, and contour detection. Following preprocessing, OCR engines can accurately extract text from the images (Kumar et al, 2022)

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## 2.5 Artificial Intelligence Chat-bot

Chatbots, also known as conversational agents, utilize dialog systems to facilitate natural language conversations with users through speech, text, or a combination of both (Zhang et al., 2020). They represent the latest advancement in replicating human conversation, integrating Artificial Intelligence (AI) technologies and natural language processing techniques (Ekanayake & Saputhanthri, 2020). These chatbots can be trained to engage in conversations with humans across various domains (Zhang et al, 2020; Ekanayake & Saputhantri, 2020).

AI chatbots differ conceptually from embodied virtual conversational agents or avatars, which focus on synthesizing multimodal signals like images, videos, and sounds to mimic face-to-face human communication. They can be accessed anytime via mobile apps, making them a versatile tool across various sectors including business, governance, education, and healthcare. The convenience of chatbots as a digital communication platform has led to their use in delivering personalized behavior change programs, leveraging their computational power to tailor interventions based on individual characteristics and behaviors. This approach not only addresses the limitations of non personalized interventions but also enhances the effectiveness of health promotion and disease prevention efforts on a large scale.(Zhang et al., 2020).

A chatbot consists of three main parts: the Responder, the Classifier, and the Graphmaster. Users interact with the Responder through the interface, which then forwards the user's input to the Classifier for processing. The Classifier filters and standardizes the input before sending it to the Graphmaster, which acts as the chatbot's core, matching user inputs with stored information and sending responses back to the Classifier. Artificial Intelligence Markup Language (AIML) is utilized to encode the Graphmaster, showcasing their corresponding intents (Ekanayake & Saputhanthri, 2020).

## 2.4 Review of Related Domain

### 2.4.1 Personal Finance

Personal finance encompasses the strategic management of an individual's financial resources to attain both short-term goals and long-term financial security (Panos et al., 2020; Munohsamy, 2015). This multifaceted discipline involves prudent decision-making regarding income allocation, expenditure prioritization, and risk mitigation to optimize financial well-being (Panos et al., 2020). It extends beyond basic money management to include broader concepts of financial literacy and behavior, such as responsible debt management and planning for significant life events (Panos et al., 2020; Munohsamy, 2015). Through the cultivation of good financial habits and informed decision-making, individuals can achieve financial independence, stability, and resilience in the face of economic challenges (Panos et al., 2020; Munohsamy, 2015). Ultimately, personal finance empowers individuals to take charge of their financial futures, enhancing overall quality of life and well-being (Panos et al., 2020; Munohsamy, 2015).

### 2.4.2 Personal Financing in the Philippines

Personal finance in the Philippines is a multifaceted domain influenced by various demographic and socioeconomic factors. According to Cortez (2023), the study on the personal financial management practices of the personnel of the Bureau of the Treasury – Central Office sheds light on the financial behaviors and attitudes prevailing among Filipino workers in a government agency. The findings underscore the importance of financial planning, money management, income and asset protection, and investments in achieving financial security amidst economic uncertainties. Cortez (2023) emphasizes that while most respondents exhibit a proactive approach to financial planning and money management, there are notable differences in attitudes based on demographic factors. For instance, younger respondents aged 25 years old and below tend to prioritize day-to-day expenses over long-term investments due to the current global crisis, reflecting a survival mindset. Additionally, differences in financial attitudes are observed based on gender, civil status, educational attainment, and employment status. These insights highlight the need for targeted financial education and intervention programs tailored to the specific needs and circumstances of different demographic groups in the Philippines, ultimately fostering greater financial resilience and well-being among its citizens.

### 2.4.3 Age Groups

#### 2.4.3.1 Age Groups According to Generations

| Generations | Born | Current Ages |
|---|---|---|

| | | |
|---|---|---|
| **Gen Z** | **1997 – 2012** | **12 – 27** |
| **Millennials** | **1981 – 1996** | **28 – 43** |
| **Gen X** | **1965 – 1980** | **44 – 59** |
| **Boomers II (a/k/a Generation Jones)*** | **1955 – 1964** | **60 – 69** |

*Figure 2.4.3.1.1 Age groups according to generations with year born*

Generational groups, like Gen Z and Millennials, are created based on when people are grown-up, usually between 18 and 21. Beresford Research (n.d.) gives a breakdown showing the birth years for each group. For instance, Generation Z covers those born from 1997 to 2012, while Millennials range from 1981 to 1996, and so on. This breakdown helps understand who belongs to which generation, which is useful for things like market research.

Debczak (2024) talks about how generational labels are changing. She mentions the rise of Generation Alpha and explains the ages of different groups. According to her, Millennials turn into Gen Z around 1997 to 2012. She also discusses the starting point of Generation Alpha, which is debated, with some saying it began around 2010 or 2012. She also mentions the impact of events like COVID-19 on shaping generational identities.

So, while Beresford Research (n.d.) gives us a clear outline of generational boundaries, Debczak (2024) adds insights about the cultural and historical context shaping these groups. Together, these sources help us understand generational dynamics in today's world.

### 2.4.3.2 Financial Activities Based on Age Group

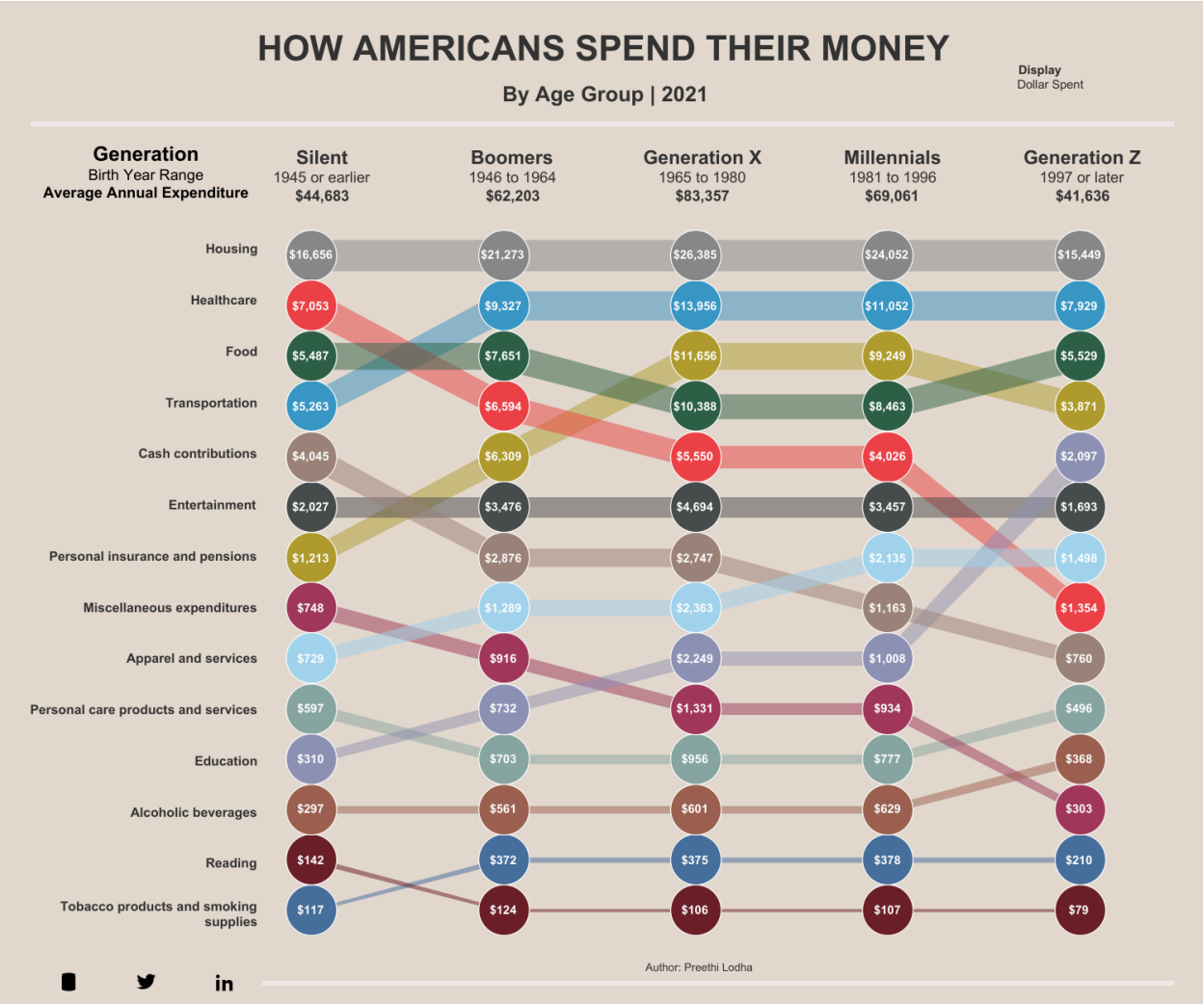



*Figure 2.4.3.2.1 Lodha, P. (2022, October 5). How different generations in the US spend their income*

Lodha (2022) delves into the diverse spending patterns across different generations in the United States, drawing upon data from the U.S. Bureau of Labor Statistics. The analysis reveals significant variations in annual expenditures among generations, with Generation X emerging as the highest spenders, allocating an average of $83,357 annually. Millennials follow closely behind, with an average annual expenditure of $69,061. Notably, Generation Z demonstrates the lowest spending habits, averaging $41,636 annually, although their expenditures are anticipated to rise as they progress in their careers. Despite these discrepancies, certain spending categories exhibit consistency across all age groups. For instance, housing consistently emerges as the most

substantial expense, constituting over 30% of total annual spending for every generation. Similarly, entertainment expenditures remain relatively stable across generations, with all age groups allocating more than 4% of their total expenditures to this category. However, healthcare spending presents significant disparities among generations, with the Silent Generation devoting a considerable portion of their budget—15.8%—to healthcare, compared to only 3.3% for Generation Z. It's important to note that while younger generations tend to spend less on healthcare, they are also less likely to be adequately insured, potentially leaving them vulnerable to substantial medical expenses in case of illness.

**Necessities**:

**Housing**: Generation Z spend the least ($15,449) likely due to factors like entering the housing market later or having smaller families. Spending increases with age, with Generation X spending the most ($26,385).

**Food**: Younger generations like Generation Z spend less on food ($5,529) compared to Generation X ($11,656), perhaps because they eat out more often.

**Healthcare**: Silent Generation spends the least ($7,053) while Millennials spend the most ($19,956).

**Transportation**:

Generation X spends the most on transportation ($10,388) possibly due to having children in the driving age or living in areas with limited public transportation options. Generation Z spends the least ($3,871) likely reflecting their use of ride-sharing services or living in walkable areas.

**Financial Planning**:

**Cash contributions**: The Boomers, likely due to having more disposable income or being more established in their careers, contribute the most to charity ($6,309). Cash contributions tend to decrease with age.

**Personal insurance and pensions**: Expenditure increases with age as older generations (Boomers at $2,876) plan for retirement and long-term care needs compared to Silent Generation who spend the least ($1,213).

**Lifestyle Choices**:

**Entertainment**: Spending on entertainment seems to decrease with age. Generation X spends the most ($4,694) while Generation Z has access to a wider variety of free or low-cost options, spending the least ($1,693).

**Miscellaneous expenditures**: It's difficult to pinpoint exactly why Generation X spends the most on miscellaneous expenditures ($2,363), but it could be due to having children in the home or spending more on hobbies.

**Personal Care**:

**Apparel and service**s: Millennials might have more disposable income or prefer name brands, reflected in their spending on apparel and services ($2,249). Silent Generation spends the least in this category ($729).

**Personal care products and services**: Generation X spends the most on personal care products and services ($1,331), possibly due to having children in the home or spending more on beauty products.

**Education**:

Millennials spend the most on education ($956) likely due to attending college or university more frequently.

**Other**:

**Alcoholic beverages**: Millennials spend the most on alcoholic beverages ($629), possibly reflecting their social stage.

**Tobacco products**: Smoking is less common among younger generations, leading to lower expenditure on tobacco products. Generation Z spends the least ($79) while Boomers spends the most ($124).

**Reading materials**: Interestingly, spending on reading materials is similar across all age groups. Except for the Silent generation which is 100 dollars below.

#### 2.4.3.3 Income Classifications According to Class

| Income Cluster | Per Capita Income | Monthly Income (for a family of 5) |
|---|---|---|
| Poor | Less than the official poverty threshold | Less than ₱12,082 |
| Low-income class (but not poor) | Between the poverty line and twice the poverty line | Between ₱12,082 and ₱24,164 |
| Lower middle-income class | Between two and four times the poverty line | Between ₱24,164 and ₱48,328 |
| Middle middle-income class | Between four and seven times the poverty line | Between ₱48,328 and ₱84,574 |
| Upper middle-income class | Between seven and 12 times the poverty line | Between ₱84,574 and ₱144,984 |
| Upper-income class (but not rich) | Between 12 and 20 times the poverty line | Between ₱144,984 and ₱241,640 |
| Rich | At least 20 times the poverty line | ₱241,640 and above |

*Figure 2.5 Understanding Social Classes in the Philippines: Where Do You Belong? by Venus Zoleta (Last updated April 12, 2023)*

In the context of the Philippines, social class refers to the stratification of society based on socioeconomic factors such as income, educational attainment, and occupational status (Zoleta, 2023). This classification system encompasses three primary categories: low-income, middle-income, and high-income, with an additional distinction between the low-income class and those considered impoverished (Zoleta, 2023).

For policy formulation and socio-economic analysis, the Philippine government employs the poverty threshold—a predetermined income level— to differentiate individuals or households above and below the poverty line (Zoleta, 2023). Those exceeding this threshold are considered non-poor by governmental standards.

Statistical data from the Philippine Statistics Office reveals that, as of 2020, approximately 43.5% of the Philippine population falls within the middle-income bracket, while 38.4% reside in the low-income category, and 16.7% are classified as the

most economically disadvantaged segment (Zoleta, 2023). A mere 1.4% are designated as high-income earners. This discrepancy in distribution is partially attributable to demographic factors, as elucidated by the Philippine Institute for Development Studies (PIDS), which notes larger family sizes among lower-income and impoverished groups (Zoleta, 2023).

In summary, social class in the Philippines serves as a framework for understanding societal stratification based on economic indicators. It informs governmental policies and interventions aimed at addressing disparities and enhancing the well-being of various segments of the population (Zoleta, 2023).

#### 2.4.2.1 In-Student Context

Personal financial literacy is a critical aspect of education, especially for students in the Philippines. According to Cipriano and Delgado Jr. (2024), Filipino students often lack essential financial knowledge, leading to detrimental financial management practices. This deficit in financial literacy poses significant challenges for students as they navigate their way through their academic years and transition into adulthood. The research highlights the importance of addressing these challenges to equip Filipino students with the necessary skills to make informed financial decisions.

One of the primary issues identified by Cipriano and Delgado Jr. (2024) is the limited access to financial education resources for Filipino students. Unlike other countries where financial education is integrated into school curricula, the Philippines lacks a formalized approach to teaching personal finance. Consequently, students often rely on their families for financial knowledge, which may not always provide comprehensive or accurate information. This reliance on familial guidance underscores the need for educational institutions to take a proactive role in offering financial education programs.

Salumintao and Cinches (2019) conducted a study focusing on the personal finance practices of millennial students in Higher Education Institutions (HEIs) in the Philippines. The researchers aimed to understand the factors influencing these practices and develop an exploratory model to explain them.

The study involved 641 students from seven HEIs in Northern Mindanao, Philippines. Through multiple survey questionnaires, in-depth interviews, and observations, data were collected to analyze the impact of financial literacy and financial socialization agents on students' personal finance practices (Salumintao & Cinches, 2019).

The findings revealed that financial literacy played a crucial role in shaping students' financial behaviors and attitudes. Students with higher levels of financial literacy tended to exhibit more responsible financial practices, including budgeting, savings, spending, borrowing, and repayment.

Moreover, the study highlighted the influence of financial socialization agents, such as parents, peers, and schools, on students' financial literacy and practices. Parents emerged as the most influential socialization agent, followed by schools and peers. Positive financial socialization experiences during childhood positively impacted students' financial practices in adulthood (Salumintao & Cinches, 2019).

Based on these findings, the study emphasized the importance of early financial education and continuous support from educational institutions and families in improving students' financial literacy and practices. Recommendations were made for HEIs to enhance their curriculum to incorporate financial education and organize regular forums for parents to enhance their financial knowledge (Salumintao & Cinches, 2019).

### 2.4.3 Financial Well-Being

Financial well-being is an integral part of an individual's total well-being. A state of being able to enjoy life while satisfying financial obligations and feeling comfortable in terms of long-term finances (Kulshreshtha et al, 2023)(Brzozowski & Visano, 2023). This concept acknowledges the various needs that money can satisfy, from the most basic needs for survival and security to more advanced needs like enjoying life as one pleases. In the last 20 years, financial well-being has been a trending topic in the academic community (Kulshreshtha et al, 2023). It refers to an individual's personal assessment of their current and future financial status involving subjective evaluations of attitudes, beliefs, and behaviors related to money management.

Income and savings are the two primary factors for determining Financial well-being. It signifies a state where an individual possesses adequate financial resources and the freedom to utilize them effectively. Three key drivers of financial well-being include active saving, exercising spending restraint, and avoiding borrowing for daily expenses (Versal et al, 2023). This subjective view means that individuals with similar incomes or debt levels may perceive their financial well-being differently (Bai, 2023). Indicators that contribute to financial well-being include punctual timely payments, self-discipline, attitudes towards impulsive spending, perceived financial efficacy, positive financial habits, risk-taking inclination in investments, and long-term financial planning. Moreover, factors such as financial experience,

knowledge, current financial standing, and marital status collectively impact an individual's financial well-being (Versal et al, 2023).

### 2.4.4 Financial Stress

Financial stress, often prevalent among individuals from lower-income households living paycheck to paycheck, poses challenges in meeting essential financial commitments. This strain, defined as the difficulty in fulfilling economic responsibilities, significantly impacts overall financial well-being, a crucial determinant of quality of life (Mukherjee et al., 2017). It is recognized as a key source of psycho-social stress due to its pervasive influence on daily activities linked to personal finances and their management, affecting both work productivity and interpersonal relationships (Bailey et al., 1998).

Studies have investigated the relationship between financial stress and financial behaviors, measuring stress through various stressors such as the inability to purchase clothing, discuss financial matters, pay utilities, save for emergencies, or cover medical bills, among others. The findings indicate a negative association between financial stress and positive financial practices, with heightened stress correlating to reduced savings, lower confidence in financial management, and increased instances of paying interest, making minimum payments, bouncing checks, and regretting purchases (Lim et al., 2017).

### 2.4.5 Financial Capabilities

Financial capability is a multifaceted concept encompassing various behavioral aspects related to resource management and decision-making. According to other researchers, financial capability serves as an interdisciplinary concept that integrates economics, psychology, and sociology, reflecting an individual's ability to take action within their available opportunities. Additional research has characterized financial capability as a composite skill, describing it as the combination of financial literacy and financial behavior to attain financial well-being. It comprises the financial knowledge, competencies, attitudes, and actions necessary for making prudent financial decisions tailored to individual situations, thereby enhancing financial well-being. Another definition suggests that financial capability encompasses individuals' comprehension of their financial situation, their capacity to grasp it, and their drive to take appropriate action (Muat et al, 2024).

### 2.4.6 Financial Behavior

Financial behavior encompasses actions related to managing money, including spending, saving, and budgeting (Acharya & Poudel, 2023; Muat et al, 2024). These behaviors are crucial for short-term goals like specific purchases and long-term commitments such as retirement savings (Muat et al, 2024). Effective management of these aspects is increasingly vital for individuals' financial well-being. Research highlights that adults with better financial

understanding and management of financial strain tend to make sounder financial decisions (Acharya & Poudel, 2023). Positive financial behaviors are associated with increased satisfaction with financial well-being (Acharya & Poudel, 2023). Financial habits like budgeting, saving, spending patterns, and credit management significantly impact young individuals' financial well-being, with a direct influence on financial stability (Muat et al, 2024). Modification of financial behaviors among young adults correlates with changes in their financial capabilities and overall well-being (Muat et al, 2024).

### 2.4.7 Financial Attitude

Financial attitude significantly shapes the financial capabilities of young adults, with positive shifts observed in those engaged in financial capability programs or education. This positive change in attitude is linked to improved financial performance. Research indicates that financial attitude strongly influences decision-making and correlates closely with financial management practices. Studies have found a relationship between financial attitudes and behavior, with enhanced financial knowledge leading to improvements in both attitudes and behaviors. This underscores the importance of addressing financial attitude in financial education programs to enhance the financial capabilities of young adults (Muat et al, 2024).

### 2.4.8 Financial Inclusion

Financial inclusion refers to the availability of a diverse array of financial services, with the proportion of adults possessing bank accounts serving as an indicator of a country's overall financial inclusivity. Several studies in the United States have investigated the financial inclusion of young adults and its correlation with their financial abilities, revealing a connection between owning bank accounts and financial capabilities. The integration of a savings account with financial education can foster positive financial behavior among young adults and enhance their financial capacities. However, solely providing financial education may not suffice to shape financial behavior and capabilities, particularly for young adults navigating uncertain economic circumstances. Scholars suggest combining financial knowledge with financial inclusion to facilitate better financial decision-making. This approach involves offering financial education alongside efforts to enhance institutional arrangements by establishing savings accounts for hands-on learning experiences (Muat et al, 2024).

### 2.4.9 Financial Socialization

Financial socialization, influenced by various agents such as family, school, peers, and media, plays a pivotal role in shaping individuals' beliefs, attitudes, and behaviors related to financial stability and well-being (Muat et al, 2024; Nguyen & Doan, 2020). This process serves as a precursor to developing favorable financial attitudes, which instill confidence in managing financial matters and contribute to overall financial well-being (Godase et al, 2023). Family-based financial socialization, particularly within foster care settings, has been shown to

impact young adults' financial capabilities positively (Muat et al, 2024). Additionally, consumer socialization, including exposure to various media sources, significantly influences financial attitudes and behaviors, with mass media playing a crucial role in shaping consumer skills and behaviors (Nguyen & Doan, 2020). However, there is a complex interplay among different socialization agents, with both positive and negative impacts observed on financial attitudes and behaviors (Godase et al, 2023).

### 2.4.10 Financial-Goals

Intrinsic goal and mindfulness are essential factors that promote well-being and conscious quality in SDT. The self-determination theory (SDT) framework was applied to FWB in this research. SDT can be employed to investigate self-motivation or self-determination of individual behavior and has been used to reveal how people improve their well-being through mental health self management. According to SDT, goals are not equal and intrinsic goals lead to subjective well-being by directly promoting basic needs satisfaction (Kasser & Ryan, 1996). For instance, in a large longitudinal study, Hope et al. (2019) demonstrated that intrinsic goals improve well-being by elevating the satisfaction of psychological needs. Further, open awareness is particularly valuable in fostering choice and engaging behaviors that align with values, interests, and basic needs of humans(Ryan & Deci, 2017). In consideration of such context, mindfulness facilitates the pursuit of intrinsic goals, leading to autonomy, competence, and relatedness needs satisfaction. This, in turn, promotes wellbeing (Ryan & Deci, 2017). Therefore, intrinsic goals and the psychological needs satisfaction were predicted to act as a chained mediating role between mindfulness and wellbeing in the current study.

#### 2.4.10.1 Intrinsic goals

Intrinsic career goals were defined as the degree to which a person’s job objectives satisfy his or her own needs and contain intrinsic motivational attributes, such as the consistent acquisition of new knowledge and skills, engaging in interesting and challenging work, and choosing jobs that contribute to society.

#### 2.4.10.2 Extrinsic goals

Extrinsic career goals are defined as the inclination to concentrate on external indicators of value in one’s career goals, which do not straight satisfy basic psychological needs, but are considered as a means to an end or substitutes to needs, such as apparent success, status in the company or society, and strong economic advantages.

#### 2.4.10.3 Goals Effects on Well-Being

Kasser and Ryan (1996) identified that the pursuit of intrinsic goals enhanced individual well-being, while the pursuit of extrinsic goals impaired well-being. Research on the workplace has suggested that people who take a relatively intrinsic goal

orientation tend to autonomously engage in work they enjoy and internalize their work as part of their self-identity. Such tendency produces positive outcomes and feelings of well-being (Deci et al., 2017; Zhang et al., 2018). Topics related to extrinsic goals are explored with regard to materialism (Ryan & Deci, 2017), which is negatively related to FWB (Helm et al., 2019).

### 2.4.11 Financial Activities and Financial Awareness

Financial activities and awareness are integral aspects of individuals' lives, shaped by their economic upbringing, knowledge, income generation, and practices related to managing, saving, borrowing, and spending money. Financial awareness, a fusion of financial knowledge and consumer behaviors and attitudes, holds significant importance in navigating the expanding accessibility of economic information and market offerings, as well as in making informed financial decisions. However, despite the increasing complexity of financial products, individuals may face challenges in selecting suitable options without adequate financial literacy. Therefore, cultivating financial awareness from an early age is essential (Smalej et al, 2022). Elevated financial awareness, acquired during youth, leads to improved decision-making and enhances overall quality of life. It encompasses knowledge, skills, attitudes, and behaviors necessary for sound financial decision-making. However, concerning deficiencies exist in financial awareness among both young individuals and adults, highlighting the need for improved financial literacy to enable wise investment, effective budget management, and market stability (Smalej et al, 2022).

### 2.4.12 Financial Planning

Financial planning is a collaborative process between an individual and a financial planner, incorporating the individual's relevant financial circumstances to help achieve their financial goals and objectives. Moreover, research indicates that financial planning involves determining financial goals, procedures, and budgets pertaining to financial activities. The financial planning process typically consists of six steps: setting goals, objectives, and priorities; collecting relevant data and information; analyzing information and assessing financial status; creating and presenting a financial plan for implementation; executing the financial plan; and monitoring the execution and reviewing of the financial plan. Two approaches to financial planning exist: individuals can either develop and manage the financial plan themselves or enlist the services of a professional financial planner. However, some studies highlight the complexity and challenges associated with individual financial planning, as many people lack the necessary knowledge to make informed financial decisions regarding investments, savings, risk management, and other aspects. Consequently, individuals may need to rely on a financial planner for financial planning services, advice, and recommendations, given the variety of financial products available in the market (Xian & Poh, 2022).

In previous generations, there was limited awareness of the importance of financial planning. However, with the proliferation of information and changing socio-economic conditions, people have begun to recognize its significance. Moreover, amidst global economic downturns, declining savings rates, and rising healthcare costs, effective financial management has become essential for achieving financial well-being. To engage in responsible financial management, particularly financial planning, individuals must possess both financial knowledge and a positive financial attitude. Those who exhibit traits such as planning, goal-setting, and frugality tend to save more and accumulate sufficient funds for emergencies and retirement compared to those who do not engage in such behaviors. Planning for the future and setting goals are linked to various positive financial behaviors, which are indicative of financial capability (Godase et al, 2023).

### 2.4.13 Financial Decision-Making

Personal finance decisions are essential aspects of individuals' lives, involving a broad spectrum of economic choices that are becoming increasingly diverse and complex due to the expanding range of financial products and services available. These decisions, ranging from everyday money management tasks like budgeting to more intricate matters such as retirement planning and estate management, remain significant throughout different life stages. The complexity of financial tasks typically escalates with age, with responsibilities often aligning with specific milestones like retirement or major life events. This progression underscores the importance of financial knowledge and expertise, which varies among individuals based on their exposure to different financial tasks over time (Hershey et al, 2015). In decision-making, individuals often prioritize expected utilities over values, with those knowledgeable about financial matters making more informed decisions for themselves and their families. Positive investment behaviors are linked to improved financial well-being, affecting various aspects of life satisfaction, psychological adjustment, and physical health throughout an individual's life cycle. Moreover, financial well-being influences individuals' evaluations and intentions regarding retirement investments (Yang et al, 2021; Virlics, 2013).

### 2.4.14 Saving

Personal savings, influenced by factors such as income levels and financial literacy, reflect an individual's financial management capability, as indicated by the life-cycle saving theory. This theory suggests that savings increase with income and are often earmarked for retirement support. Financial literacy plays a significant role in savings and household welfare, impacting retirement planning. Savings contribute to national development and economic growth by providing long-term capital and acting as economic protection during downturns. Policies supporting savings are essential for economic growth, with high savings rates indicating a flourishing economy (Nguyen & Doan, 2022). Personal saving behavior, driven by motivations and future responsibilities, demonstrates individuals' ability to balance their finances, often

measured by the regularity of saving activities to plan for uncertainties and future needs (Nguyen & Doan, 2022).

### 2.4.15 Spending

Spending is the behavior that encompasses the allocation of available funds, whether through cash, debit cards, or credit cards, to fulfill needs or desires. This behavior is dynamic and influenced by various factors including demographics like race, religion, family background, ethnicity, and geographical location. For instance, male students may move towards purchasing masculine items like expensive gadgets and watches, while female students may prefer decorative items such as makeup, cosmetics, and jewelry (Kumar et al, 2022). Numerous elements shape spending habits, including financial literacy, parental income, and peer influence. Researchers have extensively explored these dynamics. Success in life often hinges on one's capacity to manage financial behavior adeptly. Financial literacy serves as a pivotal indicator of both financial contentment and spending habits. Research indicates a significant correlation between financial management knowledge and spending behavior among university students (Kumar et al, 2022).

#### 2.4.15.1 Impulsive Spending Behavior

Impulsive buying, in general, is a behavior of buying unplanned or not well thought over purchases. It is a sudden or immediate purchase with no consideration of available better or cheaper alternatives (Jabar & Delayco, 2021) (Legaspi, Leoncio, & Galgana, 2016) (Ahn & Kwon, 2020).

Impulsive buying is also deeply rooted and related with the emotional decision-making of consumer behavior. They heavily rely on their emotions such as joy, love, hope, fear, and fantasy when making purchases and acquiring possessions. Instead of engaging in thorough research, thoughtful deliberation, and evaluation of alternatives prior to making a purchase, consumers frequently succumb to impulse buying, disregarding financial constraints and other potential consequences due to their emotional impulses. This tendency becomes more prevalent when individuals are experiencing stress or burnout from challenging circumstances, leading them to spend more impulsively (Ahn & Kwon, 2020) (Renamae et al, 2023).

Young adults may struggle with impulsive and excessive spending, even when faced with limited financial resources. Factors such as receiving allowances from parents and the lack of effective budgeting also contribute to this behavior (Labito & Supramono, 2017). Consumerism is the notion that one's degree of consumption, particularly the acquisition of material objects, has a significant impact on one's pleasure and well-being. People buy objects to satisfy needs and wants, but the moment they buy them, they are

never satisfied, so they proceed with the habit of engaging with another process of consumption (Glover, 2022).

### 2.4.16 Budgeting

Various studies have explored the multifaceted aspects of budgeting, shedding light on its significance in financial management. Labito and Supramono (2017b) emphasize the fundamental purpose of a budget, which is to regulate the flow of money. They highlight its utility in monitoring monthly expenses, enabling individuals to compare actual spending against planned allocations. Moreover, Galperti (2019) delves into the pervasive nature of personal budgeting within consumer behavior, emphasizing its practice of categorizing expenses and imposing constraints over defined time frames.

Furthermore, Aspinal et al. (2019) discuss the concept of a personal budget (PB) within the context of social care, delineating its role as a notional allocation to address an individual's care needs. This allocation can be received as a Direct Payment (DP), empowering recipients or their representatives to procure necessary services or equipment. Alternatively, individuals may opt for a managed PB, wherein the local authority arranges and pays for services on their behalf.

In addition, She et al. (2023) contribute insights into the relationship between financial knowledge, behavior, and accountability. Their study suggests that individuals with a higher level of financial acumen exhibit responsible credit card usage, timely bill payments, adherence to budgetary constraints, adept handling of financial matters, and informed decision-making regarding credit cards.

#### 2.4.16.1 Zero based budgeting

The zero-based budgeting approach is particularly suitable for individuals prone to excessive spending. It entails allocating income to various expenses each month until there's no remaining balance, ensuring every dollar has a designated purpose. If there's a shortfall in one category, adjustments must be made by redistributing funds from other areas according to priority and necessity. This method fosters discipline and encourages responsible spending. However, its effectiveness relies heavily on dedication and commitment, and it doesn't account for unexpected expenses or emergencies (Parr, 2019).

#### 2.4.16.2 Envelope System

The envelope system stands out as one of the simplest budgeting techniques, characterized by its active approach of assigning income to distinct virtual "containers." This method employs envelopes, hence its name, to allocate funds for various expenses. Each envelope represents a specific expenditure category, such as groceries, with a

predetermined amount of money allocated to it. Should an envelope be emptied, additional funds must be sourced from another envelope within the system. Notably, the envelope system serves as a practical tool for curbing overspending by imposing constraints on both individual category spending (per envelope) and overall monthly expenditure. Once an envelope is depleted, expenditure can only be drawn from other envelopes, effectively limiting monthly spending (McCaffery, 1984).

#### 2.4.16.3 50/30/20 Budget

The process involves first distributing 50% of the income proportionately among the items listed under the "needs" category. The second category, "wants," is restricted to 30% of disposable income, with the remaining 20% allocated to investments and savings. Step one entails calculating after-tax earnings, which is the portion of salary left after tax deductions. Step two involves limiting "needs" to 50% of after-tax income, prioritizing essential monthly expenses such as housing, food, medical, and transportation. Step three restricts "wants" to 30%, covering expenses that contribute to satisfaction but are not essential, such as phone, cable, internet, indulgences, and discretionary spending like trips or salon appointments. Step four allocates 20% to savings, aiming to set aside this portion of after-tax pay into a savings account for unforeseen circumstances, emergencies, or retirement (Parr, 2019).

#### 2.4.16.4 Half Payment Method

This budgeting approach is suitable for individuals receiving weekly or bi-weekly paychecks. It involves dividing expenses in half and ensuring that by the end of each month, the full amounts needed for various expenses are accumulated. This method facilitates the meeting of financial obligations while necessitating a high level of discipline (Parr, 2019).

#### 2.4.16.5 Goal-Driven Method

This technique involves taking financial actions aimed at achieving a specific goal. Developing a plan and saving with a clear objective in mind serves as a motivating factor. However, this method may not be efficient for tight budgets due to limited resources available for saving (Parr, 2019).

### 2.4.17 Investing

Investment is a strategic expenditure made in the present to yield future gains, encompassing fixed investments like buildings or machinery and monetary investments such as stocks or bonds, which are crucial for enterprise growth. Investments can be categorized as replacement, involving the replacement of physical assets, or net investments, which add new assets to existing ones. The decision to invest hinges on various factors, including profit

expectations, asset costs, financing availability, and financing methods (Virlics, 2013). Investors have a wide array of market options to align with their financial goals. Risky investments may lead to higher wealth accumulation over the long term, but they also expose individuals with long-term financial goals to risks such as inflation and longevity risk if their wealth is tied up in assets with low return rates(Mahdzan et al, 2020).

Recent studies have highlighted the significance of financial knowledge and investment literacy in shaping individuals' investment decisions. Lack of financial literacy is a prevalent issue among investors worldwide, prompting increased attention from scholars and policymakers to examine its importance in various financial decisions, including stock participation, mutual fund investment, retirement planning, credit card usage, and mortgage financing (Mahdzan et al. 2020; Yang et al. 2021). Predictors such as risk tolerance, herding behavior, and financial literacy play a crucial role in determining investment intention, which refers to an individual's inclination towards short- and long-term investment behaviors (Yang et al. 2021).

#### 2.4.17.1 Risk Tolerance

Financial risk tolerance refers to an individual's readiness to accept uncertainties in investment decisions, significantly shaping their investment behavior (Mahdzan et al. 2020; Yang et al. 2021). Investors with low risk tolerance may enter investments without fully understanding associated financial risks, impacting how they approach investments. Those with high risk tolerance often trade higher-value stocks, displaying greater flexibility, comfort with uncertainties, and a willingness to take risks, regardless of potential profits or losses (Yang et al., 2021). Risk tolerance is gauged by the highest level of volatility investors are willing to endure, with risk-averse individuals potentially refraining from entering the share market, despite the potential for higher returns. Ultimately, investors' attitudes toward risk shape their investment conduct (Mahdzan et al., 2020).

#### 2.4.17.2 Investment Behaviour

Various investors exhibit diverse investing behaviors, with some demonstrating a propensity for risk-taking by pursuing specific shares in hopes of realizing substantial gains within a short timeframe, although this approach may be perceived more as speculation than prudent investment. Conversely, others adopt a risk-averse stance, believing that no investment offers guaranteed returns over a brief period. This spectrum of investing behaviors intertwines with the realms of behavioral finance and economics, encompassing economic, cognitive, and emotional influences. Understanding these dynamics is crucial for discerning the investment decisions of Malaysian young working adults and their impact on stock prices, returns, and resource allocation. Consequently, investing behavior plays a pivotal role in shaping an investor's current and future

financial well-being, illustrating the inherent connection between investment and economics (Tat et al., 2023).

#### 2.4.17.3 Different Kinds of Investments

##### 2.4.17.3.1 Mutual Funds

Mutual funds, also called unit trusts in some areas, are favored by cautious investors over direct stock investments. They gather money from multiple investors and are overseen by professional managers. Investors can choose funds aligned with their risk tolerance and financial goals. Pooling funds enables diversification at a low cost, reducing financial risk (Mahdzan et al, 2020).

##### 2.4.17.3.2 Stocks

Common stocks, also referred to as equity securities or equities, signify ownership stakes in a company. Each common stock share grants its holder the right to vote on corporate governance issues at annual meetings and to a portion of the company's financial profits. Most major corporations offer their common stock for public trading on stock exchanges, allowing for easy buying and selling. Companies whose stocks are not publicly traded are considered closely held. The primary features of common stock investments are their residual claim on company assets and limited liability for shareholders (Brodie & Kane, 2020).

##### 2.4.17.3.3 Bonds

###### 2.4.17.3.3.1 Municipal Bonds

Municipal bonds are debt securities issued by state and local governments, resembling Treasury and corporate bonds with one significant distinction: the interest income generated from municipal bonds is not subject to federal income tax. Additionally, this interest income is typically exempt from state and local taxation within the issuing state. However, capital gains taxes may apply when these bonds mature or if they are sold at a higher price than the investor's purchase price. Municipal bonds come in two main types: general obligation bonds and revenue bonds. General obligation bonds are secured by the issuer's full faith and credit, backed by its taxing authority. On the other hand, revenue bonds are issued to fund specific projects and are secured either by the revenues generated by the project or by the specific municipal agency overseeing it. Entities such as airports, hospitals, and transportation

authorities often issue revenue bonds. While revenue bonds offer the potential for higher returns, they also carry a higher risk of default compared to general obligation bonds (Brodie & Kane, 2020).

#### 2.4.17.3.3.2 Corporate Bonds

Corporate bonds serve as a mechanism for private companies to raise funds directly from the public. Structurally, they resemble Treasury bonds, typically paying semiannual interest payments throughout their lifespan and returning the bondholder's principal investment at maturity. However, the key distinction lies in the level of risk they carry compared to Treasury bonds. Investors must consider the potential for default when investing in corporate bonds. These bonds come in several varieties: secured bonds, which are backed by specific collateral in the event of the company's bankruptcy; unsecured bonds, known as debentures, which lack collateral; and subordinated debentures, which hold a lower priority claim to the company's assets in case of bankruptcy (Brodie & Kane, 2020).

#### 2.4.17.3.4 Cryptocurrencies

Cryptocurrencies operate on a peer-to-peer basis, cutting out the need for intermediaries like banks. This means transactions can occur without the necessity of bank accounts or credit cards; instead, cryptocurrencies use digital wallets, akin to bank vaults, to facilitate exchanges. Beyond simply providing banking services to the unbanked, cryptocurrencies offer the potential for inexpensive, secure, and nearly instantaneous transactions. This opens up the world of internet commerce to billions of people, enabling them to engage in buying and selling goods or services outside of traditional banking and credit card systems. Each cryptocurrency operates under its own set of rules, with differences in consensus mechanisms, latency, and cryptographic algorithms. While cryptocurrencies lack intrinsic value, they rely on market acceptance and trust in the underlying technology rather than a central authority like a central bank. Transactions are validated by a decentralized network, with trust placed in the algorithm and blockchain network. Valid transactions must have sufficient funds, except for predetermined new issuances of the cryptocurrency (Härdle et al., 2020).

### 2.4.18 Factors Affecting Personal Finance

#### 2.4.18.1 Financial Education in the Family

Contrary to expectations, the study findings suggest that financial education provided within the family setting does not significantly influence personal financial

management among undergraduate students (Salsabilla et al., 2022). Despite the potential role of parents in imparting financial knowledge and skills to their children, the study did not find a significant correlation between family-based financial education and students' financial management practices. This indicates that factors beyond familial influences may have a more substantial impact on students' financial behaviors and decision-making processes (Novitasari et al., 2021).

#### 2.4.18.2 Financial Literacy

Financial knowledge and financial literacy are often used interchangeably, but they represent distinct aspects. Financial knowledge is one component of financial literacy, alongside financial awareness, behavior, and attitude (Muat et al, 2024). Financial literacy, which involves the competence and assurance to apply financial knowledge in decision-making, significantly influences various financial behaviors (Smalej, 2022; Xian, 2022). It encompasses understanding financial concepts, alongside financial awareness, behavior, and attitude (Xian, 2022; Muat et al, 2024). This understanding contributes to improved financial well-being by empowering individuals to make informed choices regarding financial planning, investment, risk management, and more (Xian, 2022) . Financial literacy is regarded as a crucial tool in enhancing financial decision-making, providing a structured approach to investments, instilling confidence in making rational decisions, and aiding in managing day-to-day financial tasks and emergencies (Yang et al, 2021).

Research indicates that financial literacy, encompassing comprehension of concepts like budgeting, savings, loans, insurance, and investment, plays a pivotal role in shaping the financial management practices of undergraduate students (Salsabilla et al., 2022). This underscores the notion that as students deepen their grasp of financial principles, they become better positioned to navigate financial choices with prudence and responsibility. The ramifications of enhanced financial literacy extend beyond individual financial well-being; they extend to the capacity to positively impact families and communities (Shirazian, 2018).

At its core, financial literacy equips individuals with the indispensable knowledge and competencies requisite for sound financial decision-making, encompassing proficient management of savings, investments, and overall finances (French et al., 2021). It involves a comprehensive understanding of financial concepts, principles, and tools, facilitating the formulation of robust financial plans and effective utilization of financial instruments (Yogasnumurti et al., 2021). In the contemporary global economy, the significance of financial literacy is underscored by the complexity of financial choices and the array of financial products available. A robust foundation in financial literacy

engenders a positive financial outlook and fosters overall financial well-being (Panos & Wilson, 2020). Furthermore, it empowers individuals to navigate intricate financial decisions, mitigate risks, and enhance financial health through prudent budgeting, saving, and investing practices (Bai, 2023). Ultimately, possessing financial acumen is imperative for making informed financial decisions and attaining financial stability (Panos & Wilson, 2020).

#### 2.4.18.3 Peers

Peer influence plays a crucial role in shaping students' financial attitudes and behaviors, with positive peer relationships correlating with better financial management practices (Salsabilla et al., 2022). Students who engage with their peers in financial discussions and activities exhibit improved financial management skills. Peers serve as sources of support and encouragement, fostering responsible financial behaviors and decision-making (Salsabilla et al., 2022).

#### 2.4.18.4 Hedonistic Lifestyle

A hedonistic lifestyle, characterized by prioritizing pleasure-seeking activities and indulgent spending, has a detrimental effect on personal financial management among undergraduate students (Salsabilla et al., 2022). Students who succumb to the allure of instant gratification and lavish spending tend to exhibit poorer financial management practices. This underscores the importance of promoting responsible spending habits and discouraging excessive consumption to improve students' financial resilience (Robb & Chy, 2021).

### 2.4.19 Self-Control

Self-control entails managing thoughts, emotions, and actions to achieve long-term goals and resist temptations. It involves rational thinking, impulse control, and effective money management. Studies have associated higher self-control with better financial outcomes, including increased assets, budgeting, saving, and disciplined spending. Enhanced self-control aids in financial planning and savings, contributing to financial security by encouraging saving and debt avoidance. Research indicates that individuals with greater self-control tend to make wiser investment decisions, leading to improved financial well-being. In summary, cultivating self-control is crucial for achieving financial stability and making sound financial decisions (Bai, 2023).

### 2.4.20 Financial Knowledge

Knowledge in the context of personal finance pertains to individuals' understanding of financial concepts relevant to their own financial affairs. It encompasses one's proficiency in

navigating the realm of finance. Effective management of personal finances hinges upon possessing adequate knowledge (Yogasnumurti et al., 2021).

### 2.4.21 Financial Capability

Financial capability encompasses the effective application of financial knowledge, skills, attitudes, and behaviors, leading to positive financial outcomes (French et al., 2021).

While being financially literate signifies possessing the necessary knowledge and skills, true financial capability also involves demonstrating appropriate attitudes and behaviors in financial decision-making (French et al., 2021).

This entails not only understanding financial matters but also applying this knowledge in everyday life and having the self-confidence to make prudent financial choices (French et al., 2021).

Additionally, factors such as attitudes, motivations, and biases influence financial capability, with non-impulsive individuals demonstrating better financial decision-making by prioritizing long-term financial well-being over immediate gratification (French et al., 2021).

### 2.4.22 Digital Literacy

Digital literacy, defined as the adept use of digital technologies to access, evaluate, and generate information, is crucial for fostering financially capable behaviors (French et al., 2021).

Employing tools such as budgeting and spending meters, financial goal trackers, and automated reminders can enhance money management skills and promote better financial control (French et al., 2021).

Additionally, digital technologies offer opportunities to prompt financial behavior changes through features like virtual price/product comparisons and just-in-time reminders at points of sale (French et al., 2021).

### 2.4.23 Financial Skills

Financial skills are essential for decision-making in managing personal finances, including tasks like budgeting, investing, selecting insurance, and managing credit (Yogasnumurti et al., 2021).

## 2.5 Review of Related Systems

### 2.5.1 Personal Finance Management Systems

A Personal Financial Management System, also referred to as personal finance software or a budgeting app, serves as a digital aid in effectively managing user finances. Acting as a centralized platform for all financial activities, it provides a range of features to keep users organized and well-informed. PFMS functionalities include tracking income and expenses, aids in budget creation and categorization, enabling users to identify areas for potential savings. Features such as bill reminders and payment scheduling ensure timely payments, while goal setting and tracking functionalities help users monitor progress towards financial aspirations. Some PFMS also offer investment tracking capabilities, catering to users with diversified portfolios. With this in mind, different personal finance management systems will be assessed on their functionalities and value towards adults in this section.

#### 2.5.1.1 Mint

Mint is a popular personal finance management system renowned for its user-friendly interface and robust budgeting tools. Offering accessibility at no cost, it appeals to individuals seeking an economical solution for managing their finances effectively. One of its notable strengths lies in its automatic features, streamlining the process by automatically downloading transactions from connected bank accounts and credit cards, thereby eliminating manual data entry. Additionally, Mint excels in budgeting functionalities, allowing users to categorize spending, set budget limits for various categories, and monitor progress towards financial goals. While it provides convenient bill reminders to prevent missed due dates, direct bill pay functionality is not yet available. Furthermore, although Mint offers basic credit score monitoring, its coverage may not be as comprehensive as dedicated credit monitoring services. Mint serves as an ideal choice for beginners or those seeking a straightforward and cost-free solution for tracking spending, categorizing transactions, and establishing budgets. However, users with more advanced needs, such as investment tracking or in-depth credit monitoring, may find other personal finance management system options better suited to their requirements.

#### 2.5.1.2 You Need a Budget (YNAB)

YNAB, or "You Need A Budget," offers a unique approach to personal finance management through its envelope-based budgeting system. With YNAB, every dollar earned is assigned a specific purpose, promoting mindful spending and preventing overspending in various categories such as rent, groceries, or entertainment. More than just tracking expenses, YNAB focuses on altering financial behavior by encouraging users to consider their spending decisions carefully before making purchases. While YNAB does require a subscription fee, many users find the investment worthwhile due to

its effectiveness in promoting responsible budgeting. Accessible through both a mobile app and web platform, YNAB provides convenient ways for users to manage their finances on-the-go or from the comfort of their computer. Though it lacks certain features like automatic bill pay or investment tracking, YNAB remains a powerful tool for individuals seeking to take control of their spending habits and adopt a proactive approach to budgeting and expense management.

#### 2.5.1.3 Personal Capital

Personal Capital stands out among personal finance management systems for its comprehensive budgeting functions for investors. Beyond basic budgeting and expense tracking, Personal Capital offers tools for investment tracking, net worth monitoring, and retirement planning assistance, making it a comprehensive solution for managing various aspects of financial life. Particularly valuable for active investors, Personal Capital tracks investment portfolios, providing a centralized view of stocks, bonds, and mutual funds. Advanced goal-setting features, including retirement planning calculators, aid in long-term financial planning. The platform also offers in-depth financial analysis tools such as fee analysis and investment checkups, facilitating informed decision-making and identifying areas for financial strategy improvement. Operating on a freemium model, Personal Capital provides basic features for free, while premium features like investment portfolio optimization and retirement planning require a subscription fee. Suitable for individuals seeking a holistic financial management solution, Personal Capital is ideal for those comfortable with its freemium model or willing to invest in premium features. However, users should consider the potential complexity and learning curve associated with its extensive features, which might be overwhelming for beginners. Overall, Personal Capital empowers users with the tools to manage their finances effectively, offering a powerful solution for those seeking a comprehensive view of their financial health.

#### 2.5.1.4 Quicken

Quicken is a personal finance software program renowned for its comprehensive approach to financial management. Packed with an array of features catering to diverse financial needs, Quicken extends beyond basic budgeting to encompass investment tracking, portfolio management, bill pay functionality, and tax planning tools (with additional costs). With investment tracking and management capabilities, users can monitor stocks, bonds, and mutual funds in detail, while portfolio management tools cater to those actively managing their investments. The platform's in-depth reporting and analysis functionalities enable users to generate detailed reports and charts, facilitating a deeper understanding of their financial standing and identification of trends. While

Quicken serves individuals in need of a comprehensive financial management solution, particularly those who actively manage investments and value detailed financial reporting and analysis.However, users should consider factors such as the subscription fee, learning curve associated with its extensive features.

#### 2.5.1.5 FinWise

Finwise, offers to cultivate financial literacy and decision-making skills for both children and parents. For children, the app allows them to input income, expenses, test their knowledge, and set financial goals, while also offering insights and guidance for improvement. Through this system, children can gauge their mastery of financial concepts via dynamic assessments, receiving feedback to enhance their understanding. It also promotes healthy competition by allowing children to compare their financial progress with peers, encouraging motivation. Additionally, it offers a comprehensive view of financial and non-financial data, empowering children to track their progress and reflect on their decisions. Meanwhile, parents can utilize this system to support their child's finances by sending money, creating earning activities, and overseeing home rewards requests. The Financial Activities for parents enables them to set goals, monitor progress, and provide guidance, while offering a centralized view of their children's financial performance and pending tasks, facilitating effective support. It also has features that can contribute by modifying assessment questions in its Financial Assessment, ensuring relevance and appropriateness for children's learning levels, while also monitoring assessment results to make necessary adjustments.

| **Software/System** | **Strengths** | **Weaknesses** |
|---|---|---|
| Mint | ● Reports in different categories separately, including expenses, earnings, and net worth.<br>● Enables users to frequently evaluate all of their connected financial accounts, including the ability to see exactly where they are spending their money. | ● Manual entry of expenses<br>● Limited investment tracking<br>● Mobile app not available in the Philippines<br>● For some users, it is ineffective for saving money, paying down debt, and attaining financial goals. |

| | | |
|---|---|---|
| YNAB(You Need A Budget) | • Manage users' money by allowing them to assign all of their money to anticipated bills or savings for the month.<br>• Switchable between multiple budgets without affecting other customized budgets<br>• Syncs different financial accounts<br>• teaches a new method to think about how every amount of income will be spent or saved<br>• A hands-on approach capable of saving more money than other systems. | • Manual entry of expenses<br>• Subscription based availability<br>• No investment tracking feature<br>• Lack of detailed reports<br>• There is a learning curve and it can feel overwhelming for new users. |
| Personal Capital | • Effective platform for retirement planning<br>• Free and thorough investment management tools<br>• Provides tax management strategies. | • Limited handling of transactions and budget management because it focuses largely on investment planning<br>• High management fee<br>• Completely limits manual entry of users too much<br>• Mobile app not available in the Philippines |
| Quicken | • Personalized insights for better budget planning, | • Limited features per subscription model |

| | | |
|---|---|---|
| | customizable to help set and achieve goals<br>• Investment tools offer detailed asset monitoring and return reports.<br>• Financial informations are up to date | • Mobile app not available in the Philippines |
| FinWise | • Promotes progressive development of financial literacy for children and parents<br>• Parent guidance and monitoring towards children's progress<br>• Keeps users motivated through rewards and leaderboard | • Limited investment management<br>• Limited only to parents and children<br>• Limited customization |
| Proposed System | • Customization and Personalization of financial goals, budget plans, and investment<br>• Recommendations of budget adjustments for financial goals<br>• Personalized insights on budget plan, potential overspending, expense patterns of user<br>• Optical Character Recognition technology on receipts for transaction input | • No syncing capability with financial accounts<br>• Limited tracking of investments |

#### 2.5.1.6 Synthesis

The comparison of various financial management systems highlights distinct strengths and weaknesses in each platform. Mint stands out for its detailed categorization of reports and frequent evaluation of financial accounts, though its manual entry requirement and limited investment tracking may deter some users. YNAB offers a hands-on approach to budgeting, allowing users to assign income to specific expenses or savings, yet its subscription-based model and lack of investment tracking could be drawbacks. Personal Capital is best in retirement planning and investment management, but its focus on investments overshadow transaction handling and budget management. Quicken provides personalized insights and investment tools but may feel limited due to its subscription model and lack of mobile availability in certain regions. FinWise uniquely targets financial literacy for children and parents, yet its limited scope and audience may restrict broader utility. The strengths of each system will be evaluated to see if it can be incorporated into the proposed solution. Meanwhile their weaknesses will be a potential enhancement for the proposed solution

#### 2.5.1.7 Features Table

| | MINT | YOU NEED A BUDGET (YNAB) | PERSONAL CAPITAL | QUICKEN | FINWISE | PROPOSED SYSTEM |
|---|---|---|---|---|---|---|
| AVAILABLE IN THE PHILIPPINES | ✗ | ✓ | ✗ | ✗ | ✓ | ✓ |
| EXPENSE TRACKING | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| BUDGETING | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| INVESTMENT TRACKING | ✗ | ✗ | ✓ | ✓ | ✗ | ✓ |
| PAID SUBSCRIPTION | ✗ | ✓ | ✓ | ✓ | ✗ | ✗ |
| OCR | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| FNANCIAL DASHBOARD | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| PERSONALIZED INSIGHTS | ✗ | ✗ | ✓ | ✓ | ✓ | ✓ |
| PERSONALIZATION | ✗ | ✗ | ✓ | ✗ | ✓ | ✓ |
| PERSONAL INFORMATICS | ✓ | ✗ | ✓ | ✗ | ✓ | ✓ |

*Figure 2.5.1.7.1 Features table regarding availability of different functionalities per system*

## 2.6 Review of Related Methodologies

### 2.6.1 Scrum

Scrum is an Agile methodology that is often used and well-known for managing complicated projects, mostly in the software development industry (Dyck, 2015; Ali et al., 2022). It offers an organized and adaptable method of project management that encourages cooperation, flexibility, and openness (Ali et al., 2022). Scrum is based on a specified set of roles, ceremonies, and artifacts that help teams collaborate well and produce software or products gradually that are valued (Shore & Warden, 2014).

The Product Owner, Scrum Master, and Development Team are the three main roles that form the foundation of Scrum (Ali et al., 2022). The product backlog, a dynamic list of features, improvements, and bug fixes, must be defined and prioritized by the Scrum team (Dyck, 2015). Serving as the team's servant-leader, the Scrum Master facilitates the Scrum process and removes any obstacles that stand in the way of the development(Ali et al., 2022). At the conclusion of each sprint, the cross-functional members of the Development Team are in charge of providing a potentially usable product increment (Shore & Warden, 2014).

According to Shore and Warden (2014), Scrum is implemented in fixed-length time intervals known as "sprints," which usually span two to four weeks. The Development Team aims to provide a functional, tested product increment by the conclusion of each sprint by choosing a set of items to work on from the product backlog (Ali et al., 2022). Sprints offer a time-boxed structure that encourages adaptation and focus while enabling stakeholders to regularly examine and offer input on the product (Ali et al., 2022).

Within the framework, scrum ceremonies are crucial (Shore & Warden, 2014). Team members discuss their plans, goals, and any obstacles they are encountering at the brief daily meeting known as the Daily Scrum which should only last a few minutes to keep it short (Ali et al., 2022). Each sprint begins with a meeting called "sprint planning," when the tasks to be completed for that sprint are decided (Dyck, 2015). Every sprint concludes with a Sprint Review, where the team presents its finished product to the team (Ali et al., 2022). The team can evaluate its procedures and make adjustments using the Sprint Retrospective (Shore & Warden, 2014).

Meanwhile, the product backlog, sprint backlog, and increment are part of scrum artifacts (Ali et al., 2022). The prioritized list of all the tasks that need to be completed is called the product backlog (Shore & Warden, 2014). The portion of the product backlog that the team has committed to finishing during the sprint is known as the sprint backlog (Dyck, 2015). The increment, which represents a potentially shippable product, is the total amount of work finished during the sprint (Ali et al., 2022).

With the use of these artifacts, Scrum fosters transparency and makes it possible for all team members and stakeholders to comprehend the project's priorities and status (Dyck, 2015; Ali et al., 2022). Scrum is ideally suited for projects with changing and unpredictable requirements because of its incremental and iterative character, which increases adaptation to changing requirements and client needs (Ali et al., 2022).

In conclusion, Scrum is an Agile framework that prioritizes transparency, cooperation, and adaptability while handling complicated projects (Dyck, 2015; Ali et al., 2022). It is a useful option for many software development and project management initiatives due to its organized approach, which consists of roles, ceremonies, and artifacts and gives teams a useful means to create value in a flexible and iterative manner (Shore & Warden, 2014).

### 2.6.2 Scrum vs Other Methodologies

| **Feature** | **Scrum** | **Rapid Application Development (RAD)** | **Waterfall** |
|---|---|---|---|
| **Project structure** | Iterative and incremental | Iterative and incremental | Linear and sequential |
| **Planning** | Less upfront planning, more focus on adaptive planning | Moderate upfront planning, with emphasis on rapid prototyping | Extensive upfront planning, with detailed specifications |
| **Requirements** | Evolving requirements are accommodated | Evolving requirements are accommodated | Requirements are fixed and cannot be changed |
| **Feedback** | Continuous feedback and adaptation | Feedback is incorporated throughout the development process | Feedback is primarily gathered at the end of each phase |
| **Risk management** | Risks are identified and addressed throughout the project | Risks are identified and addressed throughout the project | Risks are primarily identified and addressed during the planning phase |

| | | | |
|---|---|---|---|
| **Change management** | Changes are relatively easy to implement | Changes are relatively easy to implement | Changes are difficult and costly to implement |
| **Team collaboration** | Emphasizes close collaboration and communication | Emphasizes collaboration and communication | Emphasizes individual work and handoffs |
| **Project visibility** | High project visibility, with regular deliverables | High project visibility, with regular prototypes | Low project visibility until the end of each phase |
| **Customer involvement** | Encourages customer involvement throughout the project | Encourages customer involvement throughout the project | Limited customer involvement until the end of each phase |
| **Time to market** | Faster time to market, with early prototypes and continuous feedback | Faster time to market than Waterfall | Slower time to market than Scrum and RAD |
| **Project success rate** | Higher project success rate due to adaptability and continuous feedback | Higher project success rate than Waterfall | Lower project success rate due to rigidity and lack of flexibility |

*Figure 2.4: Scrum vs Other Methodologies Comparison*

### 2.6.3 Synthesis

Scrum is well-suited for tackling the development of a personal finance and system with integrated personal informatics aimed at alleviating issues related to financial management and data organization (Ali et al., 2022; Dyck, 2015). Primarily, the problem statement underscores the urgent need for a prompt solution to address the adverse impacts of inadequate financial management and disorganized personal data (Shore & Warden, 2014). Scrum's iterative and incremental development approach facilitates a faster time-to-market (Ali et al., 2022). It empowers teams to prioritize and address critical aspects of the problem early on, delivering functional components or modules within short timeframes (Cohn & Ford, 2010). Consequently, the personal finance and personal informatics system can be introduced gradually, offering immediate assistance to individuals while further enhancements are in progress (Ali et al., 2022).

Moreover, Scrum's flexibility is invaluable in the context of personal finance and informatics, where financial situations and data management needs can evolve rapidly (Dyck,

2015). Given the necessity for individuals to adapt to changing financial circumstances, a system developed using Scrum can readily accommodate adjustments and additional features as the project advances (Ali et al., 2022). The ability to pivot and refine based on feedback from users and financial advisors is a fundamental strength of Scrum (Ali et al., 2022).

Scrum also promotes collaboration and cross-functional teamwork, essential for projects involving multiple users (Shore & Warden, 2014). Financial advisors, software developers, data analysts, and users bring unique insights and requirements to the personal finance and informatics system (Ali et al., 2022). Scrum's framework underscores open communication and regular collaboration, ensuring that all perspectives are considered in the development process (Ali et al., 2022). Consequently, the resulting system is tailored to meet the specific needs of individuals and their financial situations, leading to improved financial management (Ali et al., 2022).

In summary, Scrum is an excellent fit for this project due to its capacity for rapid, adaptable, and collaborative development, crucial for creating a personal finance and personal informatics system to address challenges in financial management and data organization (Dyck, 2015; Ali et al., 2022). Scrum's ability to respond to evolving requirements and focus on delivering incremental value makes it a pragmatic choice for this personal finance and informatics-related initiative (Ali et al., 2022).

# CHAPTER 3

## 3.0 Scrum

Each proponent was assigned to the three team roles within the scrum framework. There was the product owner, scrum master, and development roles. The product owner was assigned to Kelsy Dionisio. She was responsible for keeping the team on track in terms of the overall goals of the project and helping the team members understand its importance. Additionally, she also ensured that the team properly built the Personal Finance Informatics System by fulfilling the target users' needs and referencing the product backlog. This ensured that all modules planned to be implemented were achieved with quality. The scrum master, on the other hand, was assigned to Charles Santos. He supported the team by making sure that the scrum process was understood by team members and ensuring that the process was implemented correctly. Besides this, he was also responsible for the effective management of the backlog. An example of this was when there were issues experienced in developing a particular module in the Personal Finance Informatics System; if this happened, he advised the team on a possible solution. The development team was composed of Reece De Padua, Kelsy Dionisio, Charles Santos, and Kyle Taguiwalo. They were assigned to build the Personal Finance Informatics System within each sprint. They collaborated and supported one another in order to achieve the project goals and follow the product backlog.

Additionally, at the beginning of each sprint, there was sprint planning to set the agenda of the sprint and the target module that needed to be accomplished. This was led by the product owner (Kelsy Dionisio) and discussed with the development team. The project was divided into sprints, and these sprints had a sprint backlog containing one module each to accomplish because the number of modules was limited to five. At the end of the sprints, the required modules in the sprint backlog needed to be accomplished by the team. Besides the module that needed to be accomplished in each sprint, there were activities that needed to be completed by the team within the sprint. These consisted of requirement gathering, analysis, designing the UI, designing the DB, designing the prototype, testing, and feedback. These had varying time frames depending on the complexity of the module that was being worked on in the sprint.

## 3.1 Work Plan

### 3.1.1 CAP-IT1 Work Timeline

| CAP-IT1 Sprint | | **Development** | April | | | May | | | | June | | | | July | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Sprint no. | Modules | Activities | 15 | 22 | 29 | 6 | 13 | 20 | 27 | 3 | 10 | 17 | 24 | 1 | 8 | 15 | 22 | 29 |
| | | Design - UI | | | | | | | | | | | | | | | | |
| | | Design - DB | | | | | | | | | | | | | | | | |
| | | Design - Prototype | | | | | | | | | | | | | | | | |
| | | Coding | | | | | | | | | | | | | | | | |
| | | Testing | | | | | | | | | | | | | | | | |
| 1 | Budget | Feedback | | | | | | | | | | | | | | | | |
| | | Design - UI | | | | | | | | | | | | | | | | |
| | | Design - DB | | | | | | | | | | | | | | | | |
| | | Design - Prototype | | | | | | | | | | | | | | | | |
| | | Coding | | | | | | | | | | | | | | | | |
| | | Testing | | | | | | | | | | | | | | | | |
| 2 | Cash Inflow | Feedback | | | | | | | | | | | | | | | | |
| | | Design - UI | | | | | | | | | | | | | | | | |
| | | Design - DB | | | | | | | | | | | | | | | | |
| | | Design - Prototype | | | | | | | | | | | | | | | | |
| | | Coding | | | | | | | | | | | | | | | | |
| | | Testing | | | | | | | | | | | | | | | | |
| 3 | Financial Goal | Feedback | | | | | | | | | | | | | | | | |
| | | Design - UI | | | | | | | | | | | | | | | | |
| | | Design - DB | | | | | | | | | | | | | | | | |
| | | Design - Prototype | | | | | | | | | | | | | | | | |
| | | Coding | | | | | | | | | | | | | | | | |
| | | Testing | | | | | | | | | | | | | | | | |
| 4 | Expense | Feedback | | | | | | | | | | | | | | | | |
| | | Design - UI | | | | | | | | | | | | | | | | |
| | | Design - DB | | | | | | | | | | | | | | | | |
| | | Design - Prototype | | | | | | | | | | | | | | | | |
| 5 | Dashboard | | | | | | | | | | | | | | | | | |

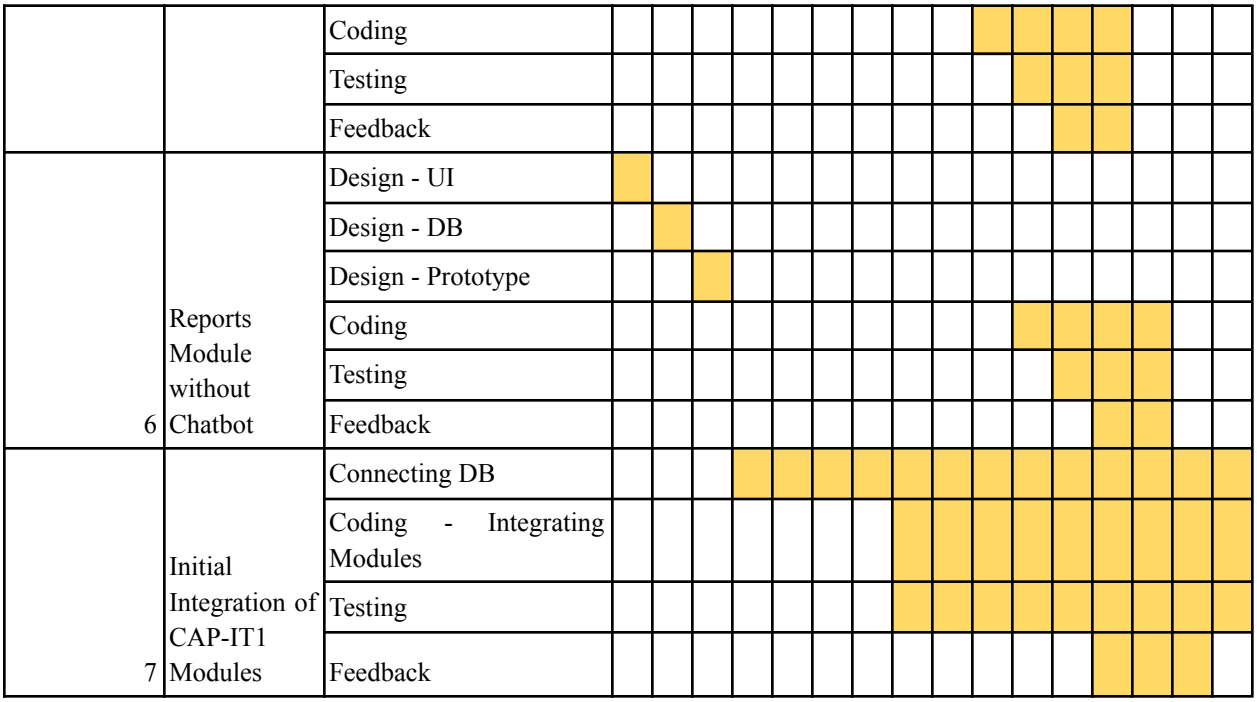



*Figure 3.7: CAP-IT1 Work Timeline*

#### 3.1.1.1 Sprint Number 1

This sprint included the budget module as part of the sprint backlog. This sprint also included activities such as designing UI, designing DB, and designing the prototype. These activities were done within three weeks. The coding, testing, and getting feedback from users were also part of this sprint. The duration of this sprint was from April 15, 2024, to May 27, 2024. The designing of UI, DB, and prototype not only covered this module but also the remaining modules, which lasted from April 15, 2024, to April 29, 2024. The coding, testing, and feedback for this module lasted from May 6, 2024, to May 27, 2024.

#### 3.1.1.2 Sprint Number 2

This sprint included the cash inflow module as part of the sprint backlog. This sprint also included activities mainly designing UI, designing DB, designing the prototype, coding, testing, and getting feedback from users. The duration of this sprint was from April 15, 2024, to June 10, 2024. The designing of UI, DB, and prototype not only covered this module but also the remaining modules, which lasted from April 15, 2024, to April 29, 2024. The coding, testing, and feedback for this module lasted from May 20, 2024, to June 10, 2024.

#### 3.1.1.3 Sprint Number 3

This sprint included the financial goal module as part of the sprint backlog. This sprint also included activities mainly designing UI, designing DB, designing the prototype, coding, testing, and getting feedback from users. The duration of this sprint was from April 15, 2024, to June 24, 2024. The designing of UI, DB, and prototype not only covered this module but also the remaining modules, which lasted from April 15, 2024, to April 29, 2024. The coding, testing, and feedback for this module lasted from June 3, 2024, to June 24, 2024.

#### 3.1.1.4 Sprint Number 4

This sprint included the expense module as part of the sprint backlog. This sprint also included activities mainly designing UI, designing DB, designing the prototype, coding, testing, and getting feedback from users. The duration of this sprint was from April 15, 2024, to July 8, 2024. The designing of UI, DB, and prototype not only covered this module but also the remaining modules, which lasted from April 15, 2024, to April 29, 2024. The coding, testing, and feedback for this module lasted from June 17, 2024, to July 8, 2024.

#### 3.1.1.5 Sprint Number 5

This sprint included the dashboard module as part of the sprint backlog. This sprint also included activities mainly designing UI, designing DB, designing the prototype, coding, testing, and getting feedback from users. The duration of this sprint was from April 15, 2024, to July 8, 2024. The designing of UI, DB, and prototype not only covered this module but also the remaining modules, which lasted from April 15, 2024, to April 29, 2024. The coding, testing, and feedback for this module lasted from June 17, 2024, to July 8, 2024.

#### 3.1.1.6 Sprint Number 6

This sprint included the reports module without a chatbot as part of the sprint backlog. This sprint also included activities mainly designing UI, designing DB, designing the prototype, coding, testing, and getting feedback from users. However, in this sprint, the chatbot was not yet included during this development. The duration of this sprint was from April 15, 2024, to July 15, 2024. The designing of UI, DB, and prototype not only covered this module but also the remaining modules, which lasted from April 15, 2024, to April 29, 2024. The coding, testing, and feedback for this module lasted from June 24, 2024, to July 15, 2024.

#### 3.1.1.7 Sprint Number 7

This sprint included the system integration as part of the sprint backlog. This sprint also included activities like connecting the DB, coding, testing, and getting feedback from users. This was the sprint where all the developed modules were integrated in order to ensure their functionality. The duration of this sprint was from May 26, 2024, to July 29, 2024. This was the last sprint that the proponents did in the sprint backlog during CAP-IT1. Any changes based on user feedback during this sprint from user testing on any of the modules were also accomplished during this sprint.

### 3.1.2 CAP-IT2 Work Timeline

| CAP-IT2 Sprint | | **Development** | August | | | | September | | | | | October | | | | November | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Sprint No. | Modules | Activities | 5 | 12 | 19 | 26 | 2 | 9 | 16 | 23 | 30 | 7 | 14 | 21 | 28 | 4 | 11 | 18 | 25 |
| 8 | Reports Module with Chatbot | Design - UI | | | | | | | | | | | | | | | | | |
| | | Design - DB | | | | | | | | | | | | | | | | | |
| | | Design - Prototype | | | | | | | | | | | | | | | | | |
| | | Coding | | | | | | | | | | | | | | | | | |
| | | Testing | | | | | | | | | | | | | | | | | |
| | | Feedback | | | | | | | | | | | | | | | | | |
| 9 | Investment | Design - UI | | | | | | | | | | | | | | | | | |
| | | Design - DB | | | | | | | | | | | | | | | | | |
| | | Design - Prototype | | | | | | | | | | | | | | | | | |
| | | Coding | | | | | | | | | | | | | | | | | |
| | | Testing | | | | | | | | | | | | | | | | | |
| | | Feedback | | | | | | | | | | | | | | | | | |
| 10 | Complete Integration of CAP-IT2 Modules | Connecting DB | | | | | | | | | | | | | | | | | |
| | | Coding - Integrating Modules | | | | | | | | | | | | | | | | | |
| | | User Acceptance Testing | | | | | | | | | | | | | | | | | |
| | | Feedback | | | | | | | | | | | | | | | | | |

*Figure 3.8: CAP-IT2 Work Timeline*

#### 3.1.2.1 Sprint Number 8

This sprint will primarily focus on the chatbot function of the reports module and integrate with the module itself. This sprint will include readjustments necessarily needed in terms of the design of the reports module to fit the chatbot. The expected duration of this sprint is from August 5, 2024 to September 9, 2024. The designing of UI, DB, and prototype will be covered during the first week of August. While coding, testing and feedback of this module will last from the second week of August up to the second week of September.

#### 3.1.2.2 Sprint Number 9

This sprint will include the investment module as part of the sprint backlog. This sprint will include activities mainly designing UI, designing DB, designing prototype, coding, testing, and getting feedback from users. The duration of this sprint is from August 5, 2024 to September 23, 2024. The designing of UI, DB, and prototype will be covered along with the chatbot function of reports module during the first week of August. While coding, testing and feedback of this module will last from August 26, 2024 to September 23, 2024.

#### 3.1.2.3 Sprint Number 10

This sprint will include the system integration as part of the sprint backlog. This sprint will include activities like connecting the DB, coding, testing, and getting feedback from users. The duration of this sprint is from the second week of August to November 5, 2024 . This will be the last sprint that the proponents will do in the sprint backlog during the CAP-IT2 for the development of the system. In this sprint, the complete integration of all modules will be ensured. Any changes based on user feedback from user testing on any of the modules will also be accomplished during this sprint.

### 3.1.3 Input

The primary input of the different sprints in terms of the different designing activities is the user requirements since this is the primary basis in designing the interface and database. Furthermore, after the designs have been developed these will be utilized as a guideline in designing a prototype in which the proponents and users will test. The module testing will be based on the test cases. Also, user testing will be conducted relying on their perception of the prototype as feedback. For every feedback, the proponents will readjust the modules necessarily.

### 3.1.4 Output

There are many outputs that will be generated based on the different activities in each sprint. In the designing activities the expected output is the different interfaces for each module, the database schema for the entire project, and the prototype. Furthermore,

once the prototype is complete another expected output needs to be generated from the point of view of users. This is their feedback regarding the prototype after user testing. While for the system integration the expected output is that modules should work together into one Personal Finance Informatics app.

#### 3.1.5 Tools/Techniques

The same tools/techniques will be used for the different sprints involving all the modules. In the designing activity tools like Figma and LucidChart will be used to create the user interfaces and the database schema. While for the development of the prototype we will use Javascript as the programming language and React as the framework. Furthermore, for managing data we will use MongoDB. While for testing the technique that the team will use is to create test cases and run through the interfaces and functions of the Personal Finance Informatics System. Lastly, for user testing we will distribute the developed modules to the user for testing.

### 3.2 Target Accomplishment for CAP-IT1 (80% of IT development)

During CAP-IT1, the proponents aimed for 80% completion across various modules and functionalities, as it was crucial to ensure significant progress while leaving room for refinement and optimization. A breakdown of what achieving 80% completion entailed for each component of the personal finance informatics system was given in this section.

The first module that the proponents wanted to accomplish was the budget module because it integrated with many modules and was crucial for the application to function properly. Budget categorization and allocation, along with the amount progression and notification of overspending, were accomplished in this stage. Meanwhile, the date and time period functionality, which was a part of other modules as well, was not targeted during this stage.

Secondly, the cash inflow module was followed by the financial goal module, and their functionalities, except for date and time filters, were targeted as well since these were significant for the budget, expense, and dashboard modules to function and to solve the aforementioned problems.

The next module that the proponents targeted to accomplish was the expense module, which also required the budget module for it to function. The main target for the expense module was to utilize Optical Character Recognition technology and capture the total amount in receipts to record expenses. Furthermore, after finishing these modules, the proponents proceeded to the dashboard. This was because all of the aforementioned modules were integrated together to form the dashboard module, which allowed the users to see their personal finances all together. Lastly,

the final module that the proponents worked on was the reports module, allowing users to visualize their financial activities for self-reflection and discovery.

### 3.3 Problems Encountered in CAP-IT1

#### 3.3.2 System

The introduction of Gradle 8.6 caused the system to become entirely non-functional, preventing any development tasks from proceeding. The build.gradle file generated errors due to package issues, and the project couldn't even run in debug mode (./gradlew assembleDebug), despite the Gradle itself functioning (./gradlew clean). React-native-reanimated was also causing the project to fail across all systems. Additionally, there were JDK incompatibilities, as react-native required JDK 17, leading to further complications. Android Studio was not being recognized despite the correct paths being set, contributing to overall incompatibility issues between outdated and latest packages. The JDK was not recognized despite having the correct path and version. To resolve these issues, the team rolled back to a stable version of Gradle, ensured the correct versions of JDK and Android Studio were installed, and systematically updated packages to compatible versions. They also reconfigured paths and environment variables to ensure proper recognition of all tools.

#### 3.3.3 Budget Module

The budget screen faced significant issues with dynamic updates, where details failed to refresh in real-time, making it difficult for users to see the latest information. Changing the user's salary to reflect projected monthly inflow was also problematic, as updates were not being processed correctly in the backend. Additionally, editing the monthly budget amount for specific categories resulted in the display of random past monthly values from the database instead of the current ones. To resolve these issues, the development team implemented real-time data binding and enhanced the backend logic to ensure accurate updates. They also corrected the database queries to fetch the current budget values consistently.

#### 3.3.4 Expenses Module

The expenses module encountered issues where recorded expenses did not update the total amount in budget categories correctly, resulting in a zero total spent display on the dashboard. The display of total expenses was not updating dynamically, causing inconsistencies in the data. Filtering expenses based on the current month negatively impacted other data, such as accounts, budgets, and the monthly allocated amount per budget. The development team solved these problems by improving the synchronization between expense entries and budget calculations, ensuring real-time updates. They also enhanced the filtering logic to maintain data integrity across different modules.

### 3.3.5 Reports Module

The reports module had several graphical issues, including limited and constantly changing colors for graphs, causing misalignment and confusion. The module was not dynamically updating properly, and filtering budget, expenses, and accounts based on the current month was malfunctioning during testing. Additionally, data fetching for current and previous month budget expenses caused errors. To resolve these issues, the team standardized the color scheme for graphs, improved the alignment and clarity of graphical details, and implemented real-time data updates. They also refined the filtering and data fetching processes to ensure accurate and timely report generation.

### 3.3.6 Dashboard and Integration of Budget, Dashboard, Reports, and Expenses Modules

Integrating accounts, expenses, and budgets within the dashboard module was highly complex due to their interconnected nature. Data refetching was malfunctioning, causing delays in data loading, and budget categories and accounts were not populating dropdown menus correctly when recording expenses. Additionally, notification handling for budget category thresholds and regular notifications at 9 PM was difficult to implement and test. Errors in data retrieval and association led to inaccuracies, and infinite loops caused by issues in useEffect and useQuery necessitated a complete system revert. To address these challenges, the team restructured the data fetching logic, implemented proper state management, and improved the synchronization of interconnected modules. They also refined the notification system and ensured accurate data association through comprehensive testing.

### 3.3.7 User (Login/Sign Up) Module

The login and signup processes faced critical issues with network errors and improper user ID handling when navigating between screens, essential for fetching data from the database. These problems hindered users from accessing the app. The development team addressed these issues by optimizing the network requests for login and signup, ensuring reliable connectivity. They also implemented robust user ID handling mechanisms to maintain data consistency across different screens.

### 3.3.8 Notifications Module

The notification system displayed alerts irregularly, with multiple notifications appearing if one budget category triggered its threshold. There was no reliable method to flag specific budget categories for their thresholds, especially when these had to be reset at the beginning of each month. The team resolved these issues by developing a more consistent notification scheduling system and implementing a reliable threshold flagging mechanism. They also ensured that notifications reset accurately at the start of each month.

### 3.3.9 Starting Screen Integration

Editing the budget often resulted in duplicating the user's salary information. Additionally, the system did not anticipate providing users with the option to add more budgets, and flagging users who completed the onboarding process was not functioning correctly. The MongoDB trigger for recording accounts and the total budget amount per month also failed to work as intended. To solve these problems, the team refined the budget editing logic to prevent duplication, incorporated user options for adding more budgets, and improved the onboarding flagging system. They also fixed the MongoDB trigger to ensure accurate recording of account and budget totals.

### 3.3.10 Cash Inflow Module

Changes to the account model, such as converting data types from integers to decimals, and updating interfaces to display decimal numbers, posed significant challenges. Adding a new input field for creating accounts required the account type to be created as a collection, and populating the account type during account creation was difficult. Additionally, passing functions between components and screens in React was problematic. The team resolved these issues by carefully updating data types and interfaces to handle decimals, improving the account creation process, and finding alternative methods for function passing in React, such as using context or state management libraries.

### 3.3.11 Financial Goal Module

The Financial Goal module encountered several issues, starting with the Object ID not working, prompting the implementation of a FinancialGoalID. However, this solution eventually failed due to package incompatibilities. Sub-goals were inconsistently displayed for each financial goal, and the FinancialGoalID duplication error further complicated matters. The amount saved was fetched incorrectly for each financial goal, and labels were not displayed properly. Additionally, the financial goal card on the dashboard failed to fetch correctly and appeared uncentered. The team resolved these issues by standardizing the use of Object IDs, ensuring compatibility with all packages. They also implemented consistent logic for displaying sub-goals and fixed the duplication error for FinancialGoalID. The amount saved per financial goal was corrected by refining the data fetching logic, and label display issues were addressed. Finally, the financial goal card on the dashboard was fixed to fetch data correctly and was properly centered for a cohesive user experience.

### 3.3.12 Account Integration

Implementing the accounts module was challenging because it affected the overall structure of the cash inflow screen. Additional states had to be added to the dashboard to support accounts, and changes to data allocation to the database were necessary. These modifications disrupted the existing workflow and required extensive adjustments. The team tackled these

challenges by restructuring the cash inflow screen to accommodate the new account states, ensuring seamless integration with the dashboard. They also optimized the data allocation process to the database, maintaining consistency and stability across the system.

**3.4 Target Accomplishment for CAP-IT2 (20% of IT development)**

During CAP-IT2 the proponents will aim to accomplish the remaining 20% that needs completion. The proponents will target to accomplish the investment module during this phase and integrate it to the application to create a cohesive personal finance informatics system. Along with the investment module is the chatbot functionality integrated in the reports module.

# CHAPTER 4

## 4.0 The Existing System

### 4.1 Organization Profile

The users of the FinSight application targets Filipino single working adults aged 18 to 65, who are not married and manage their personal finances independently. These users are employed and possess a steady income, necessitating financial management activities such as budgeting, bill payments, saving, and investing. Regardless of their income levels, these individuals aim to enhance their financial well-being due to their current financial circumstances. Users are expected to own and be proficient in using smart mobile devices, specifically Android, as the application is tailored for this platform. The app caters to those who seek to improve their financial literacy and management skills through various modules, including budgeting, expense tracking, financial goal setting, and investment monitoring.

### 4.2 Description of the Existing System

#### 4.2.1 Financial Decision Making of Single Working Adults

##### 4.2.1.1 Gender Differences in Financial Decision-Making

Research shows that single females are significantly less likely to engage in positive financial behaviors, such as maintaining emergency funds and non-retirement investments, compared to males, highlighting a gender disparity in financial preparedness among single working adults (Wagner & Walstad, 2022). This disparity can be attributed to various factors, including societal norms, education, access to financial resources, wage gaps, and career interruptions due to caregiving responsibilities. Traditional gender roles may also discourage women from actively participating in financial decision-making. Addressing these challenges requires targeted financial education and policy changes aimed at empowering women to take control of their financial futures. Financial education programs tailored to women's specific needs and circumstances can help bridge the gap, encouraging more proactive financial behaviors. By fostering an environment that supports women's financial independence, we can enhance their overall financial well-being and security.

##### 4.2.1.2 Age-Related Financial Decision-Making

Financial decision-making abilities peak in middle age, with younger adults often making riskier choices and older adults relying on accumulated experience but struggling in novel financial situations (Samanez-Larkin et al., 2014). Middle-aged individuals typically face significant financial responsibilities, such as mortgages, children's education expenses, and retirement planning, and they balance cognitive sharpness with practical experience, making

this an optimal period for financial decisions. Younger adults may lack the foresight for long-term consequences, leading to speculative investments and riskier financial behaviors, influenced by the desire for quick gains or instant gratification. Conversely, older adults might adopt an overly cautious approach, focusing on avoiding losses and preserving wealth, which can lead to missed growth opportunities. Financial advisors should consider these dynamics to provide age-appropriate guidance that maximizes financial outcomes for each group. Understanding the strengths and weaknesses of each age demographic is vital for tailoring financial advice and education, ensuring that financial planners can offer effective guidance that aligns with clients' life stages and goals.

#### 4.2.1.3 Psychological and Socio-Demographic Influences

The decision-making process in personal finances is heavily influenced by psychological factors and socio-demographic characteristics. Young adults often prioritize financial security by opting for less complex investments like real estate and commodities over more intricate financial instruments, driven by their desire for tangible assets and stability (Kaftandzieva & Cvetkoska, 2021). Psychological traits such as risk tolerance, future orientation, and financial self-efficacy play crucial roles in shaping financial decisions. Additionally, socio-demographic factors, including income level, education, and cultural background, significantly impact financial behaviors. For instance, individuals from higher socio-economic backgrounds generally have more access to financial resources and education, leading to better-informed financial decisions. Understanding these influences is essential for developing personalized financial advice and educational programs that cater to the unique needs of different demographic groups, thereby promoting better financial outcomes.

#### 4.2.1.4 Financial Behaviors and Well-Being

Positive financial behaviors, such as budgeting, saving, and investment, are crucial for financial well-being. Poor financial behaviors like impulsive spending can lead to financial instability and stress, highlighting the need for effective financial management strategies (Greenberg & Hershfield, 2018). Consistently practicing sound financial habits can build a safety net that protects against unexpected expenses and financial emergencies. Budgeting helps individuals track their income and expenditures, ensuring they live within their means. Regular saving fosters financial security and provides resources for future goals, while wise investments can grow wealth over time. Conversely, negative financial behaviors can result in debt accumulation, credit problems, and significant stress, which can adversely affect mental and physical health.

Financial education programs that teach these essential skills can greatly enhance individuals' financial stability and overall well-being.

#### 4.2.1.5 Influence of Cognitive Decline on Financial Capacity

Cognitive decline, particularly in executive functioning, affects older adults' ability to make sound financial decisions, impacting tasks such as budgeting, investment planning, and fraud detection (Strough et al., 2020). Declines in working memory, processing speed, and executive functions like planning, problem-solving, and impulse control contribute to these challenges. However, training in financial principles, ongoing education, and practical experience can help mitigate these effects. Regular cognitive exercises, simplified financial tools, and resources can support better financial decision-making. Additionally, involving trusted family members or financial advisors can provide the necessary support, ensuring older adults' financial needs are met while preserving their autonomy. Programs designed to enhance cognitive health and financial literacy can provide valuable support, enabling older adults to make informed decisions despite cognitive changes.

#### 4.2.1.6 Impact of Financial Self-Efficacy

Financial decision-making self-efficacy (FDMSE) significantly influences financial decisions. Lower FDMSE is associated with cognitive impairments and negative financial experiences, which can undermine confidence in managing finances (Bailey & Leon, 2019). Building financial self-efficacy involves enhancing individuals' beliefs in their ability to control their financial futures. This can be achieved through targeted education programs, mentorship, and practical financial planning tools that empower individuals to take charge of their financial decisions. By improving financial competence and confidence, individuals are more likely to engage in positive financial behaviors, such as budgeting, saving, and investing. Strengthening FDMSE can lead to better financial stability and well-being, as individuals feel more capable of navigating the complexities of financial management and making decisions that align with their long-term goals.

### 4.2.2 Development of Financial Decision-Making of Single Working Adults

#### 4.2.2.1 Cognitive Capacities and Financial Competence

Cognitive capacities and financial competence develop differently over the adult lifespan, influenced by basic cognitive resources, general knowledge, and domain-specific financial expertise. Younger adults tend to make more impulsive financial decisions, often driven by immediate gratification and a shorter time

horizon, due to limited experience and financial knowledge (Hershey et al., 2015). This inexperience can lead to suboptimal financial choices, such as impulsive spending or inadequate saving for future needs. As individuals age, their processing speed and memory can deteriorate, making it harder to navigate intricate financial products or strategies, though their accumulated experience and knowledge can help them make informed decisions. However, older adults face challenges such as adapting to new financial products and technologies. The interplay between cognitive development and financial expertise suggests that continuous learning and adaptation are essential for maintaining financial competence throughout life. Financial education programs should be tailored to different age groups, emphasizing risk management for younger adults and simplifying complex information for older adults, thereby enhancing cognitive resources through education and practical experience to significantly improve one's ability to make sound financial decisions and achieve better financial outcomes.

#### 4.2.2.2 Developmental Model of Financial Capability

A developmental model suggests that young adults acquire financial knowledge and behaviors gradually, influenced by their changing financial knowledge, self-beliefs about finances, and financial behaviors. This model highlights the progressive nature of financial capability, where individuals develop their skills and understanding over time (Serido et al., 2013). As young adults navigate different life stages, such as entering the workforce, starting a family, or purchasing a home, their financial knowledge and behaviors evolve. These experiences build a foundation for financial well-being, as they learn from both successes and mistakes. Financial education plays a crucial role in this developmental process, providing young adults with the tools and confidence needed to make informed decisions. By fostering positive financial habits early on, individuals can achieve greater financial stability and well-being as they progress through their lives.

#### 4.2.2.3 Psychosocial Determinants of Financial Well-Being

Factors such as financial literacy, self-beliefs, and behaviors play a significant role in determining financial well-being. Educational interventions and financial planning tools can enhance these determinants, leading to better financial outcomes (Serido et al., 2013). Financial literacy, the understanding of financial principles and concepts, empowers individuals to make informed decisions about their money. Positive self-beliefs, such as financial self-efficacy, enhance confidence in managing finances, while proactive financial behaviors,

like regular saving and investing, promote stability and growth. Psychosocial factors, including social support and emotional well-being, also influence financial behaviors. Comprehensive financial education programs that address these determinants can equip individuals with the knowledge, skills, and confidence needed to achieve financial well-being. These programs should be inclusive, considering the diverse needs and backgrounds of participants to maximize their effectiveness.

### 4.2.3 Role of Environment in Financial Decision-Making Development of Single Working Adults

#### 4.2.3.1 Impact of Economic Environment on Financial Decisions

The economic environment, encompassing elements such as financial socialization and financial security, plays a crucial role in shaping financial decision-making. For instance, students who receive grants or scholarships tend to exhibit higher levels of risk aversion compared to their peers who are supported by family finances. This distinction highlights the influence of financial responsibility and external funding sources on individual financial behaviors (Prinz et al., 2014). Students reliant on scholarships may feel a greater pressure to manage their finances prudently due to the conditional and often temporary nature of their funding, whereas those with family support might have a safety net that allows for more flexibility and risk-taking. This dynamic underscores the broader impact of economic conditions on personal financial strategies, emphasizing the need for tailored financial education that considers the diverse financial backgrounds of individuals.

#### 4.2.3.2 Role of Institutional Environment

Financial decision-making within stressful environments, such as universities experiencing financial constraints, is often governed by institutional rules and rational myths. These institutional frameworks act as filters that assign specific meanings to various decision-making strategies, thereby shaping how decisions are made under financial pressure (Martinez, 1994). For example, universities might prioritize cost-cutting measures or conservative financial planning to ensure stability, influencing the financial behaviors of students and staff. This environment can foster a culture of frugality and caution, which may extend to personal financial decisions. Understanding the role of institutional context in financial decision-making is vital for developing policies and educational programs that support sound financial practices even in challenging circumstances.

#### 4.2.3.3 Geographical and Cognitive Interactions

Financial literacy varies significantly by region, reflecting the diverse regional ecologies of finance within national and global markets. The interaction between an individual's cognitive abilities and their environment affects how they process financial information, ultimately influencing their long-term financial welfare. These regional disparities in financial literacy can reinforce existing urban and regional inequalities, with individuals in more financially literate areas having access to better resources and support (Clark, 2013). For instance, residents of urban areas with high financial literacy rates may benefit from greater access to financial education programs, investment opportunities, and advisory services, while those in less affluent regions might struggle with limited resources. Addressing these disparities requires a comprehensive approach that includes targeted financial education initiatives aimed at improving literacy in underserved areas.

#### 4.2.3.4 Influence of Stress on Financial Decisions

Exposure to acute stress can have a profound impact on financial decision-making, often leading to riskier choices. Stressful conditions can exacerbate behavioral biases, causing individuals to rely more on automatic responses rather than engaging in deliberate, thoughtful decision-making processes (Porcelli & Delgado, 2009). For example, during periods of high stress, individuals might resort to impulsive spending or making hasty investment decisions without fully considering the consequences. This tendency highlights the importance of stress management and its role in maintaining sound financial practices. Financial education programs should incorporate stress management techniques to help individuals make more rational and less emotionally-driven financial decisions.

#### 4.2.3.5 Parental Influence and Financial Socialization

Parents play a pivotal role in the financial socialization of young adults. Effective communication about financial topics and setting clear expectations significantly influence the financial independence and well-being of emerging adults (Serido et al., 2010). Positive parental involvement can lead to the development of better financial coping behaviors, such as budgeting, saving, and responsible spending. Parents who model good financial habits provide their children with practical examples of how to manage money effectively. Furthermore, open discussions about finances can demystify complex financial concepts, making them more accessible and less intimidating. Encouraging

parental engagement in financial education can thus enhance the financial literacy and competency of the next generation.

#### 4.2.3.6 Regional Financial Literacy and Inequality

Regional variations in financial literacy reflect broader socio-economic conditions, with high financial literacy regions often correlating with better financial decision-making and lower regional inequalities. Conversely, regions with lower financial literacy tend to struggle with financial decision-making, exacerbating existing disparities (Clark, 2012). For instance, areas with higher average incomes and better educational infrastructures generally exhibit higher levels of financial literacy. These regions benefit from more comprehensive access to financial information and resources, leading to more informed financial behaviors. In contrast, economically disadvantaged regions may lack the necessary infrastructure to support financial education, perpetuating cycles of poor financial decision-making and economic hardship. Addressing these regional disparities requires a multifaceted approach, including policy interventions, educational reforms, and community-based initiatives aimed at enhancing financial literacy across all regions.

### 4.2.4 Current State of Technology and Systems

#### 4.2.4.1 Traditional Methods

Pen and paper remain a straightforward and effective method for personal finance management, allowing users to manually track expenses, create budgets, and set financial goals. Despite its simplicity, this method can be time-consuming and prone to errors, making it less efficient for complex financial management (Hu & Huang, 2019). Users must diligently record every transaction, often resulting in discrepancies due to human error. However, for those who prefer a tactile approach, pen and paper provide a tangible connection to their financial activities, promoting a deeper understanding and awareness of their spending habits.

Microsoft Excel is another powerful tool for managing personal finances, enabling users to create detailed financial records, budgets, and forecasts. Excel's versatility and advanced functionalities, such as formulas and data analysis tools, make it an excellent choice for those comfortable with spreadsheets. It allows for custom financial tracking and planning, providing a more organized approach compared to pen and paper (Bhar, 2019). Interactive Excel spreadsheets further enhance the learning and application of financial management principles. These spreadsheets use dynamic formulas and feedback mechanisms to simplify

complex financial tasks, making them accessible and efficient for users (Peters & Chiu, 2022). Excel's ability to handle large data sets and perform complex calculations makes it a preferred choice for individuals looking to gain deeper insights into their financial situations through detailed and customized reports.

#### 4.2.4.2 Modern Methods

Personal finance applications offer a convenient and user-friendly way to track spending, budget, and plan investments. These mobile apps often come with features like automatic expense categorization, real-time updates, and goal-setting tools, making financial management more accessible and efficient than traditional methods (Raina et al., 2022). The integration of notifications and alerts helps users stay on top of their finances, providing immediate feedback on their spending and saving behaviors. This immediacy and accessibility foster a more proactive approach to financial management, allowing users to adjust their habits in real-time.

Automated personal finance systems, such as the "Pay-App" for students, integrate budgeting and expense management with mobile payment platforms. These systems automate the distribution of income into various budget categories, reducing the likelihood of overspending and improving financial discipline (Githuka et al., 2016). Additionally, advanced financial decision support systems leverage big data and artificial intelligence to provide personalized financial advice and manage finances efficiently. These systems can analyze vast amounts of data to offer real-time insights and predictive analytics, helping users make informed financial decisions (Tong & Tian, 2023). Online personal finance management tools offer comprehensive solutions, allowing users to track expenses, manage investments, and plan budgets from any location with internet access. These tools often come with additional features like financial goal setting, investment tracking, and financial health monitoring. The seamless integration of these modern methods into daily life simplifies financial management, making it more intuitive and less time-consuming, thereby catering to the diverse needs of today's tech-savvy users.

## 4.3 Problem Areas

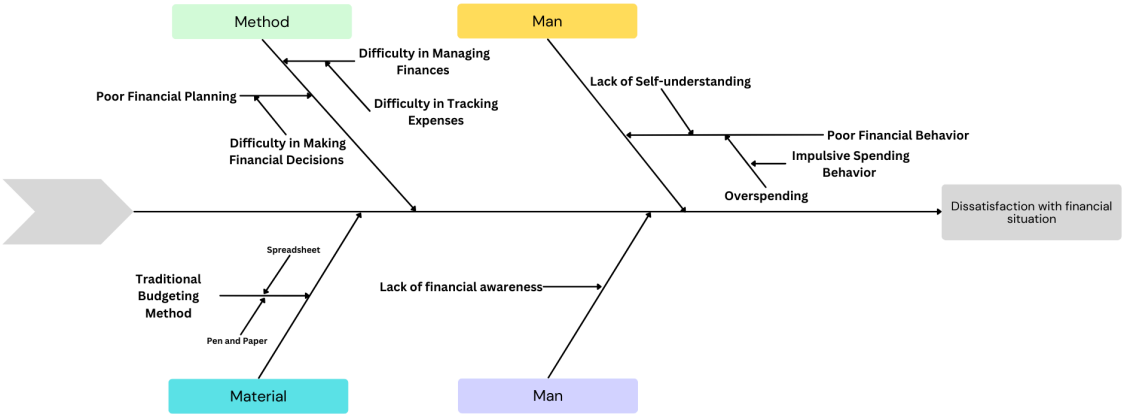



*Figure 4.3.1 Ishikawa Diagram*

### 4.3.1 Man

The "Man" aspect in financial dissatisfaction delves into individual behaviors and personal management of finances. Key issues include difficulties in managing finances, tracking expenses, and making informed financial decisions. Often, these challenges are rooted in a lack of self-understanding and financial awareness. Many individuals struggle with poor financial behavior, such as impulsive spending and chronic overspending, which can lead to a vicious cycle of debt and financial instability. This lack of discipline and knowledge underscores the importance of financial literacy and education. Developing better financial habits, such as budgeting, saving, and investing, can empower individuals to take control of their financial futures. Financial education programs, mentorship, and access to reliable financial advice can play a crucial role in helping individuals understand the long-term implications of their financial decisions and behaviors, ultimately leading to improved financial well-being and satisfaction.

### 4.3.2 Material

The "Material" aspect focuses on the tools and resources utilized for financial management. Traditional budgeting methods, such as using pen and paper or basic spreadsheets, often fall short in providing the comprehensive and dynamic solutions needed for effective financial planning. These outdated methods can be cumbersome, prone to human error, and difficult to maintain, making it challenging to keep accurate

records and efficiently track expenses. As a result, many individuals find themselves overwhelmed by the complexity of managing their finances. The advent of modern financial tools and software offers a significant improvement, providing features such as automated expense tracking, real-time financial analysis, and intuitive budgeting interfaces. By leveraging these advanced tools, individuals can gain a clearer understanding of their financial situation, identify areas for improvement, and make more informed financial decisions. Investing in better financial management materials can thus lead to more efficient and accurate financial planning, reducing stress and increasing overall financial satisfaction.

#### 4.3.4 Method

The "Method" aspect addresses the processes and strategies employed in financial planning and decision-making. Poor financial planning and reliance on traditional budgeting methods often result in difficulties in making sound financial decisions and tracking expenses accurately. Ineffective methods can stem from a lack of structured financial strategy and a reactive rather than proactive approach to financial management. This can cause individuals to feel overwhelmed and dissatisfied with their financial situation. Adopting more systematic and effective methods, such as digital budgeting tools, financial planning apps, and comprehensive financial management software, can significantly improve one's ability to manage finances. These tools offer features like automated expense tracking, predictive budgeting, and real-time financial insights, which can help individuals make informed decisions and plan for the future more effectively. Furthermore, incorporating financial planning techniques such as setting clear financial goals, creating emergency funds, and regularly reviewing financial plans can provide a structured framework for managing finances. By refining their methods, individuals can achieve better financial control, reduce anxiety related to financial management, and ultimately attain greater financial satisfaction.

### 4.4 Opportunities

#### 4.4.1 Opportunities for Single Working Adults

The Personal Finance Informatics application, FinSight, for single working adults offers a multitude of opportunities aimed at enhancing financial management and overall well-being. For single working adults, the application provides tools for effective budgeting, expense tracking, financial goal setting, and investment monitoring. These features enable users to gain better control over their finances, helping them to save more, reduce unnecessary expenditures, and make informed financial decisions. By using the app, single working adults can improve their financial literacy, achieve financial independence, and reduce financial stress, ultimately leading to a more secure and fulfilling life.

Beyond single working adults, the application can benefit other demographics as well. For instance, young professionals just starting their careers can use the app to build good financial habits early on. It can also support older adults nearing retirement in managing their savings and investments more effectively. Additionally, the app could be adapted for use by married couples or families looking to manage their household finances collectively, providing a comprehensive financial overview that supports joint decision-making and financial planning. The inclusion of modules for financial goal setting and investment tracking can cater to users with diverse financial objectives, from short-term savings to long-term investment plans .

# CHAPTER 5

## 5.0 The Proposed System

### 5.1 System Description

FinSight is a Personal Finance Management System designed specifically for single working adults. This system integrates personal informatics and optical character recognition (OCR) to streamline data entry, helping users manage their finances and achieve their financial goals. The system offers various functionalities including budgeting, expense tracking, financial goal setting, and investment monitoring. It uses a simple user interface and employs separate modules such as Budget, Cash Inflow, Financial Goal, Expense, Dashboard, Investment, and Reports, all integrated with personal informatics. The system will be developed using React Native and will run on mobile devices, available both online and offline through mobile applications. MongoDB will handle the backend database, and the system will be deployed on the Google Play Store.

### 5.2 System Objectives

#### 5.2.1 General Objective

The aim of using FinSight is to assist single working adults in the Philippines in managing their personal finances more effectively. The system addresses common challenges such as impulsive spending and the complexity of financial management, thereby helping users achieve better control over their finances. By offering functionalities like budgeting, expense tracking, and financial goal setting, the system aims to alleviate users' dissatisfaction with their financial situations and promote informed decision-making to improve their overall financial well-being.

#### 5.2.2 Specific Objectives

FinSight aims to:

1. Help users plan their monthly expenses and savings through the Budget module, which sets spending limits and tracks expenses against the budget to prevent overspending.
2. Provide a clear record of all expenditures using the Expense module, which captures data from receipts with OCR technology and automatically logs expenses to reduce manual entry and errors.
3. Assist users in setting and achieving financial goals with the Financial Goal module, enabling users to set specific savings targets, monitor progress, and stay motivated to reach their goals.
4. Track and manage all sources of income and expenditures via the Cash Inflow module for recording income and the Expense module for tracking spending, offering a comprehensive view of cash flow.

5. Help users keep track of their investments and returns using the Investment module, which allows users to input investment details and monitor performance over time.
6. Offer a summary of financial status at a glance through the Dashboard module, providing an overview of budget, expenses, goals, and investments for quick assessment of financial health.
7. Generate detailed financial reports for analysis with the Reports module, creating customizable reports on different financial aspects to help users analyze spending patterns and financial status.
8. Facilitate user interaction and provide support via a chatbot feature that assists users in navigating the system, asking questions, and retrieving information without needing to type queries.

## 5.3 System Modules

### 5.3.1 Budget Module

The Budget module empowers users to plan their monthly expenses and savings by setting specific spending limits and tracking expenses against these budgets. This structured approach aids in combating poor financial planning and decision-making difficulties by providing real-time insights, daily spending recommendations, and alerts for nearing or exceeding budgets. Additionally, it addresses poor financial behavior by promoting awareness and discipline, helping users avoid overspending and impulsive purchases. By offering a digital, organized system, the Budget module replaces traditional and cumbersome budgeting methods, such as spreadsheets or pen-and-paper, making financial planning more accessible and efficient.

### 5.3.2 Expense Module

The Expense module offers users a clear and accurate record of all their expenditures by utilizing OCR technology to capture data from receipts and automatically log expenses. This automation significantly reduces the burden of manual entry and minimizes errors, addressing the challenge of managing finances, especially in tracking expenses. It also fosters self-understanding by providing detailed records of spending habits, which is crucial for users to recognize and rectify poor financial behavior. Overall, the Expense module enhances financial management by streamlining the tracking process and ensuring accurate expenditure recording.

### 5.3.3 Financial Goal Module

The Financial Goal module assists users in setting and achieving their financial aspirations by enabling them to define specific savings targets, monitor progress, and stay motivated. This functionality helps overcome poor financial planning by creating a clear

roadmap for achieving financial goals and increasing financial awareness through education on the importance of goal setting and saving. Moreover, it promotes better financial behavior by encouraging self-awareness and disciplined saving habits, thereby aiding users in their journey towards financial stability and success.

### 5.3.4 Cash Inflow and Expense Module

The integration of the Cash Inflow and Expense modules allows users to track and manage all sources of income and expenditures, providing a comprehensive view of their cash flow. This holistic tracking system addresses the difficulty of managing finances by offering clear insights into income and spending patterns, thereby facilitating better financial planning. By ensuring users understand their financial inflows and outflows, these modules collectively help in making informed decisions and improving overall financial management.

### 5.3.5 Dashboard Module

The Dashboard module offers users a summary of their financial status at a glance by integrating data from all other modules, providing an overview of budgets, expenses, goals, and investments. This centralized display aids in making financial decisions by offering a real-time, comprehensive view of one's financial health. It enhances financial awareness by presenting the user's financial information in a clear and accessible format, allowing for quick assessment and informed decision-making. The Dashboard module thus plays a crucial role in educating users about their financial situation and empowering them to take control of their financial well-being.

### 5.3.1 Reports Module

The Reports module generates detailed financial reports, helping users analyze their spending patterns, budget adherence, and overall financial status. This in-depth analysis addresses the challenge of managing finances by providing insights and identifying areas needing improvement. It enhances financial awareness by giving users a thorough understanding of their financial habits and highlights potential overspending, thus promoting better financial discipline. Through detailed reports, users can make informed decisions, optimize their financial resources, and improve their overall financial management, contributing to their financial stability and growth.

## 5.4 System Scope

The Personal Finance Management System focuses on helping single working adults manage their finances more effectively. The system's process scope includes setting up new accounts for tracking cash inflows and allocating budgets to various expenditure categories on specific dates. Users can set financial goals and monitor their progress by allocating targeted

amounts of money, making financial decisions to achieve these goals. The system is limited to saving money, emergency funds, and specific items. Users will also follow short-term goals set by the system, contributing to their long-term financial objectives. Additional financial decisions include evaluating and understanding their current financial status through visualizations of expenditures, cash inflows, and investments.

The user scope targets Filipino single working adults aged 18 to 65 who have heightened financial concerns and are unmarried, managing their personal finances independently. These individuals should have a job and financial responsibilities such as budgeting, paying bills, saving, and investing. Income level is not a constraint, as the system aims to assist anyone looking to improve their financial well-being. Users are expected to own and operate a smart mobile device, specifically an Android device, due to the programming language limitations of the system's development.

The data and information scope involves collecting and processing personal and financial data from users. Personal information like name, username, and password is gathered during registration. Financial data, including expenses, cash inflows, investments, budget plans, financial goals, and date ranges, are collected to generate meaningful insights. This data helps users understand their financial behaviors and make informed budgeting decisions. The information is stored in an online MongoDB database, accessible through the mobile application. Users can input financial data manually or utilize Optical Character Recognition (OCR) technology for efficient data entry. The investment module is limited to tracking crypto and stocks due to the availability of applicable APIs in the Philippines.

### 5.5 System Assumptions

The system assumes that its primary users are Filipino single working adults aged 18 to 65 who are unmarried and manage their personal finances independently. The technology stack includes React Native for front-end development and MongoDB for backend database management, with the system being available as a mobile application. OCR technology will be used to capture data from receipts, streamlining expense tracking. The system is designed to function both online and offline, ensuring accessibility in various conditions. A chatbot feature will facilitate user interaction with financial data through a button-based interface, eliminating the need for typing queries. The development process follows an Agile methodology with sprints focusing on different modules and functionalities, and user feedback will be integral in refining the system. The system assumes that users need assistance in managing impulsive spending habits and making informed financial decisions. Additionally, it is assumed that users require secure handling of their financial data and personal information. These assumptions guide the system's development and deployment, ensuring it meets the specific needs and challenges faced by its target users.

## 5.6 System Feasibility

### 5.6.1 Developmental Feasibility

The developmental feasibility of the Personal Finance Management System for single working adults in the Philippines is strongly supported by the chosen technology stack and methodology. Utilizing React Native for the front end ensures cross-platform compatibility, particularly catering to the Android user base which is prevalent among the target demographic. MongoDB, as the backend database, offers scalable and flexible data management capabilities essential for handling the varied financial data inputs from users. The development process follows an Agile methodology, facilitating iterative progress and allowing for continuous user feedback and system refinement. This approach ensures that the system evolves in response to real user needs and challenges, enhancing its relevance and effectiveness. Additionally, the integration of OCR technology for data entry from receipts simplifies the process of expense tracking, addressing a common pain point for users. The system's ability to function both online and offline further enhances its usability in diverse connectivity conditions, making it accessible to a wider audience regardless of their internet access status.

### 5.6.2 Economical Feasibility

The economic feasibility of the project is grounded in its potential to significantly improve the financial well-being of its users, thereby addressing a critical need in the market. By targeting Filipino single working adults who manage their personal finances independently, the system caters to a substantial and growing segment of the population. The use of widely adopted technologies like React Native and MongoDB helps to keep development and maintenance costs manageable. Additionally, the system's capability to assist users in controlling impulsive spending and making informed financial decisions can lead to tangible savings and better financial health for users, creating a strong value proposition. The economic viability is further supported by the system's scalability, allowing for expansion and adaptation based on user demand and feedback. This scalability ensures that the initial investment in development can yield long-term benefits as the user base grows and the system's functionality is expanded.

## 5.7 Technical Requirements

### 5.7.1 For the Developers

For developing and running the React Native environment, both Node.js and npm are essential. Node.js should be version 14.x or higher, while npm needs to be version 6.x or higher. React Native is the core framework used to build the mobile application, and it is important to have the latest compatible version of the React Native CLI. Gradle, version 6.0 or higher, is necessary for building and packaging the Android application,

aligning with the Android build system requirements. Additionally, the Java Development Kit (JDK) version 17 or higher is required to run Gradle and build the Android app. Android Studio serves as the integrated development environment (IDE) for Android development, while the latest stable version of the Android SDK is essential for compatibility with the target Android OS. During development, the Android Emulator is used for testing the application.

Several third-party libraries and tools are integrated to enhance the application's functionality. React-native-reanimated is used for creating smooth animations and interactions within the app. The OCR library, TextRecognition from '@react-native-ml-kit/text-recognition', is employed for extracting text from images of receipts. MongoDB is the database utilized for storing and retrieving user data, with MongoDB Atlas being an optional choice for cloud-based database management. For building the backend API that communicates with the mobile app, Express.js is used, and JWT (JSON Web Tokens) provides authentication and secures API endpoints.

The backend infrastructure is hosted on a Node.js server, utilizing the Express.js framework to facilitate the construction of the RESTful API. An API Gateway is employed to manage API requests between the mobile app and backend services, ensuring efficient and organized communication. To guarantee secure data transmission, HTTPS is implemented, safeguarding the communication between the app and the backend.

For managing source code and facilitating collaboration among developers, Git is employed as the version control system. This ensures that changes to the codebase are tracked, and team members can work together efficiently on the project.

### 5.7.2 For the Users

To use the FinSight application on an Android device, the operating system requirements are crucial. The user's phone must be running on Android 8.0 (Oreo) or higher. This ensures that the device is compatible with the latest features and security updates provided by both Android and the FinSight app. Android 8.0 or higher versions offer improved performance, enhanced security features, and support for newer APIs that the FinSight app relies on to deliver a seamless user experience.

In terms of device requirements, the phone should have a minimum of 3GB of RAM. This amount of memory ensures that the app runs smoothly and efficiently, without lag or performance issues. Having adequate RAM is essential for handling the

app's functionalities, such as real-time data processing, animation rendering, and background operations.

Additionally, the phone must have at least 100MB of free storage space for installing and running the FinSight application. This space requirement includes the app itself and its essential data files. However, users should also have additional storage available for storing receipts and other user data captured by the app. Ensuring sufficient storage prevents potential issues with data saving and retrieval, which are critical for the app's operation.

A functional camera is also necessary for using the FinSight application, as it is required for capturing receipts to utilize the Optical Character Recognition (OCR) functionality. The OCR technology allows users to take photos of their receipts, which the app then processes to extract and digitize financial information. This feature is vital for the app's expense tracking and budgeting capabilities, making the camera an essential hardware requirement.

## 5.8 Non-Technical Requirements

### 5.8.1 Data Security and Integrity

FinSight ensures robust data security and integrity by implementing a comprehensive, multi-layered approach across its technology stack. Passwords are hashed using strong cryptographic algorithms like bcrypt before storage, ensuring they remain secure even if the database is compromised. MongoDB provides role-based access control, data encryption at rest with AES, and secure TLS/SSL protocols for data in transit. The backend, built with Node.js and Express.js, incorporates middleware for input validation, sanitization, and request rate limiting to prevent attacks, and uses modules like Helmet to set secure HTTP headers. GitHub's security features, including protected branches, access controls, security alerts, and code reviews, help maintain code integrity and address vulnerabilities promptly. The React Native mobile app secures communication with HTTPS, obfuscates and minifies code to prevent reverse engineering, and uses encrypted local storage for sensitive information. Android Studio's tools and best practices, such as the Android Keystore system and permission management, further enhance security. By combining these measures, FinSight provides a secure environment for users to manage their financial data confidently, with regular audits, code reviews, and updates maintaining its robust security posture.

### 5.8.2 System Performance

The system performance of FinSight is optimized to provide a seamless and efficient user experience through a combination of frontend and backend enhancements. On the frontend, React Native ensures fast rendering and responsive UI by leveraging virtual DOM, asynchronous processing, and optimization techniques such as code splitting and lazy loading. Caching frequently accessed data locally and supporting offline functionality further enhances performance. The backend uses well-designed RESTful APIs and load balancing to facilitate efficient communication and prevent bottlenecks. Scalability is achieved through horizontal scaling and optimized database configurations in MongoDB, which include indexing, sharding, and replication. Asynchronous processing and batch processing of data-intensive tasks prevent delays in user interactions. Real-time monitoring with tools like Prometheus and Grafana tracks system performance, while automated alerts and continuous optimization practices such as code reviews, automated testing, and user feedback integration ensure any issues are promptly addressed. The use of scalable cloud infrastructure and a CDN for static assets further reduces load times and improves reliability. By ensuring that user devices meet minimum specifications (e.g., 3GB of RAM, adequate storage, and a functional camera), FinSight maintains consistent performance across different hardware configurations. This comprehensive approach ensures that FinSight delivers a high-performing, reliable financial management tool.

### 5.8.2 User Interface

The user interface of FinSight is meticulously crafted to offer a seamless and intuitive user experience across various financial management tasks. The app's design adheres to user-centered principles, ensuring accessibility, clarity, and ease of use.

The Sign-Up and Login screens are designed to be straightforward and welcoming. They feature minimalistic layouts with clear input fields for email, password, and other necessary details. Prominent buttons for submitting information, as well as links for password recovery and account creation, are easily accessible. These screens utilize a simple color scheme to maintain focus and reduce user confusion, ensuring a smooth onboarding process.

The Cash Inflow screen is tailored for efficient income management. It includes a prominently placed add button for quick income entries and a main section displaying a list of recorded incomes. Each entry highlights essential details such as source, amount, and date, arranged chronologically. The form for adding new income is clean and straightforward, with fields for source, amount, date, and notes, enhanced by drop-down menus and auto-complete features to streamline data entry.

The Dashboard offers a comprehensive overview of the user's financial status. It features key financial metrics, visual representations like graphs and charts, and summaries of budgets, expenses, and savings goals. This screen is designed with a clear visual hierarchy to emphasize important information, ensuring users can quickly grasp their financial situation.

The Expense Tracking screen allows for both automatic and manual entry of expenses. Expenses are categorized and displayed with clear labels, and users can manually input cash expenses or adjust categorizations easily. The design ensures that users can swiftly navigate through their spending data, supported by an organized layout and responsive controls.

The Budgeting screen enables users to set and adjust budgets for various categories. It provides a detailed view of allocated amounts, spending limits, and remaining balances. Users can customize categories and receive visual feedback on their budget adherence. The layout is designed to be intuitive, making it easy for users to manage and adjust their financial plans.

The Financial Goals screen is designed to help users set, track, and achieve their financial goals. It features a clean and motivational layout where users can define goals, such as saving for a vacation or building an emergency fund. Each goal card displays the target amount, progress made, and time remaining, with visual progress bars to provide a quick overview. Users can easily add new goals, update existing ones, and monitor their progress with detailed breakdowns and insights.

The Reports screen provides detailed insights into the user's financial activities. It includes graphical representations of spending trends, income patterns, and budget performance. The design focuses on clarity and detail, allowing users to analyze their financial behavior and make informed decisions.

Overall, FinSight's user interface is designed to be user-friendly, with a focus on clarity, ease of navigation, and accessibility. Each screen is optimized for Android devices, ensuring a consistent and responsive experience across different devices and screen sizes. The cohesive design principles applied throughout the app facilitate efficient financial management and enhance user satisfaction.

## 5.9 Data Specifications

### 5.9.1 Entity Relationship Diagram

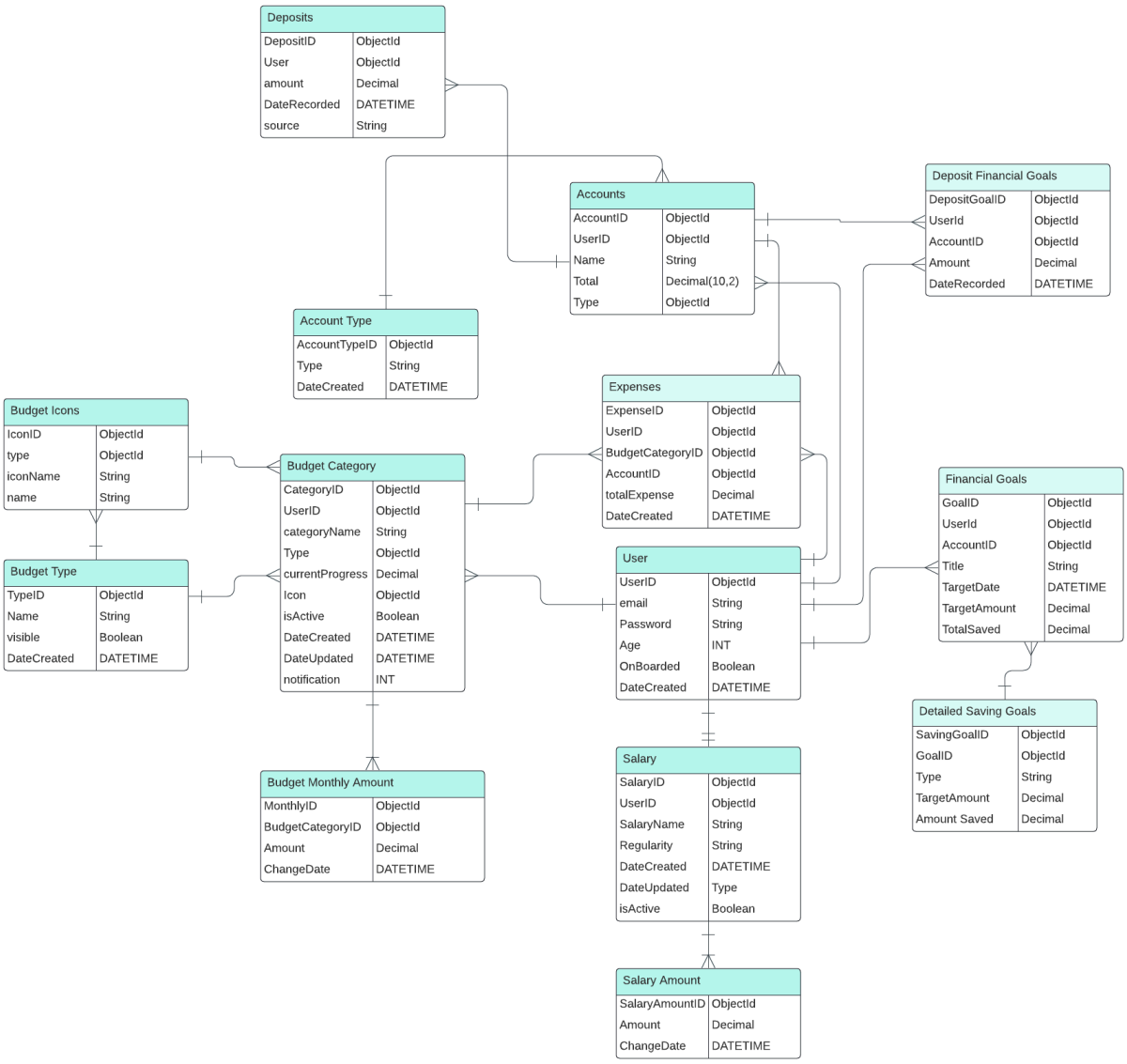



*Figure 5. Entity Relationship Diagram of FinSight*

The Entity Relationship Diagram (ERD) for the FinSight app outlines the comprehensive database structure and its interconnected entities, ensuring robust data management and a seamless user experience. The primary entity, User, is linked to several other entities, including Deposits, Accounts, Expenses, Financial Goals, Budget Categories, and Salaries. Each of these entities has unique attributes and relationships with other entities. For instance, Deposits and Expenses are linked to Accounts and Budget Categories, while Financial Goals can have multiple Deposit Financial Goals and

Detailed Saving Goals. Budget Categories are further associated with Budget Types and Budget Icons. Accounts are categorized by Account Types. The ERD ensures that users can manage their financial data efficiently, with relationships facilitating accurate tracking of deposits, expenses, financial goals, and budget management. This structure supports dynamic updates and comprehensive data retrieval, ensuring that all aspects of a user's financial data are accurately recorded and easily accessible.

### 5.9.2 Tables/Files Layout

#### 5.9.2.1 User

| Field Name | Description | Data Type |
|---|---|---|
| UserID | Unique identifier for the user | ObjectId |
| email | User’s email address | String |
| Password | User’s password | String |
| Age | User’s age | Integer |
| OnBoarded | User onboarding status | Boolean |
| DateCreated | Date when user was created | DateTime |

#### 5.9.2.2 Deposits

| Field Name | Description | Data Type |
|---|---|---|
| DepositID | Unique identifier for the deposit | ObjectId |
| User | Reference to UserID | ObjectID |
| amount | Amount deposited | Decimal |
| DateRecorded | Date when deposit was recorded | DateTime |
| source | Source of the deposit | String |

#### 5.9.2.3 Accounts

| Field Name | Description | Data Type |
|---|---|---|
| AccountID | Unique identifier for the account | ObjectId |
| UserID | Reference to UserID | ObjectId |
| Name | Name of the account | String |
| Total | Total amount in the account | Decimal(10,2) |
| Type | Type of account (reference to AccountTypeID) | ObjectId |

#### 5.9.2.4 Account Type

| Field Name | Description | Data Type |
|---|---|---|
| AccountTypeID | Unique identifier for the account type | ObjectId |
| Type | Type of account | String |
| DateCreated | Date when account type was created | DateTime |

#### 5.9.2.5 Expenses

| Field Name | Description | Data Type |
|---|---|---|
| ExpenseID | Unique identifier for the expense | ObjectId |
| UserID | Reference to UserID | ObjectId |
| BudgetCategoryID | Reference to BudgetCategoryID | ObjectId |
| AccountID | Reference to AccountID | ObjectId |
| totalExpense | Total expense amount | Decimal |
| DateCreated | Date when expense was created | DateTime |

#### 5.9.2.6 Financial Goals

| Field Name | Description | Data Type |
|---|---|---|
| GoalID | Unique identifier for the financial goal | ObjectId |
| UserID | Reference to UserID | ObjectId |
| AccountID | Reference to AccountID | ObjectId |
| Title | Title of the financial goal | String |
| TargetDate | Target date to achieve the goal | DateTime |
| TargetAmount | Target amount for the goal | Decimal |
| TotalSaved | Total amount saved towards the goal | Decimal |

#### 5.9.2.7 Deposit Financial Goals

| Field Name | Description | Data Type |
|---|---|---|
| DepositGoalID | Unique identifier for the deposit financial goal | ObjectId |
| UserID | Reference to UserID | ObjectId |
| AccountID | Reference to AccountID | ObjectId |
| Amount | Amount deposited towards financial goal | Decimal |
| DateRecorded | Date when deposit was recorded | DateTime |

#### 5.9.2.8 Detailed Saving Goals

| Field Name | Description | Data Type |
|---|---|---|
| SavingGoalID | Unique identifier for the saving goal | ObjectId |
| GoalID | Reference to GoalID | ObjectId |
| Type | Type of saving goal | String |
| TargetAmount | Target amount for the saving goal | Decimal |
| AmountSaved | Amount saved towards the goal | Decimal |

#### 5.9.2.9 Budget Icons

| Field Name | Description | Data Type |
|---|---|---|
| IconID | Unique identifier for the icon | ObjectId |
| type | Type of icon | ObjectId |
| iconName | Name of the icon | String |
| name | Display name of the icon | String |

#### 5.9.2.10 Budget Category

| Field Name | Description | Data Type |
|---|---|---|
| CategoryID | Unique identifier for the budget category | ObjectId |
| UserID | Reference to UserID | ObjectId |
| categoryType | Type of budget category | ObjectId |
| Type | Reference to TypeID | ObjectId |
| currentProgress | Current progress of the budget | Decimal |
| Icon | Reference to IconID | ObjectId |
| isActive | Status of the budget category | Boolean |
| DateCreated | Date when budget category was created | DateTime |

| Field Name | Description | Data Type |
|---|---|---|
| CategoryID | Unique identifier for the budget category | ObjectId |
| UserID | Reference to UserID | ObjectId |
| categoryType | Type of budget category | ObjectId |
| Type | Reference to TypeID | ObjectId |
| DateUpdated | Date when budget category was last updated | DateTime |
| notification | Notification settings for the budget category | Integer |

**5.9.2.11 Budget Type**

| Field Name | Description | Data Type |
|---|---|---|
| TypeID | Unique identifier for the budget type | ObjectId |
| Name | Name of the budget type | String |
| visible | Visibility status | Boolean |
| DateCreated | Date when budget type was created | DateTime |

**5.9.2.12 Budget Monthly Amount**

| Field Name | Description | Data Type |
|---|---|---|
| MonthlyID | Unique identifier for the monthly budget amount | ObjectId |
| BudgetCategoryID | Reference to BudgetCategoryID | ObjectId |
| Amount | Amount allocated for the month | Decimal |
| ChangeDate | Date when the amount was changed | DateTime |

#### 5.9.2.13 Salary

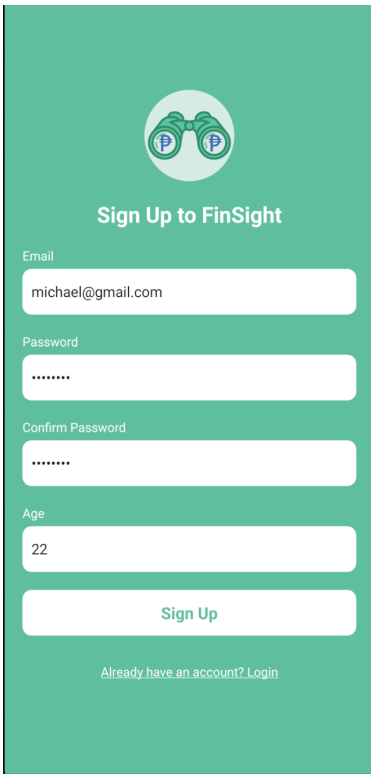
| Field Name | Description | Data Type |
|---|---|---|
| SalaryID | Unique identifier for the salary | ObjectId |
| UserID | Reference to UserID | ObjectId |
| SalaryName | Name of the salary | String |
| Regularity | Regularity of the salary (e.g., monthly, weekly) | String |
| DateCreated | Date when salary was created | DateTime |
| DateUpdated | Date when salary was last updated | DateTime |
| isActive | Status of the salary | Boolean |

#### 5.9.2.14 Salary Amount

| Field Name | Description | Data Type |
|---|---|---|
| SalaryAmountID | Unique identifier for the salary amount | ObjectId |
| Amount | Amount of the salary | Decimal |
| ChangeDate | Date when the salary amount was changed | DateTime |

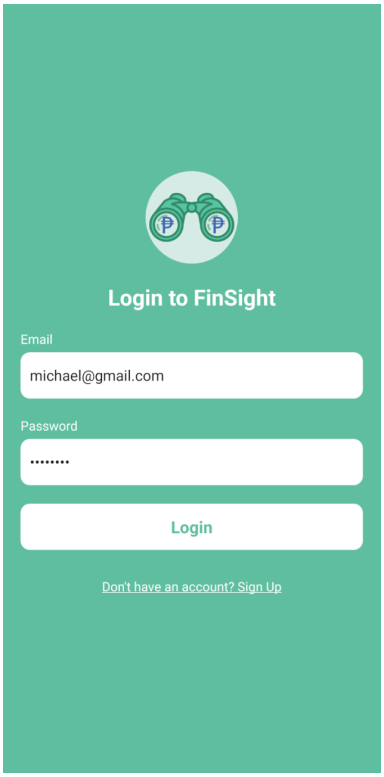
## 5.10 Screen Specifications

### 5.10.1 Sign Up Screen




The signup screen is designed to provide a smooth and straightforward registration process for new users. It features a clean and minimalistic layout, with input fields for essential information such as email, password, and username. Clear labels and placeholders guide users on what information to enter, while validation messages ensure that the data entered is correct before submission. The design includes a prominent "Sign Up" button, making it easy for users to complete the registration process. The use of a simple color scheme and intuitive navigation enhances the overall user experience.

### 5.10.2 Login Screen




The login screen focuses on simplicity and quick access to the app. It features input fields for email and password, with clear and concise labels. The overall design is user-centric, emphasizing ease of use and security. The layout is kept uncluttered to avoid distractions, and the color scheme aligns with the overall branding of the app, providing a consistent user experience.

### 5.10.3 Setup Screens

The onboarding screen welcomes new users and guides them through the initial setup of the app. It features a clean and intuitive layout with clear instructions and visuals that illustrate the app's key features. The design focuses on simplicity, with a step-by-step approach to help users understand how to navigate and utilize the app effectively. Interactive elements such as buttons and progress indicators enhance user engagement and provide a smooth onboarding experience.

### 5.10.3.1 Financial Goal Setup

Welcome To FinSight!

Set Up Your Financial Goal Now!

Get Started

Financial Goal Setup

What Is Your Financial Goal?

Vacation

How Much Does It Cost?

35000

How Much Have You Saved?

0

When Do You Want To Achieve It?

11-15-2024

Create Goal

Confirmation

Financial Goal Information

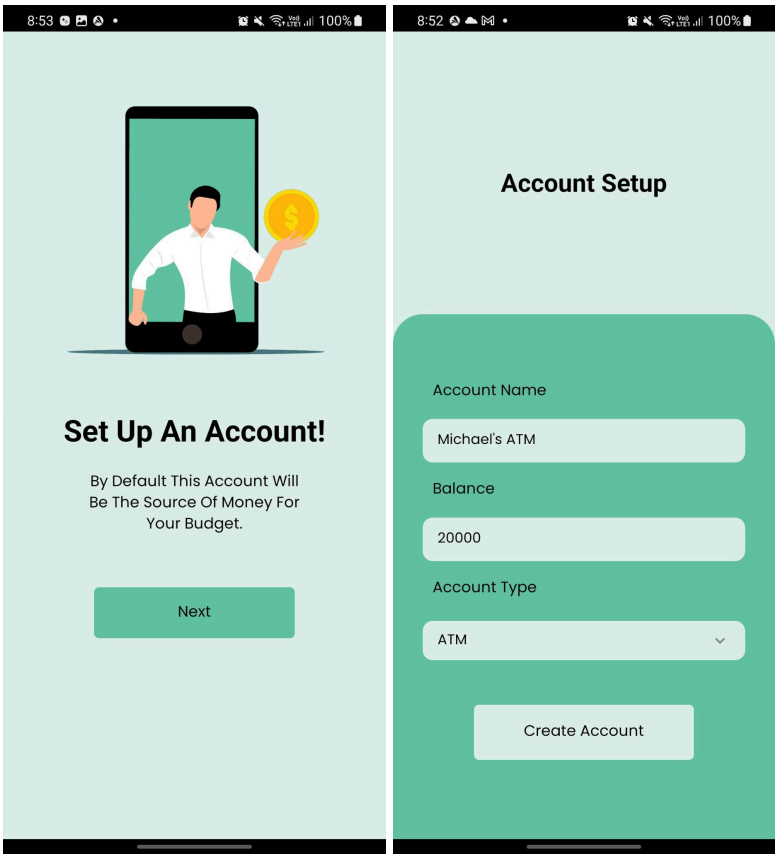
| | |
|---|---|
| **Goal Name:** | Vacation |
| **Cost:** | 35000 |
| **How Much Have You Saved?:** | 0 |
| **When Do You Want To Achieve It?:** | 11-15-2024 |

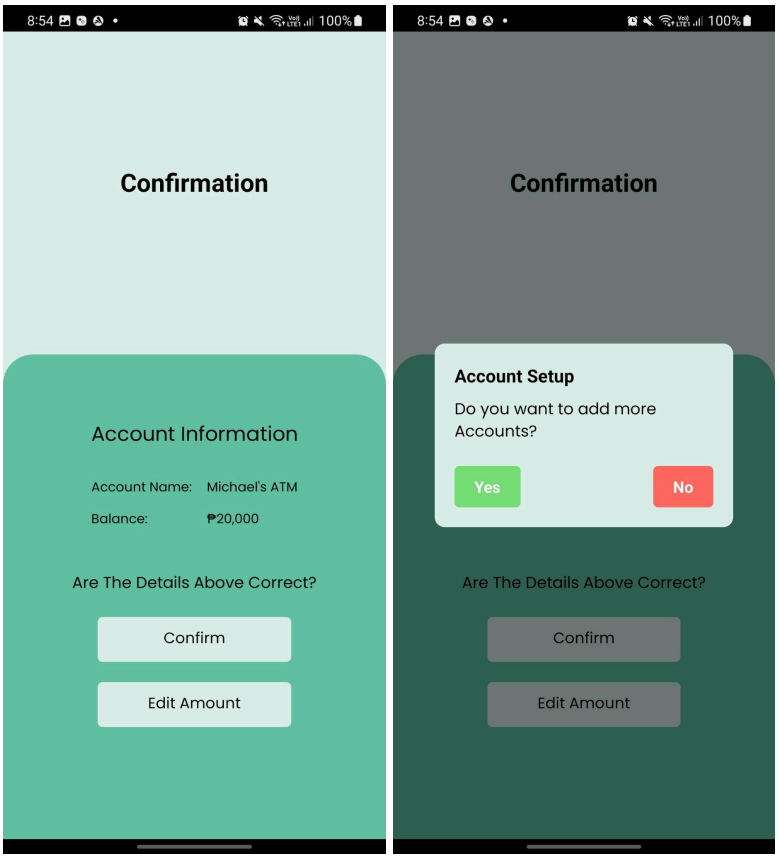
Are The Details Above Correct?

Confirm

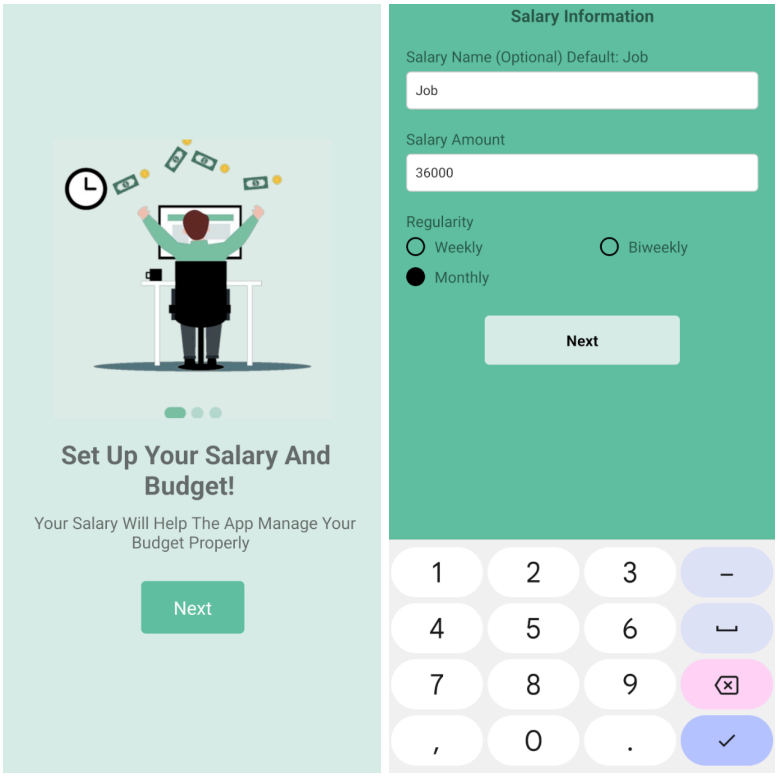
Edit Goal

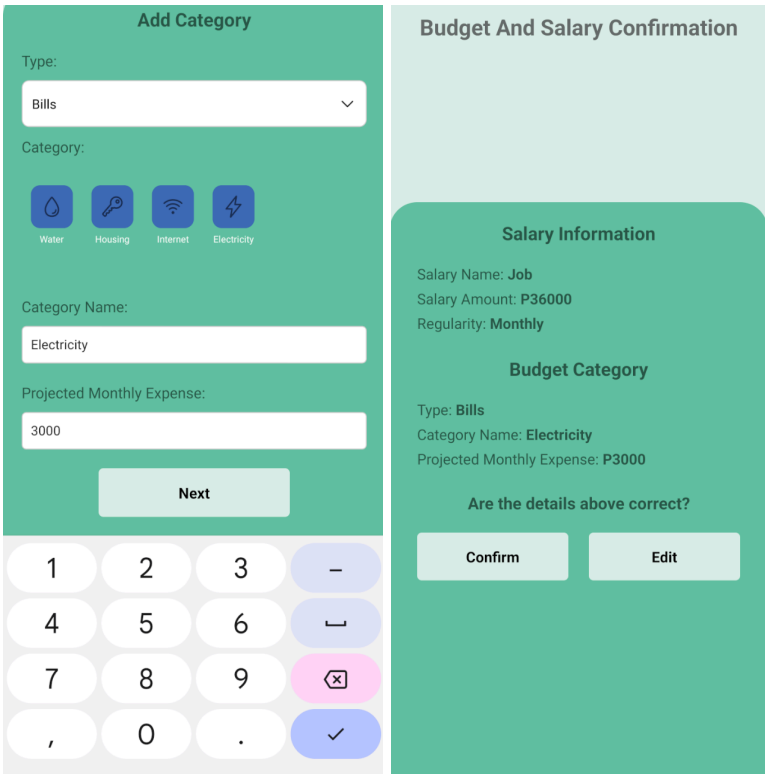
### 5.10.3.2 Account Setup

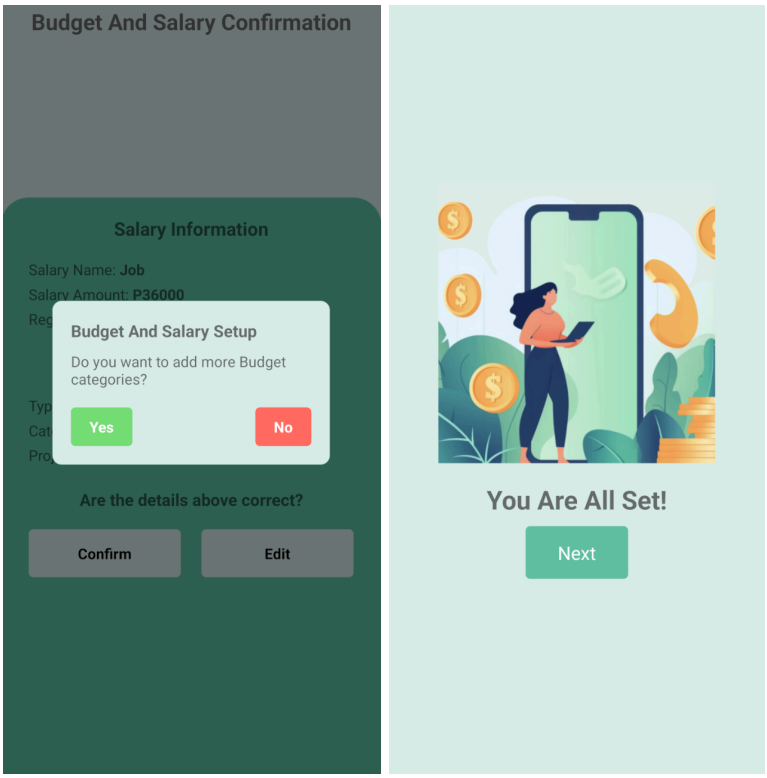



8:54
100%
Confirmation
Account Information
Account Name: Michael's ATM
Balance: ₱20,000
Are The Details Above Correct?
Confirm
Edit Amount
8:54
100%
Confirmation
Account Setup
Do you want to add more Accounts?
Yes
No
Are The Details Above Correct?
Confirm
Edit Amount

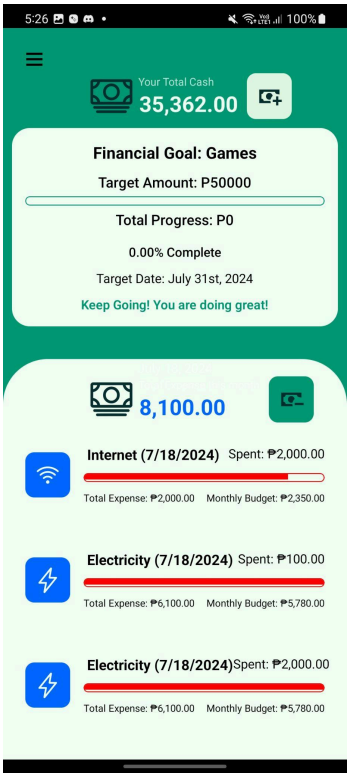
### 5.10.3.3 Salary and Budget Setup



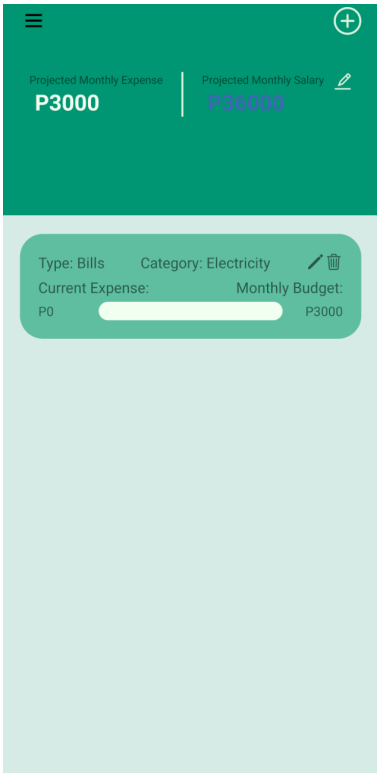

Budget And Salary Confirmation
Salary Information
Salary Name: Job
Budget And Salary Setup
Do you want to add more Budget categories?
Yes
No
Are the details above correct?
Confirm
Edit
You Are All Set!
Next

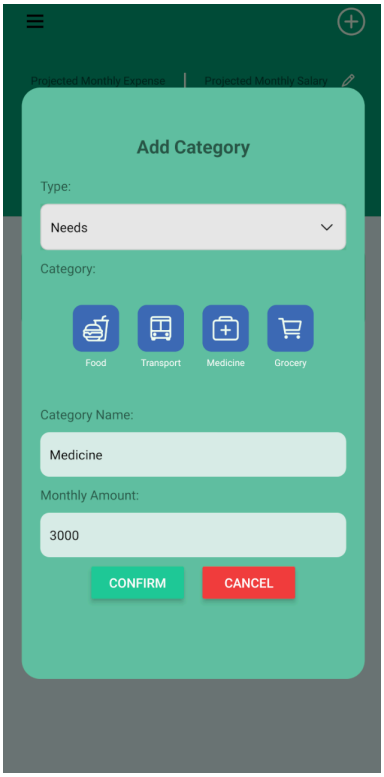
### 5.10.3 Dashboard Module

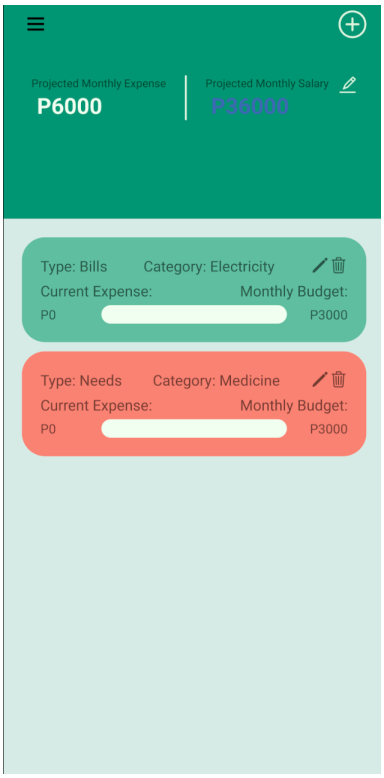


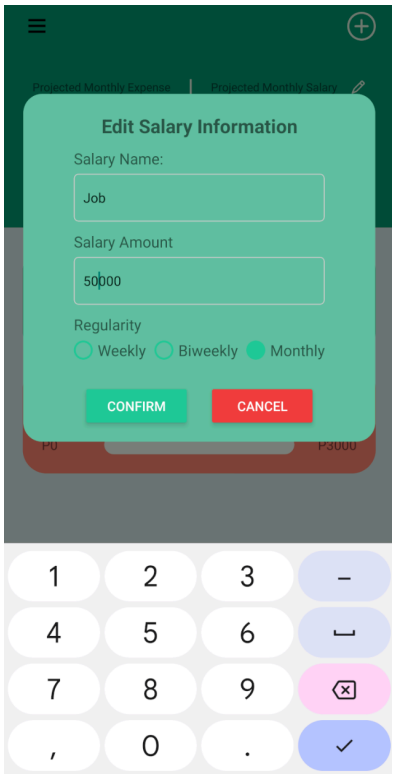

The home screen serves as the main dashboard, providing an overview of the user's financial status. It includes essential components like a summary of the current budget, recent transactions, and progress towards financial goals. The design emphasizes clarity and accessibility, using a combination of progress bars and icons to present financial data in an easily digestible format. The use of contrasting colors helps highlight critical information, while a modular layout allows users to quickly access different sections of the app.

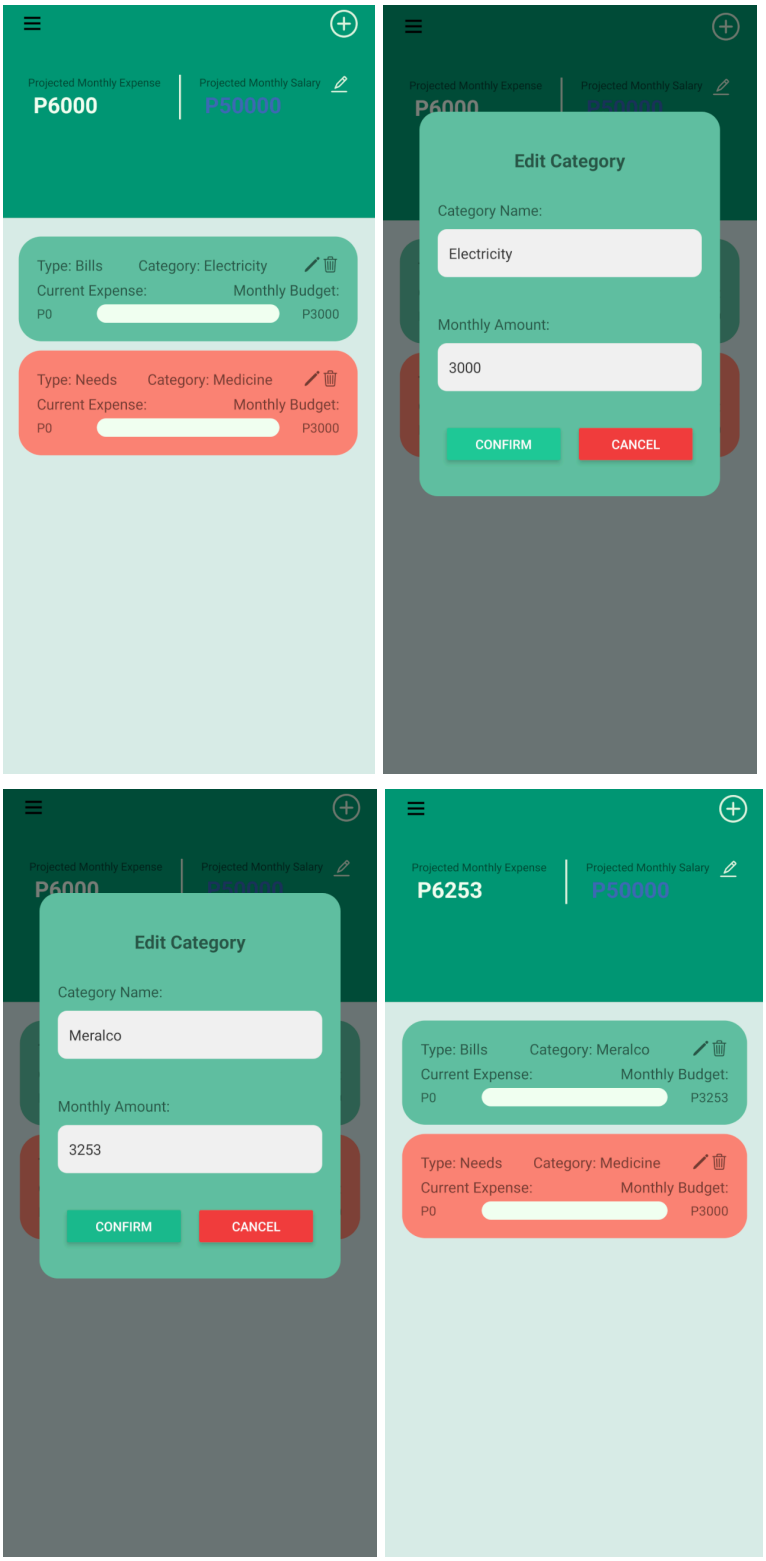
### 5.10.4 Budget Module

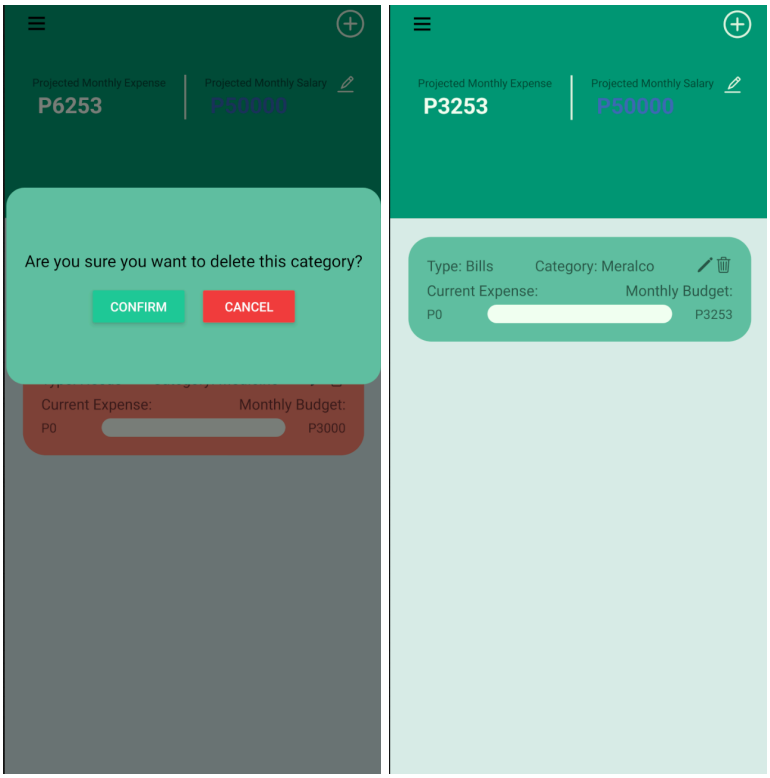


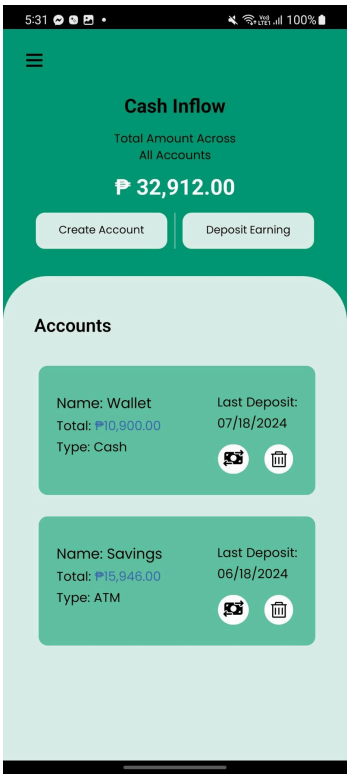



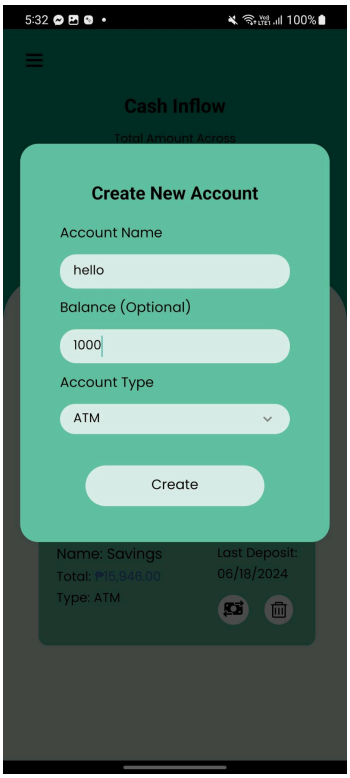

Projected Monthly Expense
P6000
Projected Monthly Salary
P50000
Type: Bills
Category: Electricity
Current Expense:
Monthly Budget:
P0
P3000
Type: Needs
Category: Medicine
Current Expense:
Monthly Budget:
P0
P3000
Edit Category
Category Name:
Electricity
Monthly Amount:
3000
CONFIRM
CANCEL
Edit Category
Category Name:
Meralco
Monthly Amount:
3253
CONFIRM
CANCEL
Projected Monthly Expense
P6253
Projected Monthly Salary
P50000
Type: Bills
Category: Meralco
Current Expense:
Monthly Budget:
P0
P3253
Type: Needs
Category: Medicine
Current Expense:
Monthly Budget:
P0
P3000

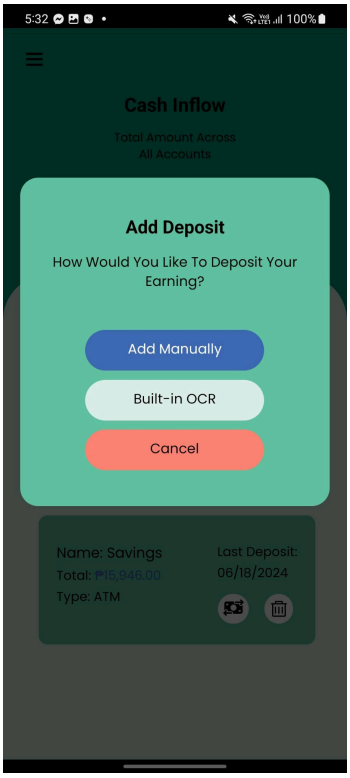


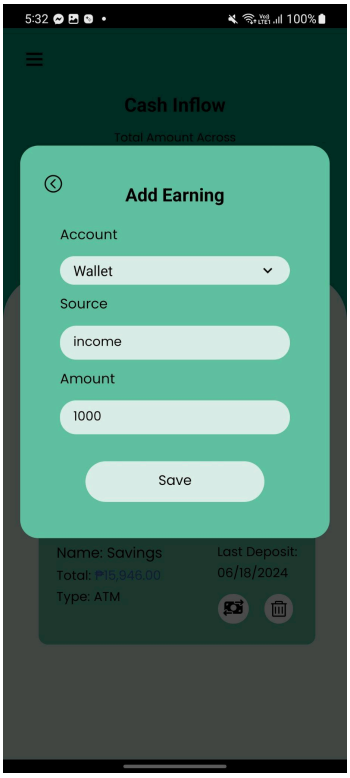

The budget screen allows users to create and manage their monthly budget. It features a detailed breakdown of budget and expenses, categorized into different spending areas such as groceries, rent, and entertainment. The design incorporates input fields, and dropdown menus for easy data entry and adjustments. Visual aids like progress bars provide a clear representation of budget allocations and spending patterns, helping users make informed financial decisions.

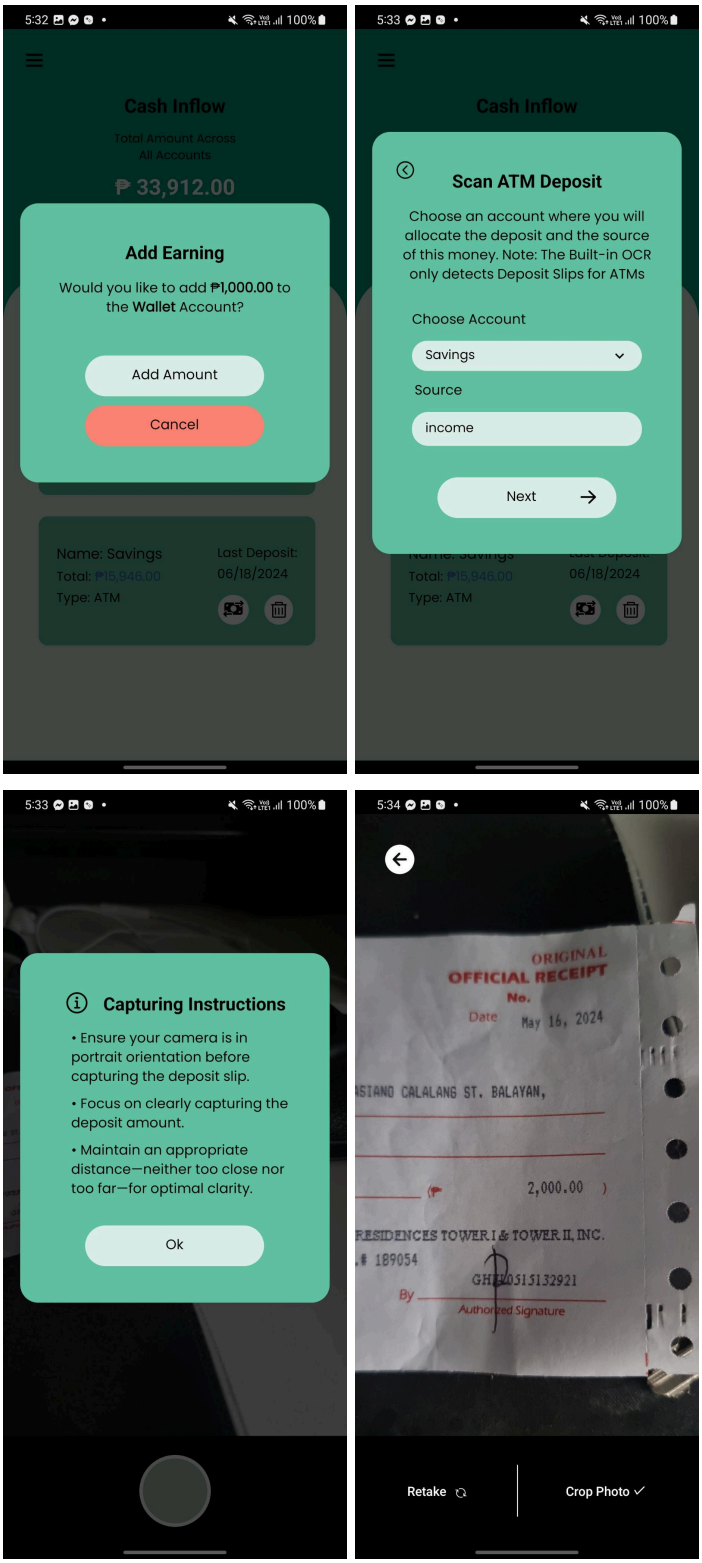
### 5.10.5 Cash Inflow Module

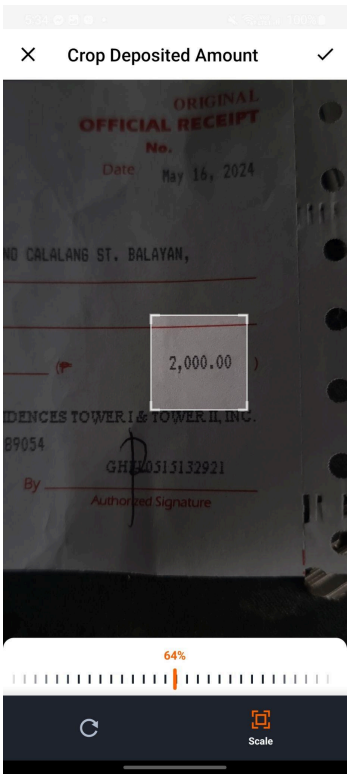


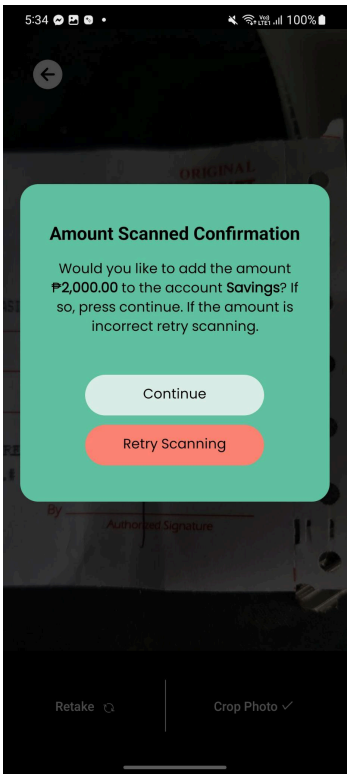



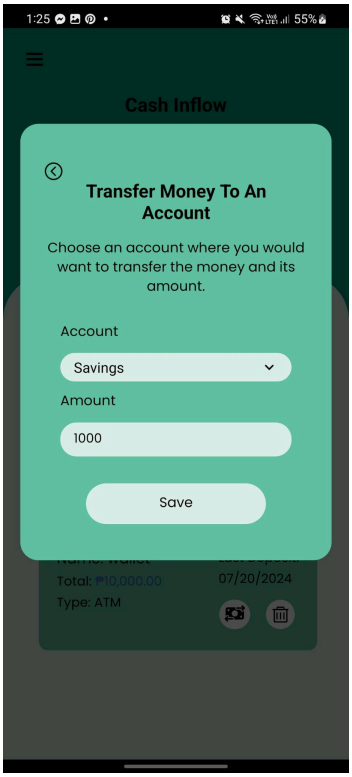



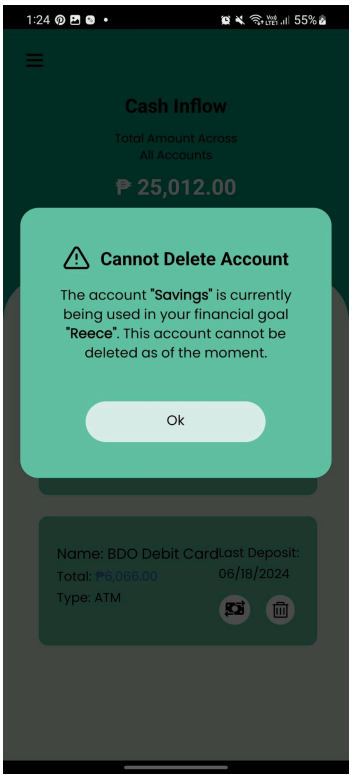

5:32
100%
Cash Inflow
Total Amount Across All Accounts
₱ 33,912.00
Add Earning
Would you like to add ₱1,000.00 to the Wallet Account?
Add Amount
Cancel
Name: Savings
Total: ₱15,946.00
Type: ATM
Last Deposit:
06/18/2024
5:33
100%
Cash Inflow
Scan ATM Deposit
Choose an account where you will allocate the deposit and the source of this money. Note: The Built-in OCR only detects Deposit Slips for ATMs
Choose Account
Savings
Source
income
Next
Total: ₱15,946.00
Type: ATM
06/18/2024
5:33
100%
Capturing Instructions
• Ensure your camera is in portrait orientation before capturing the deposit slip.
• Focus on clearly capturing the deposit amount.
• Maintain an appropriate distance—neither too close nor too far—for optimal clarity.
Ok
5:34
100%
ORIGINAL
OFFICIAL RECEIPT
No.
Date May 16, 2024
2,000.00
RESIDENCES TOWER I & TOWER II, INC.
189054
By
Authorized Signature
Retake
Crop Photo

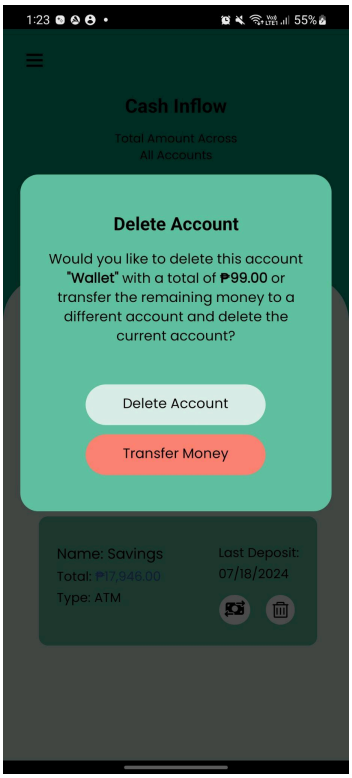

Crop Deposited Amount
ORIGINAL
OFFICIAL RECEIPT
No.
Date May 16, 2024
NO CALALANG ST. BALAYAN,
2,000.00
IDENCES TOWER I & TOWER II, INC.
89054
GHI-0515132921
By
Authorized Signature
64%
Scale

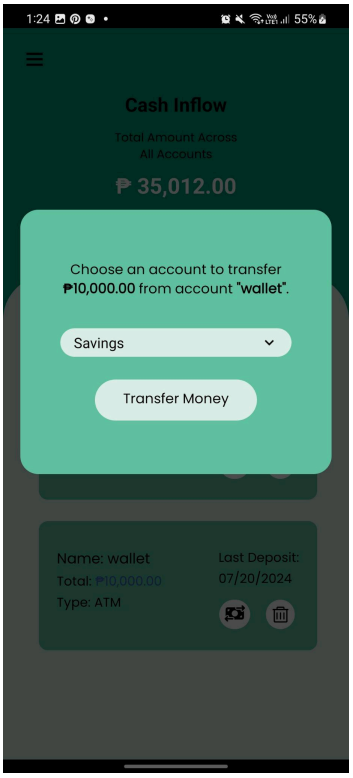

5:34
100%
Amount Scanned Confirmation
Would you like to add the amount ₱2,000.00 to the account Savings? If so, press continue. If the amount is incorrect retry scanning.
Continue
Retry Scanning
Retake
Crop Photo

5:34
100%
Cash Inflow
Wallet
Average ₱16,000.00
Total ₱11,900.00
Account Activities
Date
Deposits
Source
05/18/2024
₱2,000.00
Initial Deposit
06/18/2024
₱3,000.00
salary
07/18/2024
₱10,000.00
Account Transfer from account: Savings
07/18/2024
₱1,000.00
income


1:25
55%
Cash Inflow
Transfer Money To An Account
Choose an account where you would want to transfer the money and its amount.
Account
Savings
Amount
1000
Save
Total: ₱10,000.00
07/20/2024
Type: ATM

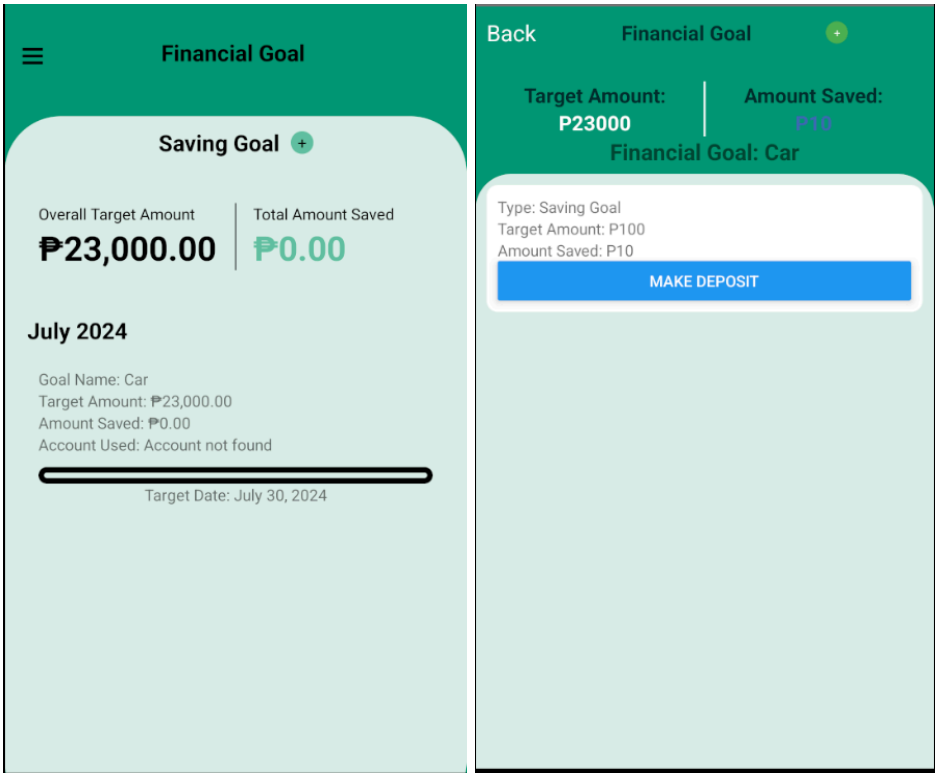

Cash Inflow
Total Amount Across All Accounts
₱ 25,012.00
Cannot Delete Account
The account "Savings" is currently being used in your financial goal "Reece". This account cannot be deleted as of the moment.
Ok
Name: BDO Debit Card
Total: ₱6,066.00
Type: ATM
Last Deposit: 06/18/2024

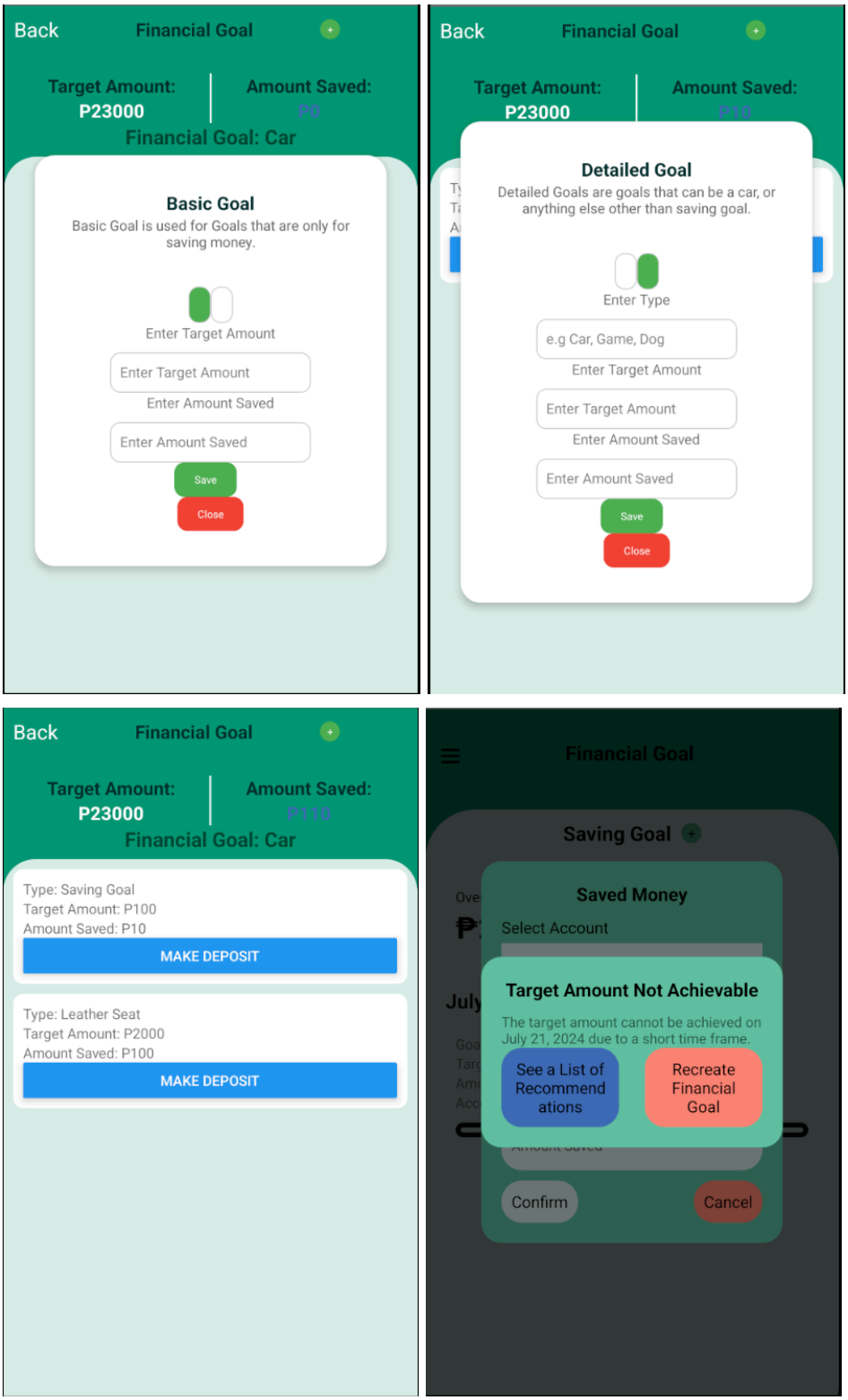

Cash Inflow
Total Amount Across All Accounts
Delete Account
Would you like to delete this account "Wallet" with a total of ₱99.00 or transfer the remaining money to a different account and delete the current account?
Delete Account
Transfer Money
Name: Savings
Total: ₱17,946.00
Type: ATM
Last Deposit: 07/18/2024

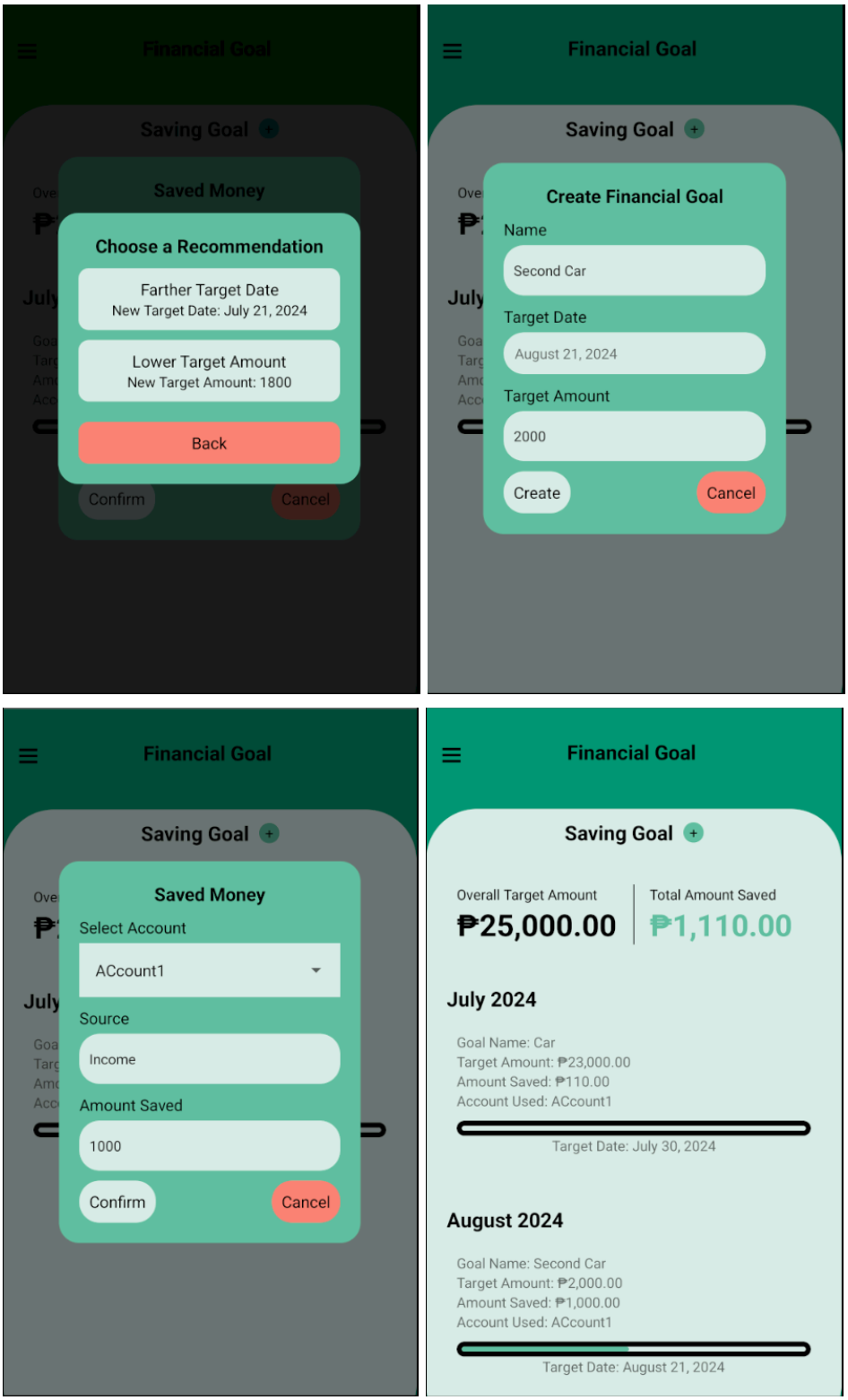

Cash Inflow
Total Amount Across All Accounts
₱ 35,012.00
Choose an account to transfer ₱10,000.00 from account "wallet".
Savings
Transfer Money
Name: wallet
Total: ₱10,000.00
Type: ATM
Last Deposit: 07/20/2024

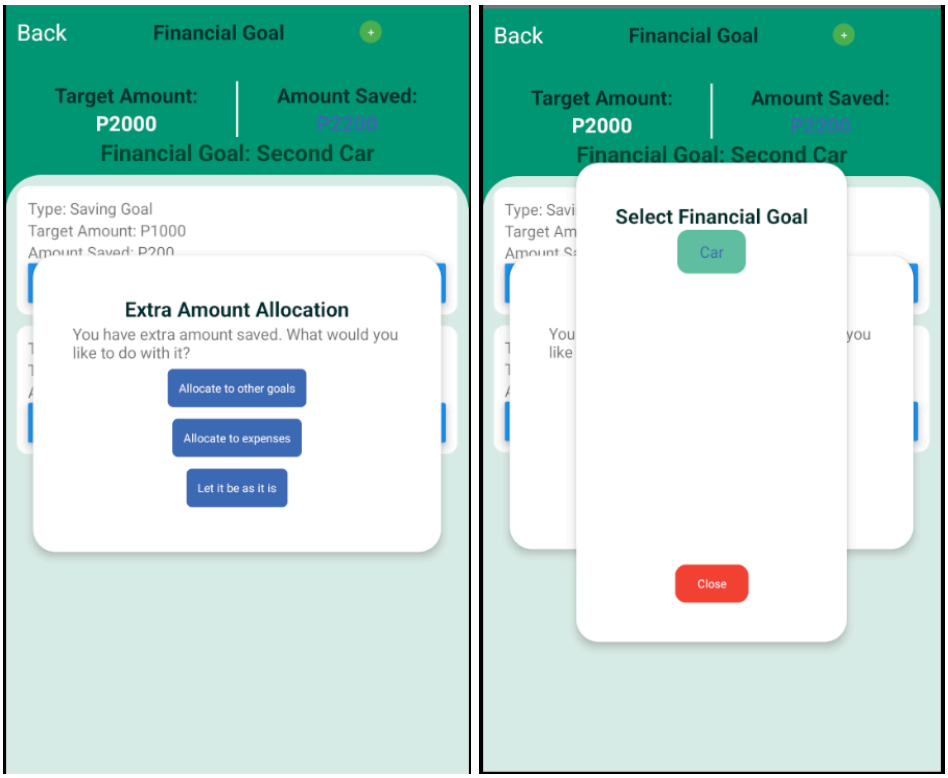
The Cash Inflow screen in FinSight is designed with user convenience and clarity in mind. It features an intuitive layout that includes a prominently placed add button for quick income entry, typically located at the top-right corner. The main section displays a list of recorded income entries in a clear and organized manner, with essential details like source, amount, and date, arranged chronologically. Adding a new income entry is facilitated through a clean form with fields for source, amount, date, and notes, supported by drop-down menus and auto-complete features to streamline data entry. Important elements, such as total cash inflow and individual income amounts, are highlighted using larger fonts and contrasting colors for quick identification. Immediate feedback through confirmation messages or alerts ensures users can promptly address any issues.

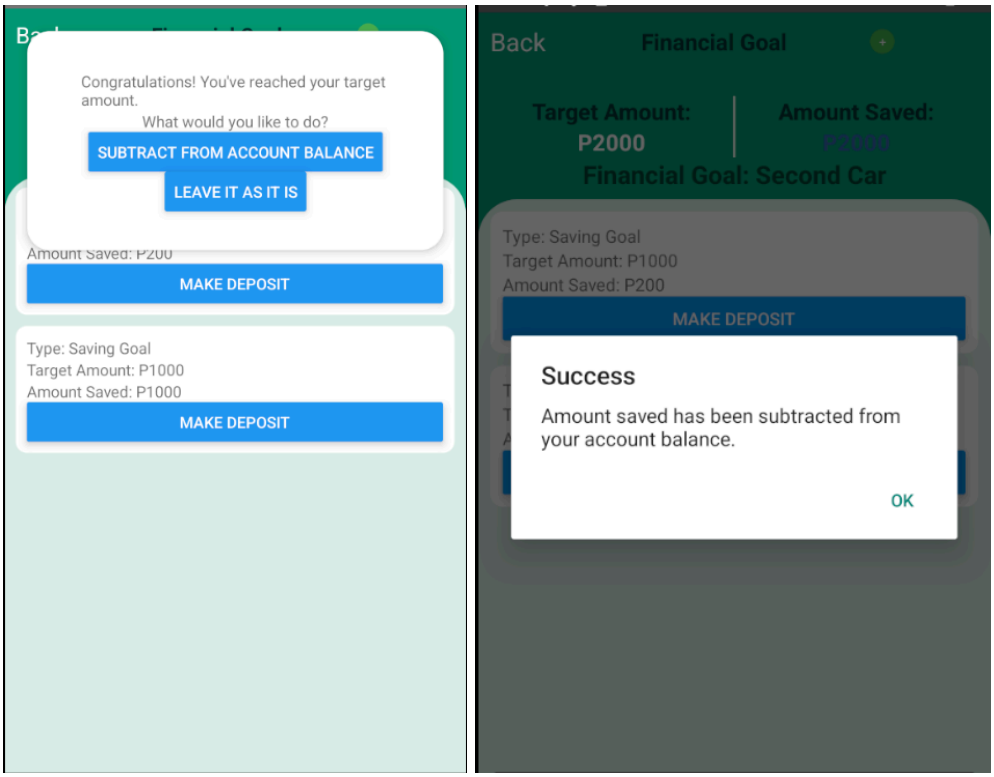
### 5.10.6 Financial Goal Module




Back
Financial Goal
Target Amount:
P23000
Amount Saved:
P0
Financial Goal: Car
Basic Goal
Basic Goal is used for Goals that are only for saving money.
Enter Target Amount
Enter Target Amount
Enter Amount Saved
Enter Amount Saved
Save
Close
Back
Financial Goal
Target Amount:
P23000
Amount Saved:
P10
Detailed Goal
Detailed Goals are goals that can be a car, or anything else other than saving goal.
Enter Type
e.g Car, Game, Dog
Enter Target Amount
Enter Target Amount
Enter Amount Saved
Enter Amount Saved
Save
Close
Back
Financial Goal
Target Amount:
P23000
Amount Saved:
P110
Financial Goal: Car
Type: Saving Goal
Target Amount: P100
Amount Saved: P10
MAKE DEPOSIT
Type: Leather Seat
Target Amount: P2000
Amount Saved: P100
MAKE DEPOSIT
Financial Goal
Saving Goal
Saved Money
Select Account
Target Amount Not Achievable
The target amount cannot be achieved on July 21, 2024 due to a short time frame.
See a List of Recommend ations
Recreate Financial Goal
Confirm
Cancel

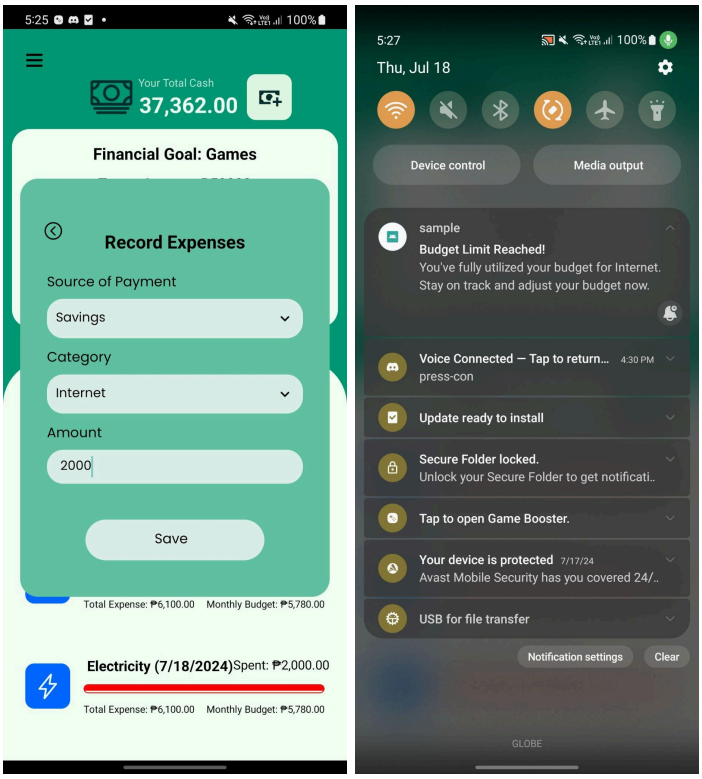

Financial Goal
Saving Goal
Saved Money
Choose a Recommendation
Farther Target Date
New Target Date: July 21, 2024
Lower Target Amount
New Target Amount: 1800
Back
Confirm
Cancel
Financial Goal
Saving Goal
Create Financial Goal
Name
Second Car
Target Date
August 21, 2024
Target Amount
2000
Create
Cancel
Financial Goal
Saving Goal
Saved Money
Select Account
ACcount1
Source
Income
Amount Saved
1000
Confirm
Cancel
Financial Goal
Saving Goal
Overall Target Amount
₱25,000.00
Total Amount Saved
₱1,110.00
July 2024
Goal Name: Car
Target Amount: ₱23,000.00
Amount Saved: ₱110.00
Account Used: ACcount1
Target Date: July 30, 2024
August 2024
Goal Name: Second Car
Target Amount: ₱2,000.00
Amount Saved: ₱1,000.00
Account Used: ACcount1
Target Date: August 21, 2024

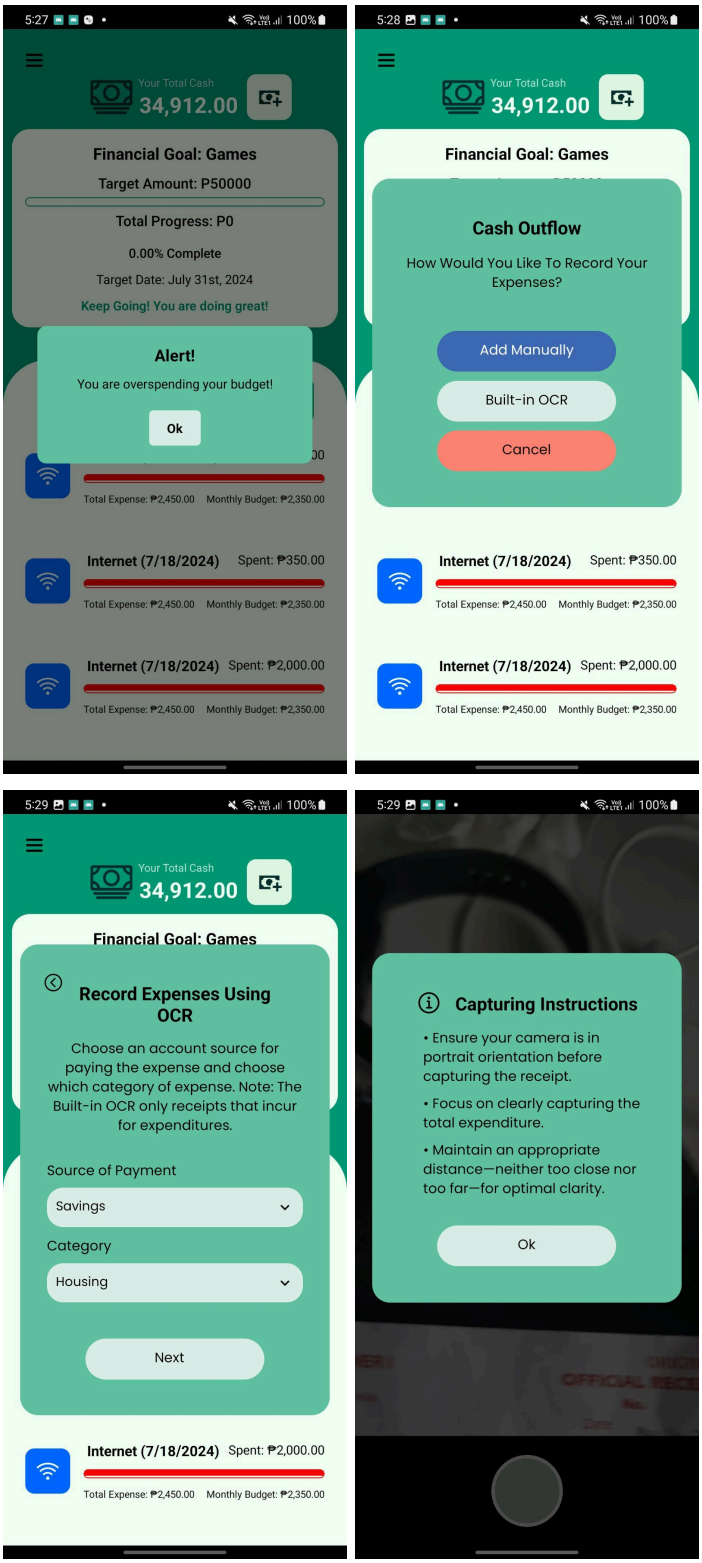

Back
Financial Goal
Target Amount:
P2000
Amount Saved:
Financial Goal: Second Car
Type: Saving Goal
Target Amount: P1000
Extra Amount Allocation
You have extra amount saved. What would you like to do with it?
Allocate to other goals
Allocate to expenses
Let it be as it is
Select Financial Goal
Car
Close

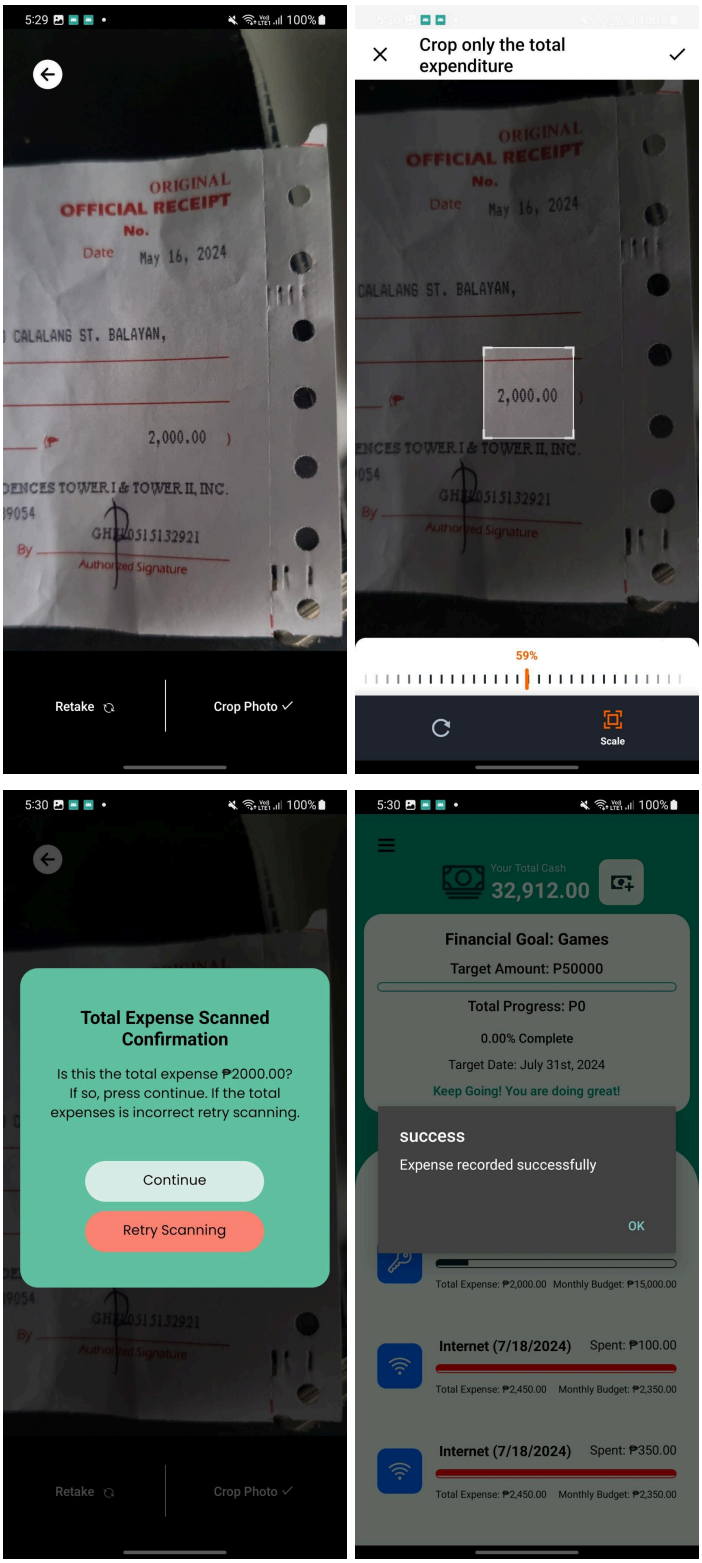

Congratulations! You've reached your target amount.
What would you like to do?
SUBTRACT FROM ACCOUNT BALANCE
LEAVE IT AS IT IS
MAKE DEPOSIT
Type: Saving Goal
Target Amount: P1000
Amount Saved: P1000
MAKE DEPOSIT
Back
Financial Goal
Target Amount:
P2000
Amount Saved:
Financial Goal: Second Car
Type: Saving Goal
Target Amount: P1000
Amount Saved: P200
MAKE DEPOSIT
Success
Amount saved has been subtracted from your account balance.
OK

The financial goals screen helps users set and track their financial objectives, such as saving for a car or a vacation or an emergency fund. The design includes input fields for goal name, target amount, and deadline, along with progress bars that visually represent how close the user is to achieving each goal. If the target amount is not achievable, FinSight lets the user recreate the financial goal or choose from a list of recommendations, which may include setting a farther target date or lowering the target amount.

### 5.10.7 Expense Module

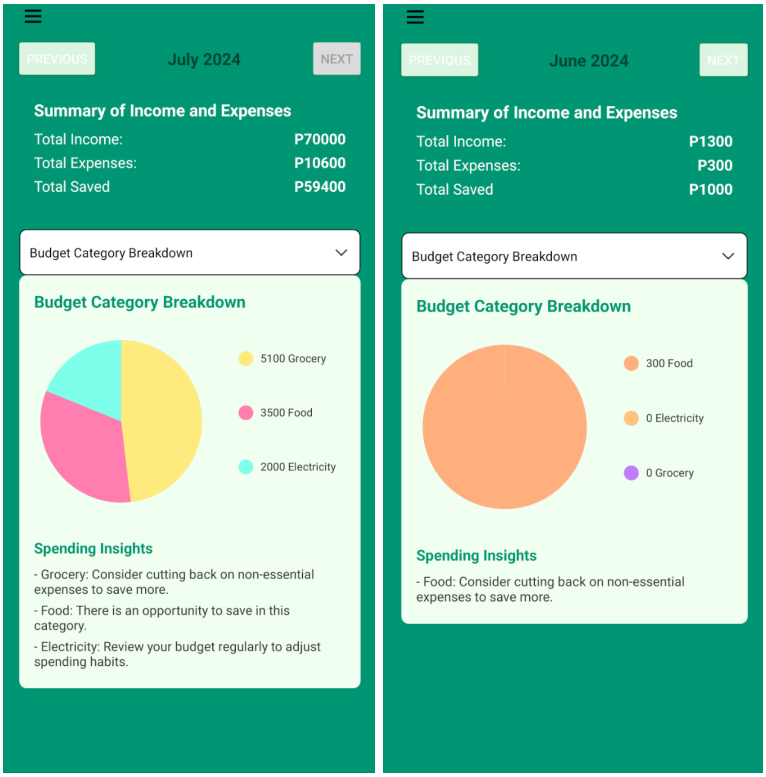


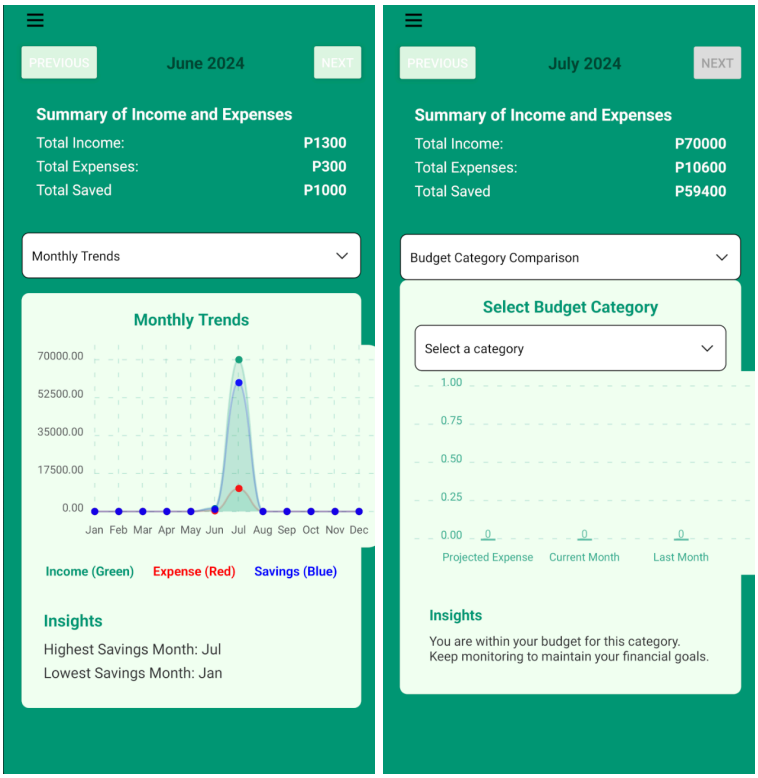

5:27
100%
Your Total Cash
34,912.00
Financial Goal: Games
Target Amount: P50000
Total Progress: P0
0.00% Complete
Target Date: July 31st, 2024
Keep Going! You are doing great!
Alert!
You are overspending your budget!
Ok
Total Expense: ₱2,450.00 Monthly Budget: ₱2,350.00
Internet (7/18/2024) Spent: ₱350.00
Total Expense: ₱2,450.00 Monthly Budget: ₱2,350.00
Internet (7/18/2024) Spent: ₱2,000.00
Total Expense: ₱2,450.00 Monthly Budget: ₱2,350.00
5:28
100%
Your Total Cash
34,912.00
Financial Goal: Games
Cash Outflow
How Would You Like To Record Your Expenses?
Add Manually
Built-in OCR
Cancel
Internet (7/18/2024) Spent: ₱350.00
Total Expense: ₱2,450.00 Monthly Budget: ₱2,350.00
Internet (7/18/2024) Spent: ₱2,000.00
Total Expense: ₱2,450.00 Monthly Budget: ₱2,350.00
5:29
100%
Your Total Cash
34,912.00
Financial Goal: Games
Record Expenses Using OCR
Choose an account source for paying the expense and choose which category of expense. Note: The Built-in OCR only receipts that incur for expenditures.
Source of Payment
Savings
Category
Housing
Next
Internet (7/18/2024) Spent: ₱2,000.00
Total Expense: ₱2,450.00 Monthly Budget: ₱2,350.00
5:29
100%
Capturing Instructions
• Ensure your camera is in portrait orientation before capturing the receipt.
• Focus on clearly capturing the total expenditure.
• Maintain an appropriate distance—neither too close nor too far—for optimal clarity.
Ok

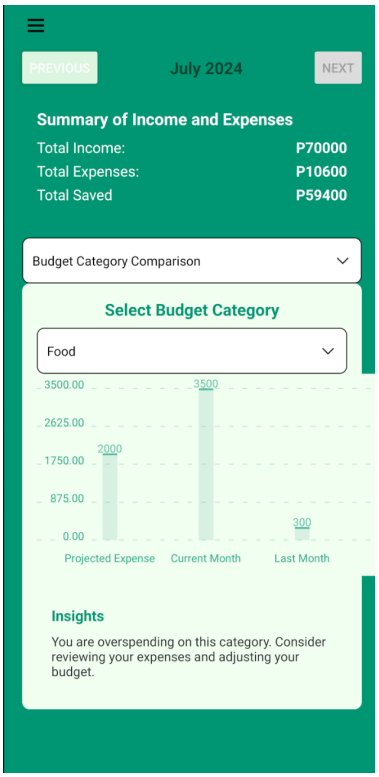

5:29
100%
ORIGINAL
OFFICIAL RECEIPT
No.
Date May 16, 2024
CALALANG ST. BALAYAN,
2,000.00
Retake
Crop Photo
Crop only the total expenditure
59%
Scale
5:30
Total Expense Scanned Confirmation
Is this the total expense ₱2000.00? If so, press continue. If the total expenses is incorrect retry scanning.
Continue
Retry Scanning
Your Total Cash
32,912.00
Financial Goal: Games
Target Amount: P50000
Total Progress: P0
0.00% Complete
Target Date: July 31st, 2024
Keep Going! You are doing great!
success
Expense recorded successfully
OK
Internet (7/18/2024)
Spent: ₱100.00
Total Expense: ₱2,450.00
Monthly Budget: ₱2,350.00
Internet (7/18/2024)
Spent: ₱350.00

The expense tracking screen is designed for users to log their daily expenses. It is located within the dashboard. It includes options for manual entry and OCR. The interface is streamlined for quick data entry, with fields for date, amount, category, and notes. The use of icons and color-coded categories helps users easily identify and categorize their expenses. Additionally, the screen supports OCR functionality, allowing users to capture receipts using their device's camera for automatic data extraction.

### 5.10.8 Reports Module




PREVIOUS
June 2024
NEXT
Summary of Income and Expenses
Total Income: P1300
Total Expenses: P300
Total Saved P1000
Monthly Trends
Monthly Trends
70000.00
52500.00
35000.00
17500.00
0.00
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Income (Green) Expense (Red) Savings (Blue)
Insights
Highest Savings Month: Jul
Lowest Savings Month: Jan
PREVIOUS
July 2024
NEXT
Summary of Income and Expenses
Total Income: P70000
Total Expenses: P10600
Total Saved P59400
Budget Category Comparison
Select Budget Category
Select a category
1.00
0.75
0.50
0.25
0.00
0
0
0
Projected Expense
Current Month
Last Month
Insights
You are within your budget for this category.
Keep monitoring to maintain your financial goals.


PREVIOUS
July 2024
NEXT
Summary of Income and Expenses
Total Income: P70000
Total Expenses: P10600
Total Saved P59400
Budget Category Comparison
Select Budget Category
Food
3500.00
2625.00
1750.00
875.00
0.00
2000
3500
300
Projected Expense
Current Month
Last Month
Insights
You are overspending on this category. Consider reviewing your expenses and adjusting your budget.

The reports screens provide detailed financial reports and insights. It features various types of graphs and charts, including line graphs, bar charts, and pie charts, to visualize income, expenses, and budget performance over time. The design focuses on clarity and precision, with interactive elements that allow users to filter data by categories and accounts. This screen helps users identify spending trends and make data-driven financial decisions.

## 5.11 Report Specifications

### 5.11.1 Monthly Income and Expense Trend Report

The Monthly Income and Expense Trend Report provides a comprehensive overview of an individual's financial activity over a month. This report tracks both income and expenses, presenting them in a visual format such as graphs to illustrate trends over time. By comparing data from previous months, users can identify patterns in their financial behavior, such as periods of higher spending or increased income. This insight helps users to better understand their financial habits, make informed decisions, and plan for future expenses. Key components of the report include total income, total expenses, net savings, and significant variations from previous months.

### 5.11.2 Monthly Budget Comparison Report

The Monthly Budget Comparison Report compares the actual financial performance against the predefined budget for the month. It highlights the differences between what was planned and what was actually spent or earned. This report is essential for evaluating how well the budget was adhered to and identifying areas where spending deviated from expectations. By presenting both the budgeted and actual amounts side by side, users can see where they overspent or underspent. This information is crucial for making adjustments to future budgets, ensuring better financial management, and achieving financial goals. The report typically includes categories such as income, essential expenses, discretionary expenses, and savings.

### 5.11.3 Budget Category Breakdown Report

The Budget Category Breakdown Report provides a detailed analysis of spending across different categories within the budget. This report breaks down expenses into various categories such as housing, food, transportation, entertainment, and others, showing the proportion of the total budget each category consumes. It helps users to see where their money is going and identify potential areas for cost-cutting or reallocation of funds. By understanding the distribution of expenses, users can make more informed decisions about their spending habits and adjust their budget to better align with their financial priorities. The report includes visual aids like pie charts to represent the data clearly.

### 5.12 System Architecture

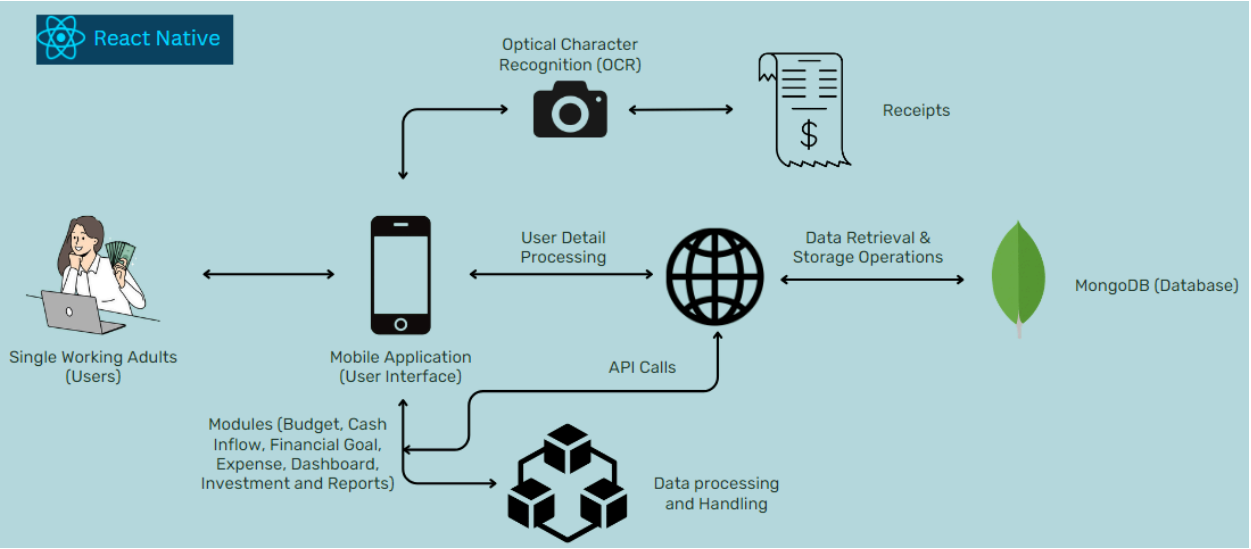



*Figure 5. System Architecture*

The FinSight system architecture provides comprehensive financial management services for single working adults through a React Native mobile app. This app includes modules for Budget, Cash Inflow, Financial Goal, Expense, Dashboard, Investment, and Reports, enabling users to manage all aspects of their finances. A key feature is Optical Character Recognition (OCR), which allows users to photograph receipts. The OCR extracts and digitizes financial information, which is then processed within the app. This data, along with other user details, is handled and communicated to backend services through API calls. The backend processes this data and stores it in MongoDB, a scalable NoSQL database. MongoDB manages data retrieval and storage, ensuring secure and efficient access. The workflow involves users capturing receipts or entering data, which the app processes and categorizes. The processed data is sent to the backend for storage and retrieval, then displayed back to users in real-time. This architecture ensures a smooth flow of data from user input to backend processing, providing users with an intuitive and reliable financial management tool. By integrating OCR and leveraging MongoDB, FinSight delivers a robust platform for effective financial management.

The FinSight system architecture was configured by setting up a comprehensive development environment that includes Node.js, npm, React Native CLI, Android Studio, and Xcode to support both Android and iOS development. The mobile app was built with essential modules like Budget, Cash Inflow, Financial Goal, Expense, Dashboard, Investment, and Reports, each connected to the backend. MongoDB was configured as the primary database, ensuring data is stored securely and retrieved efficiently. The OCR technology was integrated to allow users to capture and process data from receipts. This meticulous configuration ensures a seamless and secure financial management experience for users.

## 5.13 Infrastructure

### 5.13.1 Frontend

The frontend infrastructure of FinSight is built using React Native, tailored specifically for Android devices. This framework ensures a smooth and responsive user interface with fast rendering and native-like performance on Android. Navigation and routing within the app are handled by React Navigation, offering a fluid and intuitive user experience. The UI is designed using a component-based architecture, which facilitates the development, testing, and maintenance of individual UI components. Material Design principles are followed to ensure a consistent and visually appealing user interface, enhancing usability and accessibility. The frontend communicates with the backend via RESTful APIs, ensuring efficient data exchange and synchronization.

### 5.13.2 Backend

The backend infrastructure of FinSight is built using Node.js, providing a scalable and high-performance environment capable of handling numerous concurrent connections. Express.js is employed to create the RESTful API, offering a robust set of features for web and mobile applications, including routing, middleware integration, and HTTP request handling. MongoDB serves as the primary database, with its document-oriented structure allowing for flexible and scalable data storage. MongoDB supports indexing, sharding, and replication, ensuring efficient data retrieval and high availability. Security measures include the use of JSON Web Tokens (JWT) for authentication, ensuring that only authenticated users can access specific resources, and encryption of data both at rest and in transit using AES and HTTPS, respectively.

# CHAPTER 6

## 6.0 System Testing

### 6.1 Unit Testing

To ensure that FinSight is effectively tailored to its target audience and functions correctly across various user demographics, participants for unit testing were selected based on several criteria. The age groups of 18-24 and 25-65 were chosen to include both younger adults, who are often beginning to manage their finances independently, and older adults, who might have more complex financial situations. Gender was not restricted, allowing for a diverse participant pool to ensure the app meets the needs of users regardless of gender. Only single individuals were selected, as FinSight aims to address the financial challenges of single working adults. Participants included those who are full-time employed, part-time employed, and self-employed, capturing a wide range of financial behaviors and challenges. Comfort with using a smartphone or computer was required to ensure effective interaction with the app and relevant feedback on usability and functionality. The focus was on users who are most comfortable using Android, ensuring the app's functionality and user experience are optimized for the predominant platform among the target users. There was no restriction on whether participants regularly use personal finance applications or tools, allowing for feedback from both experienced users and those new to personal finance apps. Similarly, participants were not filtered based on their level of engagement in managing personal finances or their involvement in budgeting, saving, or investing activities, ensuring a diverse group of users with different financial behaviors and needs. Participants were recruited through online advertisements, social media, and community groups, and screened through a survey or questionnaire to ensure eligibility. Efforts were made to ensure diversity and representation within the specified criteria, considering geographic location, socio-economic background, and varying levels of financial literacy. This thorough and inclusive selection process ensures that FinSight receives diverse and actionable feedback, ultimately enhancing the app's ability to meet the needs of its intended users. FinSight was able to gather 13 respondents, 10 of whom passed the criteria to test the modules of the application.

#### 6.1.1 Demographics

According to *Appendix E.1*, most participants were single individuals aged 18-24, with only one respondent falling within the 25-65 age range, indicating a focus on younger adults. The gender distribution was relatively balanced, with 53.8% male and 46.2% female respondents. Regarding employment status, the respondents were diverse: one was a student, five were full-time employees, and seven were part-time employees. This variety suggests that the survey captured perspectives from individuals with different levels of employment commitment and potentially varying financial experiences and responsibilities. These demographics can offer valuable insights into the behaviors or

opinions being studied, particularly in areas related to employment, finance, or lifestyle choices.

### 6.1.2 Technology Proficiency

The results from *Appendix E.2*, regarding technological proficiency among respondents, indicate a high level of comfort with digital devices, as all participants reported feeling comfortable using a smartphone or computer. This suggests a baseline proficiency with technology, which is crucial for interacting with digital platforms. The preference for operating systems shows a dominant leaning towards Android, with 75% of respondents expressing greater comfort with it, while the remainder prefers iOS. Additionally, when it comes to the regular use of personal finance applications or tools, only half of the respondents reported regular usage. This split suggests that while the respondents are generally tech-savvy, there is a variance in the adoption or frequency of using digital solutions for managing personal finances, indicating potential opportunities for increasing engagement with financial technology.

### 6.1.3 Financial Behavior

The financial behavior results, indicated in *Appendix E.3*, reveal a proactive approach to personal finance management among the respondents. 70% of them actively manage their personal finances, indicating a significant level of awareness and responsibility in handling their financial affairs. This engagement is further supported by the fact that 80% of respondents participate in budgeting, saving, and investing activities, demonstrating a comprehensive approach to financial health. These activities suggest that the majority of the respondents not only manage their finances but also take deliberate steps to plan for the future and optimize their financial resources. The overlap between active management and engagement in these financial activities underscores a strong inclination toward maintaining financial stability and growth among the respondents.

### 6.1.4 Financial Goal Module

Referring to *Appendix E.4*, users found the Financial Goal Module to be a strong feature, supportive in setting and achieving financial goals. It generally received positive ratings for ease of use and functionality. However, some users suggested enhancements in design and visual appeal. Specifically, they recommended clearer labels and a more intuitive interface for modifying existing financial goals. Moreover, to cater to users who are new to budgeting software, it was suggested to include tutorials or short descriptions of financial terms and features, akin to the introductory guidance provided by apps like GCash. This would make the module more accessible and user-friendly, ensuring that users of all financial literacy levels can effectively utilize the feature.

### 6.1.5 Cash Inflow Module

Following *Appendix E.5,* the Cash Inflow Module was well-received for its support in recording income, with users finding the process of categorizing cash inflows straightforward. However, the visual design and layout were noted as areas needing improvement. Enhancements in the accuracy and clarity of the cash inflow summary were suggested to make the information more helpful and user-friendly. Some users pointed out that certain elements were too close together, causing usability issues, and recommended better spacing and distinct color differentiation for action buttons to improve intuitiveness.

### 6.1.6 Budget Module

The Budget Module was commended for its effectiveness in helping users create and manage financial budgets, as seen in *Appendix E.6*. Its budget overview and tracking features were particularly praised, with users finding the alerts on nearing or exceeding budget limits useful for preventing overspending. However, there were suggestions to improve the UI, such as adding the option to set daily budget limits and enhancing the visual appeal with better color choices and layout adjustments. Users also recommended integrating additional features, such as categorizing savings and expenses more granularly, to further aid in financial management.

### 6.1.7 Dashboard Module

Based on *Appendix E.7, t*he Dashboard Module was appreciated for presenting financial information clearly and assisting users in tracking their overall financial performance. However, users emphasized the need for a cleaner and more user-friendly design, with better spacing and color differentiation for action buttons. Enhancing visual elements and adding features like recent transaction summaries were also suggested to improve usability. Users found that a more visually appealing dashboard could significantly enhance the user experience by making it easier to assess financial health at a glance.

### 6.1.8 Expense Module

Following *Appendix E.8, t*he Expense Module was effective in allowing users to record and categorize expenses, receiving positive feedback for its expense tracking feature and overall design. Nonetheless, users pointed out that the interface could benefit from better spacing of elements and clearer visual indicators. There were suggestions to add more intuitive navigation and context for action buttons to make the module more user-friendly. For example, integrating Optical Character Recognition (OCR) technology for capturing data from receipts was seen as a valuable feature to reduce manual entry and errors.

### 6.1.9 Reports Module

As seen in *Appendix E.9, t*he Reports Module enabled users to generate and understand financial reports, which was beneficial for analyzing spending patterns and financial status. However, feedback indicated a need for more visually appealing and clear report visualizations. Users also suggested adding context and insights to the reports to help interpret the data better and make informed financial decisions. Enhancing the reports with customizable options and more detailed explanations could significantly improve their usefulness for financial planning and analysis.

### 6.1.10 General Feedback

According to *Appendix E.10*, overall, users found the app to be a valuable tool for tracking financial goals and managing budgets. However, recurring themes in the feedback included the need for a more polished UI, better visual design, and improved navigation. Adding tutorials, enhancing the onboarding experience, and providing context for action buttons and reports were emphasized to make the app more user-friendly and engaging. Some users also recommended adding features like SMS or email updates for account changes, options for dark mode and color themes, and the ability to print account summaries for added convenience.

### 6.1.11 Recommendations

#### 6.1.11.1 UI and Design Improvements

Many users noted that the visual aspects of the app, such as overlapping elements and closely spaced buttons, need refinement. Adjusting button colors for better contrast and improving the overall aesthetic with a consistent color palette were common suggestions. These UI and design improvements would enhance the app's visual appeal and usability.

#### 6.1.11.2 Enhanced Usability

Participants recommended making navigation more intuitive by considering the digital literacy of different user groups. Providing a clear user flow, especially after onboarding, and ensuring that all features are easy to find and use were highlighted as important improvements. Enhanced usability would make the app more accessible and user-friendly for a broader audience.

#### 6.1.11.3 Feature Expansion

Users suggested incorporating more features such as the ability to set daily budget limits, receive updates on financial activities, and include additional categories for expenses like SSS, Philhealth, and Pag-Ibig for more

comprehensive financial management. This feature expansion would provide users with more tools to manage their finances effectively.

#### 6.1.11.4 Educational Support

It was suggested to include tutorials or short descriptions of financial terms and features would help users who are new to personal finance apps. This educational support approach would ensure that users understand how to utilize the app's features effectively, making the app more accessible to those with varying levels of financial literacy.

## REFERENCES


Acharya, P. & Poudel, O. (2023). Interplay of Financial Socialization, Financial Behavior, and Adult Financial Well-being. Management Dynamics. 26. 33-42. 10.3126/md.v26i1.59149.

Ahn, J., & Kwon, J. (2022) The role of trait and emotion in cruise customers' impulsive buying behavior: an empirical study, Journal of Strategic Marketing, 30:3, 320-333, DOI: 10.1080/0965254X.2020.1810743

ALAG, D. T., APAWAN, D. B., PACALA, R. R. & RAMOS, J. D. (2020). Personal Financial Management Among Private Employees. SMCC Higher Education Research Journal (Business Administration Journal), 2(1).

Althnian, A. (2021). Design of a Rule-based Personal Finance Management System based on Financial Well-being. International Journal of Advanced Computer Science and Applications, 12(1). https://doi.org/10.14569/ijacsa.2021.0120122

Aspinal, F., Stevens, M., Manthorpe, J., Woolham, J., Samsi, K., Baxter, K., ... & Ismail, M. (2019). Safeguarding and personal budgets: the experiences of adults at risk. The Journal of Adult Protection, 21(3), 157-168.

Bai, R. (2023). Impact of financial literacy, mental budgeting and self control on financial wellbeing: Mediating impact of investment decision making. PLOS ONE, 18(11), e0294466. https://doi.org/10.1371/journal.pone.0294466

Bailey, P., & Leon, T. (2019). FINANCIAL DECISION MAKING SELF-EFFICACY IN COGNITIVELY AND FINANCIALLY VULNERABLE OLDER ADULTS. Innovation in Aging, 3, S477 - S478. https://doi.org/10.1037/pag0000368.

Bailey, W.C., Woodiel, D., Turner, M.J., & Young, J. (1998). The Relationship of Financial Stress to Overall Stress and Satisfaction.

Bennett, J. (2022). Beyond Inflation Numbers: Shrinkflation and Skimpflation. PAGE ONE Economics

Bhar, A. (2019). Design & implementation of a personal Cash flow program using Microsoft Excel®. Global Journal of Business, Economics and Management: Current Issues. https://doi.org/10.18844/GJBEM.V9I1.4032.

Bodie, Z., & Kane, A. (2020). Investments.

Brzozowski, M., & Visano, B.S. (2023). Canadian Consumer Financial Vulnerability, Stress, and Well-Being. Canadian Public Policy 49(2), 114-135. https://www.muse.jhu.edu/article/908206.

Carlhenricsdotter, E. (2022). The future design of personal informatics : Potentials of designing personal informatics with a focus on the phenomenological matrix using theories of the self and storytelling (Dissertation). Retrieved from https://urn.kb.se/resolve?urn=urn:nbn:se:uu:diva-480804

Cipriano, N. Y., & Delgado Jr., E. C. (2024). An analysis of the financial literacy of graduating students in the College of Business: Background of the study. Retrieved from https://www.researchgate.net/publication/377494105

Clark, G. (2013). Mapping financial literacy: cognition and the environment. Geografiska Annaler: Series B, Human Geography. https://doi.org/10.1111/geob.12013.

Cortez, D. D. (2023). Personal financial management practices among selected personnel of the Bureau of the Treasury – Central Office. Polytechnic University of the Philippines, Sta. Mesa, Manila, Philippines. Retrieved from https://philarchive.org/archive/CORPFM-2

Cuttone, A., Petersen, M.K., Larsen, J.E. (2014). Four Data Visualization Heuristics to Facilitate Reflection in Personal Informatics. In: Stephanidis, C., Antona, M. (eds) Universal Access in Human-Computer Interaction. Design for All and Accessibility Practice. UAHCI 2014. Lecture Notes in Computer Science, vol 8516. Springer, Cham. https://doi.org/10.1007/978-3-319-07509-9_51

Di Domenico SI, Ryan RM, Bradshaw EL, Duineveld JJ. Motivations for personal financial management: A Self-Determination Theory perspective. Front Psychol. 2022 Sep 20;13:977818. doi: 10.3389/fpsyg.2022.977818. PMID: 36204768; PMCID: PMC9530992.

Dudley, C. (2021).Personal Informatics Systems and the Integration of Data from Novel Sensor Technologies. University of BATH

Ekanayake, J., & Saputhanthri, L. (2020). E-AGRO: Intelligent chat-bot. IoT and artificial intelligence to enhance farming industry. *AGRIS on-Line Papers in Economics and Informatics, 12*(1), 15-21. doi:https://doi.org/10.7160/aol.2020.120102

Ekhtiar, T., Karahanoğlu, A., Gouveia, R., & Ludden, G. (2023). Goals for Goal Setting: A Scoping Review on Personal Informatics. https://doi.org/10.1145/3563657.3596087
https://www.researchgate.net/publication/221516793_A_stage-based_model_of_personal_informatics_systems

Elisabeth C. Brüggen, Jens Hogreve, Maria Holmlund, Sertan Kabadayi, Martin Löfgren, Financial well-being: A conceptualization and research agenda, Journal of Business Research, Volume 79, 2017, Pages 228-237, ISSN 0148-2963,https://doi.org/10.1016/j.jbusres.2017.03.013.

Elisabeth T. Kersten-van Dijk, Joyce H.D.M. Westerink, Femke Beute & Wijnand A. IJsselsteijn (2017) Personal Informatics, Self-Insight, and Behavior Change: A Critical Review of Current Literature, Human–Computer Interaction, 32:5-6, 268-296, DOI: 10.1080/07370024.2016.1276456

Eloriaga, E. F., Roxas, E. R. F., Cuaresma, P. J. R., & Cabauatan, R. R. (2022). A Study on Financial Literacy and Financial Behavior of Young Professionals in Metro Manila. International Journal of Research in Engineering, Science and Management, 5(1). Retrieved from https://www.ijresm.com | ISSN (Online): 2581-5792

Epstein, D. A. (2018). *Everyday Personal Informatics*. ProQuest Dissertations Publishing.

EveryDollar | Get Premium Features. (2021). Ramsey Solutions. https://www.ramseysolutions.com/ramseyplus/everydollar/mvelopes

Ferrer, Jerick. (2017). Caught in a Debt Trap? An Analysis of the Financial Well-being of Teachers in the Philippines. The Normal Lights Journal on Teacher Education. 11. 297-324. 10.56278/tnl.v11i2.538.

Flood, Mark D. “Embracing Change: Financial Informatics and Risk Analytics.” Quantitative finance 9.3 (2009): 243–256. Web.

French, D., McKillop, D., & Stewart, E. (2021). Personal finance apps and low‐income households. Strategic Change, 30(4), 367-375.

Galperti, S. (2019). A theory of personal budgeting. Theoretical Economics, 14(1), 173–210. https://doi.org/10.3982/te2881

Garcia, M. & Claour, J. (2021). Mobile Bookkeeper: Personal Financial Management Application with Receipt Scanner Using Optical Character Recognition. 10.1109/OT4ME53559.2021.9638794.

Githuka, G., Jeddidah, T., & Edward, W. (2016). Automated Personal Finance System (Pay App) for College and University Students in Developing Countries. A Case for Egerton University. , 4, 14-19. https://doi.org/10.12691/JCSA-4-1-3.

Glover, Brianna C., "The Consumption and Saving Habits of Young Adults During Their Transitional Period to Adulthood" (2022). Senior Projects Spring 2022. 151. https://digitalcommons.bard.edu/senproj_s2022/151

Godase, Radnyi & P, Jyothi & Supriya, M Lalitha. (2023). Financial planning propensity in working adults: exploring the role of media. Managerial Finance. 50. 10.1108/MF-04-2023-0253.

Greenberg, A., & Hershfield, H. (2018). Financial decision making. Consumer Psychology Review. https://doi.org/10.1002/ARCP.1043.

Halladay, S. D. (2013). Using predictive analytics to improve decisionmaking. The Journal of Equipment Lease Financing, 31(2), B1-B6.

Härdle, W. K., Harvey, C. R., & Reule, R. C. (2020). Understanding cryptocurrencies. Journal of Financial Econometrics, 18(2), 181-208.

Herrala, J., Vartiainen, T., & Koskelainen, T. (2023). How personal finance management systems emancipate and oppress young people. ResearchGate. https://www.researchgate.net/publication/368219967_How_Personal_Finance_Management_Systems_Emancipate_and_Oppress_Young_People/citations

Hershey, Douglas & Austin, James & Kiso, Helen. (2015). Financial Decision Making across the Adult Life Span. 10.1016/B978-0-12-417148-0.00016-9.

Hu, S., & Huang, M. (2019). Teaching of Personal Finance Management Course Based on Multimedia. . https://doi.org/10.1007/978-981-15-3250-4_91.

Jabar, M. A., & Delayco, M. L. C. (2021). Impulse Buying and Financial Literacy Among Public Elementary and High School Teachers in the Philippines. DLSU Business & Economics Review, 31(1), 42–54.

Kadre, S., & Konasani, V. R. (2021). Machine learning and deep learning using Python and TensorFlow. McGraw-Hill Education.

Kaftandzieva, T., & Cvetkoska, V. (2021). FINANCIAL DECISION MAKING AMONG YOUNG ADULTS: AN AHP APPROACH. International Journal of the Analytic Hierarchy Process. https://doi.org/10.13033/ijahp.v13i3.864.

Khoo, Y. Y., Yusof, R., & Abe, Y. (2022). Integrating financial literacy into economics courses through digital tools: The finlite app. Journal of International Education in Business, 15(2), 331-350. doi:https://doi.org/10.1108/JIEB-06-2021-0068

Kołodziej, S. (2014). The relation between financial knowledge and economic decisions. Problems of Education in the 21st Century, 59, 59-67. Retrieved from https://www.proquest.com/scholarly-journals/relation-between-financial-knowledge-economic/docview/2343800977/se-2

Kumar, Darshini & Sudin, Hafizah & Othman, Jaizah & Salehuddin, Sharullizuannizam. (2022). The Influence of Spending Behaviour Among University Students in Malaysia. Asian Journal of Accounting and Finance. 10.55057/ajafin.2022.4.3.3.

Kumar, V., & Garg, M. L. (2018). Predictive analytics: a review of trends and techniques. *International Journal of Computer Applications*, *182*(1), 31-37.

Labito, A. M., & Supramono, S. (2017). Can allowance, personal budgeting and self control as mediating role manage compulsive buying behavior among college students? International Journal of Social Sciences & Educational Studies, 4(3). https://doi.org/10.23918/ijsses.v4i3p1

Legaspi, J. L., Leoncio, I., & Galgana, O. (2016). Impulsive Buying Behavior of Millennials on Online Shopping. Paper presented at the 4th National Business and Management Conference, Ateneo de Davao University, July 22-23, 2016.

Lim, H., Heckman, S. J., Letkiewicz, J. C., & Montalto, C. P. (2014). Financial stress, self-efficacy, and financial help-seeking behavior of college students. *Journal of Financial Counseling and Planning, 25*(2), 148–160.

Li, Ian & Dey, Anind & Forlizzi, Jodi. (2010). A stage-based model of personal informatics systems. Conference on Human Factors in Computing Systems - Proceedings. 1. 557-566. 10.1145/1753326.1753409.

Lodha, P. (2022, October 5). Here's how different generations in the US spend their income. Retrieved from https://www.weforum.org/agenda/2022/10/americans-spend-their-money-by-generation/

Mahdzan, Nurul & Zainudin, Rozaimah & Yoong, Siew-Chan. (2020). Investment Literacy, Risk Tolerance and Mutual Fund Investments: An Exploratory Study of Working Adults in Kuala Lumpur. International Journal of Business and Society. 21. 111-133. 10.33736/ijbs.3230.2020.

Martinez, N. (1994). Myth and ceremony in financial decision making under stress: Case studies from Mexican universities. Higher Education, 27, 297-312. https://doi.org/10.1007/BF01432072.

McCaffery, J. (1984). Canada's Envelope Budgeting System. American Review of Canadian Studies, 14(1), 45-62.

Mint. (n.d.). Managing money, made simple. https://mint.intuit.com/

Moumita Mukherjee, Lyle McKinney, Linda Serra Hagedorn, Agustina Purnamasari & Franco Santiago Martinez (2017) Stretching Every Dollar: The Impact of Personal Financial Stress on the Enrollment Behaviors of Working and Nonworking Community College Students, Community College Journal of Research and Practice, 41:9, 551-565, DOI: 10.1080/10668926.2016.1179602

Muat, Susnaningsih & Mahdzan, Nurul & Edil, Mohd. (2024). What shapes the financial capabilities of young adults in the US and Asia-Pacific region? A systematic literature review. Humanities and Social Sciences Communications. 11. 10.1057/s41599-023-02588-9.

Munohsamy, T. (2015). Personal Financial Management. Research. https://www.researchgate.net/publication/279198054

Neng, H. Z. (2022). *Automated scanned receipt processing with optical character recognition and machine learning* (Order No. 30594363). Available from ProQuest Dissertations &

Theses A&I; ProQuest Dissertations & Theses Global. (2830209549). Retrieved from https://www.proquest.com/dissertations-theses/automated-scanned-receipt-processing-with-optical/docview/2830209549/se-2

Novitasari, D., Juliana, J., Asbari, M., & Purwanto, A. (2021). The Effect of Financial Literacy, Parents' Social Economic and Student Lifestyle on Students Personal Financial Management. Economic Education Analysis Journal, 10(3), 522-531. https://doi.org/10.15294/eeaj.v10i3.50721

Nguyen, V. T., & Doan, M. D. (2020). The correlation between financial literacy and personal saving behavior in vietnam. *Asian Economic and Financial Review, 10*(6), 590-603. doi:https://doi.org/10.18488/journal.aefr.2020.106.590.603

Ohlin, F. & Olsson, C. (2015). Intelligent Computing in Personal Informatics. https://www.researchgate.net/publication/283102665_Intelligent_Computing_in_Personal_Informatics_Key_Design_Considerations

Panos, G. A., & Wilson, J. O. S. (2020). Financial literacy and responsible finance in the FinTech era: capabilities and challenges. The European Journal of Finance, 26(4–5), 297–301. https://doi.org/10.1080/1351847X.2020.1717569

Parr, M. (2019). Budgeting for the cost of living: A comparative study of monthly expenses in three Oklahoma cities.

Pascual, M. P. (2022). Low-Income Earners of Nueva Ecija: Their Budgeting and Decision-Making before and during the Covid-19 Pandemic. Open Journal of Social Sciences, 10(13), 81–90. https://doi.org/10.4236/jss.2022.1013008

Personal Capital | LinkedIn. (2019, March 5). Linkedin.com. https://jo.linkedin.com/company/personalcapital?trk=public_profile_profile-section-card_subtitle-click

Peters, M., & Chiu, C. (2022). INTERACTIVE SPREADSHEETING: A LEARNING STRATEGY AND EXERCISES FOR CALCULATIVE MANAGEMENT ACCOUNTING PRINCIPLES. Issues in Accounting Education. https://doi.org/10.2308/issues-2020-093.

Porcelli, A., & Delgado, M. (2009). Acute Stress Modulates Risk Taking in Financial Decision Making. Psychological Science, 20, 278 - 283. https://doi.org/10.1111/j.1467-9280.2009.02288.x.

Prinz, S., Gründer, G., Hilgers, R., Holtemöller, O., & Vernaleken, I. (2014). Impact of personal economic environment and personality factors on individual financial decision making. Frontiers in Psychology, 5. https://doi.org/10.3389/fpsyg.2014.00158.

Qian, Ruicheng. (2023). Management of Personal Finances and Investment Decisions. Advances in Economics, Management and Political Sciences. 64. 202-212. 10.54254/2754-1169/64/20231532.

Quicken Personal Finance & Money Management Software. (2021, July). Quicken. https://www.quicken.com/canada/?cc_bypass

Raina, O., Mishra, S., & Pote, P. (2022). Personal Finance Application- GROW MORE. International Journal for Research in Applied Science and Engineering Technology. https://doi.org/10.22214/ijraset.2022.42166.

Rapp, A., & Cena, F. (2016). Personal informatics for everyday life: How users without prior self-tracking experience engage with personal data. International Journal of Human-Computer Studies, 94, 1–17. https://doi.org/10.1016/j.ijhcs.2016.05.006

Rapp, A., & Tirassa, M. (2017). Know Thyself: A Theory of the Self for Personal Informatics. Human-Computer Interaction, 32(5–6), 335–380. https://doi.org/10.1080/07370024.2017.1285704

Renamae D. Azul, Ken Drexel Q. Cadano, Ryan Joseph M. Guiao, Norishi Maranga, Carmencita H. Salonga, and Melody N. Usita (2023); The Impulse Buying Behavior and Financial Well-Being Profile of Young Professionals in the Philippines; International Journal of Scientific and Research Publications (IJSRP) 13(03) (ISSN: 2250-3153), DOI: http://dx.doi.org/10.29322/IJSRP.13.03.2023.p13537

Robb, C., & Chy, S. (2021). Undergraduate Financial Knowledge, Attitudes, and Behaviors: The Impact of Financial Life Skills Course on College Students. AccessLex Institute Research Paper Forthcoming. Retrieved from https://arc.accesslex.org/cgi/viewcontent.cgi?article=1063&context=grantee

Safira, Dinda & Meilani, Ananda & Oktaviani, Wulan & Adinugraha, Hendri & Sholihah, Ria & Achmad, Dliyaudin. (2023). Socialization of Personal Financial Management through Basic Accounting Principles among Santriwati Islamic Boarding School Griya Santri Mahabbah. Pengabdian: Jurnal Abdimas. 1. 10.55849/abdimas.v1i4.203.

Salsabilla, S. I., Tubastuvi, N., Purnadi, P., & Innayah, M. N. (2022). Factors Affecting Personal Financial Management. Jurnal Manajemen Bisnis, 13(1), 168-184. https://doi.org/10.18196/mb.v13i1.13489

Salumintao, M. T. G., & Cinches, M. F. C. (2019). Personal Finance Practices of Millennial Students: An Exploratory Model. Journal of International Research in Social and Environmental Sciences, 17(1). Retrieved from https://www.seaairweb.info/journal/articles/JIRSEA_v17_n01/JIRSEA_v17_n01_Article 13.pdf

Samanez-Larkin, G., Hagen, T., & Weiner, D. (2014). Financial Decision Making Across Adulthood. The Psychological Science of Money. 121-135. 10.1007/978-1-4939-0959-9_6.

Serido, J., Shim, S., & Tang, C. (2013). A developmental model of financial capability. International Journal of Behavioral Development, 37, 287 - 297. https://doi.org/10.1177/0165025413479476.

She, L., Rasiah, R., Weissmann, M. A., & Kaur, H. (2023). Using the theory of planned Behaviour to explore predictors of financial behaviour among working adults in Malaysia. FIIB Business Review, 13(1), 118–135. https://doi.org/10.1177/23197145231169336

Shirazian, Z. (2018). Investigation Effect of Financial Literacy and Money Management on Personal Financial Management of Tehran stock exchange investors. Financial Knowledge of Securities Analysis, 11(38), 145-163. Retrieved from https://jfksa.srbiau.ac.ir/article_12562.html?lang=en

Singh, Tanay, "Preparing for the Future: The Effects of Financial Literacy on Financial Planning for Young Professionals" (2020). Senior Theses. 465. https://scholarcommons.sc.edu/senior_theses/465

Smalej, O., Kawczyńska-Butrym, Z., Mącik, R., & Pantyley, V. (2022). Financial activity and financial awareness of high-school students - case study of the lublin region. *Horyzonty Polityki, 13*(45), 159-182. doi:https://doi.org/10.35765/HP.2251

Strough, J., Wilson, J., & Bruin, W. (2020). Aging and Financial Decision Making. Psychological Perspectives on Financial Decision Making. https://doi.org/10.1007/978-3-030-45500-2_8.

Sue Nugus, Chapter 5 - Regression Analysis, Editor(s): Sue Nugus, In CIMA Professional Handbook, Financial Planning Using Excel (Second Edition), CIMA Publishing, 2009, Pages 59-74, ISBN 9781856175517, https://doi.org/10.1016/B978-1-85617-551-7.00005-7.

Sue Nugus, Chapter 6 - Time Series Analysis, Editor(s): Sue Nugus, In CIMA Professional Handbook, Financial Planning Using Excel (Second Edition), CIMA Publishing, 2009, Pages 75-84, ISBN 9781856175517, https://doi.org/10.1016/B978-1-85617-551-7.00006-9.

Sunderaraman P, Chapman S, Barker MS, Cosentino S (2020) Self-awareness for financial decision-making abilities in healthy adults. PLOS ONE 15(7): e0235558. https://doi.org/10.1371/journal.pone.0235558

Tat, Wong & Ting, Cherish & Vasudevan, Asokan & Fong, Wong & Cheng, Chuah & Sam, Toong Hai & RuiTeng, Xue. (2023). Factors Affecting Financial Distress among Malaysian Young Working Adults In Malaysia. 13. 1570-1584

The ultimate get started guide: Ynab. NAB. (n.d.). https://www.ynab.com/guide/the-ultimate-get-started-guide

Tong, D., & Tian, G. (2023). Intelligent financial decision support system based on big data. Journal of Intelligent Systems, 32. https://doi.org/10.1515/jisys-2022-0320.

Tovanich, N., Centellegher, S., Seghouani, N. B., Gladstone, J., Matz, S., & Lepri, B. (2021). Inferring psychological traits from spending categories and dynamic consumption patterns. EPJ Data Science, 10(1). https://doi.org/10.1140/epjds/s13688-021-00281-y

Virlics, Agnes. (2013). Investment Decision Making and Risk. Procedia Economics and Finance. 6. 169–177. 10.1016/S2212-5671(13)00129-9.

Versal, Nataliia & Honchar, Ihor & Balytska, Mariia & Erastov, Vasyl. (2023). How Do Savings And Personal Budgeting Matter On Financial Literacy And Well-Being. Business Management and Economics Engineering. 21. 190-203. 10.3846/bmee.2023.19062.

V. Kumar, P. Kaware, P. Singh, R. Sonkusare and S. Kumar, "Extraction of information from bill receipts using optical character recognition," *2020 International Conference on Smart Electronics and Communication (ICOSEC)*, Trichy, India, 2020, pp. 72-77, doi: 10.1109/ICOSEC49089.2020.9215246.

Wagner, J., & Walstad, W. B. (2023). Gender Differences in Financial Decision-Making and Behaviors in Single and Joint Households. The American Economist, 68(1), 5-23. https://doi.org/10.1177/05694345221076004

Windfried, I. (2017).Investigating the Importance of Creating a Personal Budget among Millenials, their Perception and Satisfaction: A Case Study of Eastern Mediterranean University

Xian, T.K; Poh, R. (2022). Factors Influencing Working Adults' Awareness of Individual Financial Planning, GATR-Global J. Bus. Soc. Sci. Review, 10(3), 148–162. https://doi.org/10.35609/gjbssr.2022.10.3(4)

Yang, Marvello & Mamun, Abdullah & Mohiuddin, Muhammad & Al-shami, Samer & Zainol, Noor. (2021). Predicting Stock Market Investment Intention and Behavior among Malaysian Working Adults Using Partial Least Squares Structural Equation Modeling. 9. 1-16. 10.3390/math9080873.

Yogasnumurti, R. R., Sadalia, I., & Irawati, N. (2021). The effect of financial, attitude, and financial knowledge on the personal finance management of college collage students. In Proceedings of the 2nd Economics and Business International Conference-EBIC (pp. 649-657).

Yu, H., Luo, Z., Wang, L., Ding, X., Wang, S. (2023). Improving the Accuracy of Flood Susceptibility Prediction by Combining Machine Learning Models and the Expanded Flood Inventory Data. Remote Sensing. 15. 3601. 10.3390/rs15143601.

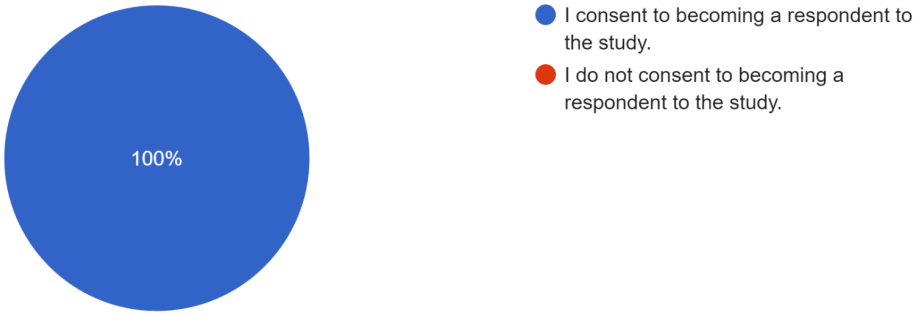
Zhang, W., Hu, B., Liu, Y., Zhang, X., Li, Z. (2023). Urban Flood Risk Assessment through the Integration of Natural and Human Resilience Based on Machine Learning Models. Remote Sensing. 15. 3678. 10.3390/rs15143678.

# APPENDICES

## Appendix A. Budget Management User Requirements Survey

Statement of Consent: I have read and understood the data privacy and consent form. I consent to the collection, use, and storage of my data as des...at my data will be kept confidential and anonymous

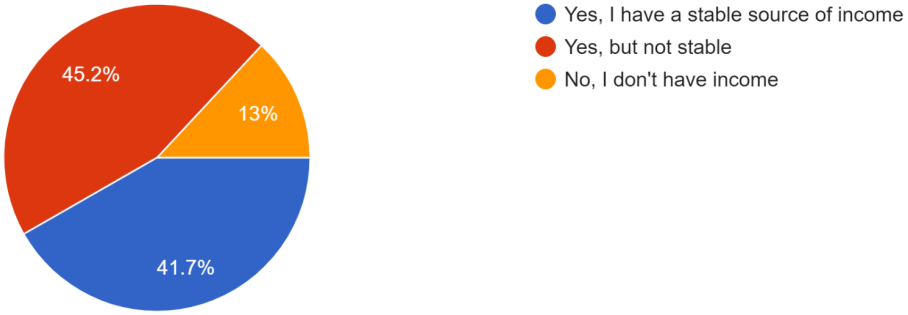
115 responses




### A.1. Personal Information

Do you currently have a source of income?

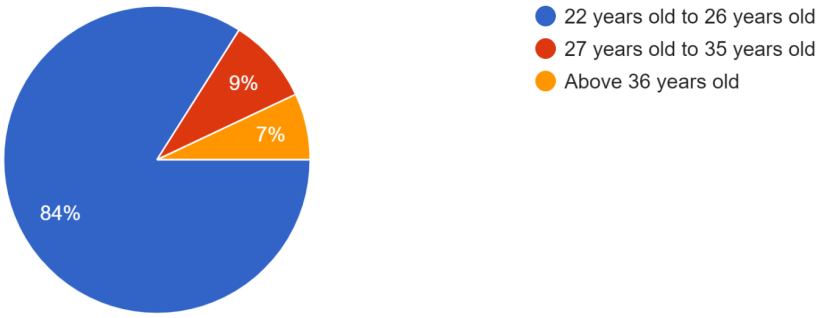
115 responses



## A.2. Demographics

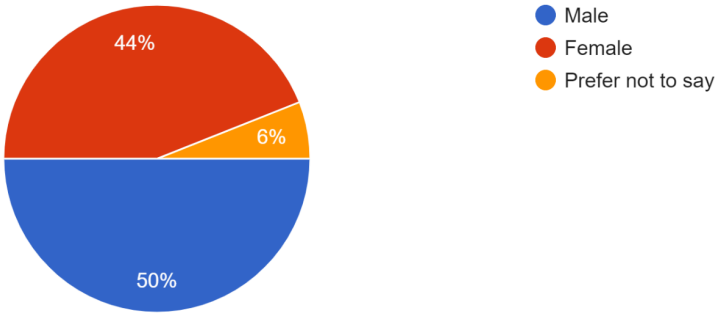
Age

100 responses



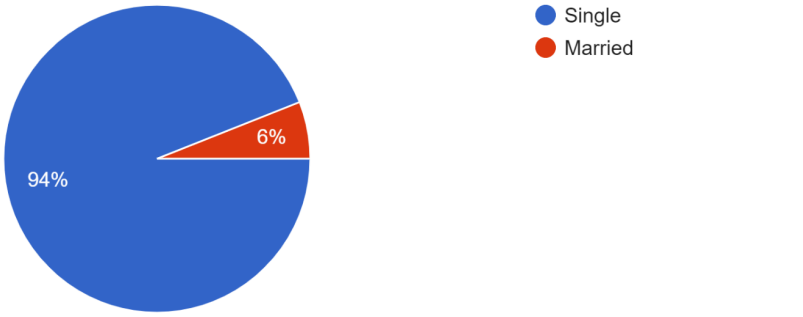

Gender

100 responses



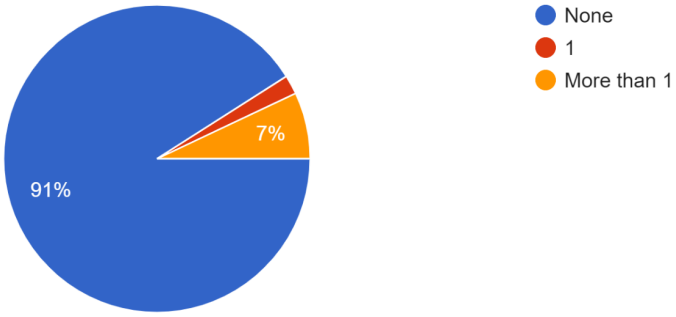
Marital Status

100 responses



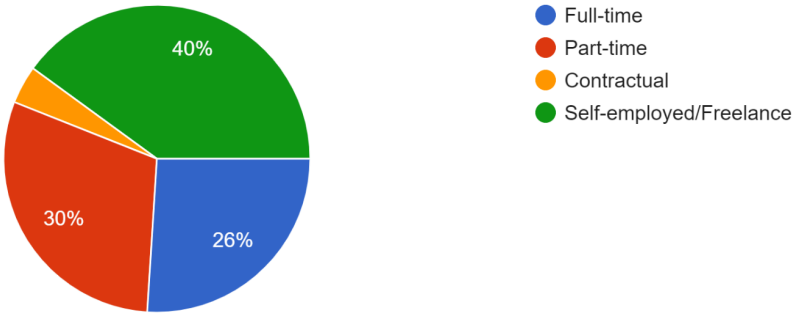

How many children do you have?

100 responses




What type of employment do you have?

100 responses



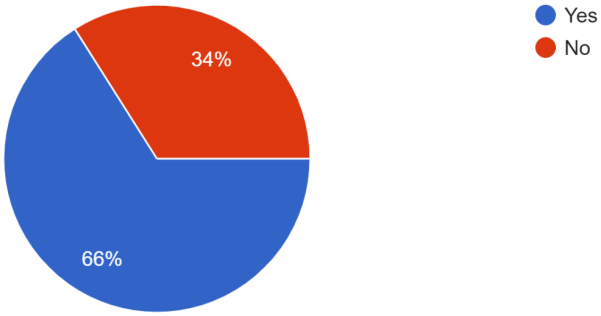
How much is your monthly income

100 responses

### A.3. Savings and Expenses

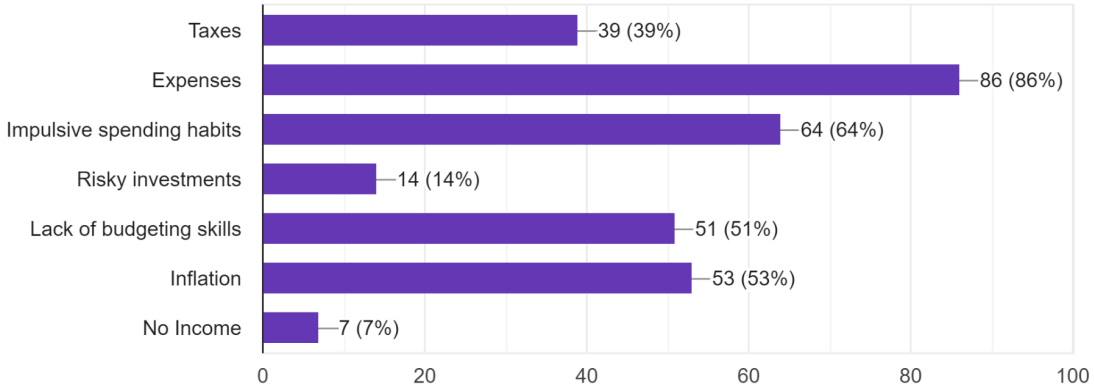
Are you able to save money for yourself from your income?

100 responses



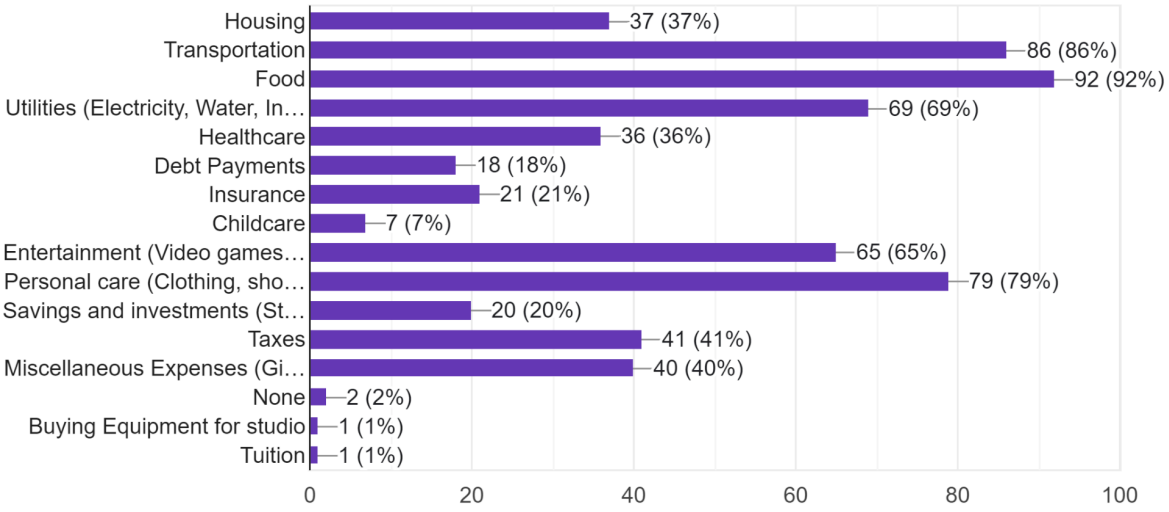
What are the reasons that hinders you from saving enough money? (Check all that apply)

100 responses



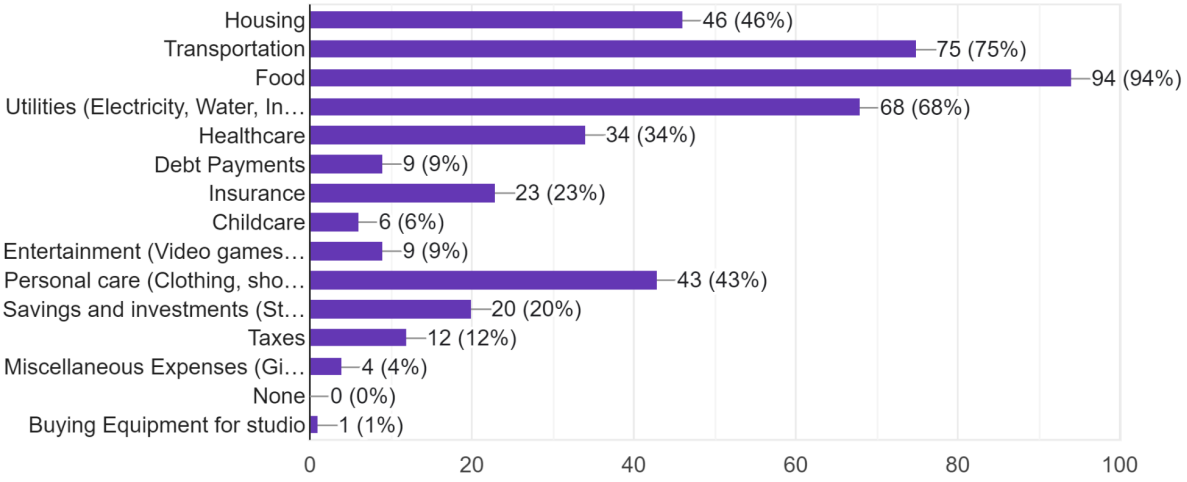

What do you include as part of your expenses? (Check all that apply)

100 responses



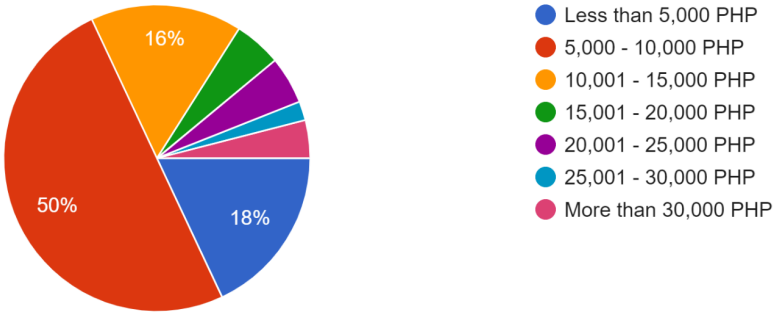
What do you consider as your essential expenses? (Check all that apply)

100 responses



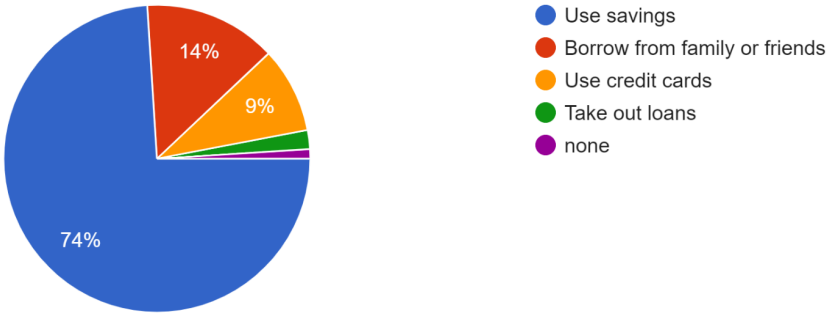

On average, how much do you spend per month on essential expenses such as rent, utilities, groceries, and transportation?

100 responses



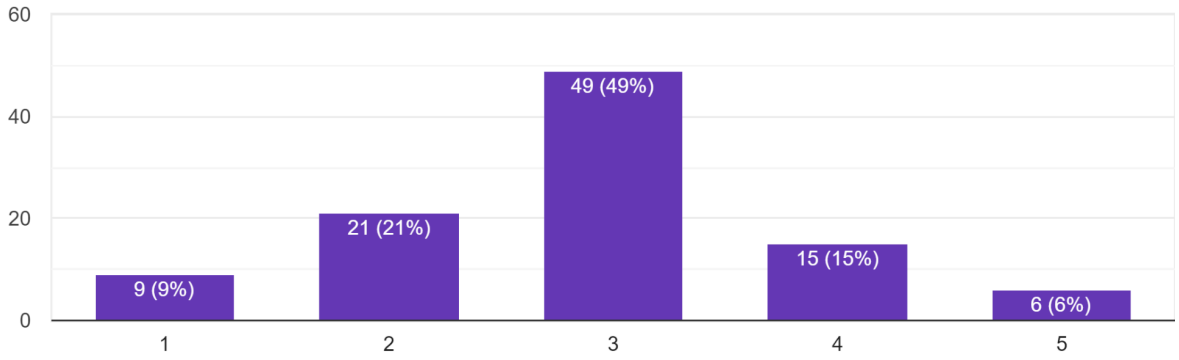
How do you manage unexpected expenses?

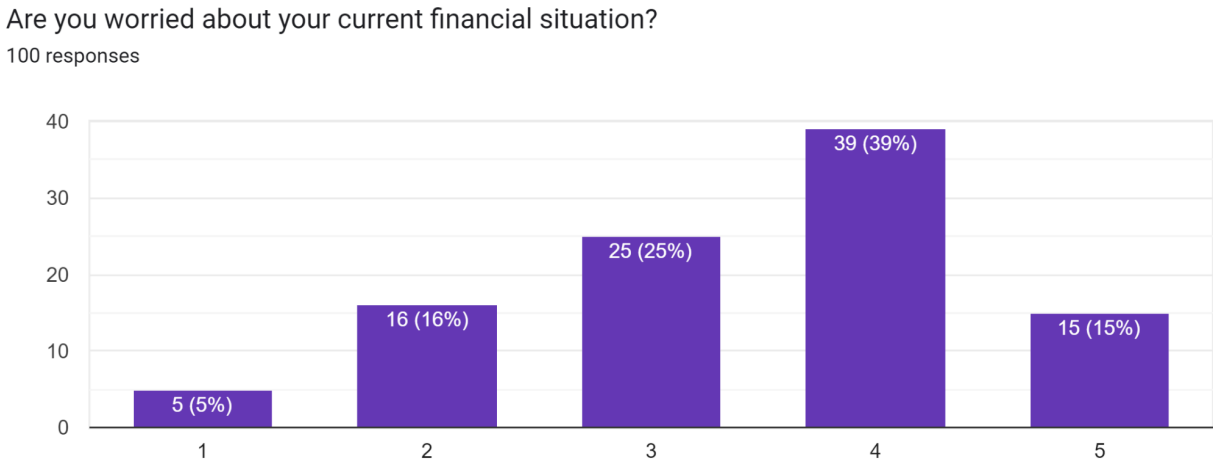
100 responses



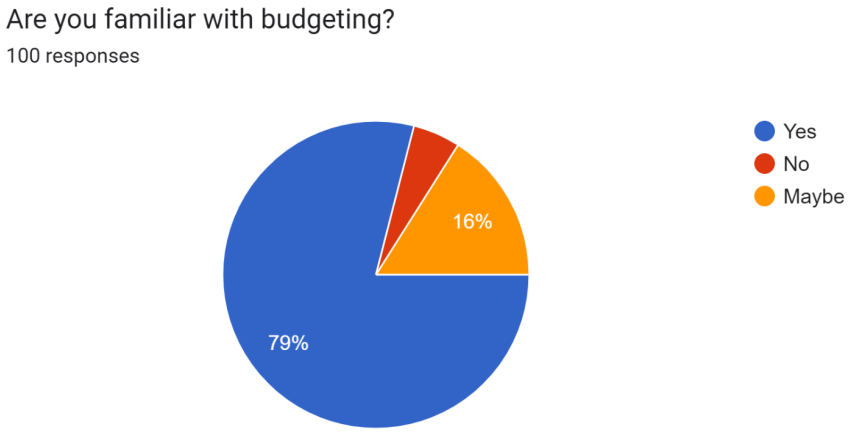

How do you feel about your current financial situation?

100 responses





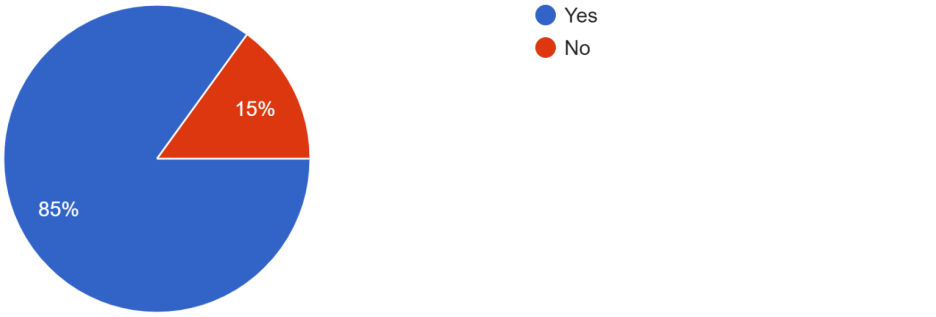

### A.4. Budgeting




How would you define “budgeting”?

Do you currently budget your money?

100 responses

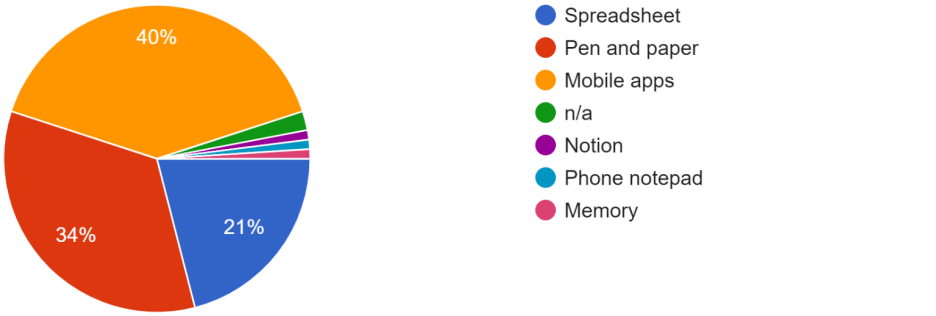



What budgeting plan/technique do you usually follow?

100 responses

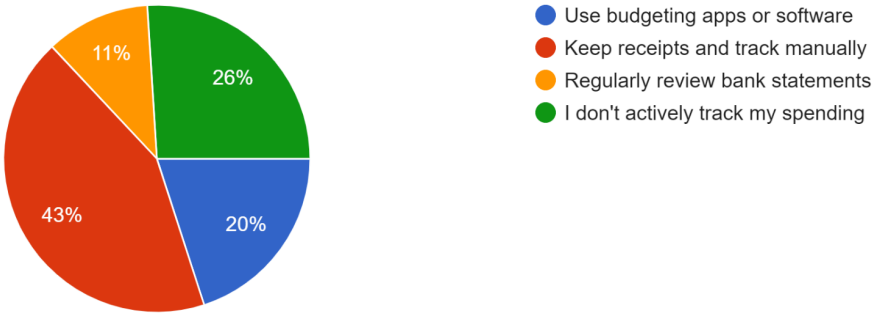
Which budgeting method do you commonly use?

100 responses



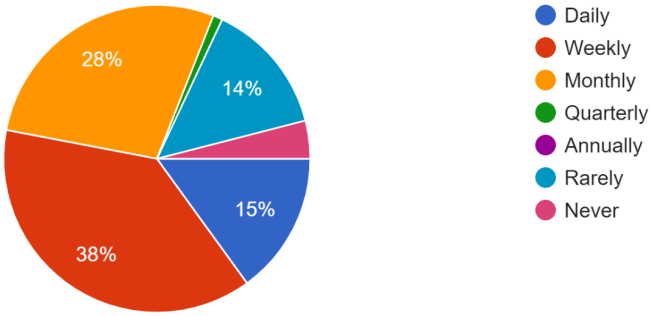
How do you usually track your spending habits?

100 responses



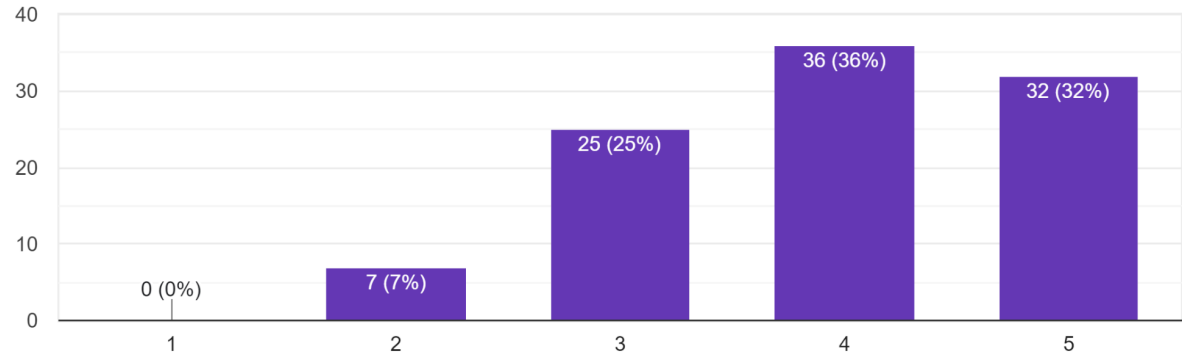

How frequently do you update or review your budget and expenses?

100 responses



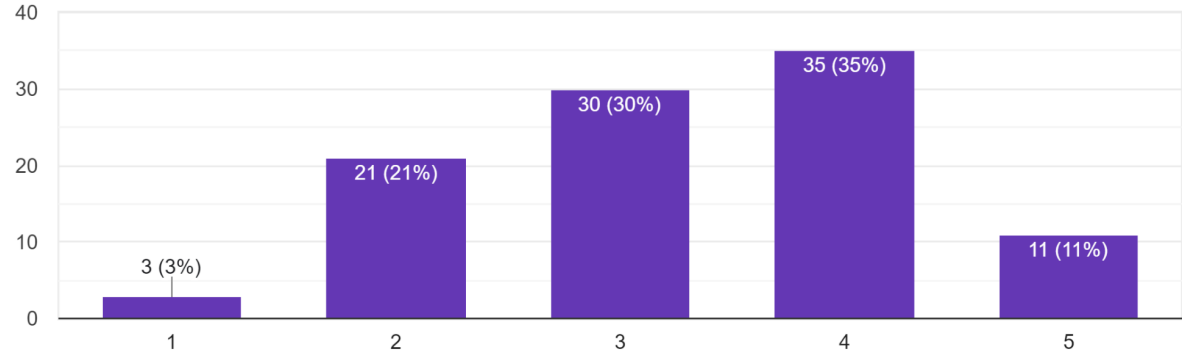
How often do you see the need to build a budget?

100 responses



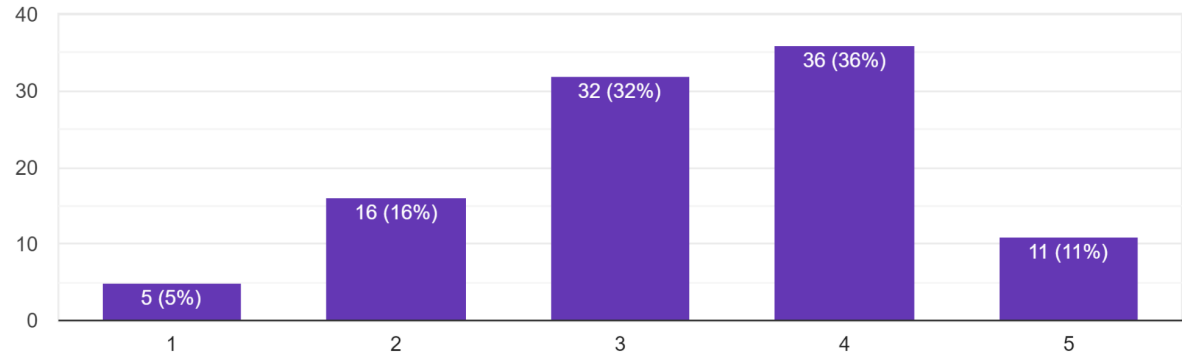

How often do you create a budget?

100 responses



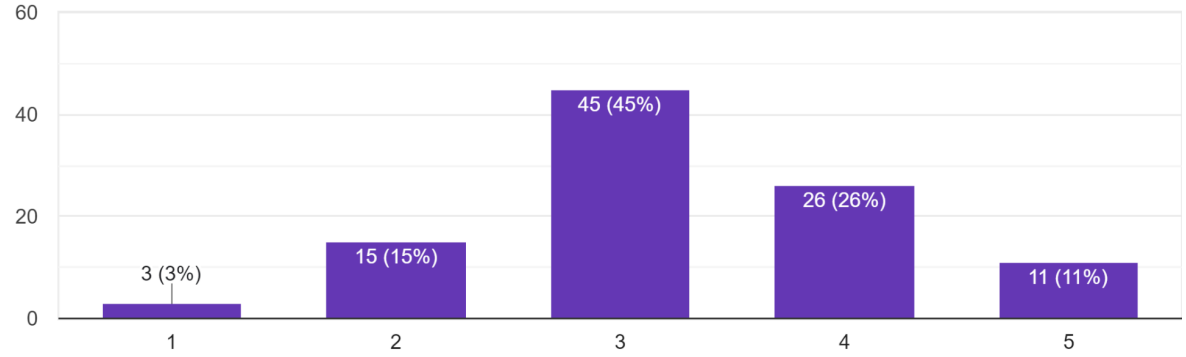
How often do you stick to your budget?

100 responses



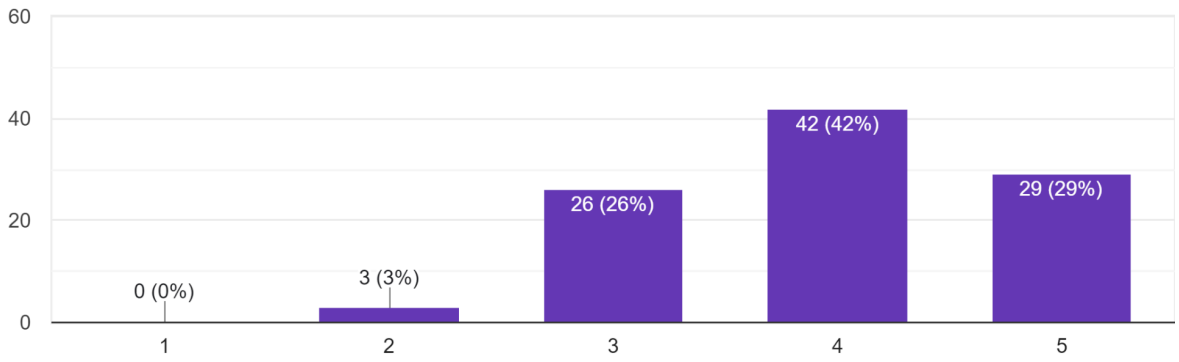

How committed are you when it comes to tracking your expenses?

100 responses



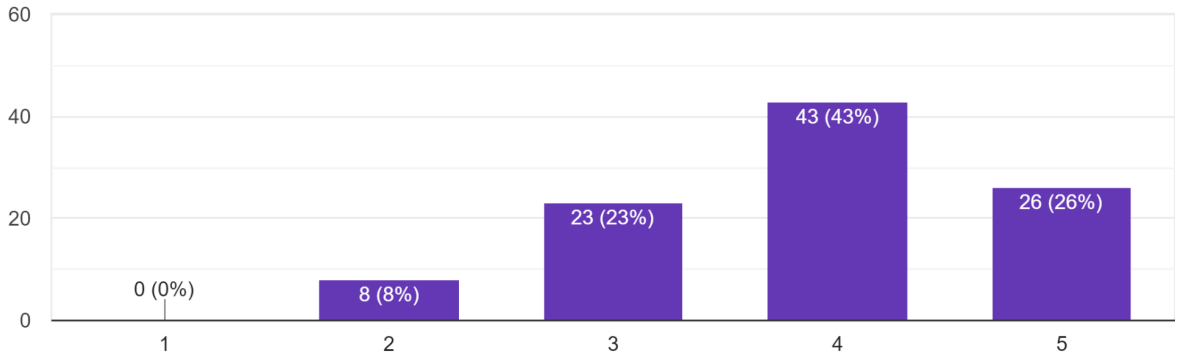
Does your financial situation improve when sticking to your budget?

100 responses



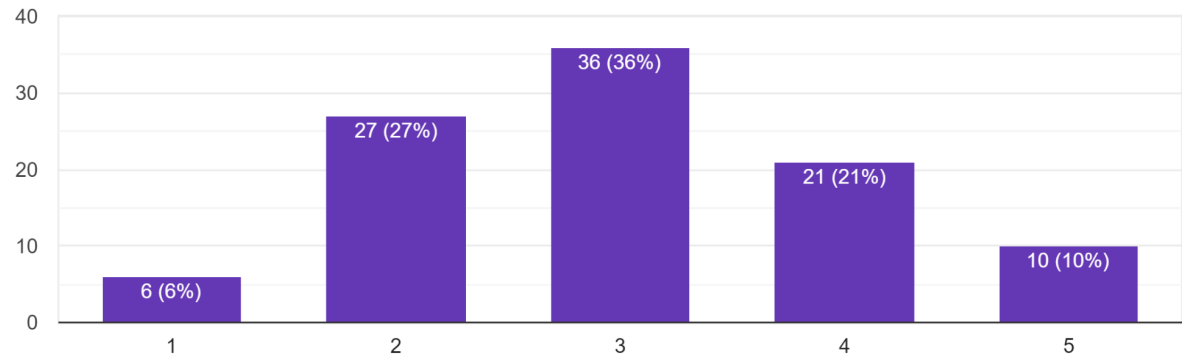

Are you satisfied with how your financial situation improves whenever you stick to your budget?

100 responses



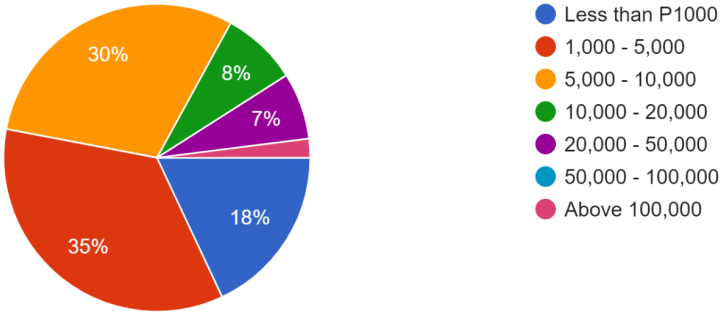
Are you confident with your budgeting skills?

100 responses



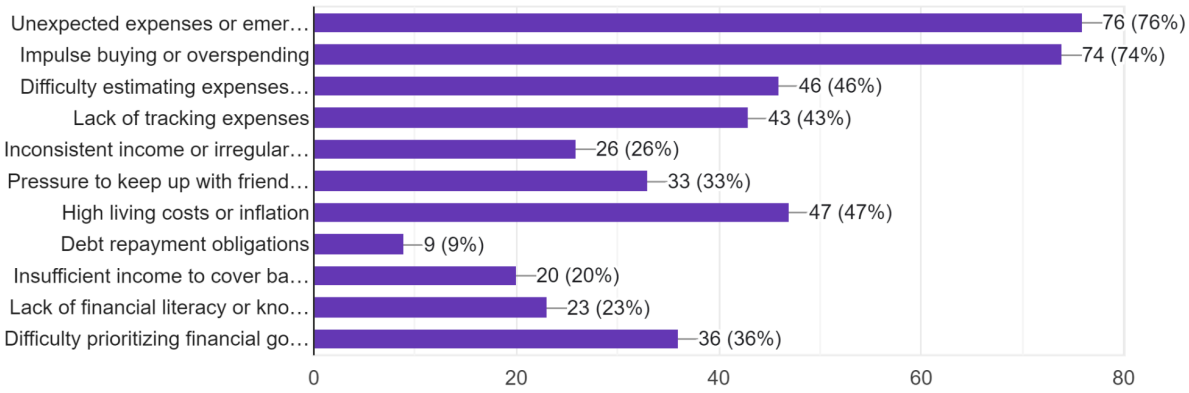

Based on your estimate, how much do you usually save monthly from budgeting your money?

100 responses



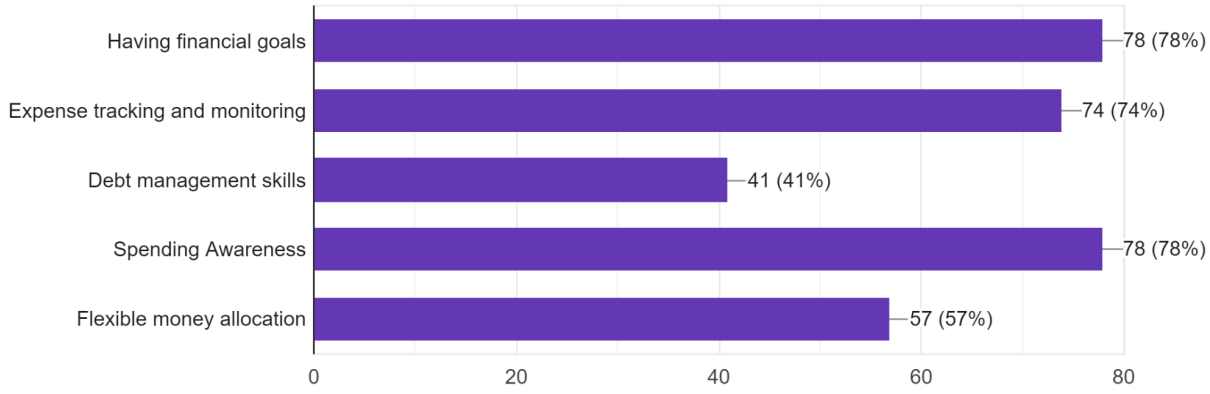
What challenges have you faced when trying to stick to your budget? (Check all that apply)

100 responses



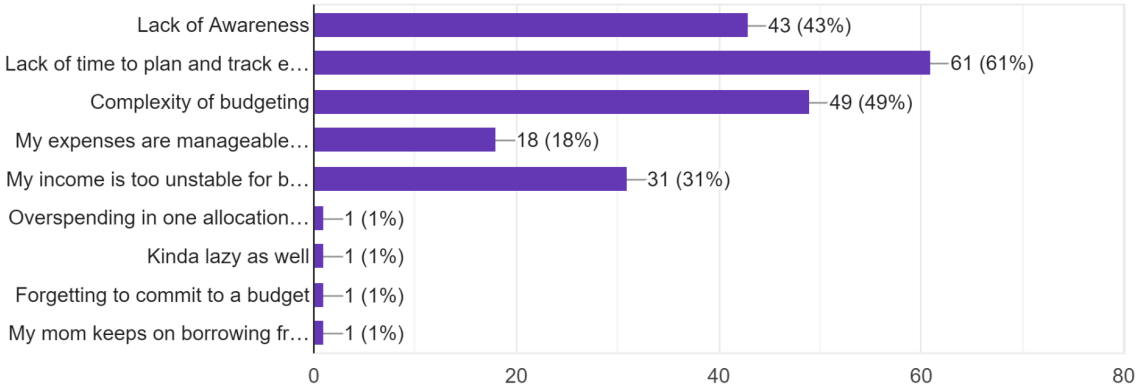

What factors do you consider to have a successful budget? (Check all that apply)

100 responses



What are the reasons that prevents you from budgeting your money? (Check all that apply)

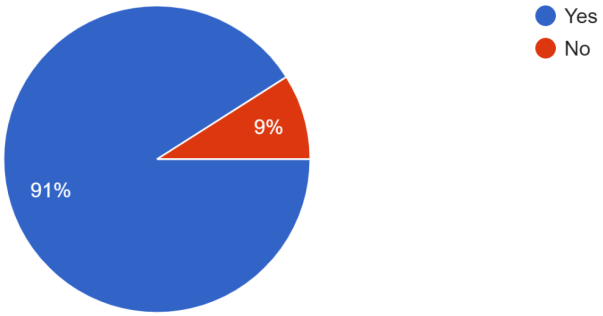
100 responses




### A.5. Goal-Setting

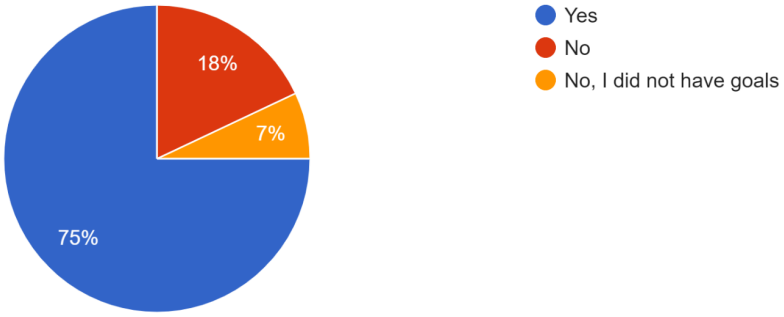
Have you ever had a financial goal like buying an item/saving an expected amount of money/paying debts?

100 responses



Did you ever achieve your previous goal?

100 responses

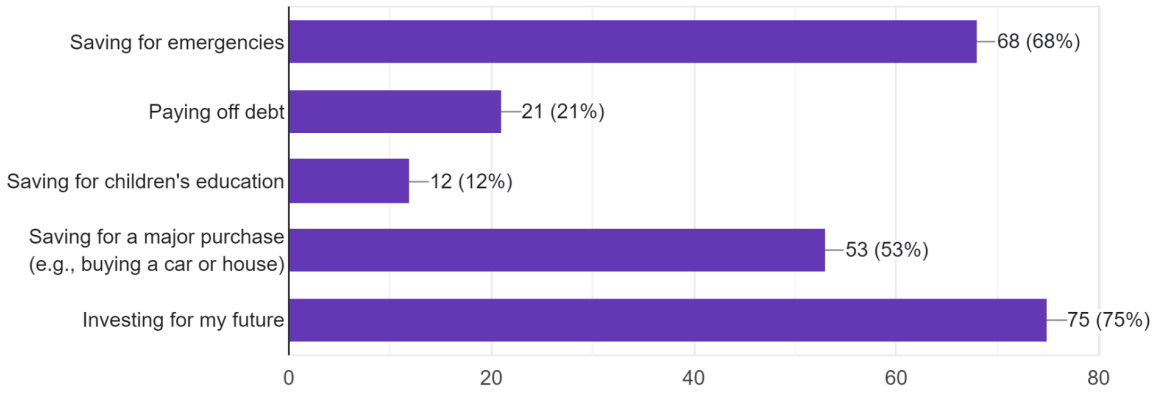



How did you achieve your goal?

100 responses

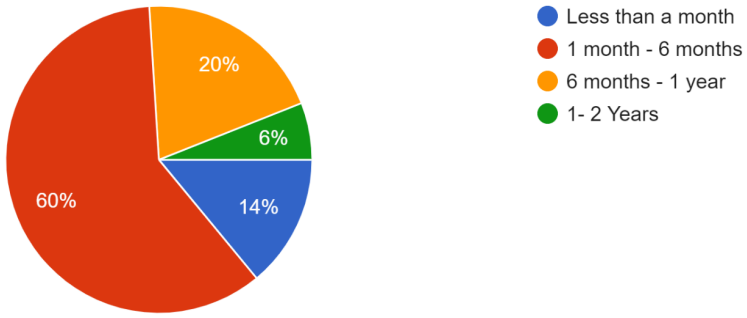
What do you consider as your common primary financial goals?

100 responses



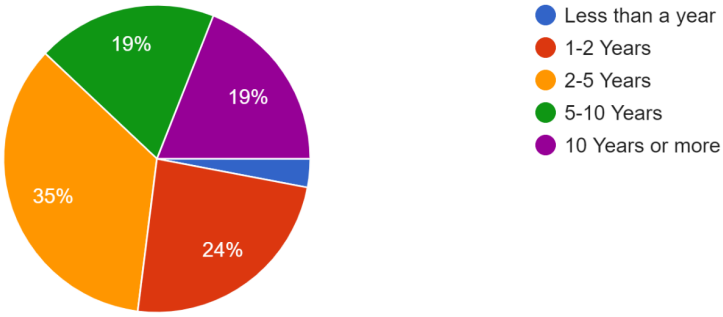

How long do you think it will take you to achieve a short-term financial goal?

100 responses



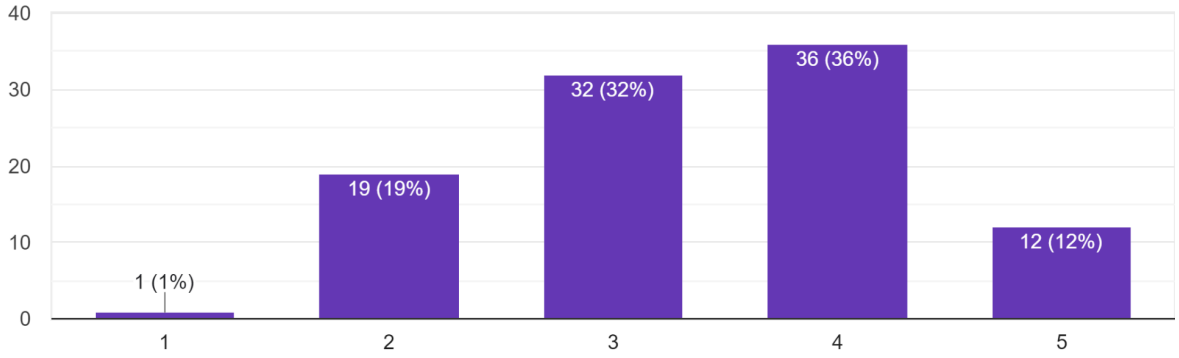
How long do you think it will take you to achieve a long-term financial goal?

100 responses



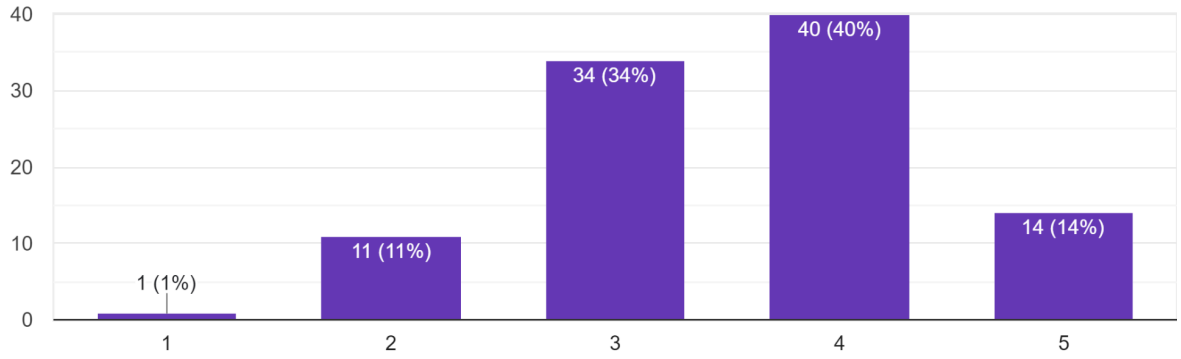

On a scale of 1 to 5, how confident are you in your ability to prioritize your financial goals effectively?

100 responses



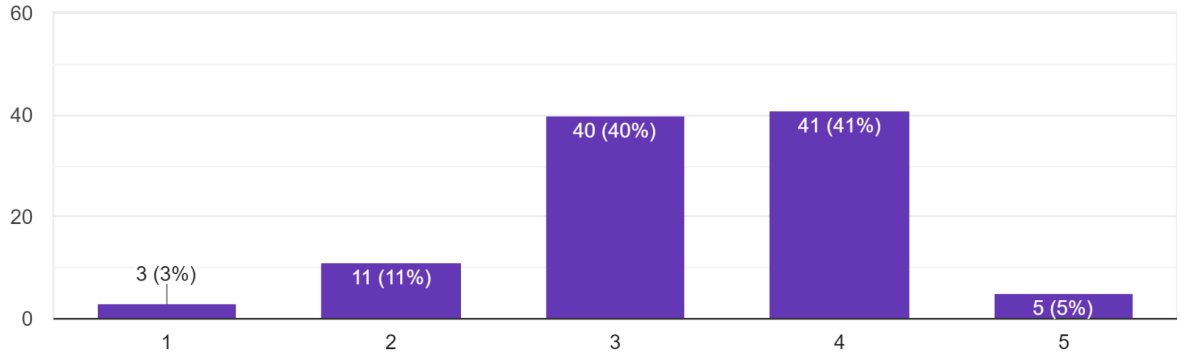
On a scale of 1 to 5, how challenging do you find it to overcome obstacles in achieving your financial goals?

100 responses



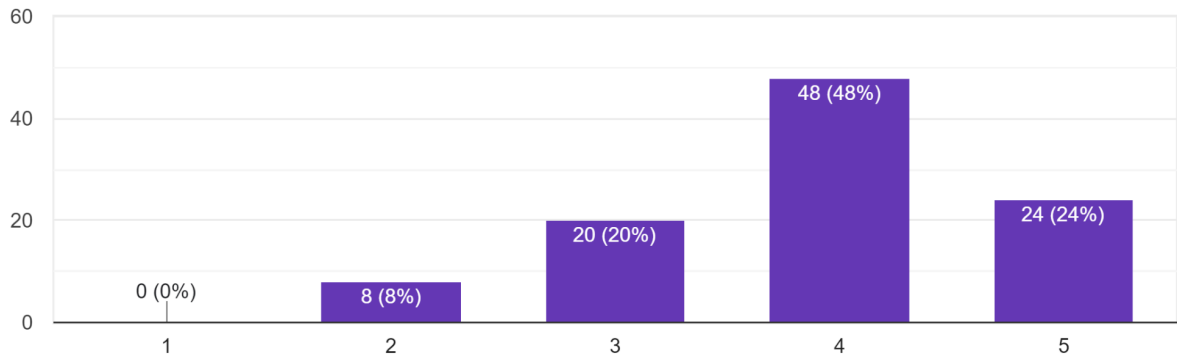

On a scale of 1 to 5, how flexible are you in adjusting your financial goals in response to changes in income, expenses, or other circumstances?

100 responses



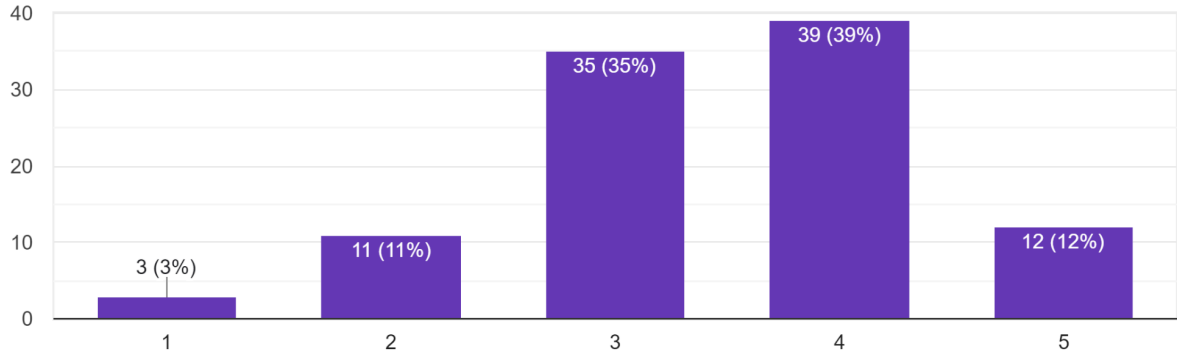
On a scale of 1 to 5, how motivated are you to achieve your long-term financial goals?

100 responses



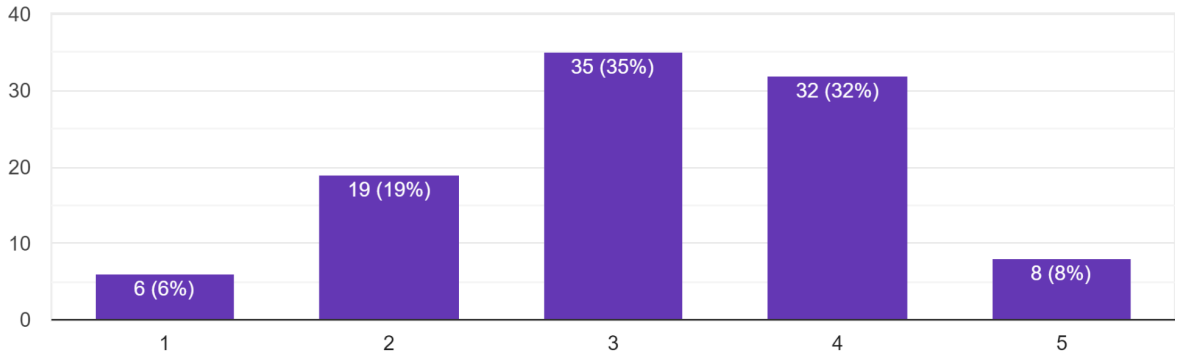

On a scale of 1 to 5, how satisfied are you with your progress towards achieving your financial goals?

100 responses



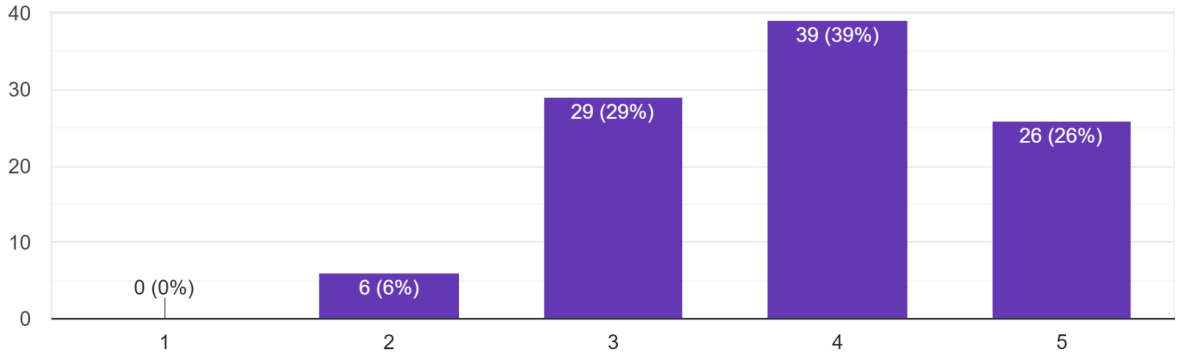
On a scale of 1 to 5, how likely are you to seek professional financial advice or guidance in setting and achieving your financial goals?

100 responses




On a scale of 1 to 5, how helpful do you find resources or tools in setting and tracking your financial goals?

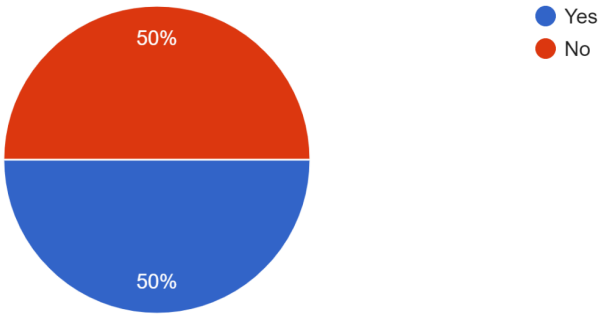
100 responses



## A.6. Investments

Do you have any experience in investing?

100 responses

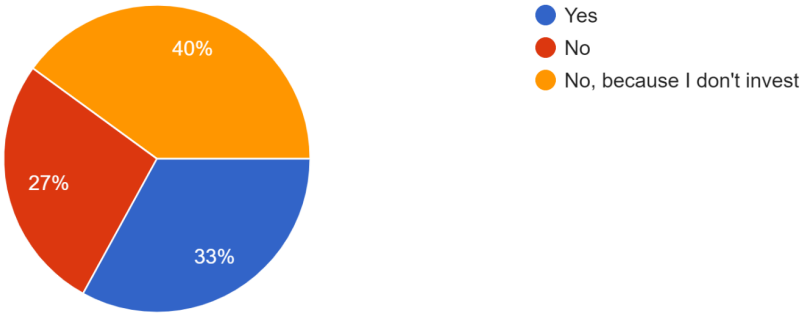



How frequent do you invest?

100 responses

Do you have troubles tracking your investments?

100 responses




How much have you invested in total?

100 responses

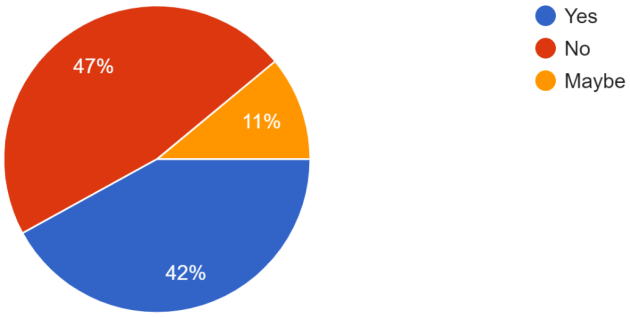
How much do you think would you be willing to invest as a first timer?

100 responses

## A.7. Personal Budget Management System

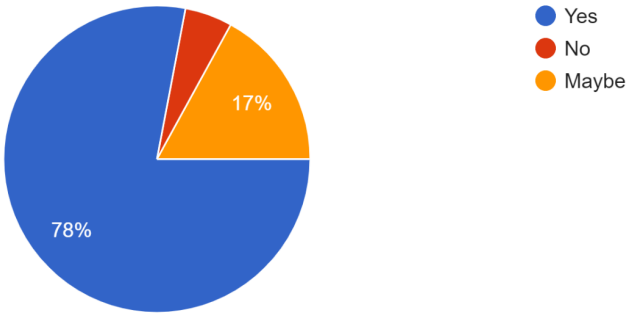
Have you ever used a personal budget management system in which helps you budget your money by tracking your income and expenses?

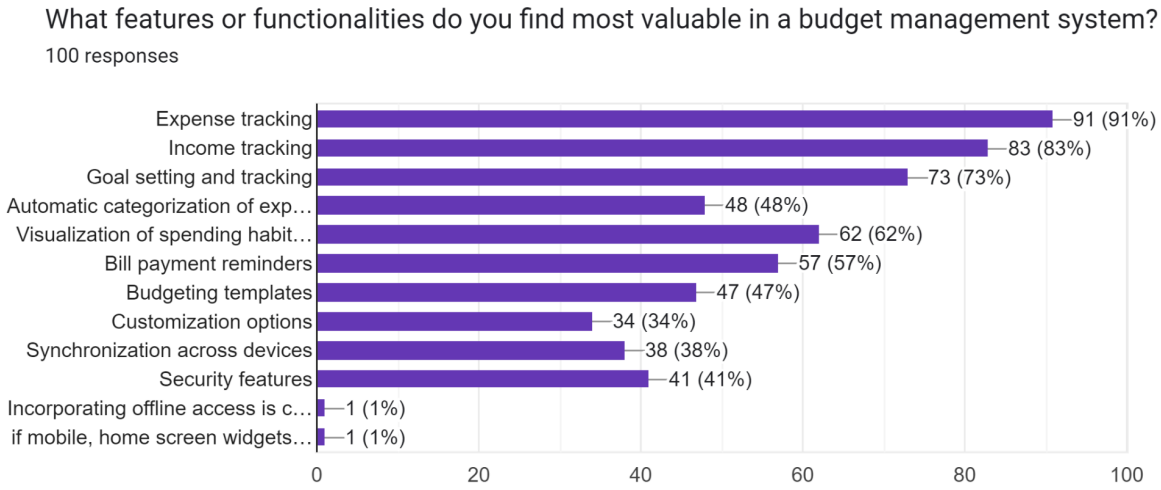
100 responses




Do you believe that using a budget management system can help you improve your financial habits and decision-making?

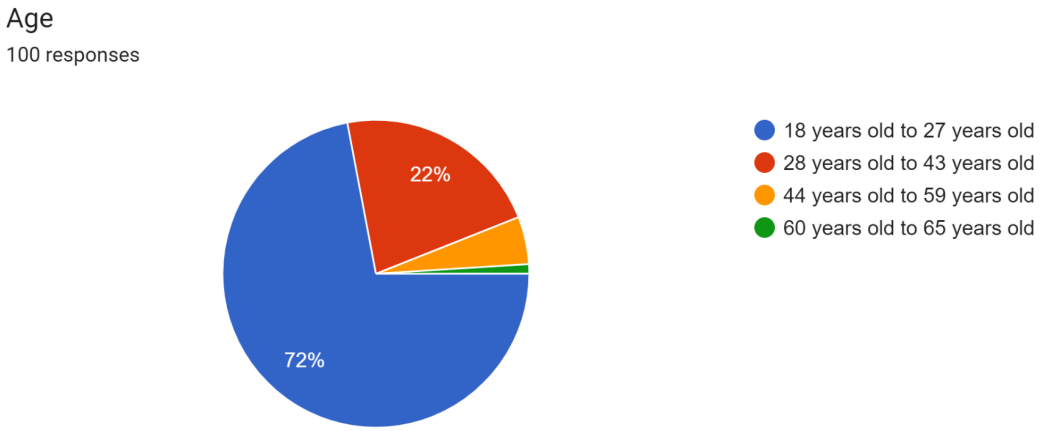
100 responses





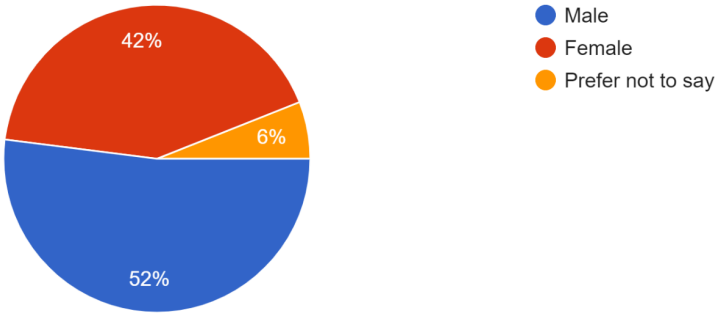

## Appendix B. Personal Finances Among Single Working Adults Survey

### B.1. Personal Information



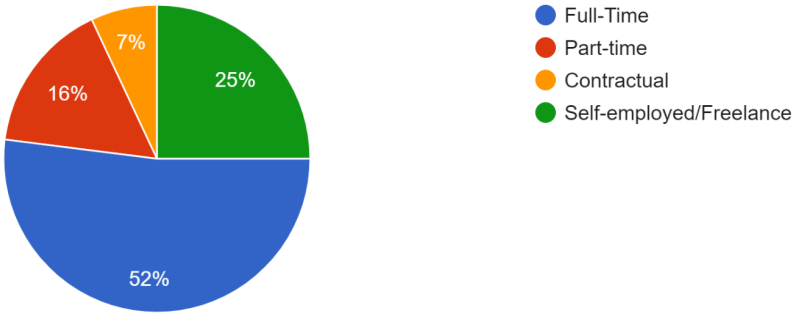
Gender

100 responses




Type of employment

100 responses



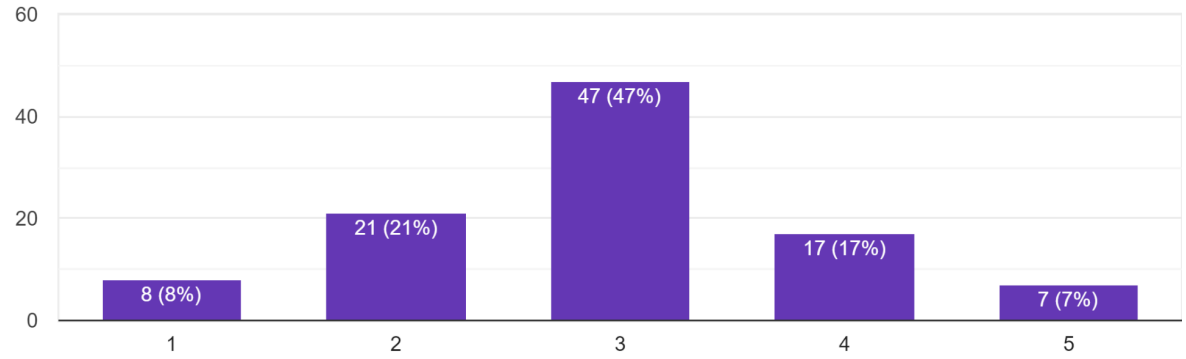

How much is your monthly income

100 responses

### B.2. Financial Well-Being

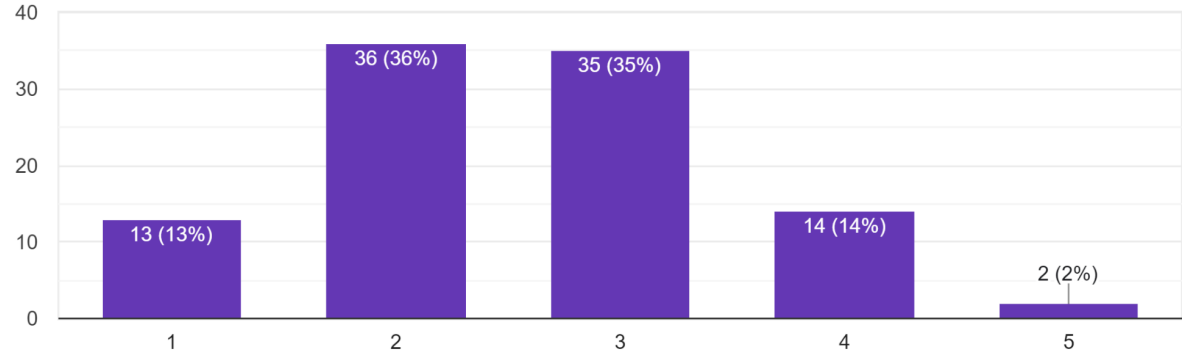
1. How would you rate your current financial well-being?

100 responses



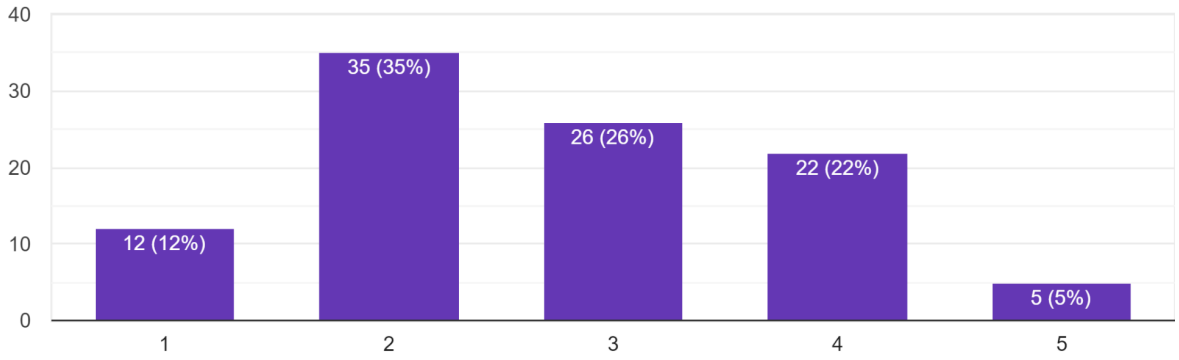

2. How satisfied are you with your current level of financial stability?

100 responses



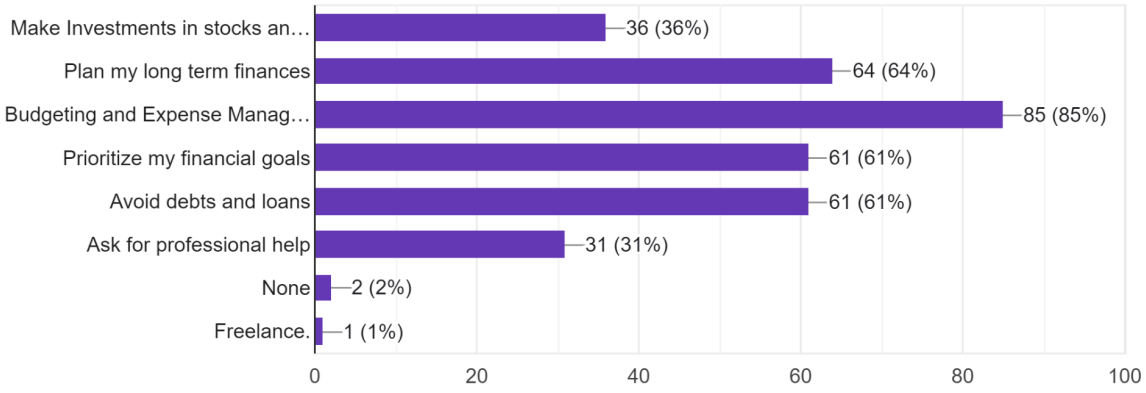
3. How confident are you in your ability to handle unexpected expenses or financial emergencies?

100 responses



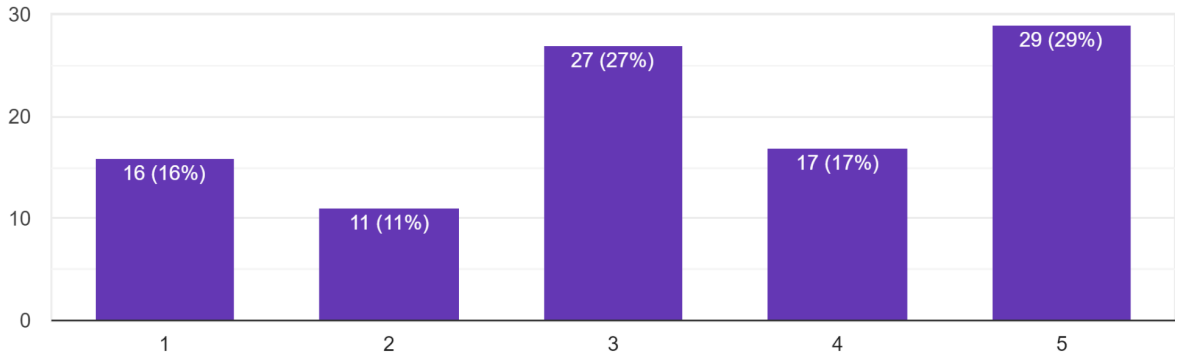

4. What actions do you currently take to ensure your long-term financial security and well-being? (Check all that apply)

100 responses



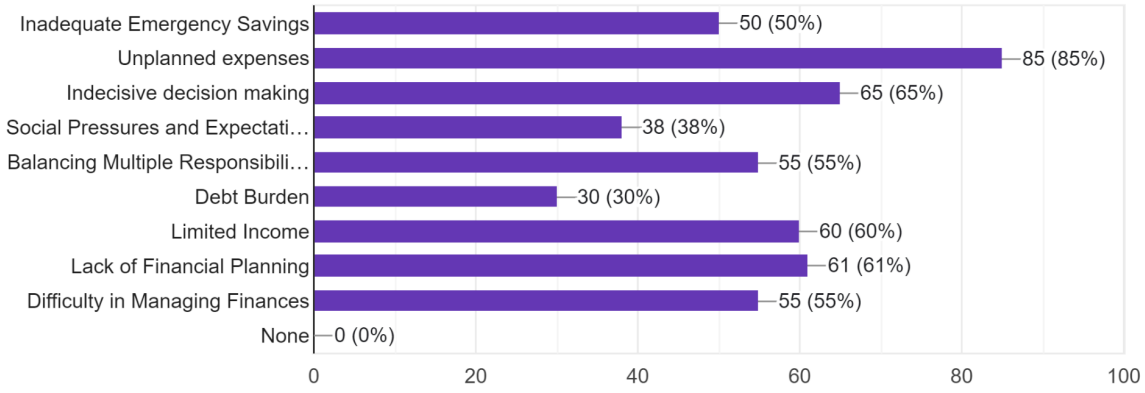
5. How comfortable are you with your current level of debt, if any?

100 responses



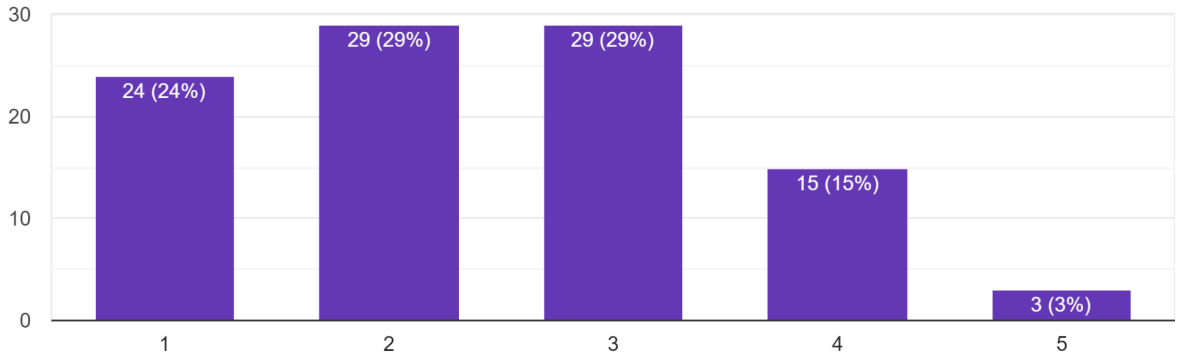

6. In your opinion, what are the most significant barriers to achieving financial stability and well-being? (Check all that apply)

100 responses



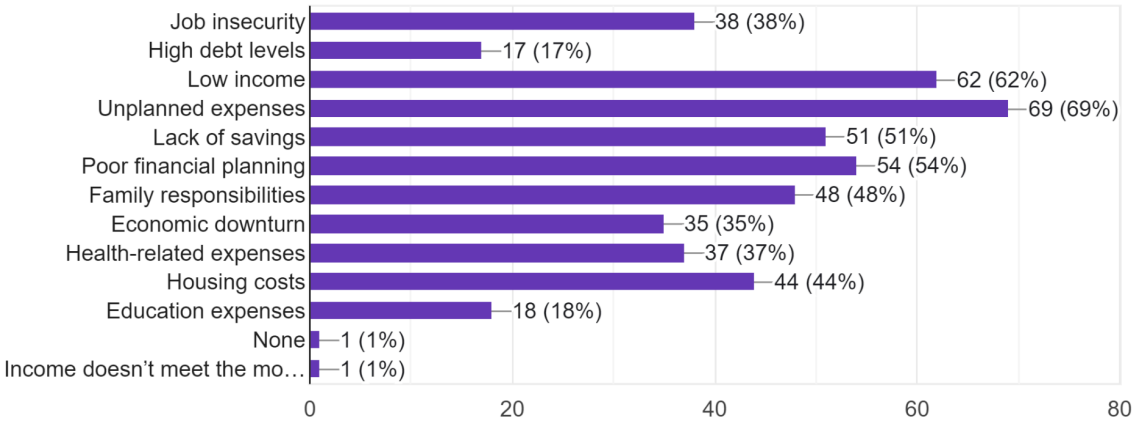
7. How do you feel when managing your finances?

100 responses




8. What factors, if any, contribute to your feelings of stress regarding your finances? (Check all that apply)

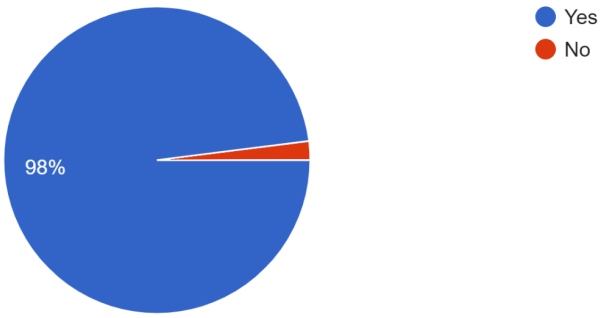
100 responses



### B.3. Financial Goal

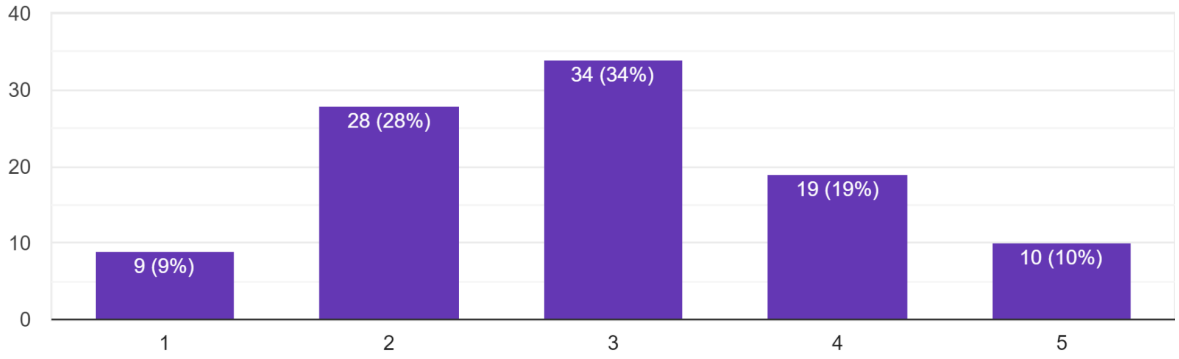
9. Do you currently have any financial goals like buying an item, saving an expected amount of money for vacation, or making investment plans?

100 responses



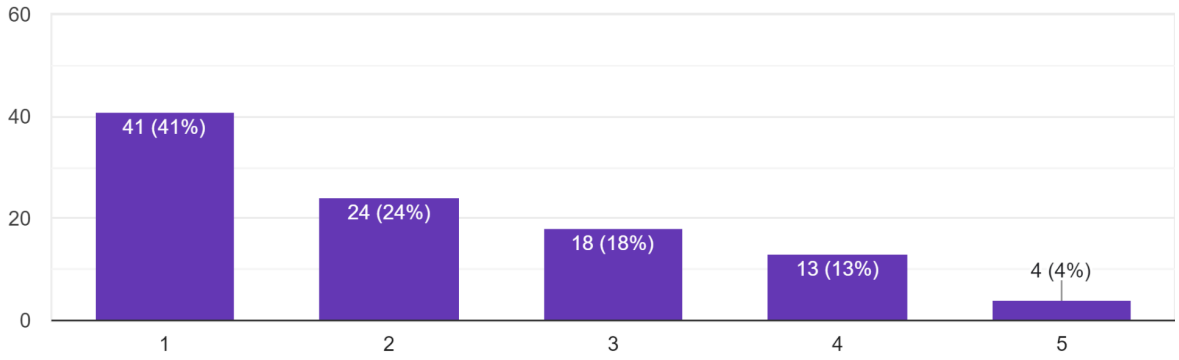

10. How satisfied are you with your current income level in meeting your financial needs and goals?

100 responses



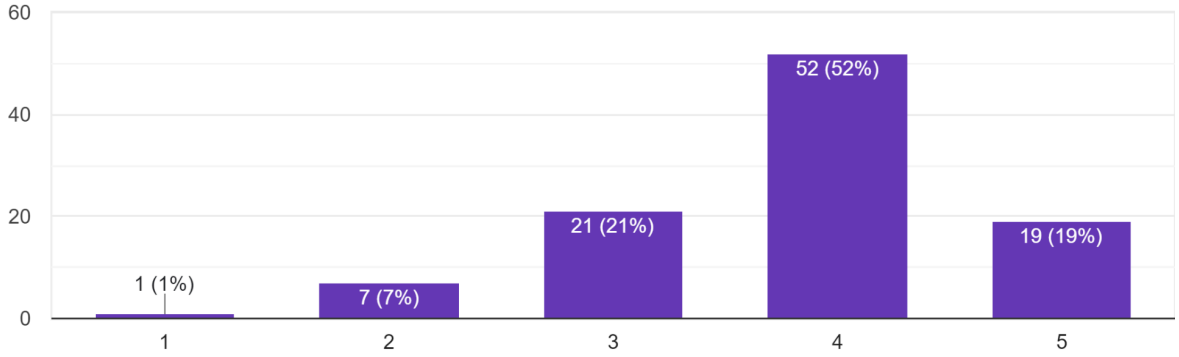
11. How likely are you to take loans in order to achieve your financial goals faster

100 responses



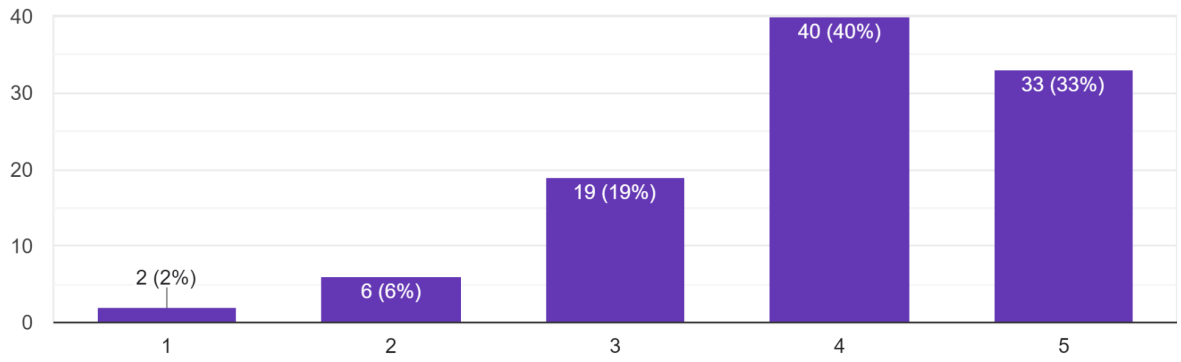

12. How much priority do you place on your financial goals?

100 responses



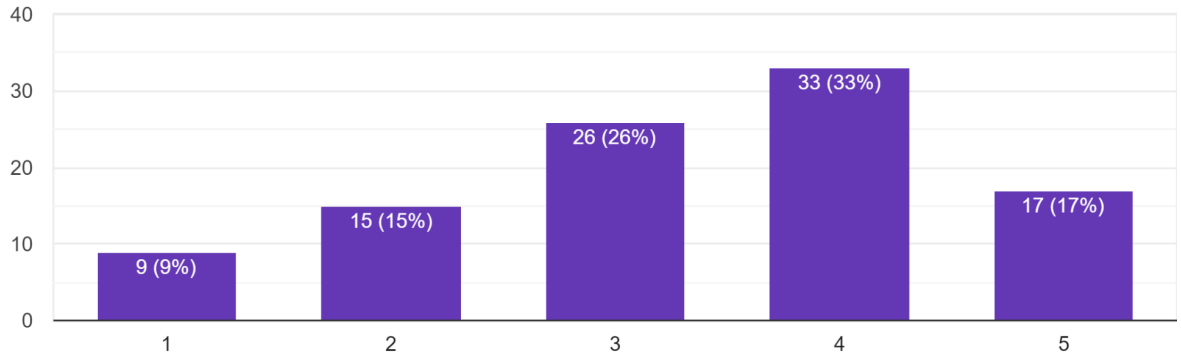
13. How motivated are you to achieve your financial goals?

100 responses



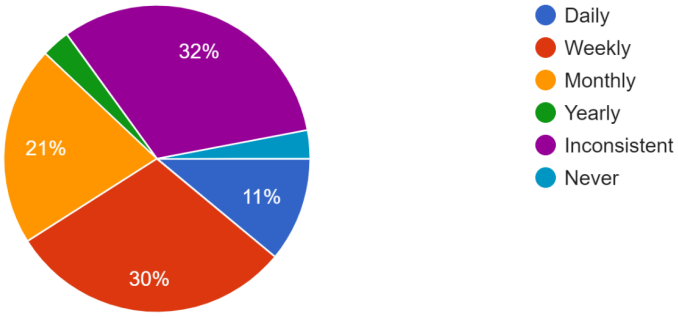

14. How confident are you in your ability to achieve your financial goals?

100 responses



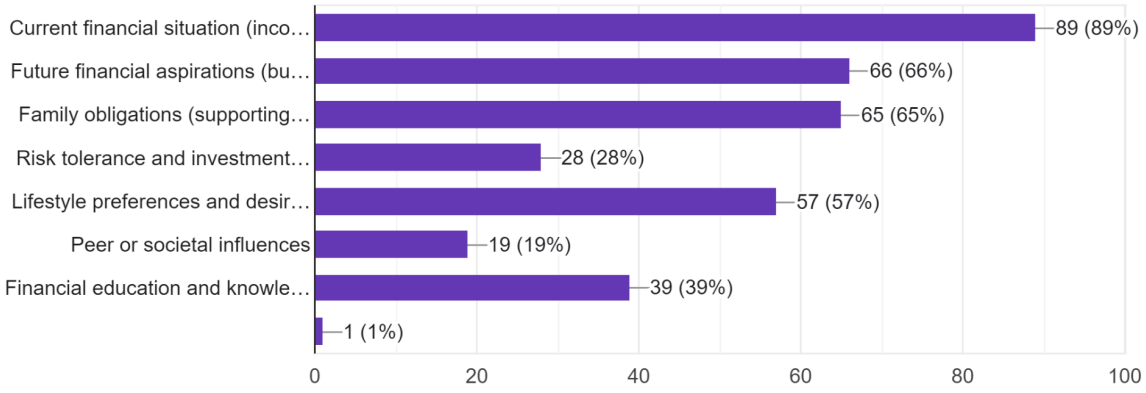
15. How often do you review your progress towards your financial goals?

100 responses



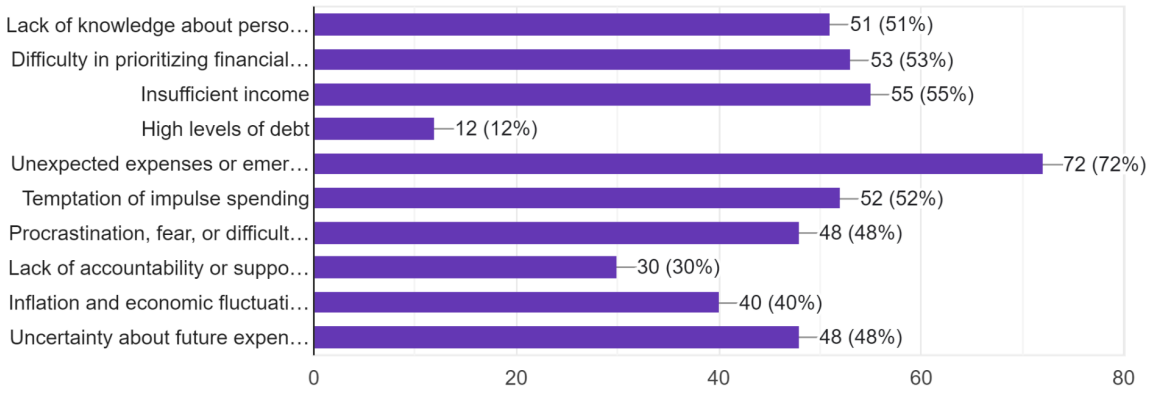

16. What factors influence your decision-making process when setting financial goals? (Check all that apply)

100 responses



17. What are some challenges you face when it comes to financial planning and goal setting?

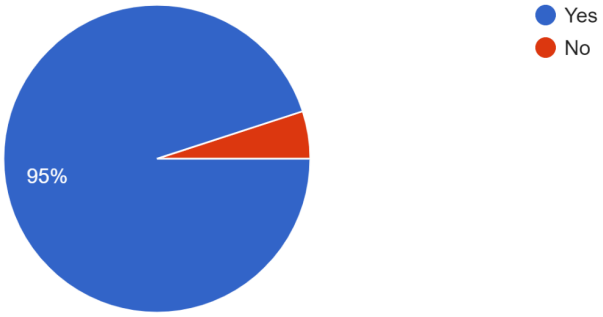
100 responses




## B.4. Savings

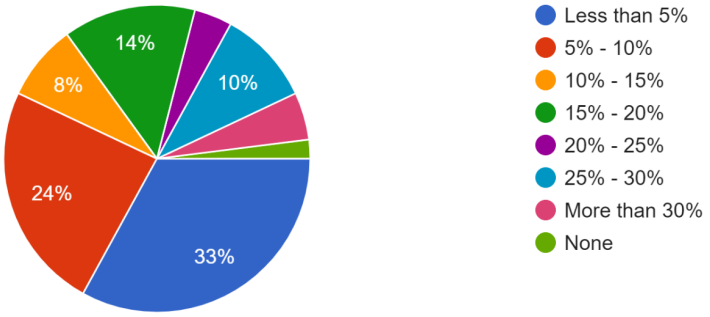
18. Do you save money for yourself?

100 responses



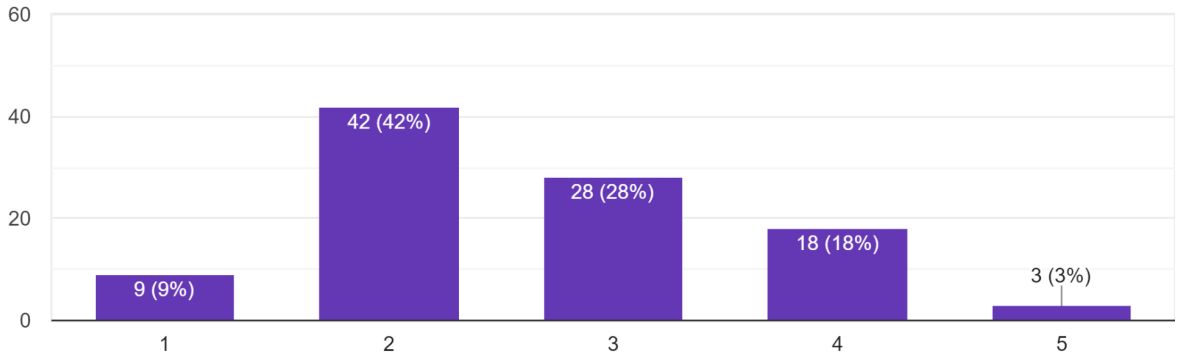
19. What percentage of your monthly income do you typically allocate towards savings?

100 responses



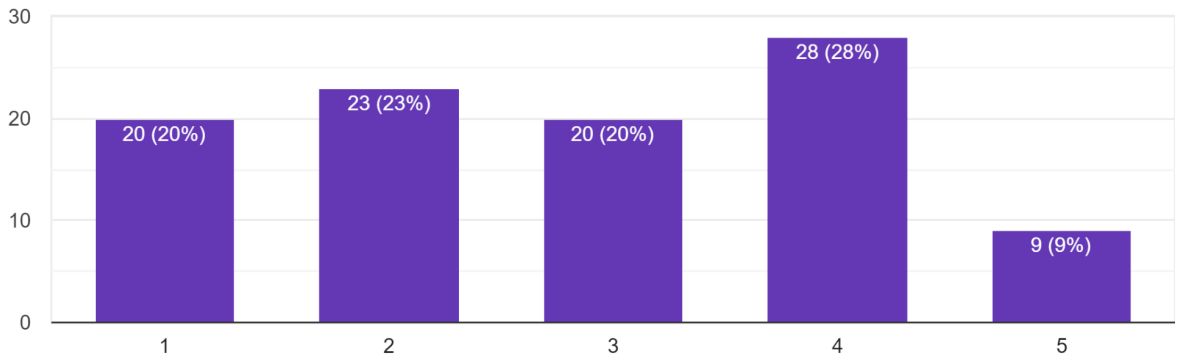

20. How satisfied are you with the amount you currently save each month?

100 responses



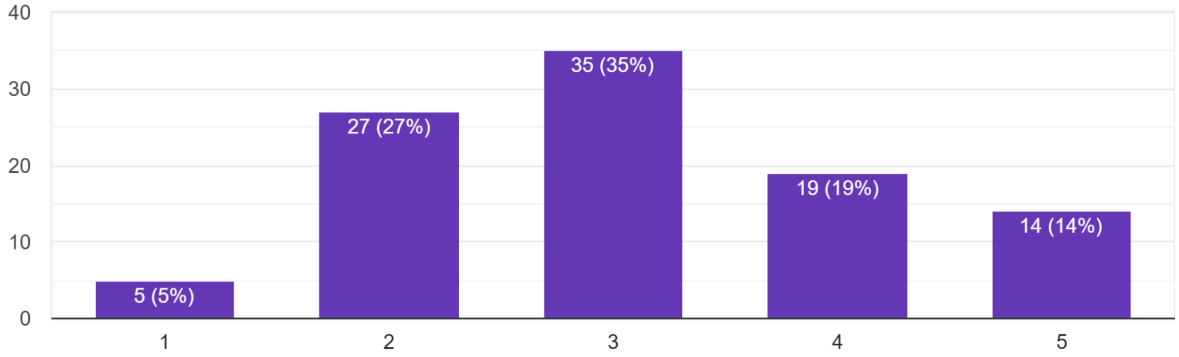
21. How confident are you in your ability to consistently save a portion of your income?

100 responses



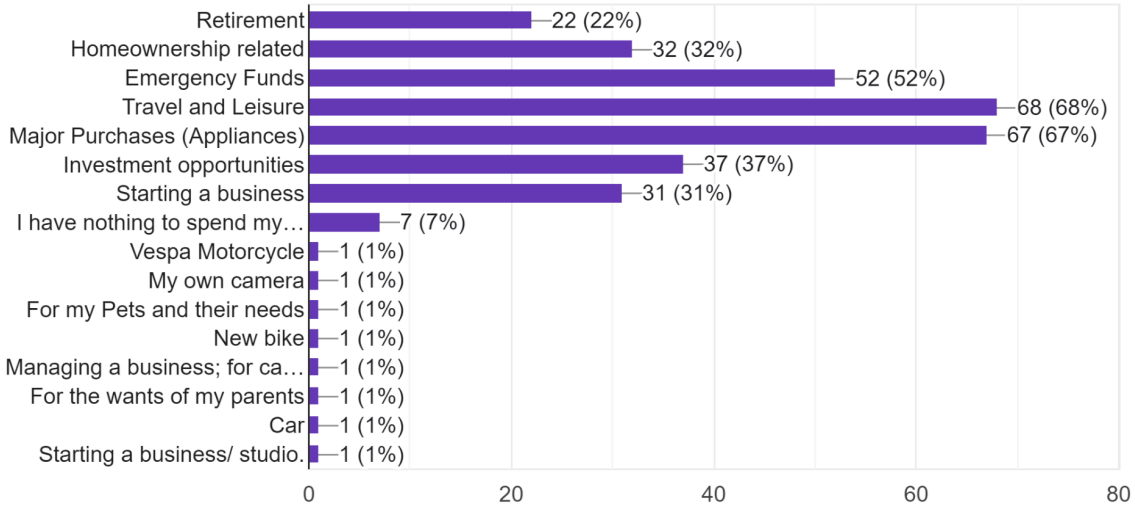

22. How often do you review your savings progress?

100 responses



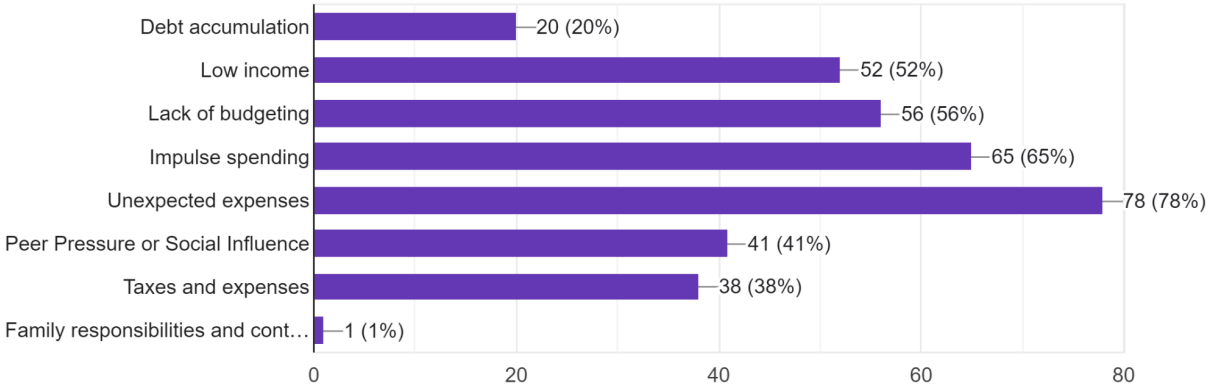
23. What is your purpose for saving? (Check all that apply)

100 responses




24. In your experience, what hindered you from saving enough money for yourself? (Check all that apply)

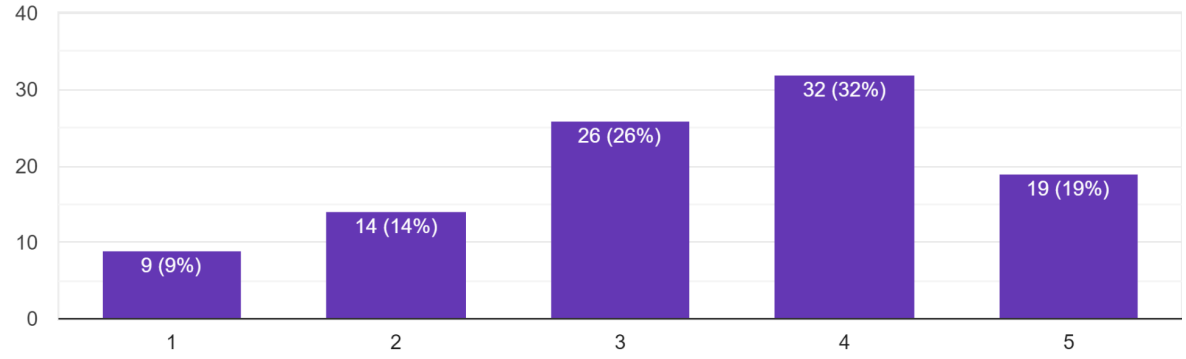
100 responses



### B.5. Expenses

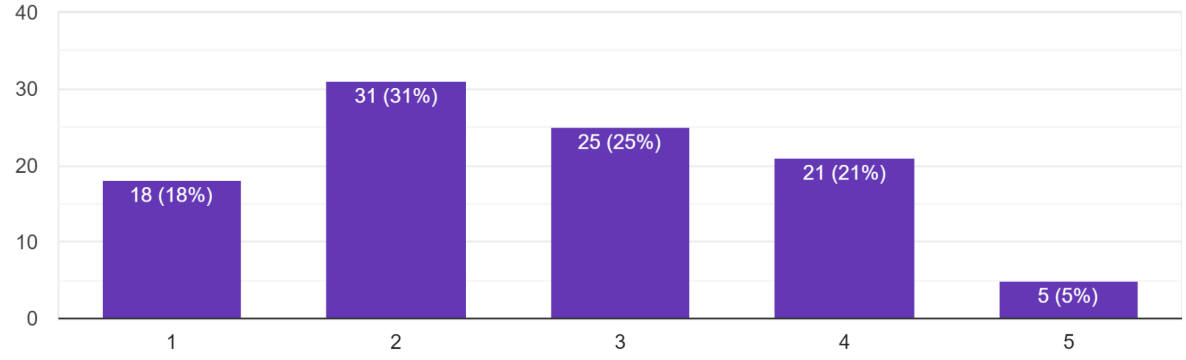
25. How well do you feel you understand the impact of your spending habits on your overall financial health?

100 responses



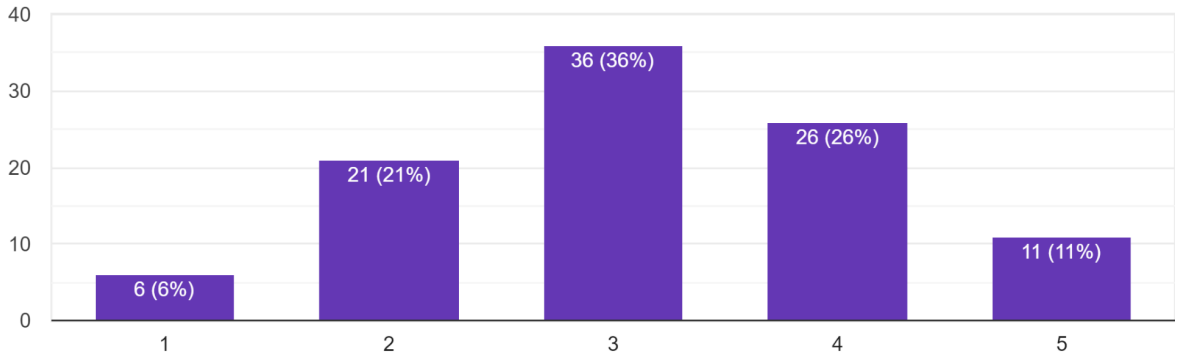

26. How satisfied are you with the level of control you have over your expenses?

100 responses



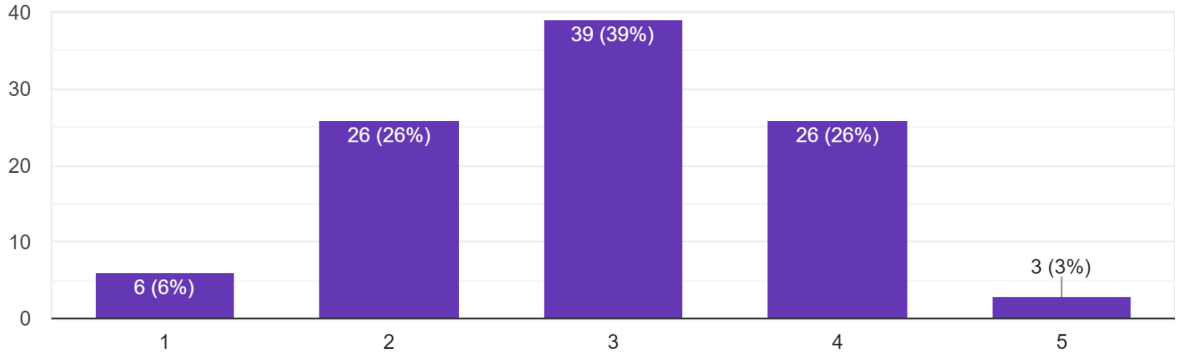
27. How often do you review your expenses to identify areas where you can cut back or save more?

100 responses



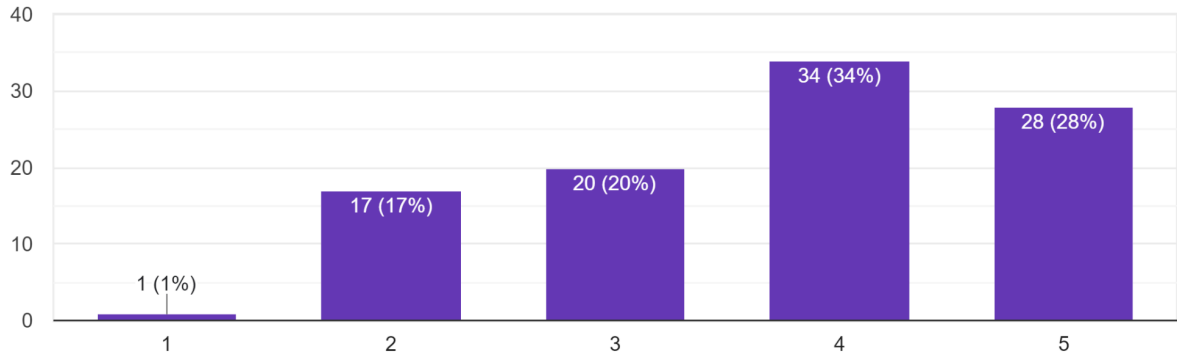

28. To what extent do you feel your expenses align with your financial goals and priorities?

100 responses



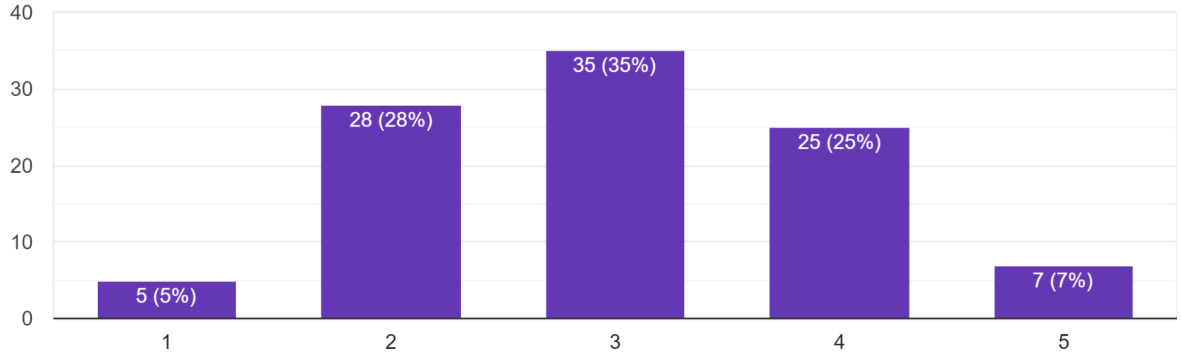
29. How frequently do you find yourself making purchases without prior planning?

100 responses



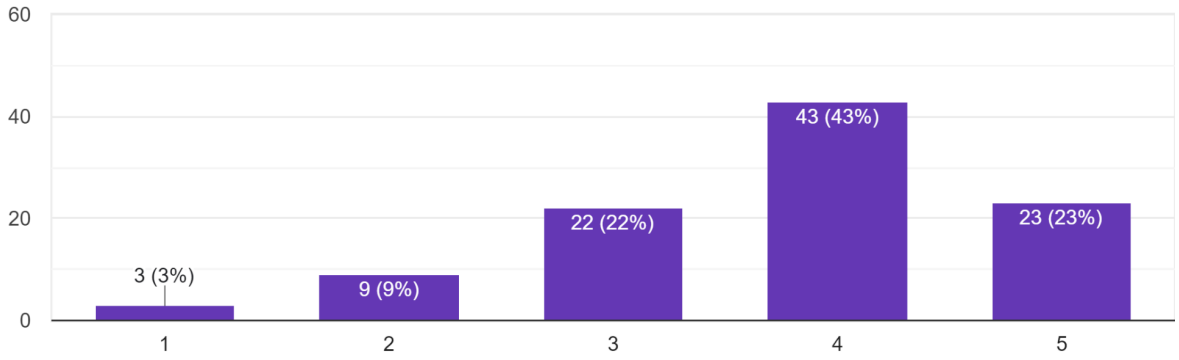

30. How often do you track your expenses against your budget to ensure you are staying on track?

100 responses



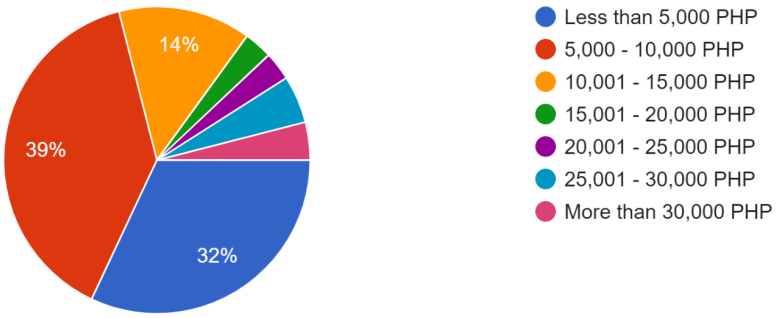
31. How often do you find yourself exceeding your budget due to unplanned expenses?

100 responses



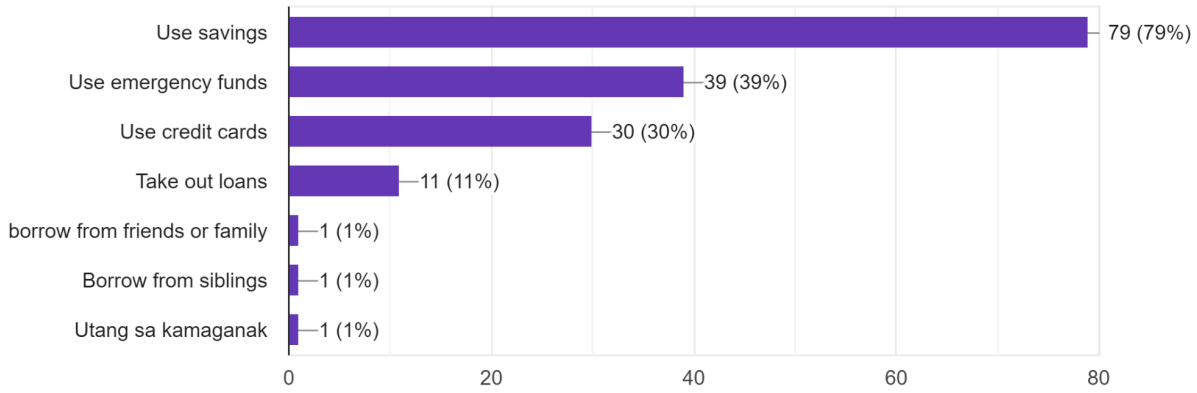

32. On average, how much do you spend per month on essential expenses such as rent, utilities, groceries, and transportation?

100 responses



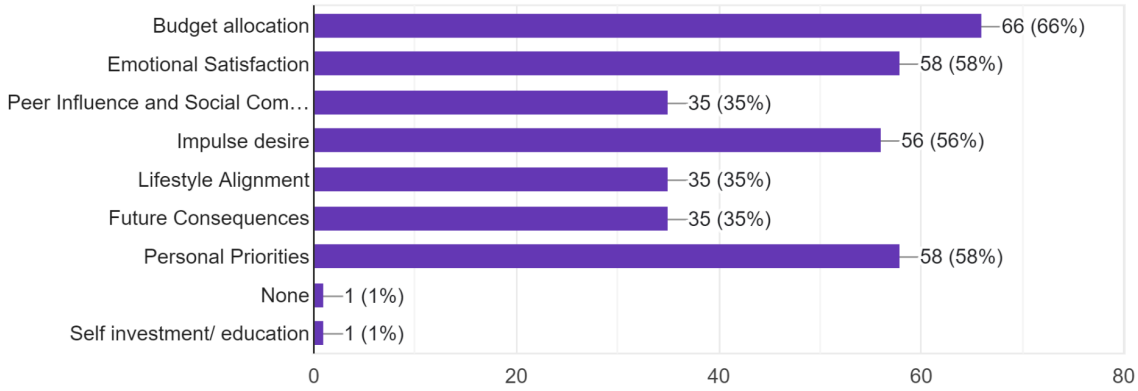
33. How do you manage unexpected expenses? (Check all the apply)

100 responses




34. What factors do you consider when making decisions about spending money on non-essential items or experiences?

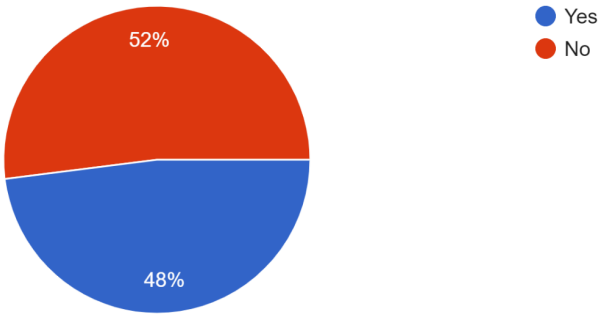
100 responses



## B.6. Investment

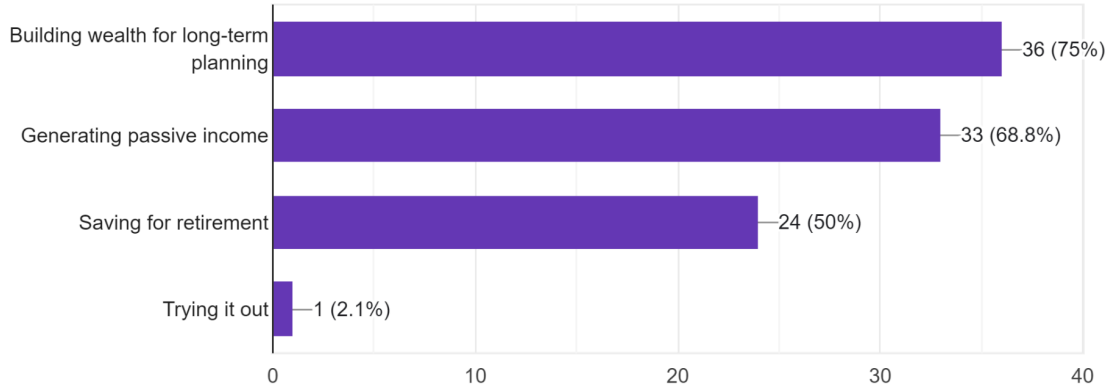
35. Do you invest your money?

100 responses



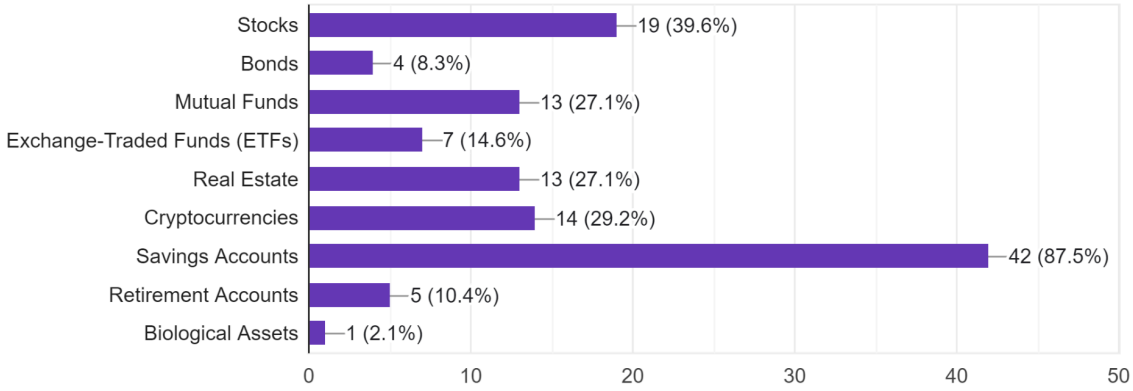

36. What is your primary motivation for investing

48 responses



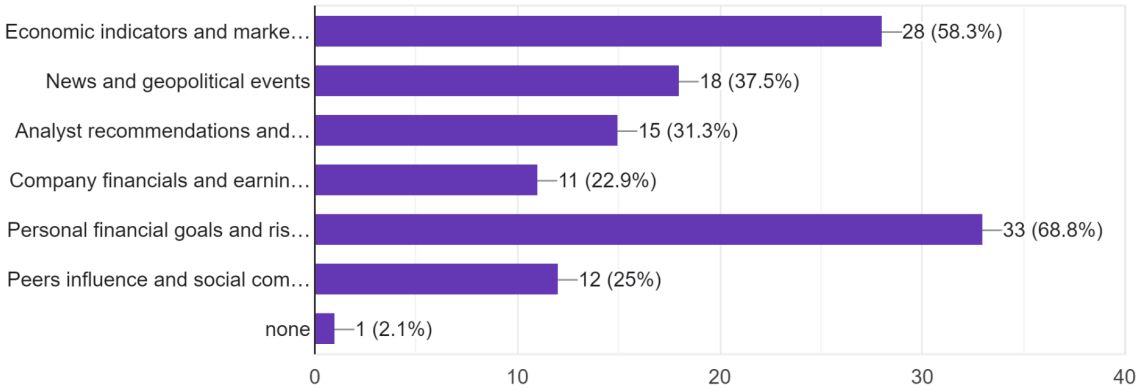
37. Where do you invest your money?

48 responses



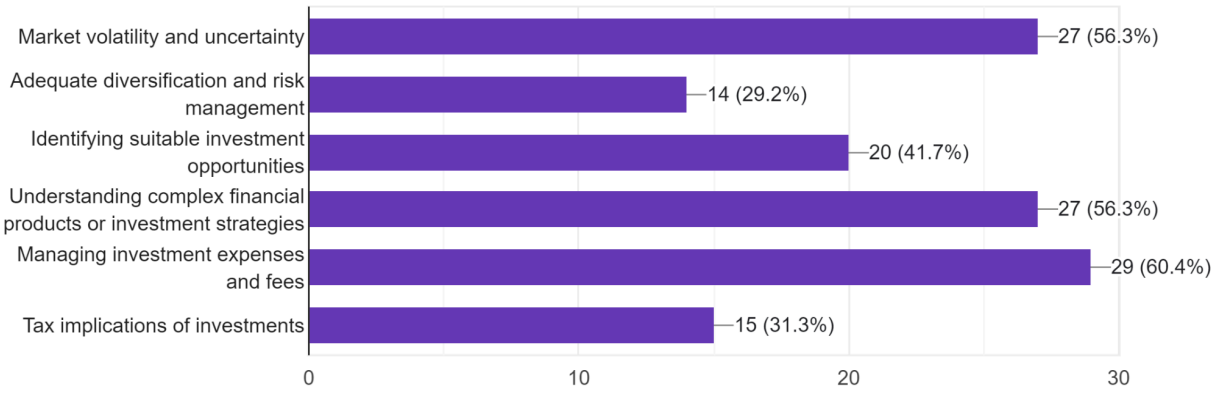

38. What factors influence your decision to buy or sell investments?

48 responses



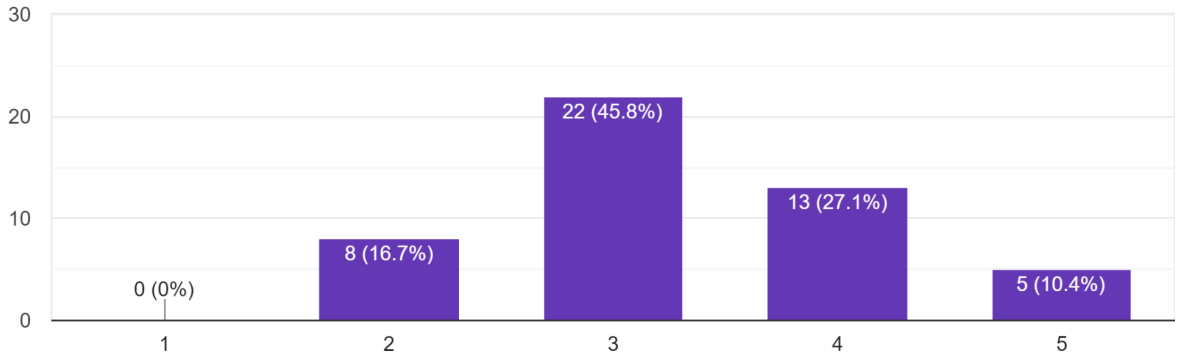
39. What are your main concerns or challenges when it comes to investing?

48 responses



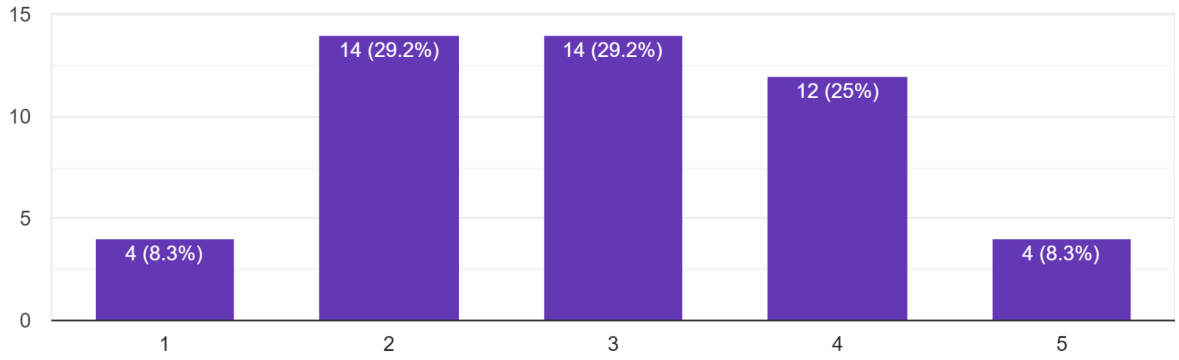

40. How comfortable are you with the level of risk associated with your investments?

48 responses



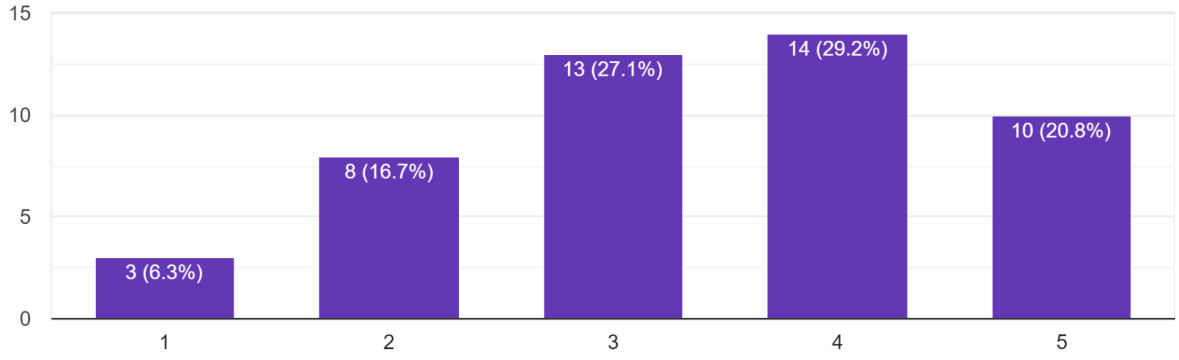
41. How confident are you in your understanding of different investment options?

48 responses



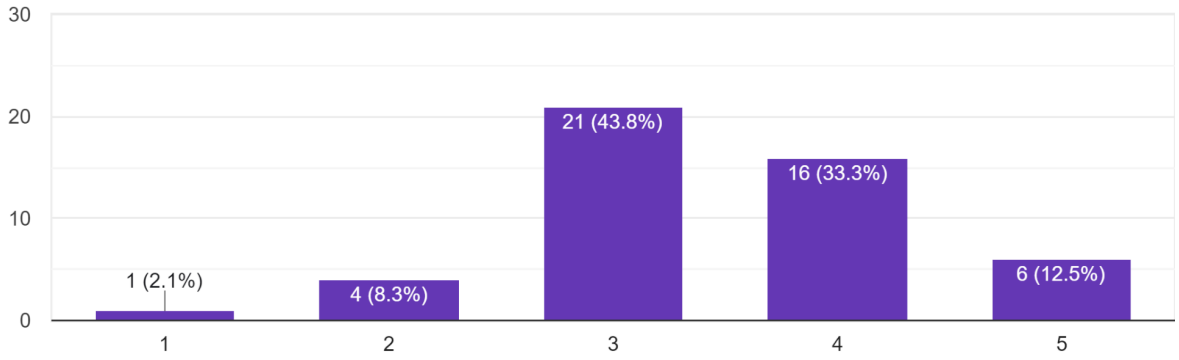

42. To what extent do you actively seek out information or advice before making investment decisions?

48 responses



43. How well do you feel your investments align with your risk tolerance and financial objectives?

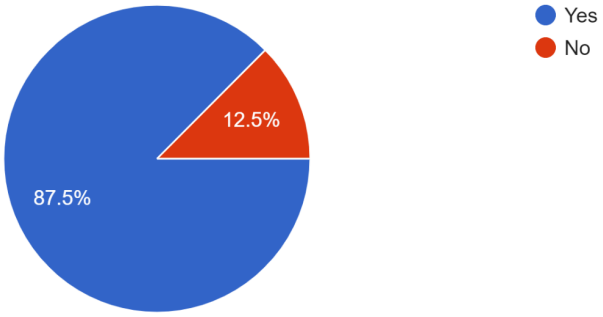
48 responses




### B.7. Budget

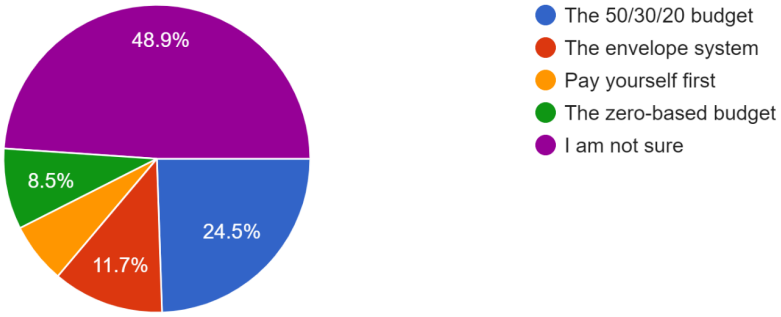
44. Do you budget your money?

48 responses



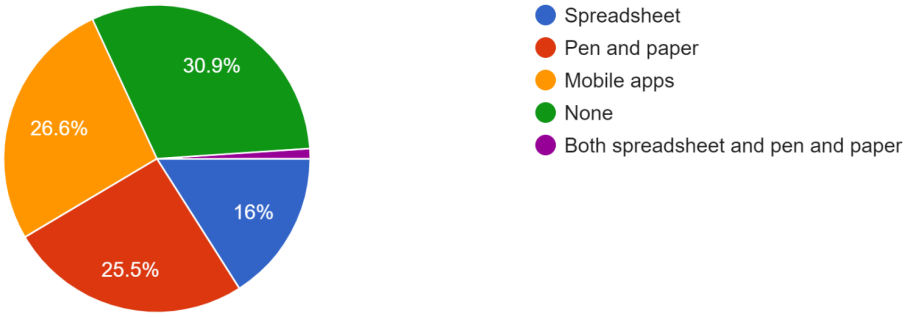
45. What budgeting plan/technique do you usually follow?

94 responses



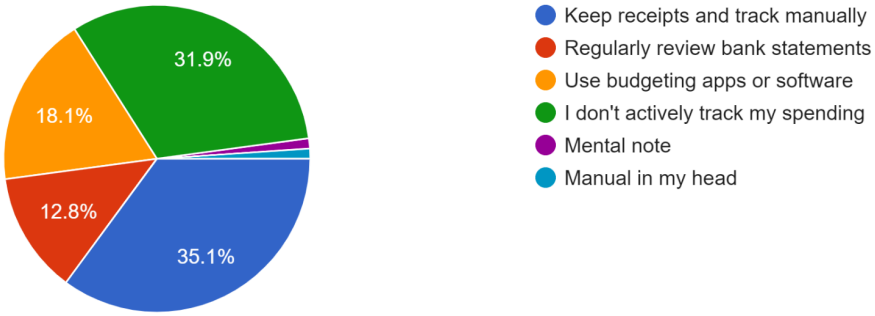

46. Which budgeting method do you commonly use?

94 responses



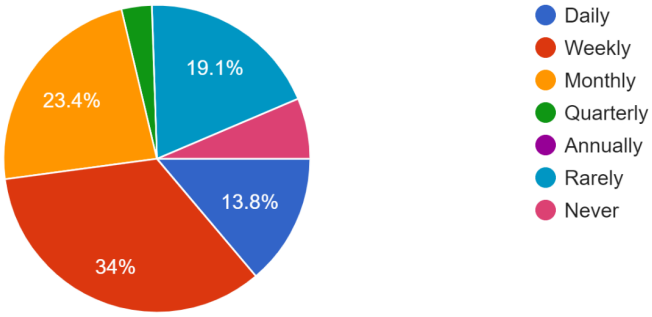

47. How do you usually track your spendings?

94 responses



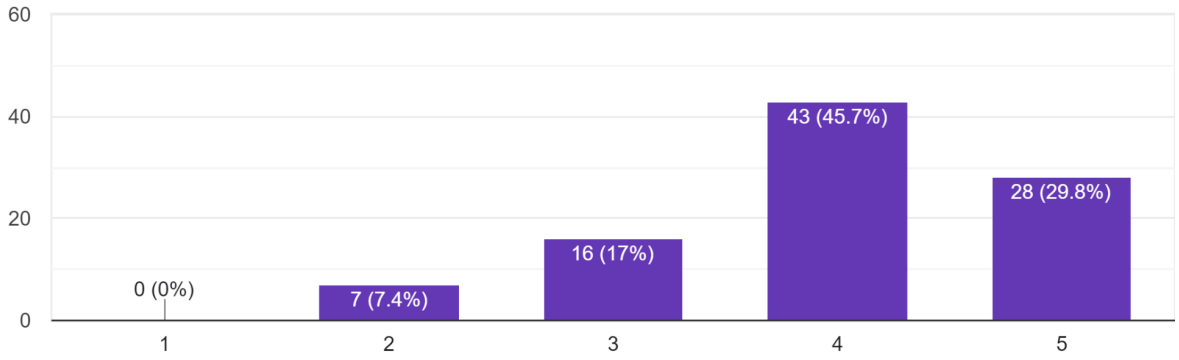
48. How frequently do you update or review your budget and expenses?

94 responses



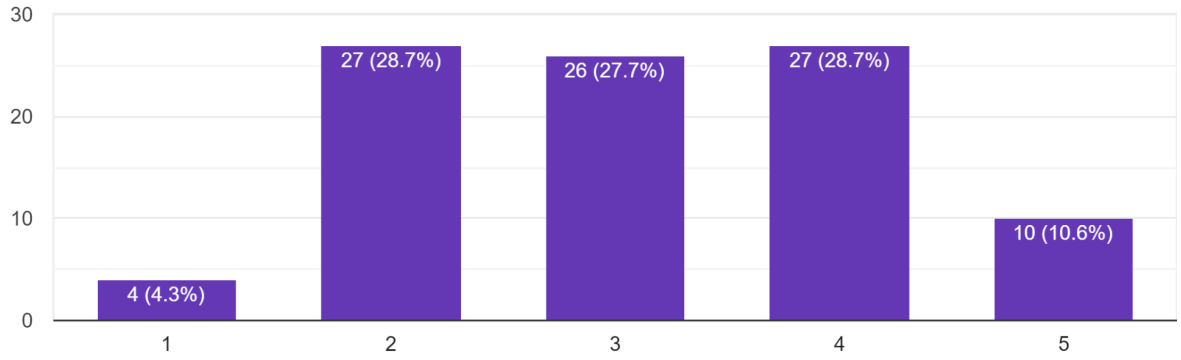

49. How often do you see the need to build a budget?

94 responses



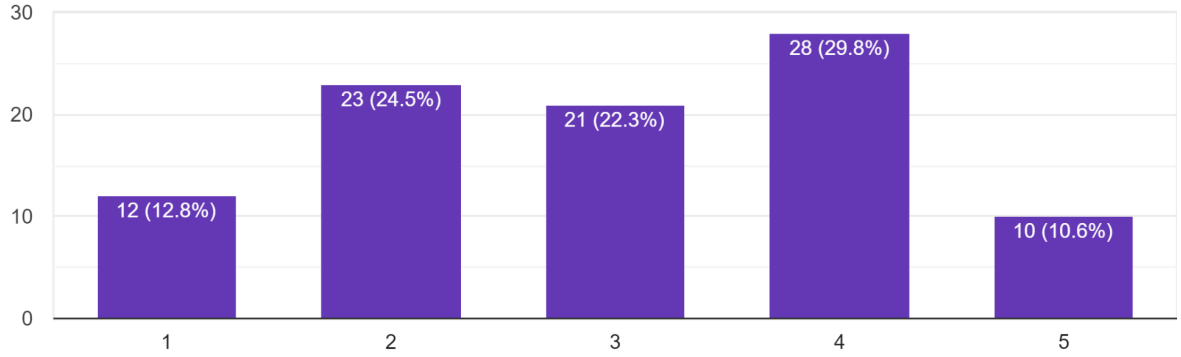
50. How often do you stick to your budget?

94 responses



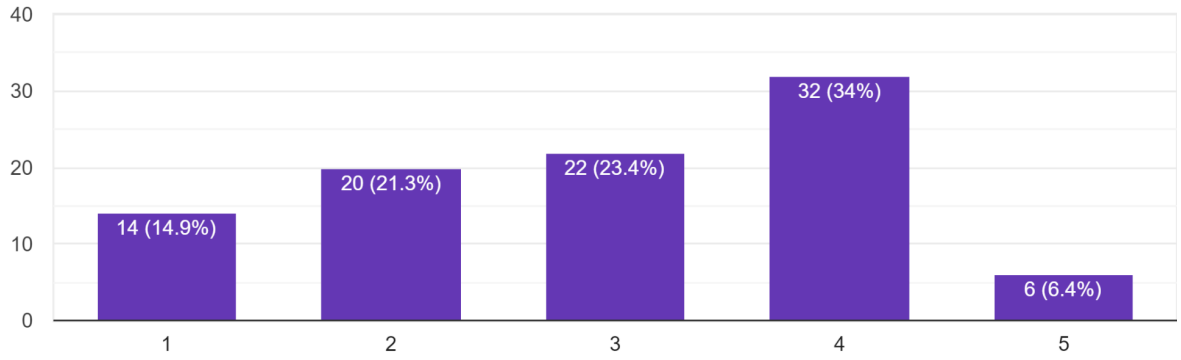

51. How confident are you in your ability to create a realistic and effective budget?

94 responses



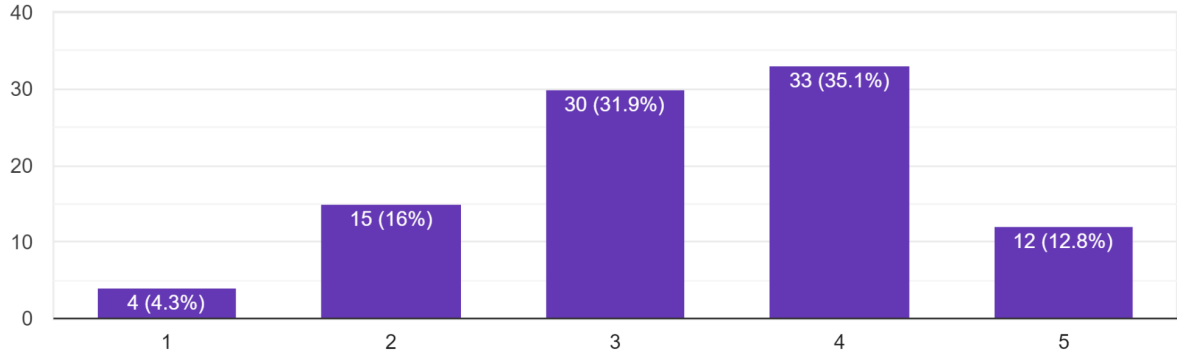
52. How satisfied are you with the effectiveness of your budget in managing your finances?

94 responses



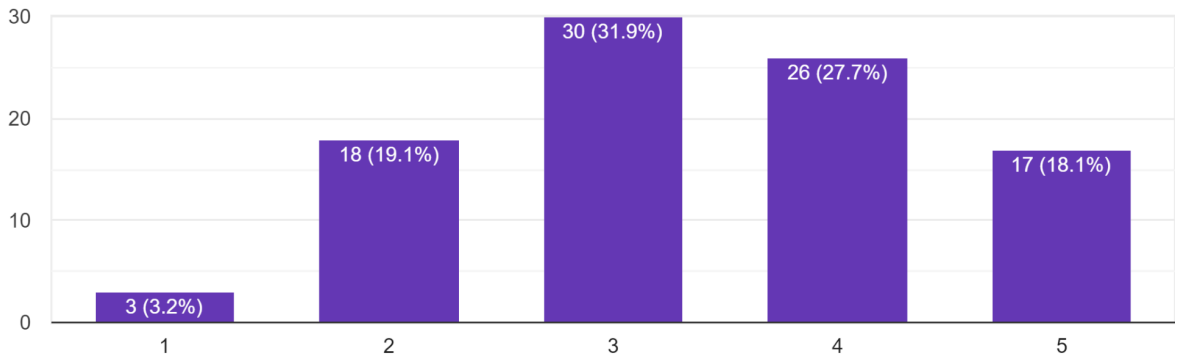

53. How well do you feel your budget helps you achieve your financial goals?

94 responses



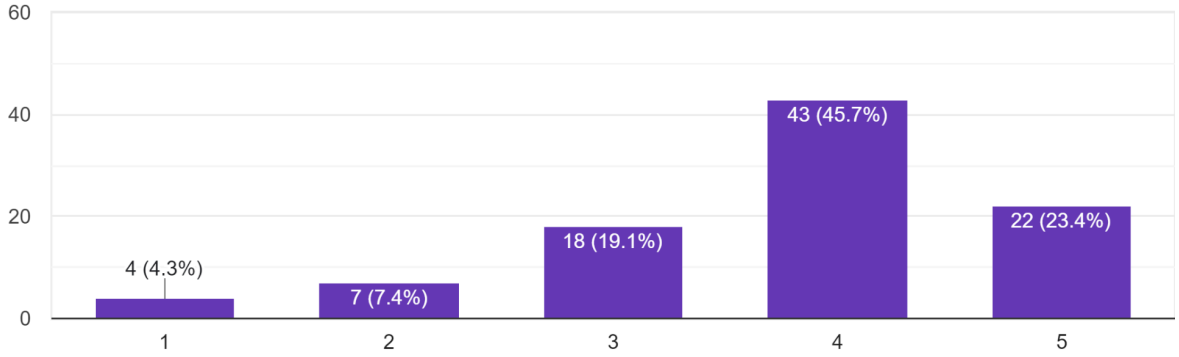
54. How committed are you when it comes to tracking your expenses?

94 responses



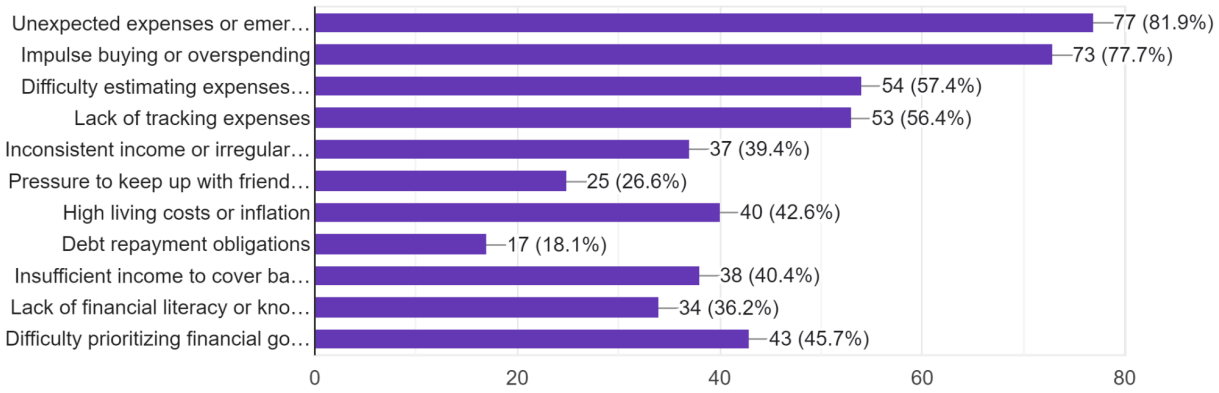

55. Does your financial situation improve when sticking to your budget?

94 responses



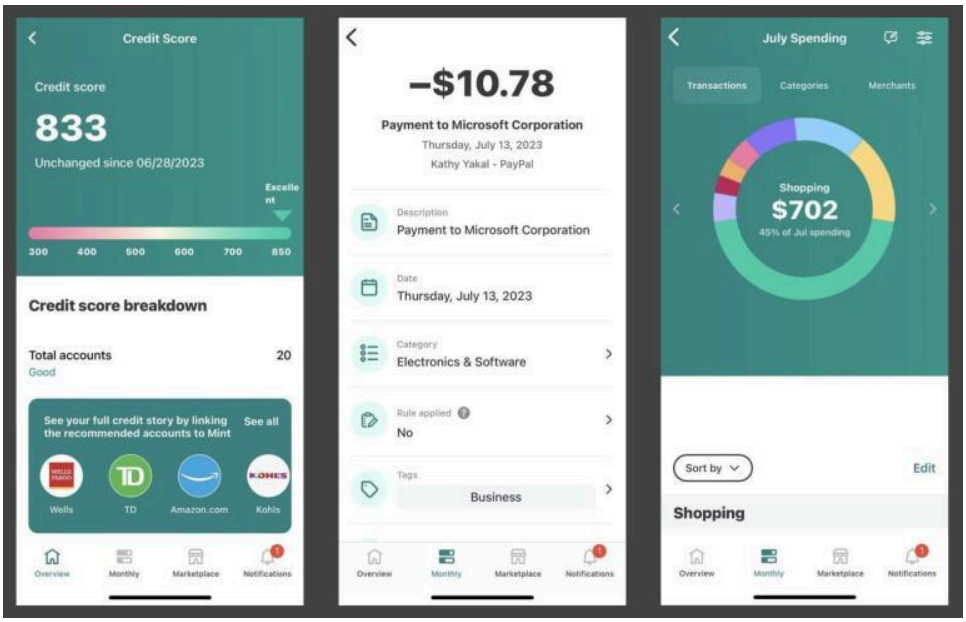
56. What challenges have you faced when trying to stick to your budget? (Check all that apply)

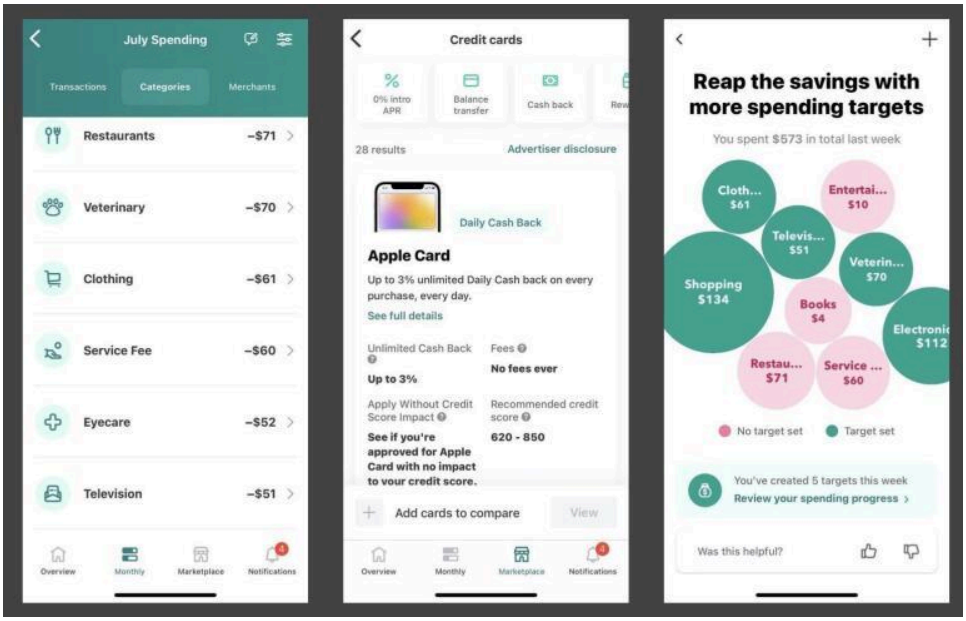
94 responses



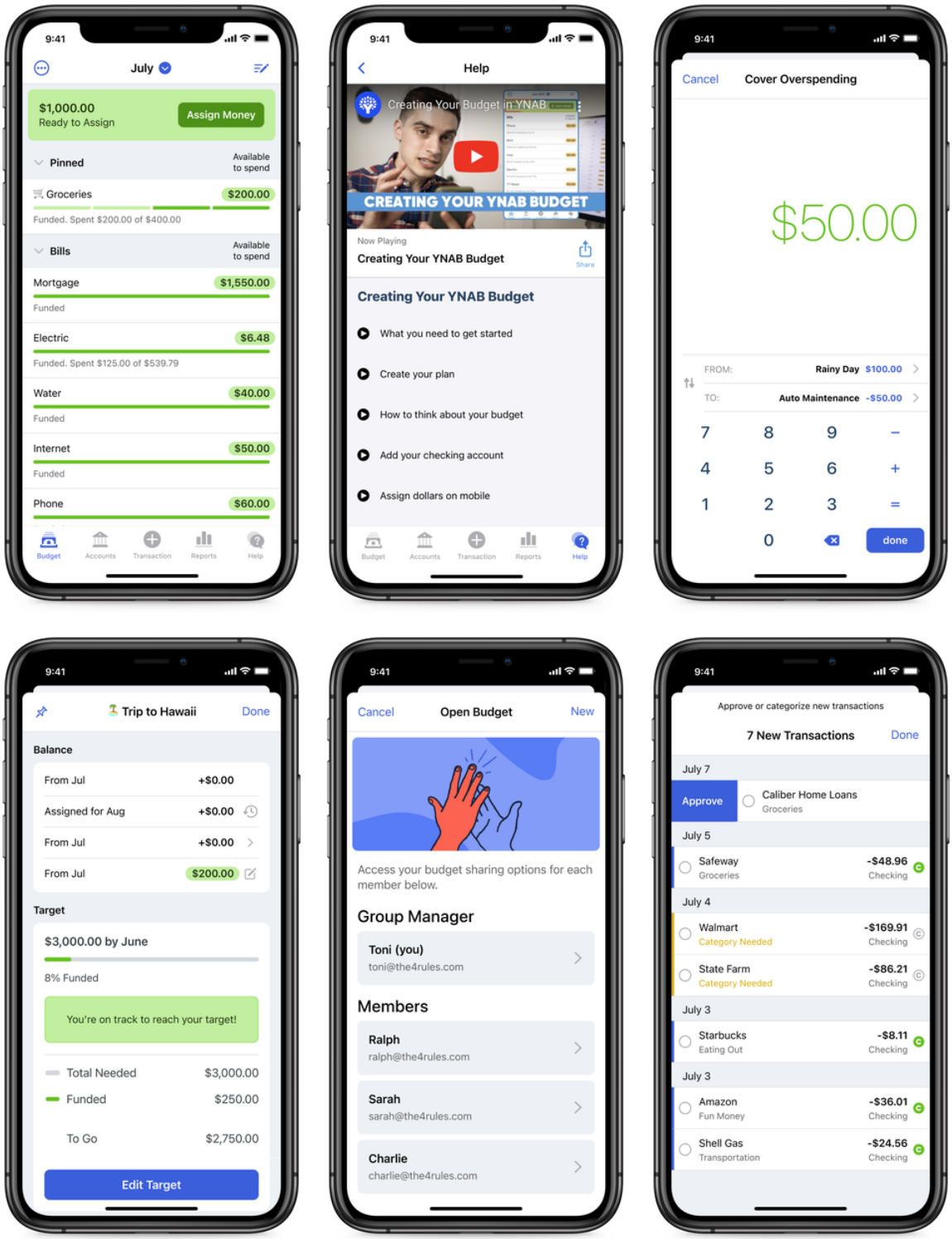
## Appendix C. Review of Related Systems

### C.1. Mint

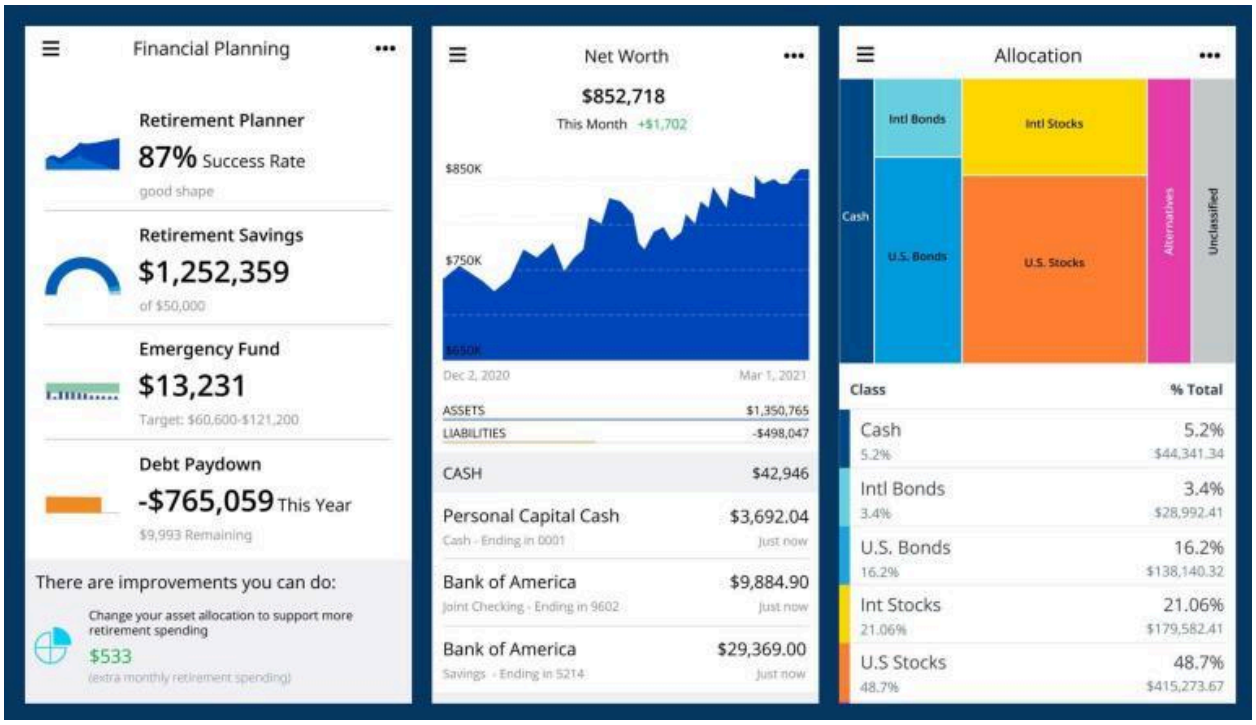


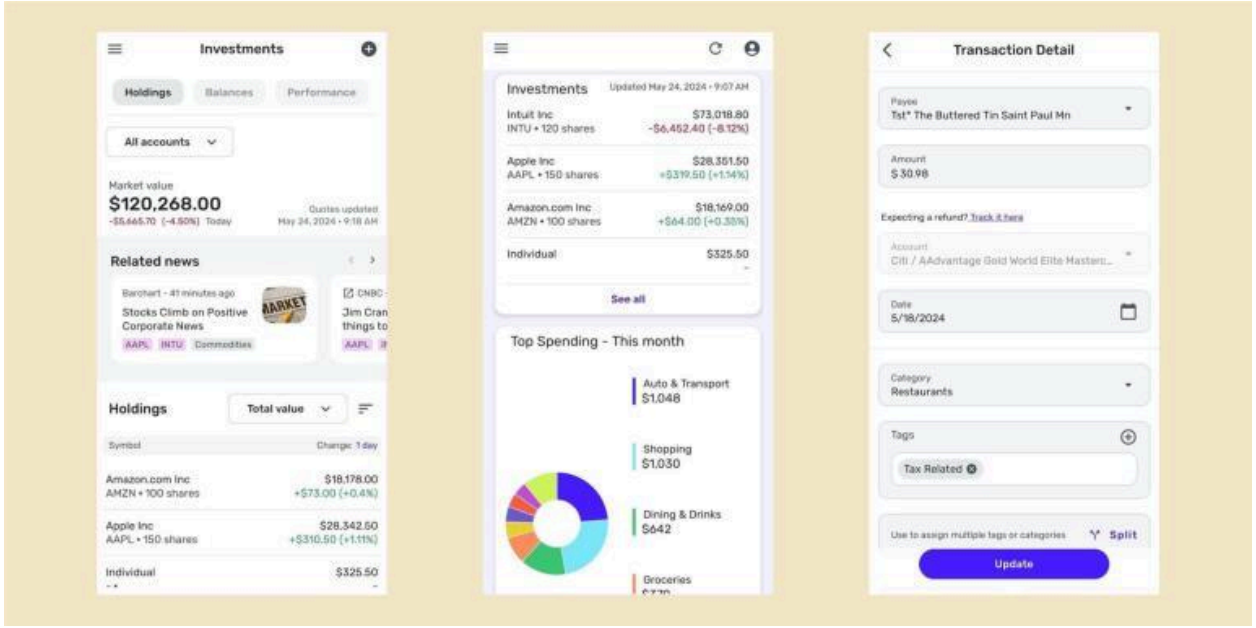
## C.2. You Need a Budget (YNAB)



### C.3. Personal Capital

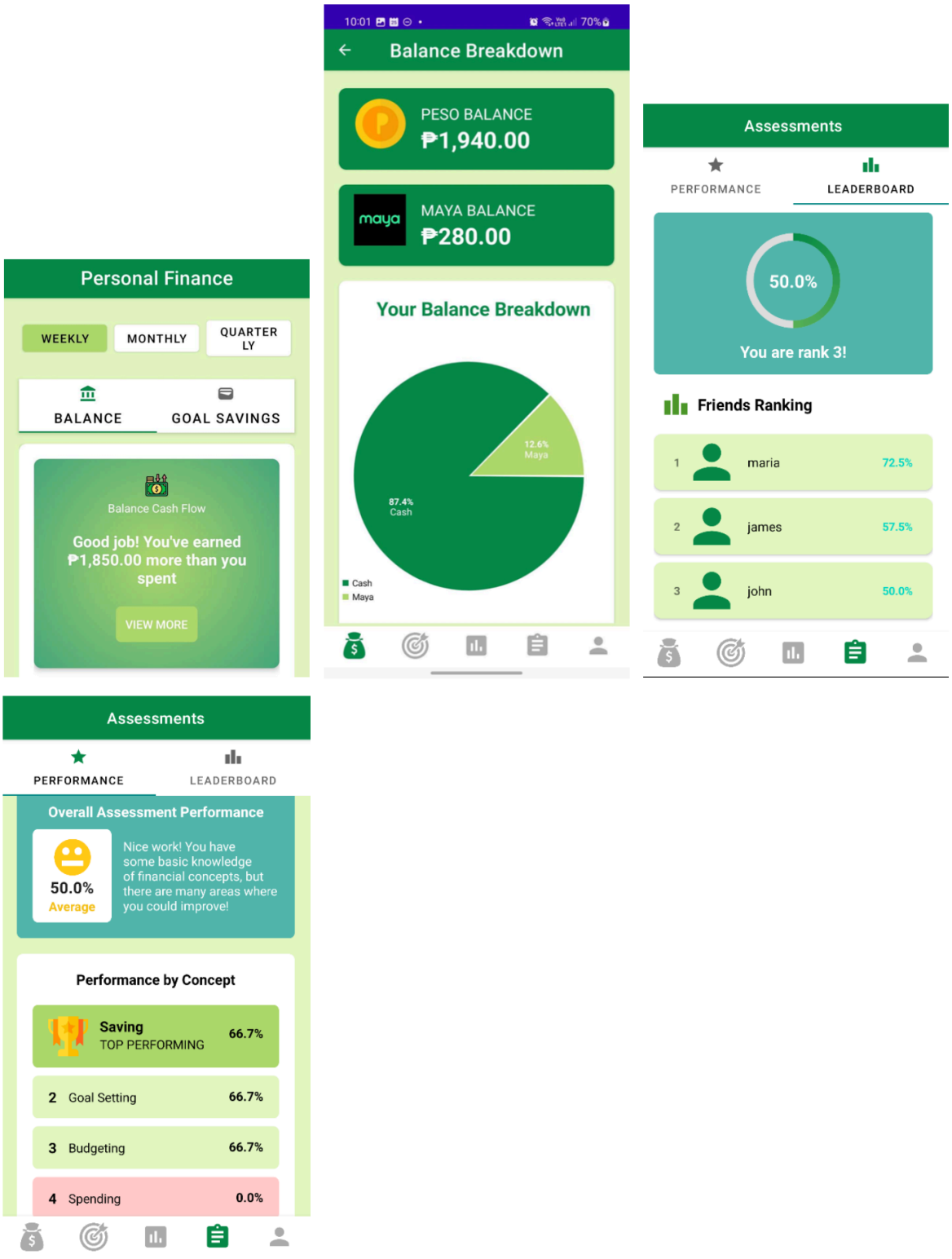



### C.4. Quicken



### C.5. FinWise

FinWise
LOGIN
REGISTER
Personal Finance
BALANCE
₱2,320.00
Income
Expense
Goal Deposit
Earning
Transactions
Personal Finance Performance
Excellent!
Your income is much more than your expenses, and you're saving a large amount of money. You're doing an amazing job!
Transaction History
ALL
INCOME
EXPENSE
Put Away Laundry +₱30.00 03/25/2023
Fold Laundry +₱40.00 03/25/2023
Chips -₱100.00 03/25/2023
Deposit to 'Deposit to -₱50.00 03/25/2023
Monday Allowance +₱50.00 03/25/2023
Chips -₱50.00 03/25/2023
initial +₱2,000.00 03/22/2023
Personal Finance
Income
Expense
Goal Deposit
Earning
Transactions
Personal Finance Performance
Excellent!
Your income is much more than your expenses, and you're saving a large amount of money. You're doing an amazing job!
VIEW EXPLANATION
Scroll down to see more!
WEEKLY
MONTHLY
QUARTERLY
Transaction History
ALL
INCOME
EXPENSE
Put Away Laundry +₱30.00 03/25/2023
Fold Laundry +₱40.00 03/25/2023
Monday Allowance +₱50.00 03/25/2023
initial +₱2,000.00 03/22/2023
Transaction History
ALL
INCOME
EXPENSE
Chips -₱100.00 03/25/2023
Deposit to 'Deposit to -₱50.00 03/25/2023
Chips -₱50.00 03/25/2023


Personal Finance
WEEKLY
MONTHLY
QUARTERLY
BALANCE
GOAL SAVINGS
Balance Cash Flow
Good job! You've earned ₱1,850.00 more than you spent
VIEW MORE
10:01
70%
Balance Breakdown
PESO BALANCE
₱1,940.00
maya
MAYA BALANCE
₱280.00
Your Balance Breakdown
12.6% Maya
87.4% Cash
Cash
Maya
Assessments
PERFORMANCE
LEADERBOARD
50.0%
You are rank 3!
Friends Ranking
1 maria 72.5%
2 james 57.5%
3 john 50.0%
Assessments
PERFORMANCE
LEADERBOARD
Overall Assessment Performance
50.0%
Average
Nice work! You have some basic knowledge of financial concepts, but there are many areas where you could improve!
Performance by Concept
Saving
TOP PERFORMING
66.7%
2 Goal Setting 66.7%
3 Budgeting 66.7%
4 Spending 0.0%

## Appendix D. Unit Testing Form

### D.1. Demographics

1. Age
    a. Under 18
    b. 18 - 24
    c. 25 - 65
    d. Above 65
2. Gender
    a. Male
    b. Female
    c. Prefer not to say
    d. Other:
3. Marital Status
    a. Single
    b. Married
4. Employment Status
    a. Full time
    b. Part time
    c. Self-employed
    d. Unemployed
    e. Student
    f. Retired

### D.2. Technology Proficiency

1. Are you comfortable using a smartphone or computer?
    a. Yes
    b. No
2. Which mobile operating system are you most comfortable using?
    a. iOS (Apple)
    b. Android
    c. Other:
3. Do you regularly use personal finance applications or tools?
    a. Yes
    b. No

### D.3. Financial Behavior

1. Do you actively manage your personal finances?
    a. Yes
    b. No

2. Do you engage in budgeting, saving, or investing activities?
    a. Yes
    b. No

**D.4. Financial Goal Module**

1. How well does the Financial Goal Module support you in setting and achieving goals?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
2. How well does the app assist you in tracking your progress in financial activities?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
3. How easy is it to use the features in the Financial Goal Module?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
4. How intuitive is the design for modifying existing financial goals?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well

**D.5. Cash Inflow Module**

1. How well does the Cash Inflow Module support you in recording your income?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well

    e. 5 - Extremely Well
2. How easy is it to categorize your cash inflows in the Cash Inflow Module?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
3. How accurate and helpful is the cash inflow summary provided in this module?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
4. How effectively does the design allow users to view and track recorded cash inflows?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well

**D.6. Budget Module**

1. How well does the app assist in creating and managing your financial budget?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
2. How easy is it to adjust your budget in the Budget Module?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
3. How useful is the budget overview and tracking provided by this module?
    a. 1 - Not Well At All
    b. 2 - Not That Well

c. 3 - Neutral
d. 4 - Well
e. 5 - Extremely Well

4. How well does the design support adjustments and modifications to the budget?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
5. How clear and useful are the budget visualizations provided by the design?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well

**D.7. Dashboard Module**

1. How well does the Dashboard present information about your financial activities?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
2. How well does the Dashboard assist you in tracking and understanding your overall financial performance?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
3. How well does the app support recording your income and expenses, and tracking your financial goals?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well

    e. 5 - Extremely Well
4. How useful is the information displayed on the Dashboard for your financial management?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well

**D.8. Expense Module**

1. How well does the Expense Module allow you to record and categorize your expenses for financial activities?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
2. How effectively does the design allow users to view and track recorded expenses?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
3. How helpful is the expense tracking feature in this module?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well

**D.9. Reports Module**

1. How well does the Reports Module allow you to generate and understand reports?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well

2. How well does the Reports Module assist you in understanding your current financial situation?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
3. How well does the module assist you in identifying improvement points and acting upon them?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
4. How well does the module support you in making better financial decisions?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
5. How useful are the reports for your financial planning and analysis?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well

**D.10. General**

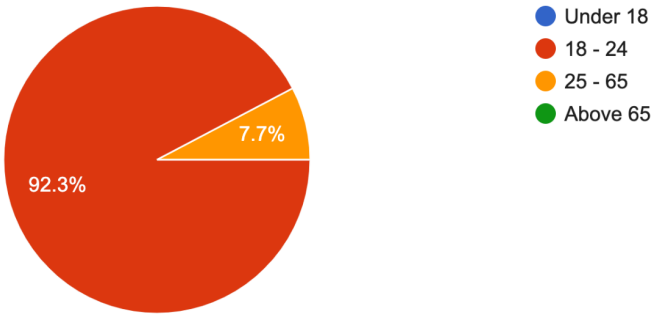
1. How well does the design and flow of the app help you use it?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral
    d. 4 - Well
    e. 5 - Extremely Well
2. How well protected and secure did you feel with the data in the app?
    a. 1 - Not Well At All
    b. 2 - Not That Well
    c. 3 - Neutral

   d. 4 - Well
   e. 5 - Extremely Well
3. How well does the app prevent you from inputting the wrong data?
   a. 1 - Not Well At All
   b. 2 - Not That Well
   c. 3 - Neutral
   d. 4 - Well
   e. 5 - Extremely Well
4. How well does the overall application meet your needs?
   a. 1 - Not Well At All
   b. 2 - Not That Well
   c. 3 - Neutral
   d. 4 - Well
   e. 5 - Extremely Well
5. How easy is it to navigate and use the application as a whole?
   a. 1 - Not Well At All
   b. 2 - Not That Well
   c. 3 - Neutral
   d. 4 - Well
   e. 5 - Extremely Well
6. How visually appealing and user-friendly is the design of the application?
   a. 1 - Not Well At All
   b. 2 - Not That Well
   c. 3 - Neutral
   d. 4 - Well
   e. 5 - Extremely Well
7. Comments & Suggestions

## Appendix E. Unit Testing Form Results
### E.1. Demographics

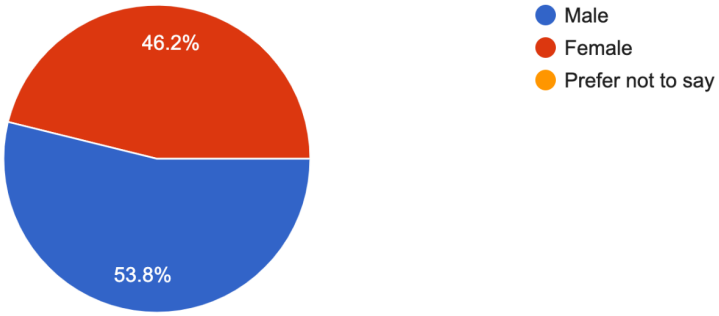
Age

13 responses



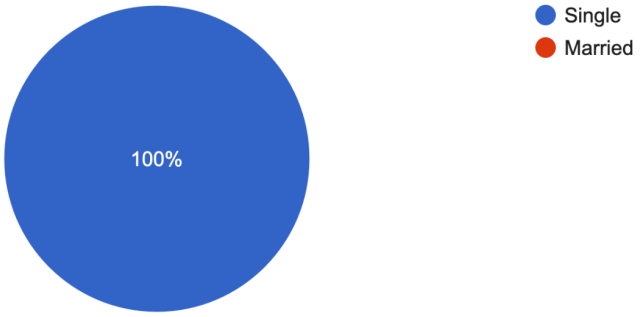

Gender

13 responses



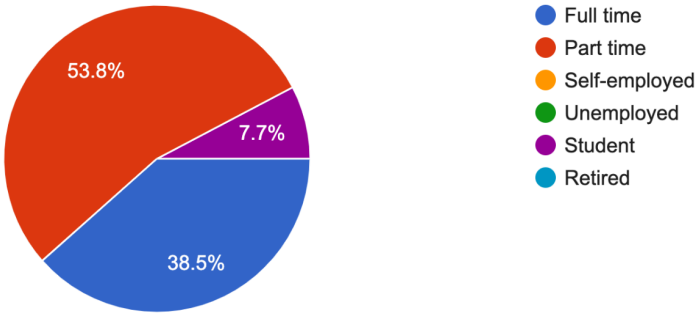
## Marital Status

13 responses




## Employment Status

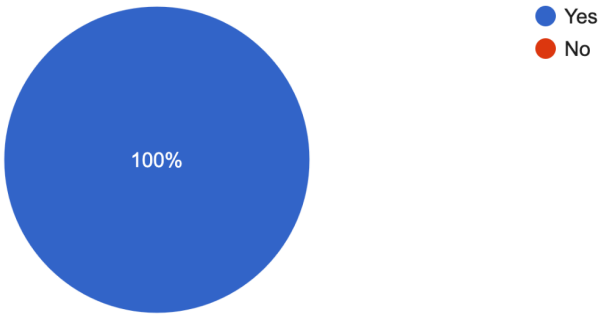
13 responses



## E.2 Technology Proficiency

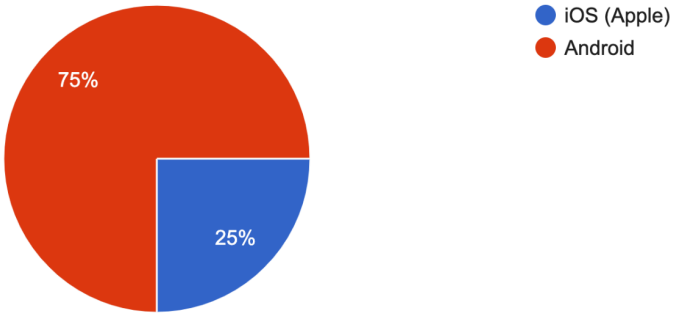
Are you comfortable using a smartphone or computer?

12 responses



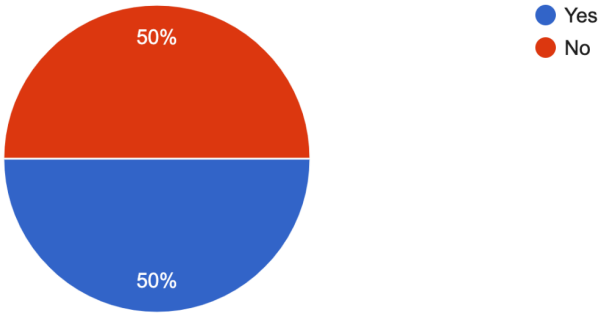

Which mobile operating system are you most comfortable using?

12 responses



Do you regularly use personal finance applications or tools?

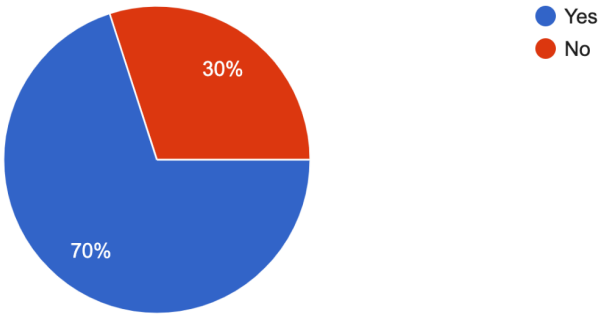
12 responses



### E.3 Financial Behavior

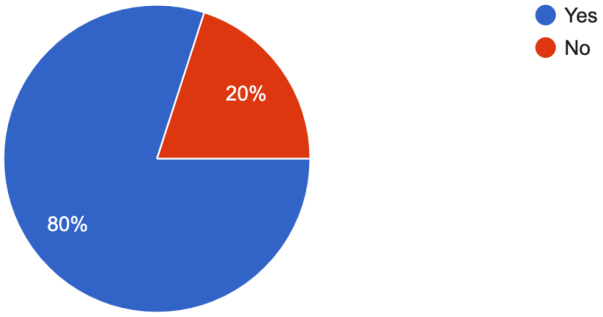
Do you actively manage your personal finances?

10 responses




Do you engage in budgeting, saving, or investing activities?

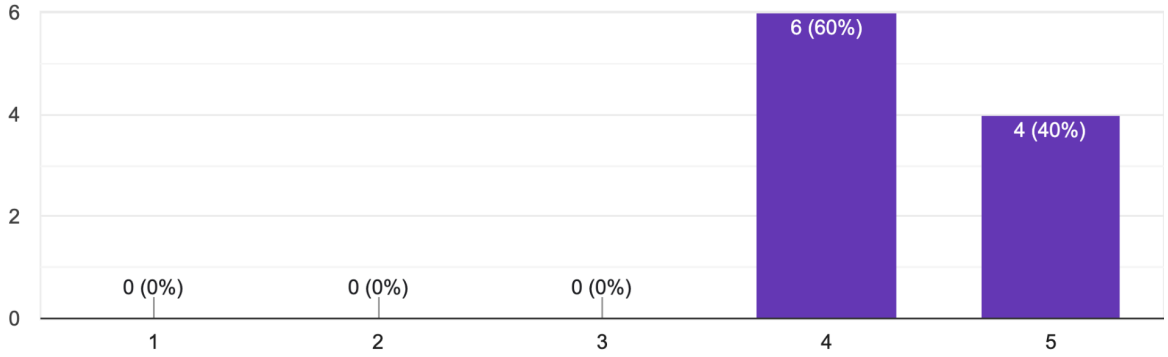
10 responses



## E.4 Financial Goal Module

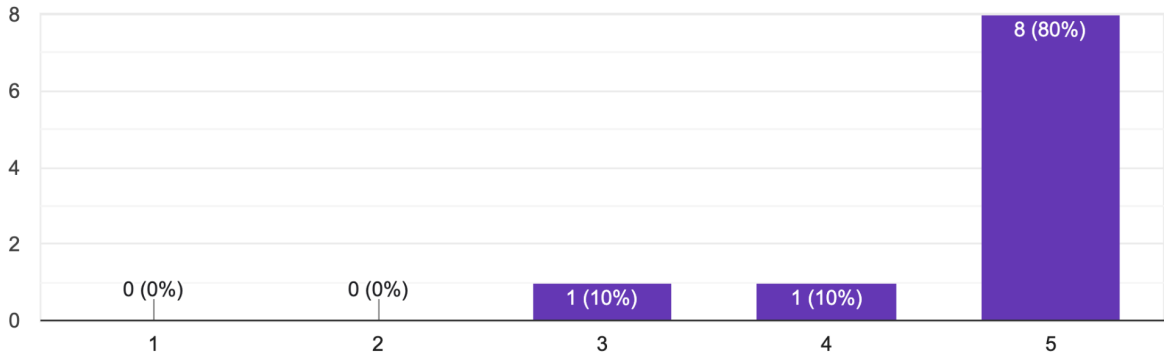
How well does the Financial Goal Module support you in setting and achieving goals?

10 responses



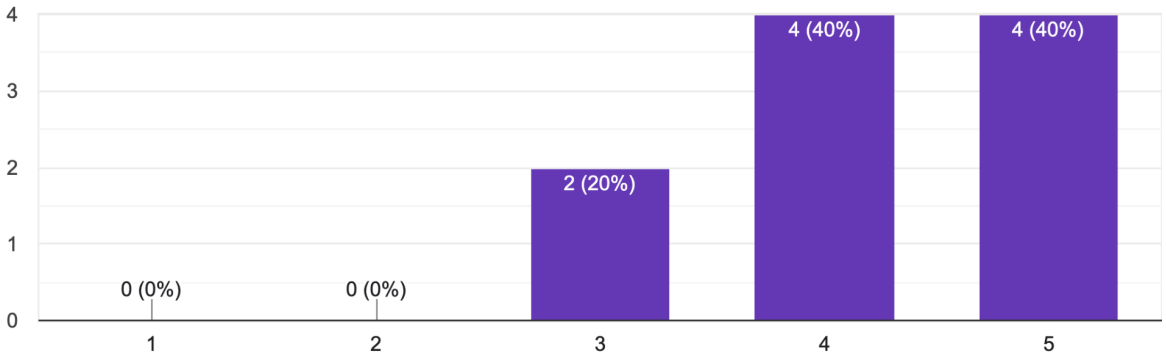

How well does the app assist you in tracking your progress in financial activities?

10 responses



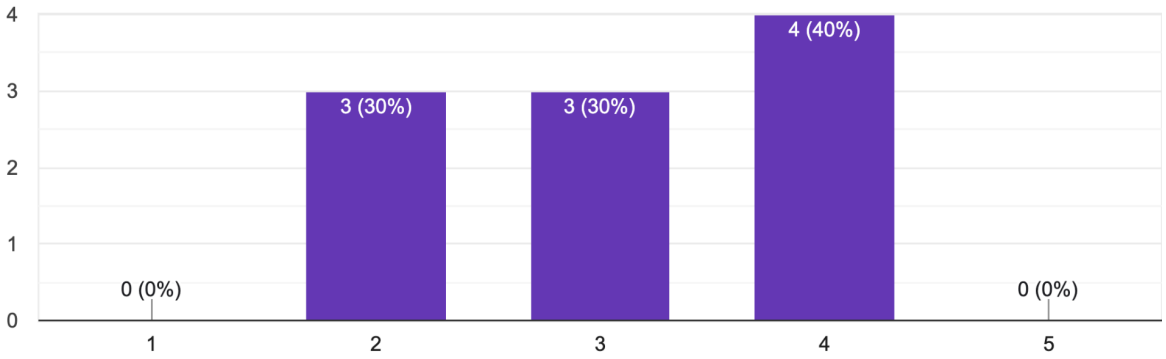
How easy is it to use the features in the Financial Goal Module?

10 responses




How intuitive is the design for modifying existing financial goals?

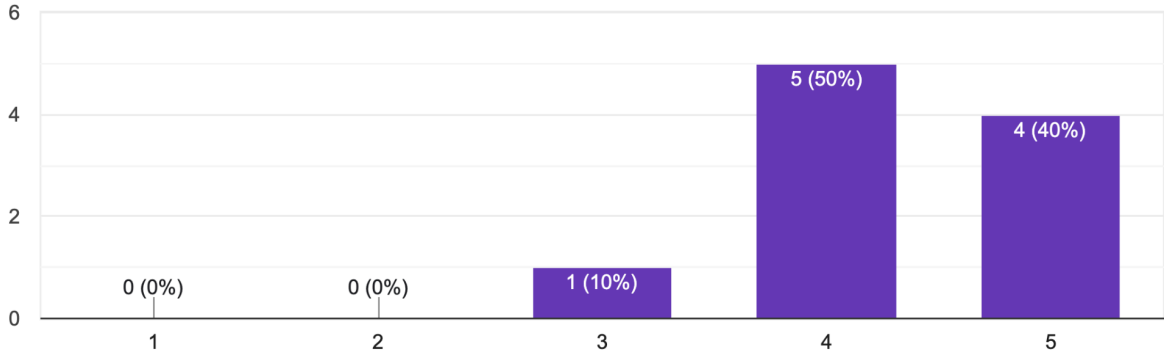
10 responses



## E.5 Cash Inflow Module

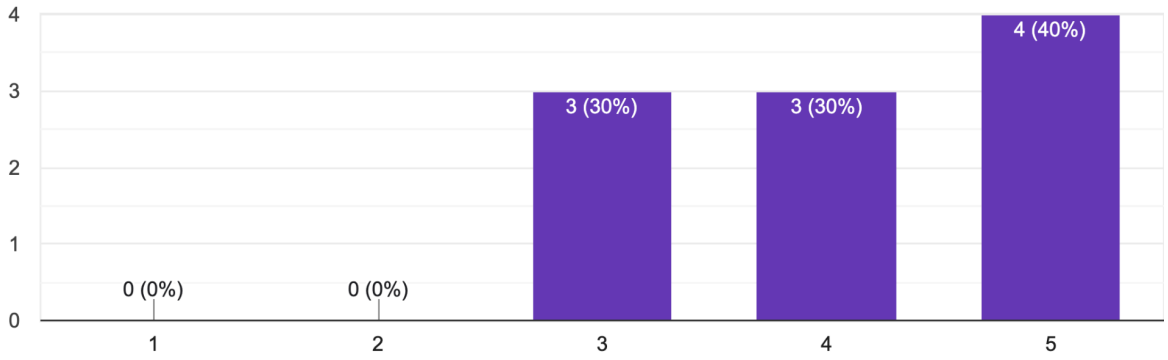
How well does the Cash Inflow Module support you in recording your income?

10 responses



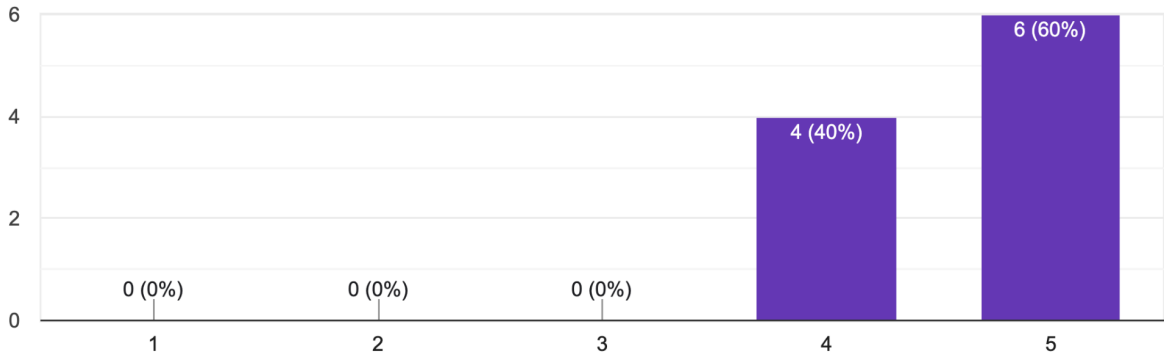

How easy is it to categorize your cash inflows in the Cash Inflow Module?

10 responses



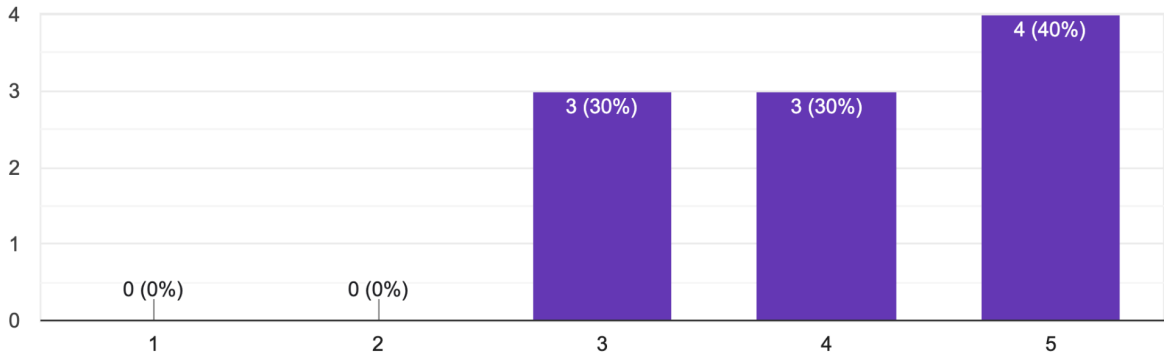
How accurate and helpful is the cash inflow summary provided in this module?

10 responses




How effectively does the design allow users to view and track recorded cash inflows?

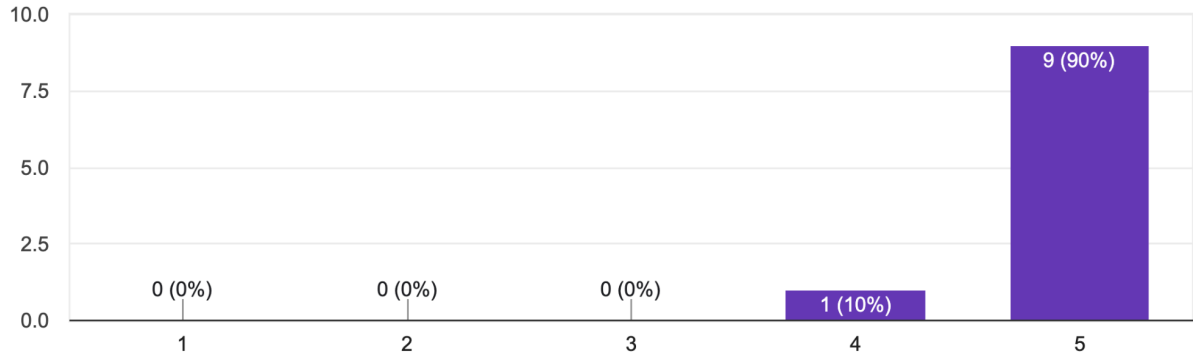
10 responses



## E.6 Budget Module

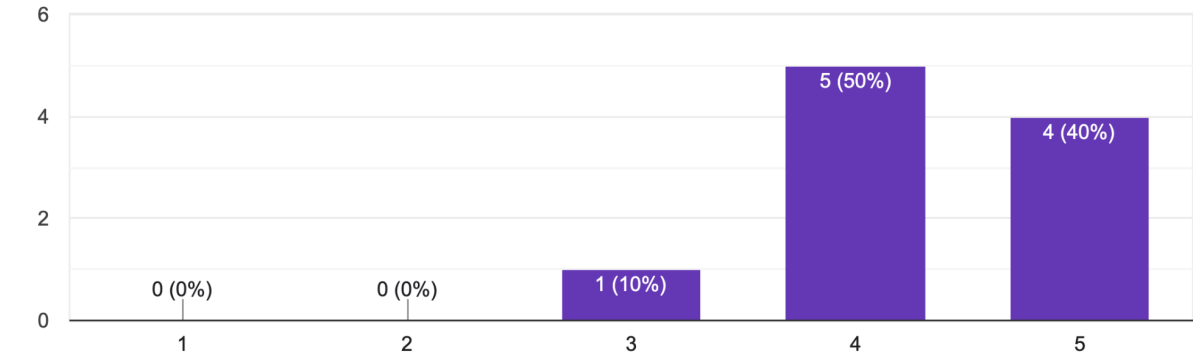
How well does the app assist in creating and managing your financial budget?

10 responses



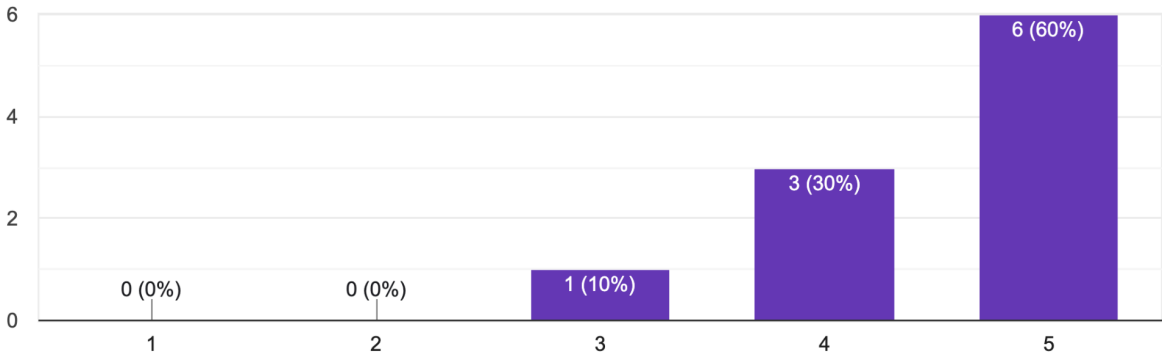

How easy is it to adjust your budget in the Budget Module?

10 responses



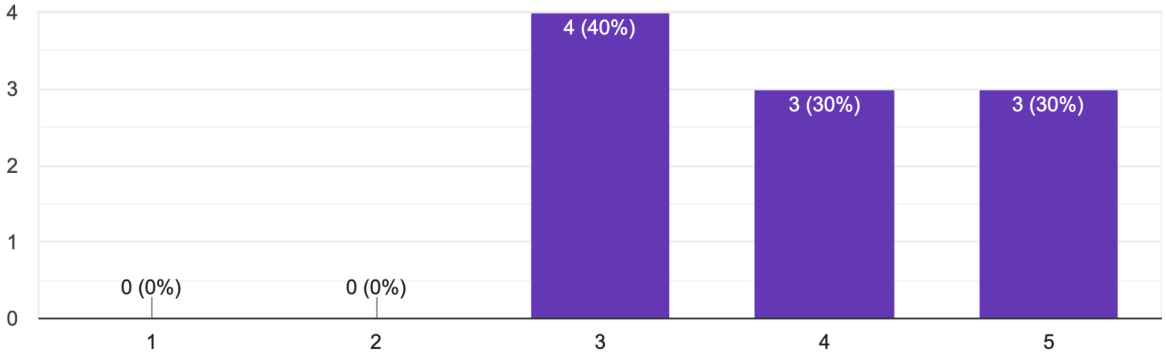
How useful is the budget overview and tracking provided by this module?

10 responses



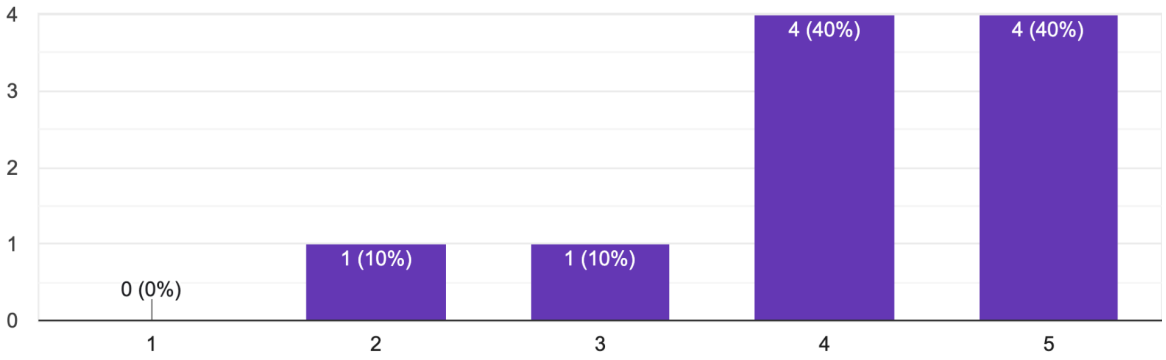

How well does the design support adjustments and modifications to the budget?

10 responses



How clear and useful are the budget visualizations provided by the design?

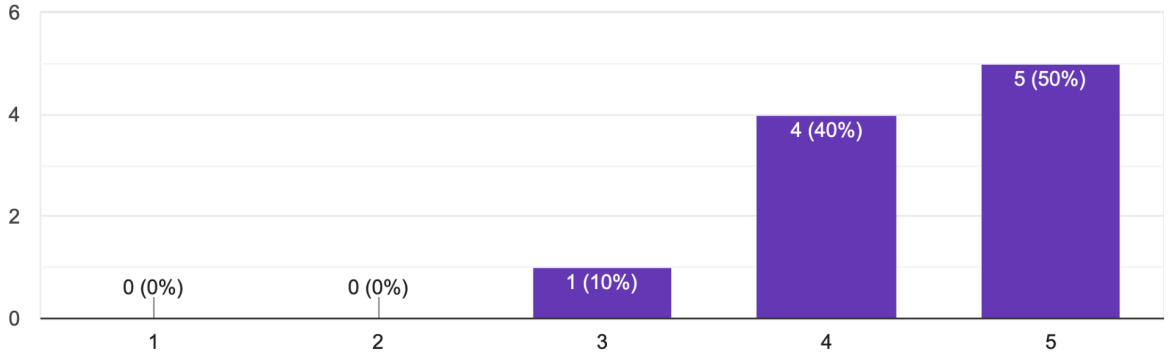
10 responses



### E.7 Dashboard Module

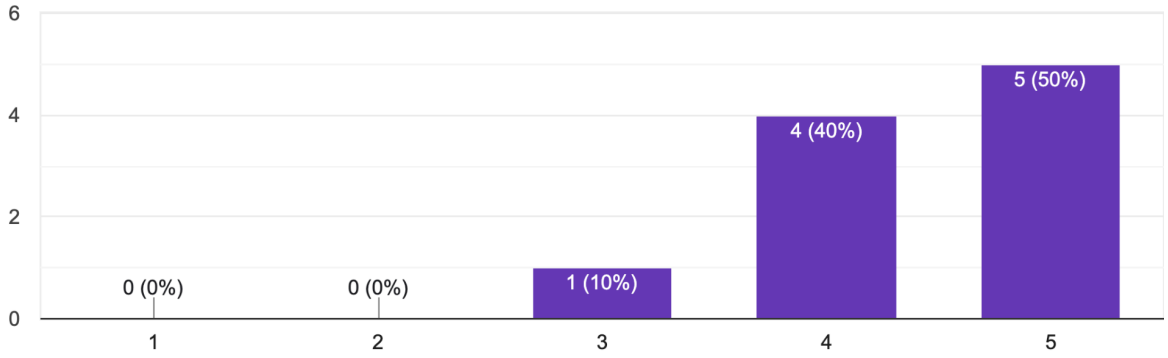
How well does the Dashboard present information about your financial activities?

10 responses



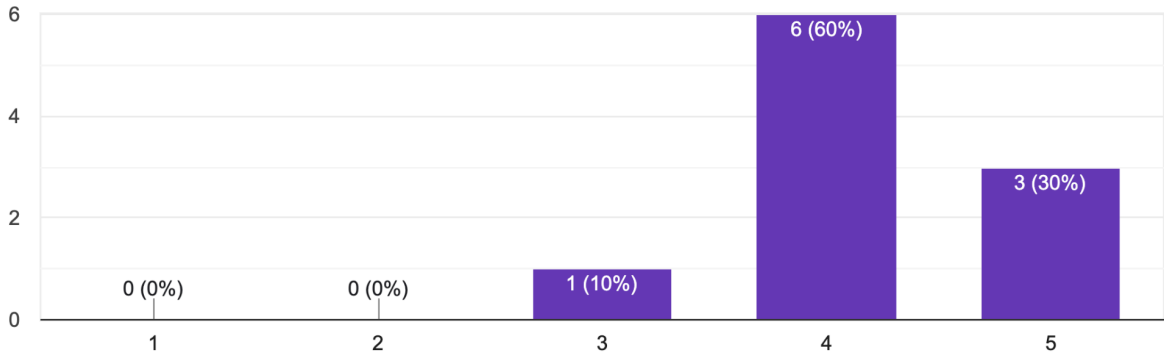

How well does the Dashboard assist you in tracking and understanding your overall financial performance?

10 responses



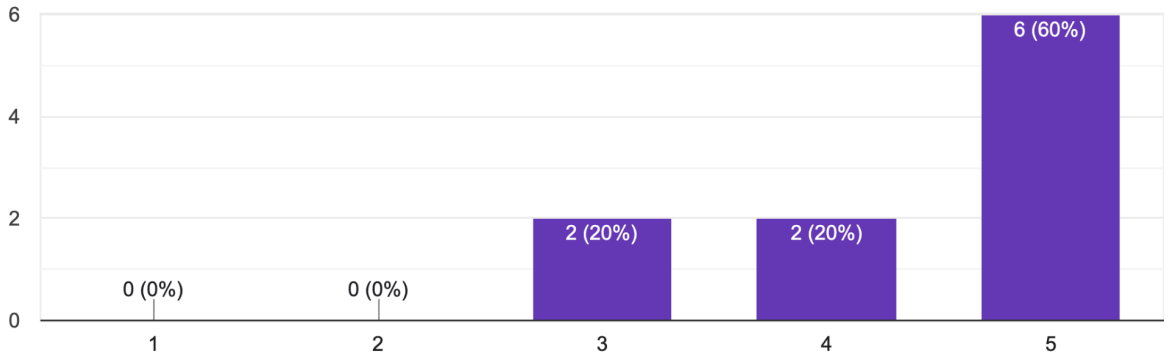
How well does the app support recording your income and expenses, and tracking your financial goals?

10 responses




How useful is the information displayed on the Dashboard for your financial management?

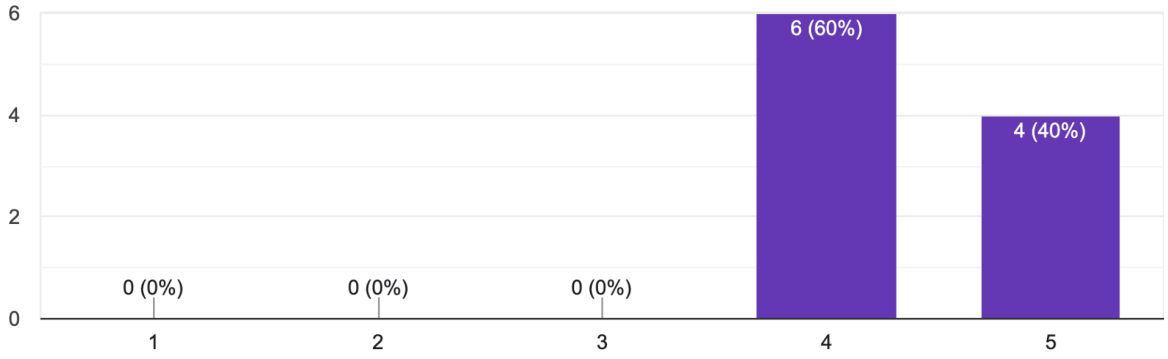
10 responses



### E.8 Expense Module

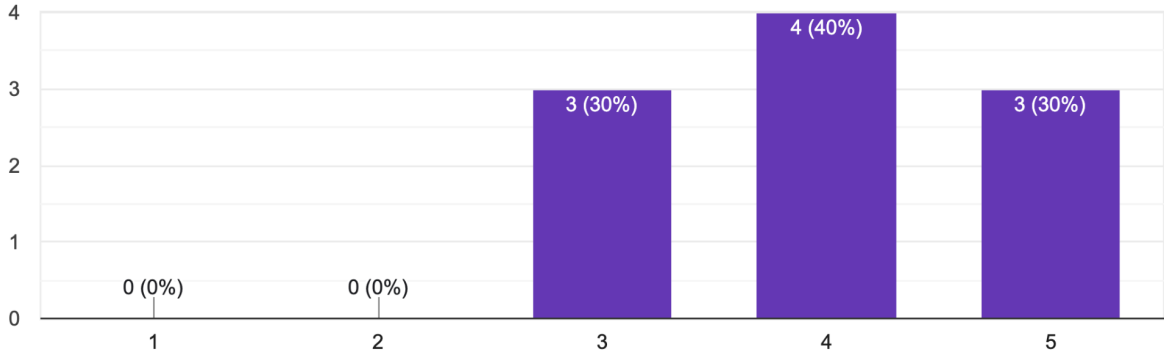
How well does the Expense Module allow you to record and categorize your expenses for financial activities?

10 responses



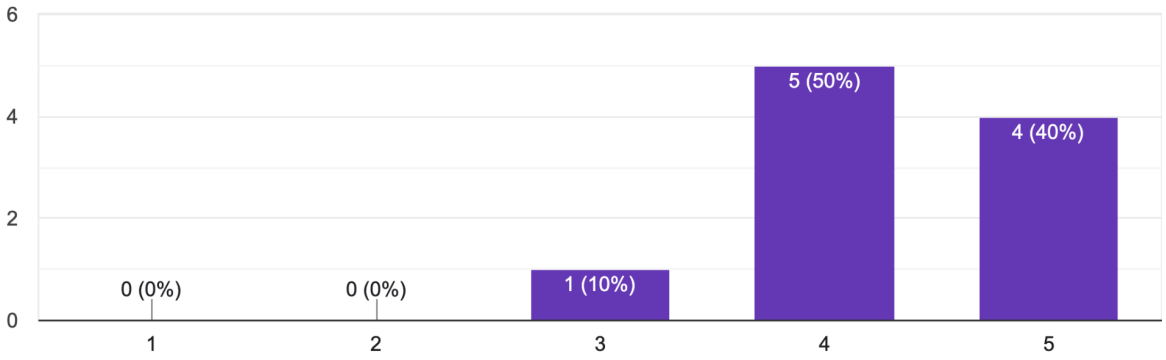

How effectively does the design allow users to view and track recorded expenses?

10 responses



How helpful is the expense tracking feature in this module?

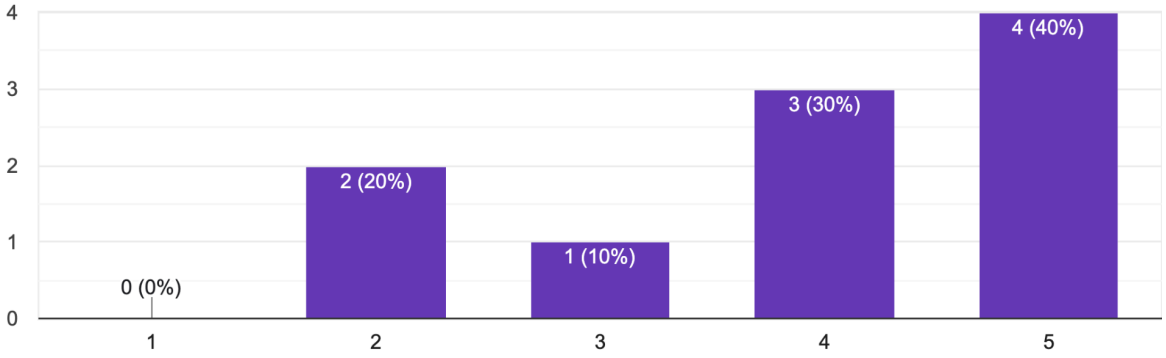
10 responses




### E.9 Reports Module

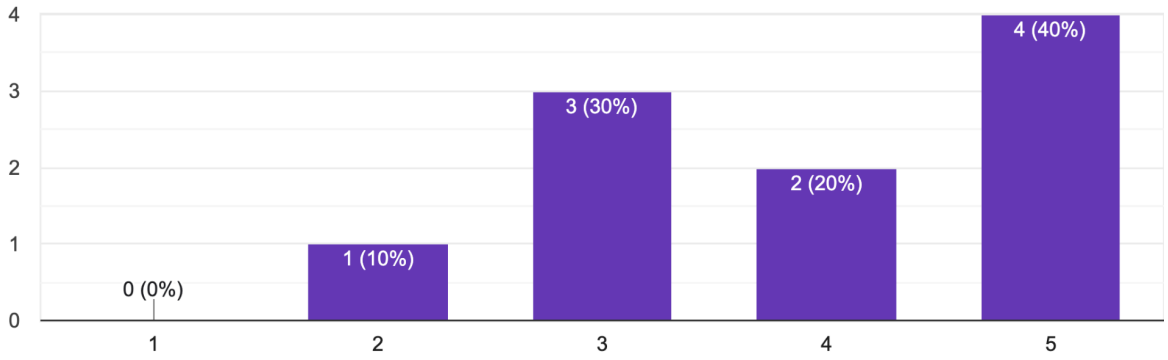
How well does the Reports Module allow you to generate and understand reports?

10 responses



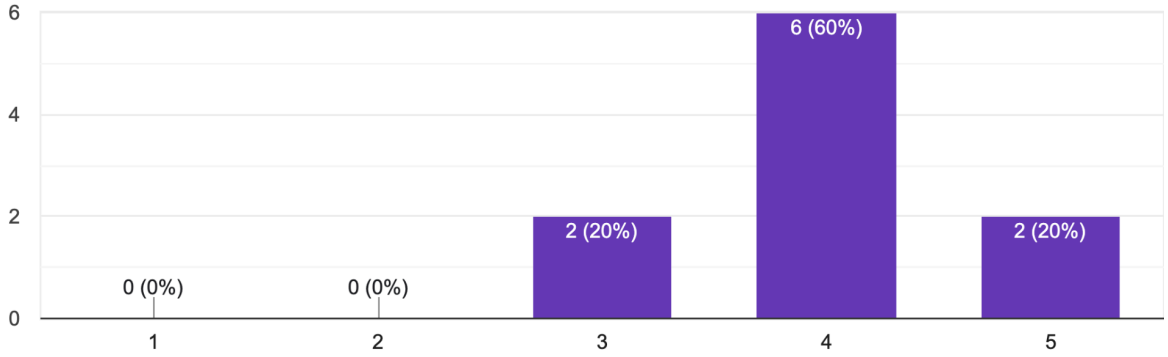
How well does the Reports Module assist you in understanding your current financial situation?

10 responses



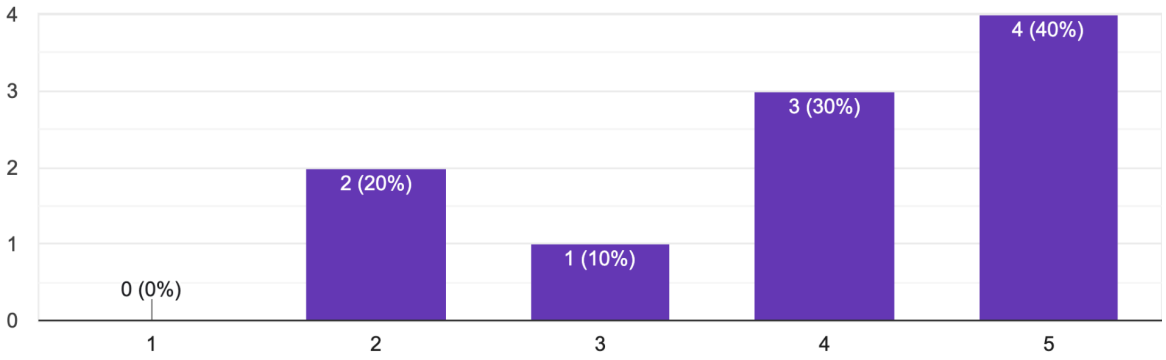

How well does the module assist you in identifying improvement points and acting upon them?

10 responses



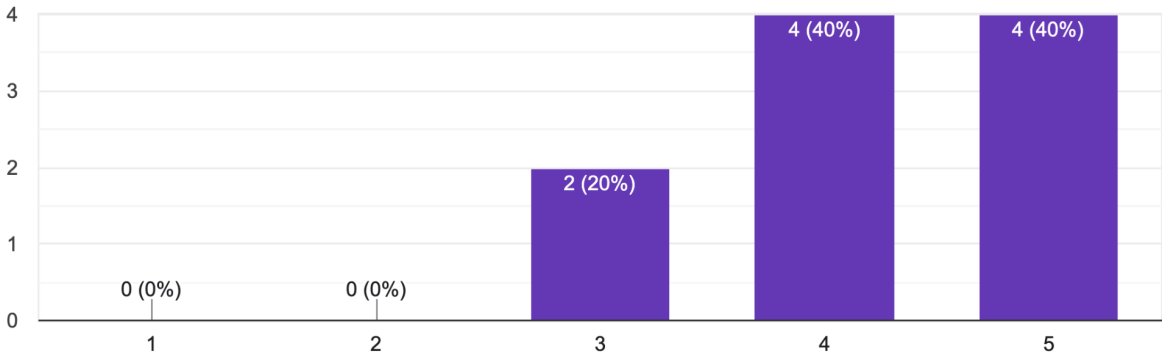
How well does the module support you in making better financial decisions?

10 responses




How useful are the reports for your financial planning and analysis?

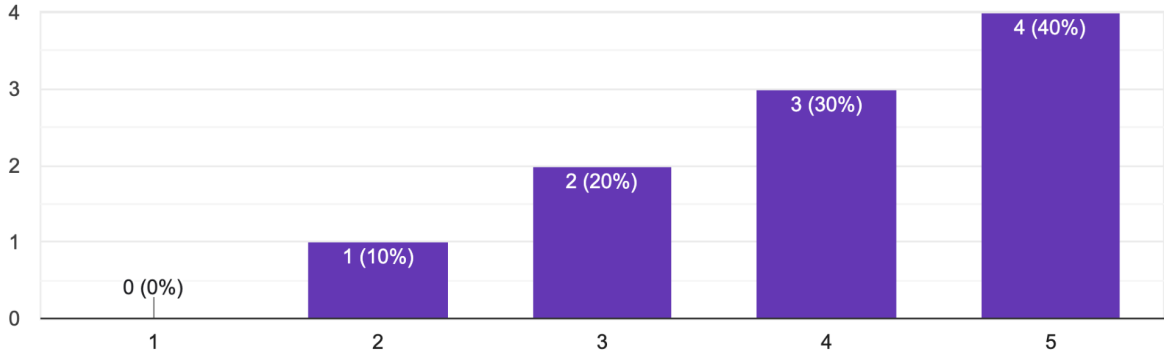
10 responses



## E.10 General

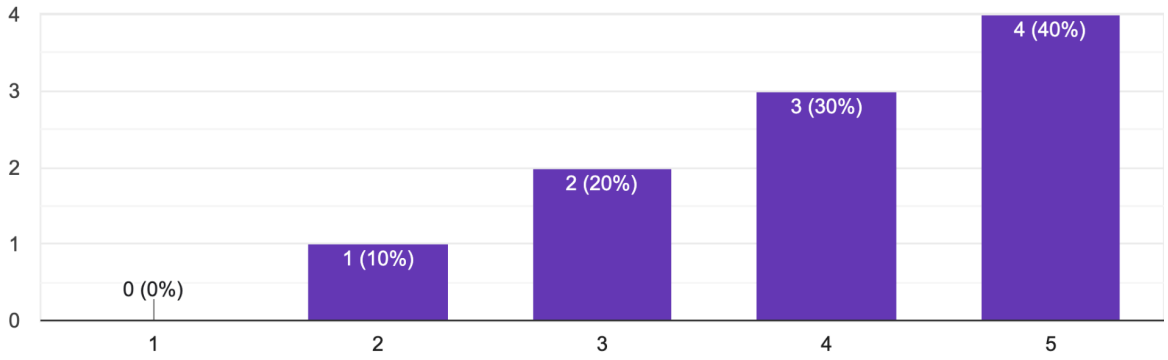
How well does the design and flow of the app help you use it?

10 responses



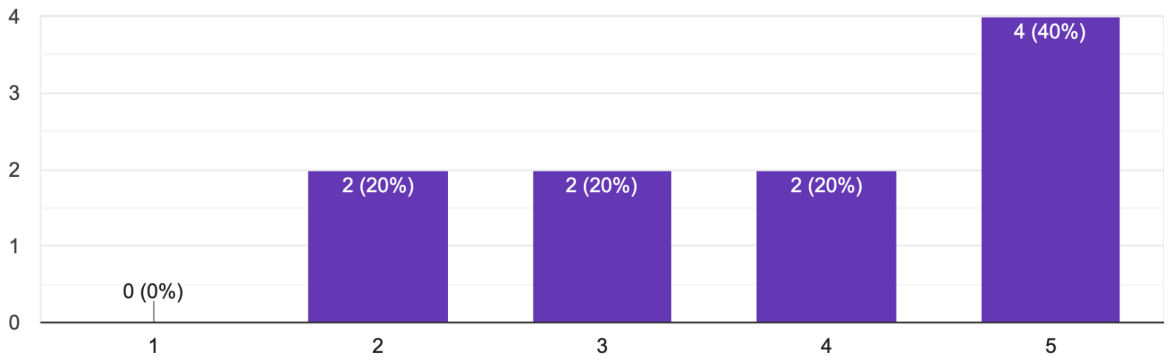

How well protected and secure did you feel with the data in the app?

10 responses



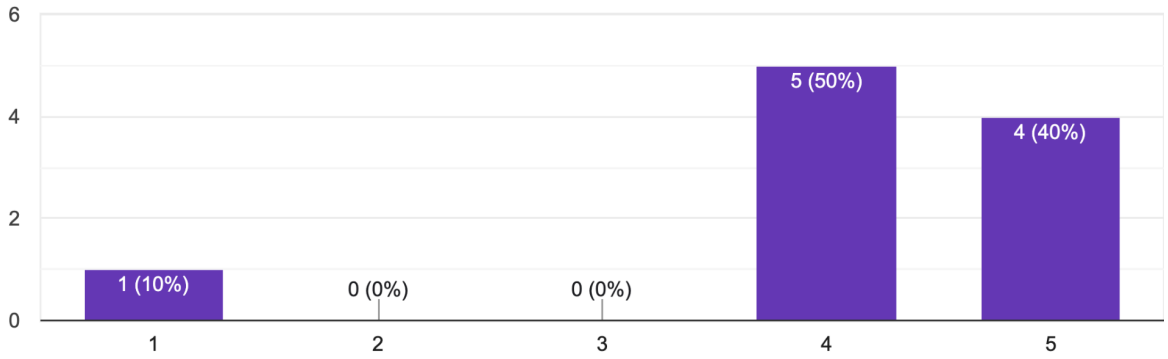
How well does the app prevent you from inputting the wrong data?

10 responses



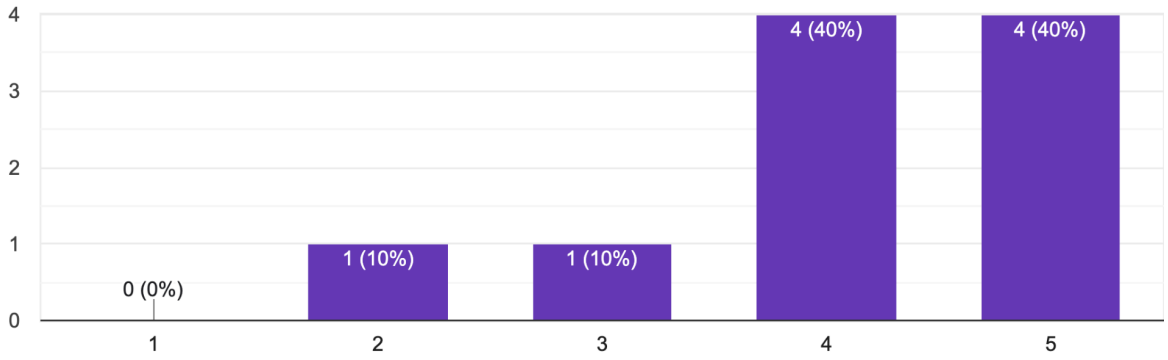

How well does the overall application meet your needs?

10 responses



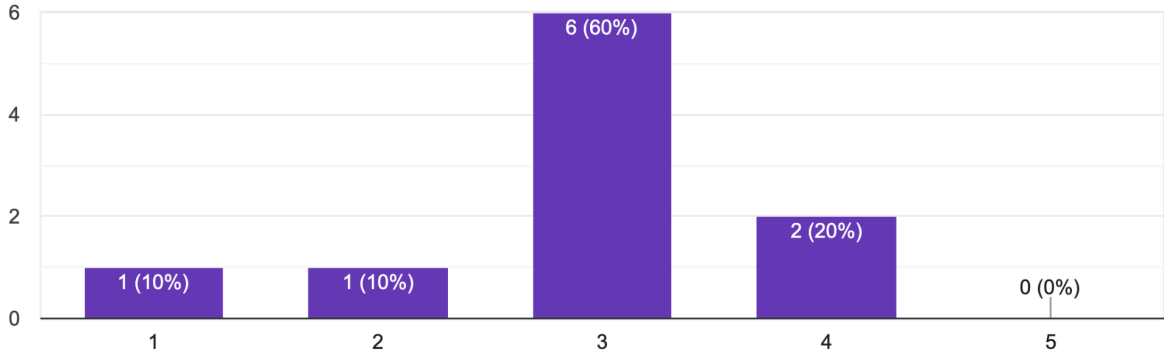
How easy is it to navigate and use the application as a whole?

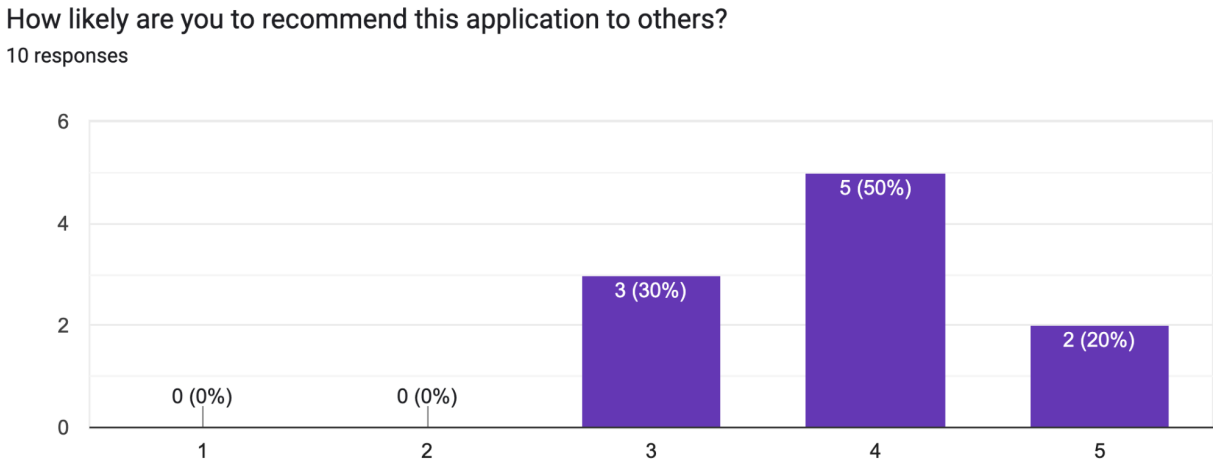
10 responses




How visually appealing and user-friendly is the design of the application?

10 responses






**E.11. Comments & Suggestions**

1. Pls add sa acct features yung pfp siguro para may distinction yung acct bukod sa names. Also sa app settings naman yung option na dark mode and i think may options of colors din dapat ng interface para mas nakakacatch ng attention ng user.Then sa summary ng account info pwede din da features yung print option ng account details para bukod sa mobile may hardcopy din. In creating din ng account, much better if valid din na email yung gagamitin. Saka maganda din if may sms updates or email updates (if possible) na rereceive everytime na may updates din sa accounts. Sa pag log in din, pwede nyo i-add yung hide and view option sa password. Lagay din bukod sa housing at other billings yung for SSS, Philhealth, and PagIbig and etc. since yung target nyo is hindi lang regular employees.
2. Visually, the colors need a bit of work. I personally have sensitive wyes and I find it difficult to view the entire content's design as I find it a bit messy. I recommend sticking to a single color palette with 3-5 colors in different values to maximize the simplicity of the software while keeping the professional concept and pleasant to the eyes.

   The terms used in the app should consider those who are new to budgeting softwares as some have little to no idea what the terms are or are simply not familiar with such words. Think of the Gcash E-wallet app and the way they introduce their features to new users. Whether old or young, they

have a short description of the terms they use or the features they offer so the user has an idea what they're doing while using the app. I think it could also work for your software and would do great if there was a short introduction or description or even have a tutorial for new users. That's all, great job!

3. Tweak the UI: I noticed some overlapping elements and a few buttons and input fields are too close together. Spacing these out would improve usability.

   Adjust Button Colors: The "confirm" button blends in with the app's background color, which could be confusing for users. Differentiating it will make it more intuitive.

   Enhance Visuals in the Reports Tab: It would be great if we could make the visualizations in the reports tab more appealing and clear.

   The overall design can look cleaner and more user-friendly with careful color choices, better element spacing, and a few adjustments.
4. Improve UI and maybe add a recent transaction feature in accounts and cash inflow
5. Overall, it's a good app when it comes to tracking your current and future financial goals. I just think that there should be an option where the savings and expenses amount can be set to daily. The UI can also be improved and be more organized.
6. Basic Goal or Detailed Goal option - Make them more recognizable
   Small "Target Amount" -> rename to something else
   Add option to ignore (when setting "unrealistic" goal)
   Navigation - consider digital literacy of other people
7. I don't think its something that I would openly invite my other peers to use but I do find it useful. At the start, I feel lost. I think it needs an overall tutorial of the application. Although, I find it easy to use once I get the hang of it after a few minutes. At the dashboard, I didn't know I can move the financial goals around. I also initially didnt know the buttons and what it did. The camera option to scan is nice, but I think it will still be exhaustive. The account module is pretty straightforward but again, i had to experiment. The budget module, I do think I get it but I believe that I wasn't able to maximize it. As for the financial goal, it's also straightforward and easy to use. For the reports, I like the graphs but I don't see how I can understand the graph. I think it lacks meaning as to

what the graphs mean. Although there are insights which can be a guide to understand, I'd prefer if it was told to me directly. There are some buttons that are not visually pleasing such as the buttons in the financial goal. There's also a set of buttons that is shown on top of the app instead of the center. The notification doesnt show the app details, I only knew about the notification because I was informed about it.

8. User interface needs improvement. Overall flow and aesthetic of the app needs to be improved for users to be attracted. But the app is a great idea for someone who wants to budget their savings and track his or her expenses.
9. Design: Straightforward but not easy. Labels are misleading. Could choose a better color.
   Navigation: Navigation is straightforward.
   Action Buttons: Lack of context.
   Reports: Missing context, hard to completely grasp what the datas mean.
   Lack of user flow after the onboarding: You lack the guide that can be used by your users. It has a tendency to make them lose interest in the app.

   All in all, app would suffice as a budgeting app.
10. Overall app is okay. Lacks other inputs such as other recurring transactions. UI could be a lot better. Reports need more information about the data being presented.